Studies in Algebra and Group Theory
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Contents

Lecture 1.

Set Theory

1.1 Ordered Pairs

We begin with a discussion of ordered pairs. Consider a set of numbers $\{1, 2, 3, 4\}$. For any set there is no particular order that triumphs over another, i.e., $\{1, 2, 3, 4\} = \{1, 3, 2, 4\} = \{3, 4, 2, 1\} = \ldots$, an obvious truth. Say we want $\{1, 2, 3, 4\}$ to be in the order of $\{3, 2, 1, 4\}$ for some arbitrary reason; we may then consider the collection,

$$\mathfrak{C} = \{\{3\}, \{3, 2\}, \{3, 2, 1\}, \{3, 2, 1, 4\}\}.$$

It becomes obvious what is happening here: the method by which we get order is dependent on the number of iterations in which a particular element appears in any number of sets in a given collection (in comparison to the other elements in the other sets of the respective collection).

Then consider this process but for a nested collection (a collection within a collection) which contains two sets, namely the arbitrary elements a, b in opposing orders, and notice that it gives us the definition of a 2-tuple (ordered pair),

$$\mathfrak{C}_1 = \{\{a\}, \{a, b\}\} \coloneqq (a, b) \text{ and } \mathfrak{C}_2 = \{\{b\}, \{b, a\}\} \coloneqq (b, a).$$

Note:-

This form of generating or representing ordered pairs is called the Kuratowski of ordered pairs.

This, of course, naturally extends to greater n-tuples. There are some obvious consequences of this that will not be gone over here.

The **Cartesian product** is a set of ordered pairs which can be given by taking power sets.

Definition 1.1.1: Cartesian Product

The Cartesian product is the cross product of two sets given by

$$A \times B = \{(a, b) \mid a \in A, b \in B\},$$

where, by the Kuratowski encoding,

$$(a, b) \coloneqq \{\{a\}, \{a, b\}\} \in \mathcal{P}(\mathcal{P}(A \cup B)).$$

1.2 Relations

The review or ordered pairs leads us to our next discussion on relations. A relation in more colloquial terms is a dynamic or characteristic of a given element shared with another given element such that they are arranged in the form of ordered pairs. Think of equality, then think of equality as equating $(2 + 3, 1 + 4)$, or perhaps something more interesting, the set of all (x, y) such that x is a man, y is an enby (slay), and x is married to y (many such cases). We can then think of relations as a set of ordered pairs $R \subset A \times B$. Consider the following examples:

1. $R \coloneqq \{(x, y) \in \mathbb{N}^2 \mid y - x > 0\} \iff xRy \equiv y > x$, the relation of greater than;
2. $R \coloneqq \{(a, B) \in A \times \mathcal{P}(A) \mid a \in B\} \iff aRB \equiv a \in B$, the relation of belonging (being an element of);
3. $R \coloneqq \{(c, d) \in \{\text{all cats}\} \times \{\text{all dogs}\} \mid c, d \text{ live in the same house}\}$.

Naturally, there are an infinite number of possible relations, however there is a particular class of these which we are interested in for the time being—that which relates two equivalent elements (the definition of what equivalence means is, for now, notably ambiguous).

If a relation R exists s.t. xRx for any $x \in X$, then we say R is reflexive. If $xRy \iff yRx$, it is symmetric, and if xRy, $yRz \implies xRz$, it is transitive. If a given relation R satisfies all three of these properties (reflexivity, symmetry, transitivity), then we say it is an **equivalence relation**.

Definition 1.2.1: Equivalence relations

An equivalence relation, usually denoted $\sim$, is a relation that is reflexive, symmetric, and transitive $\forall(a, b) \in A \times B$.

Question 1

Consider each of the three qualifiers (axioms) for the existence of an equivalence relation. Find an explicit relation that has one property but not the other for each of the three axioms.

Solution

I apologize for the terse abstraction involved in this first example, but let

$$R \coloneqq \{(1,1),(2,2),(3,3)(1,2),(2,3) \in \{1,2,3\}^2\},$$

notice that $(1,1),(2,2),(3,3)$ is true for R, thus reflexivity holds; however, it is quick to see that $(1,2)$ exists in R but $(2,1) \notin R$, thus symmetry cannot hold. For transitivity, notice that we have $(1,2)$ and $(2,3)$, so we can set up the first step, but $(1,3) \notin R$! Thus transitivity fails. This example is a reminder that we need not think of concrete properties that can be applied in front of our eyes (shapes, dogs, even equality!) in order to have or manipulate some sort of mathematical relation. Next, consider a the relation $R \coloneqq \{(a,b) \mid a \neq b\}$. First, aRa is categorically untrue thus reflexivity means nothing to us. Symmetry, however, tells us that if aRb then bRa, which is true! Transitivity may seems sneakily true at first, but notice that aRb, $bRa \implies aRa$ is not true (we simply replaced c with another instance of a). Finally, transitivity as a singular property is pretty straightforward. Consider $x > y$, notice that $x > x \implies x < x$ is categorically untrue since x can never be greater or less than itself, additionally, $x > y$ never implies $y > x$. For transitivity however, notice that $x > y$, $y > z \implies x > z$ is true!

Returning to our discussion of relations and partitions we can have this thing called an induced relation. Let $R \coloneqq X/\mathfrak{C}$, we say that if $x, y \in \mathfrak{C}$ and $xX/\mathfrak{C}y$, then $X/\mathfrak{C}$ is the relation induced by the partition $\mathfrak{C}$.

Proposition 1.2.1 Relationship between equivalence relations, classes, and set partitions

Explicitly the relationship between the two is given as follows: if R is an equivalence relation on X, then the set of equivalence classes form a partition of X that induces the relation R, and if $\mathfrak{C}$ is a partition of X, then the induced relation is an equivalence relation whose set of equivalence classes is exactly $\mathfrak{C}$.

1.3 Functions

First I want to remind us of some notation and terminology things (mathematicians are **very** specific):

Note:-

To say a function is on X into Y, means that the function is from X to Y

Definition 1.3.1: Functions

{We all know what functions are at some level, but here were are going to be extraordinarily explicit (though you likely have already heard it in this form, regardless). A function on X into y is a relation (R) given by a letter like $f, g, h, \varphi, \ldots$, such that:

1. $\operatorname{dom} f = X$;
2. if $(x, y) \in f$ and $(x, z) \in f$, then $y = z$ (uniqueness of y for each x); additionally, $f(x) = y$ is conventional notation (instead of $(x, y) \in f$ or xfy);
3. y is called the **value** that f **assumes** at the **argument** x; i.e., f **maps sends/ transforms/** x to/ into y. We often use the terms map, mapping, transformation, correspondence, operator, and function interchangeably.

We denote this correspondence via
$$f : X \to Y.$$

Finally, we note that the set of all functions from X to Y is a subset of the power set $\mathcal{P}(X \times Y)$ and is denoted Y^X.

Something to note is that a function is not an active entity despite action-invoking words such as transforms; it merely is. The use of the word function to describe the undefined object that is somehow active in relation to a graph is nevertheless a static set, similar to that of a directory of names that correspond to addresses. Reminder that while the $\operatorname{dom} f = X$, it need not be the case that $\operatorname{ran} f = Y$, instead the range is reserved for the subset of all $y \in Y$ that have a corresponding x such that $f(x) = y$, i.e., is mapped to by some x. We call this property being **onto** and we say f maps X **onto** Y; more commonly however we say that f is **surjective**—we explicitly write this as $\forall y \in Y, \exists x \in X : f(x) = y$—and we denote this surjectivity via
$$f : X \twoheadrightarrow Y,$$
and if $A \subset X$, we may consider all $y \in Y$ such that there is an $x \in A$ such that $f(x_A) = y$ (here I use x_A as crude shorthand for $f(x)$ s.t. $x \in A$, where Halmos uses the more primitive $f(A)$), the set of those $y \in Y$ is known as the **image** of f under A. If it is the case that $X \subset Y$, the function f defined by $f(x) = x$, $\forall x \in X$, we call this the **inclusion map embedding/ injection/** or we say f is **one-to-one**; we explicitly write this as $\forall a \neq b$, $f(a) \neq f(b)$. We denote injectivity via
$$f : A \hookrightarrow Y.$$

A special case of injectivity comes when we inject $X \hookrightarrow X$, relation-wise this is equality in X, language-wise, we call this the **identity map** on X.

Proposition 1.3.1 Restriction and extensions

Consider a function $f : Y \to Z$ and a set $X \subset Y$. We say that there is a **natural way** of constructing a new function $g : X \to Z$; let $g(x) = f(x), \forall x \in X$, we then say g is a **restriction** of f to X and f is the **extension** of g to Y. We denote this as $g = f|X$ where we can express the definition as $(f|X)(x) = f(x) \forall x \in X$ and $\operatorname{ran}(f|X) = f(x_X)$.

There are a few other terminologies that follow from all that has been mentioned so far, and I will go through them rather quickly:

Definition 1.3.2: Projection

f (also pr is a **projection** from $X \times Y$ onto X if if $f(x, y) = x$; similarly, if $R = X \times Y$, then instead of being a projection of R onto X, it is now the range of the projection f (ran(pr)).

Definition 1.3.3: Bijection

Expounding on our notion of inclusion mappings and surjections from earlier, **bijectivity**—one-to-one correspondence-ness—is a function that is both injective and surjective.

Bijections are derived from an equivalence relation R in X with $aRb \iff f(a) = f(b)$ for $a, b \in X$; so given a function/ set $g(y) = \{x \in X \mid f(x) = y, \forall y \in Y\}$, the equivalence relation R tells us that $g(y)$ is an equivalence class of the relation R; that is to say, we define the function as $g : Y \twoheadrightarrow X/R$, notice that for g, if $c \neq d$, $c, d \in Y$, due to the nature of equivalence classes (remember, all the classes are disjoint), $g(c) \neq g(d)$; notice this is the exact explicit definition of injectivity, thus this is a surjective-injective, i.e., bijective mapping. Although there is no universially agreed on notation for bijection, we shall denote this mapping as $g : Y \leftrightarrow X/R$.

Definition 1.3.4: Characteristic functions

If $A \subset X$ the characteristic function of A is the function$\mathcal{X} : X \to 2$ given by

$$\mathcal{X}(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \in X - A \end{cases}$$

, however, we may write $\mathcal{X}_A$ to indicate that it depends on A. The function that assigns to each $A \subset X$, i.e., $\mathcal{P}(X)$, the characteristic function of $A \in 2^X$ is a bijection, i.e., $f : \mathcal{P}(X) \leftrightarrow 2^X$.

Question 2

Concerning bijectivity (point iii.), the examples have all the inclusion maps as one-to-one, however, except in a few trivial special cases, the projections are not. What are these special cases?

Solution

Consider a function's projection as the cannonical map $\gamma : X \to X/R$, the following cases are trivial examples of non-injective projection maps:

1. $X = \emptyset$, vacuously fulfilled (it matters not what R is);
2. $R \coloneqq \{(x, x) \mid x \in X\}$, so every equivalence class is a singleton.

Question 3

Prove:

1. $Y^{\emptyset}$ has exactly one element, namely $\emptyset$, regardless of if Y is empty or not.
2. If X is non-empty, then $\emptyset^{X}$ is empty.

Solution

1. Given that $Y^{\emptyset} \subset \mathcal{P}(\emptyset \times Y)$, and the definition of Cartesian sets has **and** as a qualifier, i.e., $A \times B = \{(a, b) \mid a \in A \textbf{ AND } b \in B\}$, we can see that $\emptyset \times Y$ contains in fact no elements which satisfy this, and thus, it is the empty set. So $Y^{\emptyset} \subset \mathcal{P}(\emptyset) = \{\emptyset\}$, thus the only element that can be in $Y^{\emptyset}$ is the singleton $\{\emptyset\}$. This is a vacuous satisfaction and is non-intuitive; some intuition, perhaps, may be gained but thinking "<u>the only thing nothing can map to is nothing</u>," or by taking the cardinality of these two sets with each other given by $|\emptyset| \cdot |Y| = 0|Y| = 0$ (though this is mainly for intuition).
2. $\emptyset^{X}$ means that something must map to the emptyset, meaning there must be an element within the emptyset, but this cannot be true! Thus, by definition, the set must be empty.}

1.4 Families

Forget literally everything you know about notational convention! Why? Idk, that's just how mathematicians decided to approach this topic—the family (<u>awwwwww</u>). Consider a function x (yes, this is strange) s.t. $x : I \to X$. We call I the <u>**index set**</u>, and thereby each $i \in I$ is an <u>**index**</u>; the range of the function is then called the <u>**indexed set**</u> (why, this is so stupid?!); the function has a special name, being called a <u>**family**</u>, and the value of x at a given i is called a <u>**term**</u> of the family (denoted x_i). Other even more bewildering notation may be used such as $\{x_i\}$ being a family in X or, more horrifically, a family $\{A_i\}$ of subsets of X, being a function $A : I \to \mathcal{P}(X)$ (now that I think of it, this notation, sickeningly, is becoming somewhat preferable).

Given that $\operatorname{ran}(\{A_i\} = \bigcup_{i \in I} A_i)$, we notice that every arbitrary collection of sets is in fact the range of some arbitrary family: consider I and X equal to $\mathfrak{C}$ s.t. $\{A_i\} : \mathfrak{C} \to \mathfrak{C}$. I am realizing it is the terminology which is so obtuse, that what I do not like.

We may think of $\bigcup_{i \in I} A_i$ as $\bigcup_{i=1}^{n} A_i$ (similar to what we have seen for summations and products), where n is the order of i and we assume 1 is the first element in i.

Example 1.4.1

For some index set $i = \{1, 2, 3\}$, and an indexed set A_i we have $\bigcup_{i=1}^{3} = A_1 \cup A_2 \cup A_3$. Another example may be given as follows: let $I = \mathbb{N}$, and $A_i = \{-i, 0, i\}$, that is to say for every i we have

$A_1 = \{-1, 0, 1\}, A_2 = \{-2, 0, 2\}, \ldots$, and so on. Notice that the infinite union of A_i produces $\mathbb{Z}$ and the infinite intersection produces $\{0\}$.

Example 1.4.2

Let us now consider an uncountably infinite indexing set it a plane $I = [-1, 1], A_i = \{i\} \times [0, 1] \subset \mathbb{R}^2$. Notice this is going to be a subset of ordered pairs, i.e., $\{(i, y) \mid 0 \leqslant y \leqslant 1\}$. Geometrically, the union over all $i \in I$ of A_i gives us a rectangle in $\mathbb{R}^2$ defined by $[-1, 1] \times [0, 1]$; similarly the intersection of these sets, notice, are intersections of finite parallel lines, thus there exists no element in common with these sets, thus it is $\emptyset$.

Theorem 1.4.1 Associative law of unions

Let $\{I_j\} : J \to K \mid K = \bigcup_{j \in I}$ and $\{A_k\} : K \to X$, then

$$\bigcup_{k \in K} A_k = \bigcup_{j \in J} \left(\bigcup_{i \in I_j} A_i \right).$$

Question 4

Prove the theorem above.

Sorry! I am leaving this one as an exercise for the reader (sorrryyyyyy:))

De Morgan's laws apply as standard for index/ indexed sets and their unions. From such, it is easy to see that if $\{A_i\}, \{B_i\}$ are families of sets, then $\bigcup_i A_i \cap \bigcup_j B_j = \bigcup_{i,j}(A_i \cap B_j)$, a similarly for $\bigcap_{i,j}(A_i \cup B_j)$.

Let us consider a special type of family: let $I = \{a, b\}$, $a \neq b$ and let Z be the set of all families z indexed by I such that $z_a \in X, z_b \in Y$. Now consider $f : Z \leftrightarrow X \times Y$ such that $f(z) = (z_a, z_b)$. This is the **generalized Cartesian product**! There is further detail one can go into about families of sets and sets of all given families of sets, and even sets of those sets (including their Cartesian products), but these are so disgusting as to induce spontaneous retching in even the most tough-stomached mathematicians. So we will avoid it, if possible.

1.5 Inverses and Composites

Consider a function $f : X \to Y$, a natural mapping that might arise (when thinking on power sets) is some function $F : \mathcal{P}(X) \to \mathcal{P}(Y)$; just as each element of X is mapped to an element Y under f, each subset of X maps to some subset of Y under F, so if $A \subset X$ then $\exists(A \mapsto \mathrm{Im}(f_A))\forall A \in \mathcal{P}(X)$. Cool? I guess? What is to be done about this? In all honesty, I am not sure this factoid has much use other than being a pedagogical tool (though I am almost certainly wrong and I am sure someone will rush to correct me, wherein I might toil and languish over my incorrectness and stupidity, suffering immensely—but probably not, because I simply do not care), but for our cases we use it to introduce its behavior in the inverse.

Note:-

Regarding notation—as we all too often are—$\mathrm{Im}(f_A)$ is the same thing as $f(A)$ when regarding images of subsets. I prefer the former notation as it feels a bit more natural to me.

Definition 1.5.1: Inverses and inverse images

Given a function $f : X \to Y$ we say f^{-1} is the **inverse** of f given by $f^{-1} : \mathcal{P}(Y) \to \mathcal{P}(X)$ such that if $B \subset Y$, then

$$\mathrm{Im}(f_B^{-1}) = \{x \in X \mid f(x) \in B\},$$

is the **inverse image** of B under f

This definition results in some immediate necessary and sufficient conditions: first, for $f : X \to Y$ to exist, the inverse image under f of each non-empty subset of Y must be a non-empty subset of X; second, for f to be injective is that the inverse image under f of each singleton in the range of f be a singleton in X.

Note:-

We often use f^{-1} for another purpose: for some $f : X \to Y$, $f^{-1} : \mathrm{ran}(f) \to X \implies f^{-1}(y) = x \iff f(x) = y$. This will be the most common use case for f^{-1} as the inverse function $g : Y \to X$

There are some connections between images and inverse images which I will quickly enumerate, but I will not provide proof for (for the proofs see Halmos,~pp. 59): Let f be a function such that $f : X \to A$ for each of the following,

1. $B \subset Y \implies f(f^{-1}(B)) \subset B$;
2. $f : X \twoheadrightarrow Y$, $B \subset Y \implies f(f^{-1}(B)) = B$;
3. $A \subset X \implies A \subset f^{-1}(f(A))$;
4. $f : X \hookrightarrow Y$, $A \subset X \implies f^{-1}(f(A)) = A$;
5. If $\{B_i\}$ is a family of subsets of Y, then $f^{-1}(\bigcup_{i\in I} B_i) = \bigcup_{i\in I} f^{-1}(B_i)$ and $f^{-1}(\bigcap_{i\in I} B_i) = \bigcap_{i\in I} f^{-1}(B_i)$;
6. $f^{-1}(Y - B) = X - f^{-1}(B)$ for each $B \subset Y$.

We now move onto **composites**.

Definition 1.5.2: Composition

Let $f : X \to Y$ and $g : Y \to Z$ it is clear that we can make a map from $X \to Z$ since the range of f can be the domain of g. We can represent this new function—i.e., the **composite function**—as either $h : X \to Z$ or $g \circ f$, and often simply just gf. We read this right to left.

Notice that function composition is not always commutative, nor need it be. Also notice that this remains true under the special case that $f : X \to Y$ and $g : Y \to X$; functional composition, however, is always associative. There will be a strong proof for this later on (in Groups). Notice that if $g : X \to Y$ for standard f, if it is bijective we have an inverse function and identity map given by h. Functional composition under inverses appears something like this: if $f : X \to Y$ and $g : Y \to Z$, and $f^{-1} : \mathcal{P}(Y) \to X$ and $g^{-1} : \mathcal{P}(Z) \to \mathcal{P}(Y)$. Then $g \circ f$ has an inverse given by $f^{-1} \circ g^{-1}$. Notice this is not true for $g^{-1} \circ f^{-1}$.

1.5.1 Inverse/ converse and composite relations

The main idea of this comes out of our exploration of inverse functions. Functions are relations after all. Consider the **converse relation** of R on $X \times Y$ s.t. xRy and $yR^{-1}x$. Similar things can be done for composites by generalizing composition to relations, observe: let S be a relation on $Y \times Z$ and R be a relation on $X \times Y$ such that xRy and ySz. Now let T over $X \times Z$ be given by $S \circ R$ such that xTz exists $\iff \exists y \in Y$ such that xRy and ySz. Congrats! You made a $\{$*composite relation*$\}$. If it is the case that R and S are functions s.t. $R(x) = y$, $S(y) = z$ and $S(R(x)) = z \iff xTz$ which implies that functional composition is a special case of a greater abstraction called the $\{$*relative product*$\}$(??).

Lecture 2.

Group Theory Basics

Groups are the central object of abstract algebra, capturing the essence of symmetry and invertible operations. The definition is deceptively simple—three axioms—yet groups appear everywhere: in geometry (symmetries), number theory (modular arithmetic), topology (fundamental groups), physics (gauge theories), and even chemistry (molecular symmetries). This lecture develops the foundations rigorously.

2.1 The group axioms

Definition 2.1.1: Groups

A **group** is a set G together with a map

$$m : G \times G \to G$$

called a **law of composition** (also **binary operation**). Writing ab for $m(a, b)$, the law of composition must satisfy three axioms:

1. **Associativity:** $\forall a, b, c \in G$, $(ab)c = a(bc)$.
2. **Identity:** $\exists e \in G$ such that $ea = ae = a$ for all $a \in G$.
3. **Inverses:** $\forall a \in G$, $\exists b \in G$ such that $ab = ba = e$.

We write (G, m) or simply G for the group, and call $|G|$ the **order** of G.

The axioms have immediate but important consequences that we now prove carefully.

Proposition 2.1.1 Uniqueness of identity

The identity element e is unique.

Proof: Uniqueness of identity

Suppose e and e' are both identities. Then $e = ee'$ (since e' is an identity) and $ee' = e'$ (since e is an identity). Thus $e = e'$. □

Proposition 2.1.2 Uniqueness of inverses

For each $a \in G$, the inverse is unique. We denote it a^{-1} (or $-a$ in additive notation).

Proof: Uniqueness of inverses

Suppose b and c are both inverses of a, so $ab = ba = e$ and $ac = ca = e$. Then:

$$b = be = b(ac) = (ba)c = ec = c.$$

□

Proposition 2.1.3 Cancellation laws

In a group G:

1. **Left cancellation:** $ab = ac \implies b = c$.
2. **Right cancellation:** $ba = ca \implies b = c$.

Proof: Proof of cancellation

For left cancellation: if $ab = ac$, multiply both sides on the left by a^{-1}: $a^{-1}(ab) = a^{-1}(ac)$, so $(a^{-1}a)b = (a^{-1}a)c$, giving $eb = ec$, i.e., $b = c$. Right cancellation is similar. □

Proposition 2.1.4 Inverse of products

For all $a, b \in G$: $(ab)^{-1} = b^{-1}a^{-1}$. Note the reversal of order!

Proof: Proof of inverse formula

We verify $(ab)(b^{-1}a^{-1}) = e$: $(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = aa^{-1} = e$. Similarly for the other side. □

Proposition 2.1.5 Inverse of inverse

For all $a \in G$: $(a^{-1})^{-1} = a$.

Proof: Proof of double inverse

By definition, a^{-1} satisfies $a \cdot a^{-1} = e$. This says a is an inverse of a^{-1}, so by uniqueness, $(a^{-1})^{-1} = a$. □

Note:-

It is customary to use additive notation $(+, 0, -a)$ for abelian groups and multiplicative notation $(\cdot$, 1 or e, $a^{-1})$ in general. This does not mean that multiplication implies non-commutativity!

2.1.1 Group variants and related structures

Groups sit in a hierarchy of algebraic structures with fewer or more axioms.

Definition 2.1.2: Abelian groups

A group G is **abelian** (or **commutative**) if $ab = ba$ for all $a, b \in G$.

Definition 2.1.3: Monoid

A **monoid** is a set with an associative binary operation and an identity element, but inverses are not required. Example: $(\mathbb{N}, +)$ with identity 0, or $(\mathbb{Z}, \times)$ with identity 1.

Definition 2.1.4: Semigroup

A **semigroup** is a set with an associative binary operation—no identity or inverses required. Example: $(\mathbb{Z}_{>0}, +)$.

Note:-

There's a pattern here: Group = Monoid + Inverses = Semigroup + Identity + Inverses. Category theory provides a unifying framework: a group is a category with one object where every morphism is invertible; a monoid is a category with one object (morphisms need not be invertible).

2.2 Fundamental examples

2.2.1 Integers and modular arithmetic

Example 2.2.1 (The integers)
The set $(\mathbb{Z}, +)$ is an abelian group with identity 0 and inverse $-n$ for each n. This is the prototypical infinite cyclic group.

Example 2.2.2 (Integers mod n)
$\mathbb{Z}/n\mathbb{Z} = \{0, 1, 2, \ldots, n-1\}$ with addition mod n:

$$a + b := \begin{cases} a+b & \text{if } a+b < n \\ a+b-n & \text{if } a+b \geqslant n. \end{cases}$$

Identity is 0, inverse of a is $n - a \pmod{n}$. This is an abelian group of order n.

Proposition 2.2.1 Units mod n
The set $(\mathbb{Z}/n\mathbb{Z})^\times = \{a \in \mathbb{Z}/n\mathbb{Z} : \gcd(a, n) = 1\}$ forms a group under multiplication mod n. Its order is $\varphi(n)$, Euler's totient function.

Proof: Proof that units form a group

Closure: If $\gcd(a, n) = \gcd(b, n) = 1$, then $\gcd(ab, n) = 1$ (if a prime p divides both ab and n, it divides a or b, contradicting coprimality).

Identity: 1 is coprime to n.

Inverses: If $\gcd(a, n) = 1$, then by Bézout's identity, $\exists r, s$ with $ra + sn = 1$. Thus $ra \equiv 1 \pmod{n}$, so r is a multiplicative inverse. □

2.2.2 Multiplicative groups of fields

Example 2.2.3 (Multiplicative groups)
$\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$, $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$, and $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ are abelian groups under multiplication, with identity 1 and inverse $1/x$.

Example 2.2.4 (The circle group)
The unit circle $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ is a group under multiplication. It is isomorphic to $\mathbb{R}/\mathbb{Z}$ via $t \mapsto e^{2\pi i t}$. This is a compact abelian group and plays a role in Fourier analysis and representation theory.

Example 2.2.5 (Roots of unity)
For each $n \geqslant 1$, the nth roots of unity $\mu_n = \{z \in \mathbb{C} : z^n = 1\} = \{e^{2\pi i k/n} : k = 0, 1, \ldots, n-1\}$ form a cyclic group of order n under multiplication. We have $\mu_n \cong \mathbb{Z}/n\mathbb{Z}$.

2.2.3 Symmetric and permutation groups

Definition 2.2.1: Permutation group

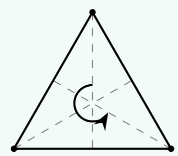
A **permutation** of a set X is a bijection $\sigma : X \to X$. The set of all permutations of X forms a group $\mathrm{Sym}(X)$ under composition, with identity id_X and inverse σ^{-1}.

Definition 2.2.2: Symmetric group

The **symmetric group** on n letters is $S_n = \mathrm{Sym}(\{1, 2, \ldots, n\})$. It has order $n!$ and is non-abelian for $n \geqslant 3$.

Example 2.2.6 (Symmetries of a triangle)

S_3 can be visualized as symmetries of an equilateral triangle: 3 rotations (including identity) and 3 reflections.



Labeling vertices $1, 2, 3$, rotations give permutations $e, (123), (132)$ and reflections give $(12), (13), (23)$. Thus $|S_3| = 6$.

2.2.4 Matrix groups

Definition 2.2.3: General linear group

The **general linear group** $\mathrm{GL}_n(\mathbb{F})$ is the group of invertible $n \times n$ matrices over a field $\mathbb{F}$, under matrix multiplication. A matrix is invertible iff its determinant is nonzero.

Definition 2.2.4: Special linear group

The **special linear group** $\mathrm{SL}_n(\mathbb{F}) = \{A \in \mathrm{GL}_n(\mathbb{F}) : \det(A) = 1\}$ is a subgroup of $\mathrm{GL}_n(\mathbb{F})$.

Note:-

Matrix groups are our first examples of Lie groups—groups that are also smooth manifolds. They are, in my opinion, the most exciting part of representation theory: a representation of G is a homomorphism $\rho : G \to \mathrm{GL}_n(\mathbb{F})$, making G act on $\mathbb{F}^n$ by linear transformations.

Example 2.2.7 (Non-abelian example)

In $\mathrm{GL}_2(\mathbb{R})$, let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Then:

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = BA.$$

So $\mathrm{GL}_2(\mathbb{R})$ is non-abelian.

2.3 Constructions: Products and sums

How do we build new groups from existing ones?

2.3.1 Direct products

Definition 2.3.1: Direct product

Given groups G and H, the **direct product** $G \times H$ is the Cartesian product with componentwise operation:

$$(a, b) \cdot (a', b') = (aa', bb').$$

Identity is (e_G, e_H), inverse of (a, b) is (a^{-1}, b^{-1}).

Proposition 2.3.1 Direct product is a group

The direct product $G \times H$ satisfies all group axioms. Moreover, $|G \times H| = |G| \cdot |H|$ (for finite groups).

Proof: Verification of group axioms for $G \times H$

Associativity: $((a, b)(a', b'))(a'', b'') = (aa', bb')(a'', b'') = (aa'a'', bb'b'')$. Similarly, $(a, b)((a', b')(a'', b'')) = (a, b)(a'a'', b'b'') = (aa'a'', bb'b'')$. Equal by associativity in G and H.

Identity: $(e_G, e_H)(a, b) = (e_G a, e_H b) = (a, b)$, and similarly on the right.

Inverses: $(a, b)(a^{-1}, b^{-1}) = (aa^{-1}, bb^{-1}) = (e_G, e_H)$. $\square$

Example 2.3.1

We have $\mathbb{Z}/2 \times \mathbb{Z}/3 \cong \mathbb{Z}/6$ (since $\gcd(2, 3) = 1$), but $\mathbb{Z}/2 \times \mathbb{Z}/2 \not\cong \mathbb{Z}/4$ (the former has no element of order 4).

This generalizes to arbitrary (even infinite) collections of groups.

Definition 2.3.2: Arbitrary direct products

For a family of groups $\{G_i\}_{i \in I}$, the **direct product** is:

$$\prod_{i \in I} G_i = \{(a_i)_{i \in I} : a_i \in G_i\}$$

with componentwise operations.

Definition 2.3.3: Direct sum

The **direct sum** (for abelian groups) is the subgroup:

$$\bigoplus_{i \in I} G_i = \{(a_i)_{i \in I} \in \prod_{i \in I} G_i : a_i = e_{G_i} \text{ for all but finitely many } i\}.$$

Note:-

For finite index sets, $\prod = \bigoplus$. For infinite sets, they differ crucially. "O-plus" notation will become important when we move onto linear algebra.

Example 2.3.2 (Power series vs polynomials)

Taking $G_i = (\mathbb{R}, +)$ for all $i \in \mathbb{N}$:

$$\prod_{i=0}^{\infty} \mathbb{R} \cong \mathbb{R}[[x]] \quad \text{(formal power series under addition)}$$

$$\bigoplus_{i=0}^{\infty} \mathbb{R} \cong \mathbb{R}[x] \quad \text{(polynomials under addition)}$$

A power series $\sum_{i=0}^{\infty} a_i x^i$ can have infinitely many nonzero terms; a polynomial cannot.

Question 1

1. Prove that $\mathbb{Z}/m \times \mathbb{Z}/n \cong \mathbb{Z}/mn$ if and only if $\gcd(m, n) = 1$.
2. Find all groups of order 4 up to isomorphism.
3. Prove that $G \times H \cong H \times G$ for any groups G, H.

Solution

(1) ($\Leftarrow$) Assume $\gcd(m, n) = 1$. The element $(1, 1) \in \mathbb{Z}/m \times \mathbb{Z}/n$ has order $\operatorname{lcm}(m, n) = mn$ (since the order of (a, b) is $\operatorname{lcm}(\operatorname{ord}(a), \operatorname{ord}(b))$). Thus $\mathbb{Z}/m \times \mathbb{Z}/n$ contains an element of order $mn = |G|$, so it's cyclic, hence $\cong \mathbb{Z}/mn$.
($\Rightarrow$) If $d = \gcd(m, n) > 1$, then for any $(a, b) \in \mathbb{Z}/m \times \mathbb{Z}/n$, we have $\frac{mn}{d} \cdot (a, b) = (\frac{mn}{d} a, \frac{mn}{d} b) = (0, 0)$ since $m \mid \frac{mn}{d}$ and $n \mid \frac{mn}{d}$. So every element has order dividing $\frac{mn}{d} < mn$, meaning no element has order mn. But $\mathbb{Z}/mn$ has an element of order mn, so they're not isomorphic.
(2) Let G have order 4. Every element has order dividing 4, so orders are 1, 2, or 4.
Case 1: G has an element a of order 4. Then $G = \{e, a, a^2, a^3\} = \langle a \rangle \cong \mathbb{Z}/4$.
Case 2: Every non-identity element has order 2. Then $G = \{e, a, b, c\}$ with $a^2 = b^2 = c^2 = e$. What is ab? It's not e (else $a = b^{-1} = b$), not a (else $b = e$), not b (else $a = e$). So $ab = c$. Similarly $ba = c$ (check: $(ab)(ba) = a(bb)a = a^2 = e$, so $ba = (ab)^{-1} = ab$ since ab has order 2). So G is abelian, and $G \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (Klein four-group).
(3) Define $\varphi : G \times H \to H \times G$ by $\varphi(g, h) = (h, g)$. This is clearly a bijection. It's a homomorphism: $\varphi((g_1, h_1)(g_2, h_2)) = \varphi(g_1 g_2, h_1 h_2) = (h_1 h_2, g_1 g_2) = (h_1, g_1)(h_2, g_2) = \varphi(g_1, h_1)\varphi(g_2, h_2)$.

2.4 Subgroups

Definition 2.4.1: Subgroup

A **subgroup** of a group G is a nonempty subset $H \subseteq G$ such that:

1. **Closure:** $a, b \in H \implies ab \in H$;
2. **Inverses:** $a \in H \implies a^{-1} \in H$.

We write $H \leqslant G$ (or $H < G$ if $H \neq G$, a **proper subgroup**).

Proposition 2.4.1 Subgroup criterion
A nonempty subset $H \subseteq G$ is a subgroup iff $a, b \in H \implies ab^{-1} \in H$.

Proof: One-step subgroup test
($\Rightarrow$) If $H \leqslant G$ and $a, b \in H$, then $b^{-1} \in H$ (inverses) and $ab^{-1} \in H$ (closure).

($\Leftarrow$) Assume the condition. Since $H \neq \emptyset$, pick $a \in H$. Then $e = aa^{-1} \in H$. For any $b \in H$, $b^{-1} = eb^{-1} \in H$. For $a, b \in H$, we have $b^{-1} \in H$, so $ab = a(b^{-1})^{-1} \in H$. ☐

Example 2.4.1 (Subgroups of $\mathbb{Z}$)

Every subgroup of $(\mathbb{Z}, +)$ has the form $n\mathbb{Z} = \{\ldots, -2n, -n, 0, n, 2n, \ldots\}$ for some $n \geqslant 0$. These form a chain: $\mathbb{Z} = 1\mathbb{Z} \supset 2\mathbb{Z} \supset 4\mathbb{Z} \supset 8\mathbb{Z} \supset \cdots$ (not all subgroups are comparable, though: $2\mathbb{Z}$ and $3\mathbb{Z}$ are incomparable).

Theorem 2.4.1 Subgroups of $\mathbb{Z}$

Let H be a subgroup of $(\mathbb{Z}, +)$. Then either $H = \{0\}$ or $H = n\mathbb{Z}$ where n is the smallest positive element of H.

Proof: Classification of subgroups of $\mathbb{Z}$

If $H = \{0\}$, we're done. Otherwise, H contains some nonzero element a; then $-a \in H$ too, so H contains a positive element. Let n be the smallest positive element of H.

We show $H = n\mathbb{Z}$. Clearly $n\mathbb{Z} \subseteq H$ since H is closed under addition: $n \in H \implies 2n = n + n \in H \implies 3n \in H \implies \ldots$, and $-n \in H \implies -2n \in H \implies \ldots$

For $H \subseteq n\mathbb{Z}$: take any $m \in H$. By division algorithm, $m = qn + r$ with $0 \leqslant r < n$. Then $r = m - qn \in H$ (since $m, qn \in H$). By minimality of n, we must have $r = 0$, so $m = qn \in n\mathbb{Z}$. ☐

Proposition 2.4.2 Greatest common divisor

Let $a, b \in \mathbb{Z}$, not both zero, and let $d = \gcd(a, b)$. Then:

1. $\mathbb{Z}a + \mathbb{Z}b = \{ra + sb : r, s \in \mathbb{Z}\} = d\mathbb{Z}$;
2. d divides both a and b;
3. If n divides both a and b, then n divides d;
4. $\exists r, s \in \mathbb{Z}$ such that $d = ra + sb$ (Bézout's identity).

Proof: Proof of GCD properties

The set $S = \mathbb{Z}a + \mathbb{Z}b$ is a subgroup of $\mathbb{Z}$ (verify: closed under addition and negation). By the theorem, $S = d\mathbb{Z}$ for some $d \geqslant 0$. Since $a = 1 \cdot a + 0 \cdot b \in S$ and $b \in S$, and $S = d\mathbb{Z}$, we have $d \mid a$ and $d \mid b$. Since $d \in S$, we have $d = ra + sb$ for some r, s. If $n \mid a$ and $n \mid b$, then $n \mid (ra + sb) = d$. These properties characterize $d = \gcd(a, b)$. ☐

Corollary 2.4.1 Coprimality criterion

a and b are coprime (i.e., $\gcd(a, b) = 1$) iff $\exists r, s \in \mathbb{Z}$ with $ra + sb = 1$.

Corollary 2.4.2 Euclid's lemma

If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof: Proof of Euclid's lemma

Suppose $p \nmid a$. Since p is prime, $\gcd(p, a) = 1$. By Bézout, $\exists r, s$ with $rp + sa = 1$. Multiply by b: $rpb + sab = b$. Since $p \mid rpb$ and $p \mid ab$, we have $p \mid b$. ☐

2.4.1 Generated subgroups and cyclic groups

Definition 2.4.2: Subgroup generated by a set

For $S \subseteq G$, the **subgroup generated by** S, denoted $\langle S\rangle$, is the smallest subgroup of G containing S. Explicitly:

$$\langle S\rangle = \{s_1^{\epsilon_1} s_2^{\epsilon_2} \cdots s_k^{\epsilon_k} : k \geqslant 0,\ s_i \in S,\ \epsilon_i \in \{1, -1\}\}.$$

Definition 2.4.3: Cyclic groups

A group G is **cyclic** if $G = \langle a\rangle$ for some $a \in G$. The element a is called a **generator**.

Definition 2.4.4: Order of an element

The **order** of $a \in G$, denoted $\operatorname{ord}(a)$ or $|a|$, is the smallest positive integer n such that $a^n = e$, or ∞ if no such n exists.

Proposition 2.4.3 Structure of cyclic groups

Let $G = \langle a\rangle$.

1. If $\operatorname{ord}(a) = \infty$, then $G \cong \mathbb{Z}$ via $n \mapsto a^n$.
2. If $\operatorname{ord}(a) = n < \infty$, then $G = \{e, a, a^2, \dots, a^{n-1}\}$ and $G \cong \mathbb{Z}/n\mathbb{Z}$.

Proof: Proof of cyclic group structure

Define $\varphi : \mathbb{Z} \to G$ by $\varphi(k) = a^k$. This is a homomorphism: $\varphi(k + l) = a^{k+l} = a^k a^l = \varphi(k)\varphi(l)$. It's surjective since $G = \langle a\rangle$.

Case 1: $\operatorname{ord}(a) = \infty$. If $\varphi(k) = \varphi(l)$ with $k > l$, then $a^{k-l} = e$, contradicting infinite order. So φ is injective, hence an isomorphism $\mathbb{Z} \xrightarrow{\sim} G$.

Case 2: $\operatorname{ord}(a) = n$. Then $\ker(\varphi) = \{k \in \mathbb{Z} : a^k = e\}$. Since $a^n = e$, we have $n\mathbb{Z} \subseteq \ker(\varphi)$. Conversely, if $a^k = e$, write $k = qn + r$ with $0 \leqslant r < n$. Then $a^r = a^{k-qn} = a^k(a^n)^{-q} = e$. By minimality of n, $r = 0$, so $k \in n\mathbb{Z}$. Thus $\ker(\varphi) = n\mathbb{Z}$, and $G \cong \mathbb{Z}/n\mathbb{Z}$ by the First Isomorphism Theorem (this will be introduced later but to oversimplify: surjectivity is guaranteed iff the kernal only contains the identity). □

Proposition 2.4.4 Subgroups of cyclic groups

Every subgroup of a cyclic group is cyclic. If $G = \langle a\rangle$ has order n, the subgroups of G are exactly $\langle a^d\rangle$ for divisors $d \mid n$, and $|\langle a^d\rangle| = n/d$.

Proof: Subgroups of cyclic groups

Let $H \leqslant \langle a\rangle$. If $H = \{e\}$, it's cyclic. Otherwise, let d be the smallest positive integer with $a^d \in H$. We claim $H = \langle a^d\rangle$.

Clearly $\langle a^d\rangle \subseteq H$. For the reverse: take $a^k \in H$. Write $k = qd+r$ with $0 \leqslant r < d$. Then $a^r = a^k(a^d)^{-q} \in H$. By minimality of d, $r = 0$, so $a^k = (a^d)^q \in \langle a^d\rangle$.

If $|G| = n$, then a^d has order $n/\gcd(d, n)$. The subgroups correspond to divisors via $d \leftrightarrow \langle a^{n/d}\rangle$ of order d. □

Question 2

1. Find all subgroups of $\mathbb{Z}/12\mathbb{Z}$ and draw the subgroup lattice.
2. Prove that $\mathbb{Z}/p\mathbb{Z}$ has no proper nontrivial subgroups for p prime.

3. How many generators does $\mathbb{Z}/n\mathbb{Z}$ have?

4. Prove: if G is a group where every element satisfies $x^2 = e$, then G is abelian.

Solution

(1) Divisors of 12: 1, 2, 3, 4, 6, 12. Subgroups: $\langle 0\rangle = \{0\}$, $\langle 6\rangle = \{0, 6\}$, $\langle 4\rangle = \{0, 4, 8\}$, $\langle 3\rangle = \{0, 3, 6, 9\}$, $\langle 2\rangle = \{0, 2, 4, 6, 8, 10\}$, $\langle 1\rangle = \mathbb{Z}/12$.
Lattice (larger subgroups higher):

$$\begin{tikzcd}
& \mathbb{Z}/12 \arrow[dl, dash] \arrow[dr, dash] & \\
\langle 2\rangle \arrow[d, dash] \arrow[drr, dash] & & \langle 3\rangle \arrow[d, dash] \arrow[dll, dash] \\
\langle 4\rangle \arrow[dr, dash] & & \langle 6\rangle \arrow[dl, dash] \\
& \{0\} &
\end{tikzcd}$$

(2) By Lagrange's theorem, the order of any subgroup divides p. So subgroups have order 1 or p, meaning $\{0\}$ or $\mathbb{Z}/p$.
(3) a generates $\mathbb{Z}/n$ iff $\operatorname{ord}(a) = n$ iff $\gcd(a, n) = 1$. The count is $\varphi(n)$, Euler's totient.
(4) For any $a, b \in G$: $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$ (using $x^{-1} = x$ since $x^2 = e$). So G is abelian.

2.5 Homomorphisms

Definition 2.5.1: Homomorphism

A **homomorphism** $\varphi : G \to H$ between groups is a map satisfying

$$\varphi(ab) = \varphi(a)\varphi(b) \quad \text{for all } a, b \in G.$$

Proposition 2.5.1 Basic properties of homomorphisms
Let $\varphi : G \to H$ be a homomorphism.

1. $\varphi(e_G) = e_H$.
2. $\varphi(a^{-1}) = \varphi(a)^{-1}$ for all $a \in G$.
3. $\varphi(a^n) = \varphi(a)^n$ for all $n \in \mathbb{Z}$.
4. If $K \leqslant G$, then $\varphi(K) \leqslant H$.
5. If $L \leqslant H$, then $\varphi^{-1}(L) \leqslant G$.

Proof: Proofs of homomorphism properties

(1) $\varphi(e_G) = \varphi(e_G e_G) = \varphi(e_G)\varphi(e_G)$. Multiply both sides by $\varphi(e_G)^{-1}$: $e_H = \varphi(e_G)$.

(2) $\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e_G) = e_H$. So $\varphi(a^{-1})$ is the inverse of $\varphi(a)$.

(3) For $n > 0$: induction. For $n = 0$: part (1). For $n < 0$: $\varphi(a^n) = \varphi((a^{-1})^{-n}) = \varphi(a^{-1})^{-n} = (\varphi(a)^{-1})^{-n} = \varphi(a)^n$.

(4) For $\varphi(a), \varphi(b) \in \varphi(K)$ (with $a, b \in K$): $\varphi(a)\varphi(b)^{-1} = \varphi(a)\varphi(b^{-1}) = \varphi(ab^{-1}) \in \varphi(K)$ since $ab^{-1} \in K$.

(5) For $a, b \in \varphi^{-1}(L)$: $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in L$ (since $L \leqslant H$), so $ab^{-1} \in \varphi^{-1}(L)$. $\square$

Note:-

The homomorphism condition can be expressed as a commutative diagram: the two paths $G \times G \xrightarrow{m_G} G \xrightarrow{\varphi} H$ and $G \times G \xrightarrow{\varphi\times\varphi} H \times H \xrightarrow{m_H} H$ give the same result.

$$\begin{tikzcd} G \times G \arrow[r, "\varphi\times\varphi"] \arrow[d, "m_G"'] & H \times H \arrow[d, "m_H"] \\ G \arrow[r, "\varphi"'] & H \end{tikzcd}$$

Definition 2.5.2: Types of homomorphisms

Let $\varphi : G \to H$ be a homomorphism.

1. φ is a **monomorphism** (or **embedding**) if it's injective.
2. φ is an **epimorphism** if it's surjective.
3. φ is an **isomorphism** if it's bijective. We write $G \cong H$.
4. An isomorphism $G \to G$ is called an **automorphism**. The set of all automorphisms of G, denoted $\mathrm{Aut}(G)$, forms a group under composition.

Example 2.5.1 (Important(ish) homomorphisms)

1. **Reduction mod n:** $\pi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $\pi(a) = a \mod n$. Surjective, kernel $n\mathbb{Z}$.
2. **Quotient mod n:** If $n \mid m$, then $\mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $a \mapsto a \mod n$. Example: last-two-digits to last-digit, $\mathbb{Z}/100 \to \mathbb{Z}/10$.
3. **Determinant:** $\det : GL_n(\mathbb{R}) \to \mathbb{R}^*$. Homomorphism since $\det(AB) = \det(A)\det(B)$. Kernel is $SL_n(\mathbb{R})$.
4. **Exponential:** $\exp : (\mathbb{R}, +) \to (\mathbb{R}_{>0}, \times)$, $x \mapsto e^x$. Isomorphism with inverse ln.
5. **Complex exponential:** $\exp : (\mathbb{R}, +) \to (S^1, \times)$, $t \mapsto e^{2\pi i t}$. Surjective, kernel $\mathbb{Z}$. Induces $\mathbb{R}/\mathbb{Z} \cong S^1$.

2.5.1 Isomorphisms and invariants

Definition 2.5.3: Isomorphic groups

Groups G and H are **isomorphic**, written $G \cong H$, if there exists an isomorphism $\varphi : G \to H$. Isomorphic groups are "the same" for all group-theoretic purposes.

What properties are preserved by isomorphisms? These are called invariants.

Proposition 2.5.2 Isomorphism invariants

If $G \cong H$, then:

1. $|G| = |H|$ (same order);
2. G is abelian iff H is abelian;
3. G is cyclic iff H is cyclic;
4. For each n, G and H have the same number of elements of order n;
5. For each n, G and H have the same number of subgroups of order n.

Proof: Why these are invariants

Let $\varphi : G \xrightarrow{\sim} H$ be an isomorphism.

(1) Bijections preserve cardinality.

(2) If G is abelian and $h_1, h_2 \in H$, write $h_i = \varphi(g_i)$. Then $h_1h_2 = \varphi(g_1)\varphi(g_2) = \varphi(g_1g_2) = \varphi(g_2g_1) = \varphi(g_2)\varphi(g_1) = h_2h_1$.

(3) If $G = \langle a \rangle$, then $H = \varphi(G) = \varphi(\langle a \rangle) = \langle \varphi(a) \rangle$.

(4) $\mathrm{ord}(\varphi(a)) = \mathrm{ord}(a)$ since $\varphi(a)^n = \varphi(a^n) = e_H \iff a^n = e_G$.

(5) φ induces a bijection between subgroups of G and subgroups of H. $\square$

Example 2.5.2 (Non-isomorphic groups of same order)

$\mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$ both have order 4, but:

- $\mathbb{Z}/4$ has an element of order 4 (namely, 1);
- $\mathbb{Z}/2 \times \mathbb{Z}/2$ has no element of order 4 (all non-identity elements have order 2).

So they're not isomorphic. These are the only groups of order 4 up to isomorphism.

Example 2.5.3 (All groups of order 2 are isomorphic)

If $|G| = 2$, write $G = \{e, x\}$. Then $x \neq e$, so $x^2 \neq x$ (else $x = e$). Thus $x^2 = e$, and $G = \langle x \rangle \cong \mathbb{Z}/2$.

Question 3

1. Prove that $\mathrm{Aut}(\mathbb{Z}/n\mathbb{Z}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.
2. Find $\mathrm{Aut}(\mathbb{Z})$.
3. Prove that if $\varphi : G \to H$ is an isomorphism, then $\varphi^{-1} : H \to G$ is also an isomorphism.
4. Prove that "is isomorphic to" is an equivalence relation on groups.

Solution

(1) An automorphism φ of $\mathbb{Z}/n$ is determined by $\varphi(1)$, since $\varphi(k) = k\varphi(1)$. We need $\varphi(1)$ to generate $\mathbb{Z}/n$, i.e., $\gcd(\varphi(1), n) = 1$, i.e., $\varphi(1) \in (\mathbb{Z}/n)^\times$. Conversely, any $a \in (\mathbb{Z}/n)^\times$ defines an automorphism $\varphi_a : k \mapsto ka$. The map $a \mapsto \varphi_a$ is an isomorphism $(\mathbb{Z}/n)^\times \to \mathrm{Aut}(\mathbb{Z}/n)$ (check it's a homomorphism and bijection).
(2) An automorphism of $\mathbb{Z}$ is determined by $\varphi(1)$. We need $\langle \varphi(1) \rangle = \mathbb{Z}$, so $\varphi(1) = \pm 1$. Thus $\mathrm{Aut}(\mathbb{Z}) = \{\mathrm{id}, -\mathrm{id}\} \cong \mathbb{Z}/2$.
(3) φ^{-1} is a bijection. For $h_1, h_2 \in H$, let $g_i = \varphi^{-1}(h_i)$. Then $\varphi^{-1}(h_1h_2) = \varphi^{-1}(\varphi(g_1)\varphi(g_2)) = \varphi^{-1}(\varphi(g_1g_2)) = g_1g_2 = \varphi^{-1}(h_1)\varphi^{-1}(h_2)$.
(4) Reflexive: $\mathrm{id}_G : G \to G$ is an isomorphism. Symmetric: if $\varphi : G \to H$ is an isomorphism, so is $\varphi^{-1} : H \to G$ by part (3). Transitive: if $\varphi : G \to H$ and $\psi : H \to K$ are isomorphisms, so is $\psi \circ \varphi : G \to K$.

2.6 The order of an element and its properties

We've introduced order; now we develop its theory more fully.

Proposition 2.6.1 Powers and order

Let $a \in G$ have finite order n. Then:

1. $a^k = e \iff n \mid k$.

2. $a^i = a^j \iff n \mid (i-j)$.

3. The elements $e, a, a^2, \ldots, a^{n-1}$ are all distinct.

Proof: Properties of order

(1) ($\Leftarrow$) If $k = mn$, then $a^k = (a^n)^m = e^m = e$. ($\Rightarrow$) Write $k = qn + r$ with $0 \leqslant r < n$. Then $e = a^k = a^{qn+r} = (a^n)^q a^r = a^r$. By minimality of n, $r = 0$, so $n \mid k$.

(2) $a^i = a^j \iff a^{i-j} = e \iff n \mid (i-j)$ by (1).

(3) If $a^i = a^j$ for $0 \leqslant i < j < n$, then $a^{j-i} = e$ with $0 < j-i < n$, contradicting minimality of n. □

Proposition 2.6.2 Order of powers

Let $\operatorname{ord}(a) = n$. Then $\operatorname{ord}(a^k) = \frac{n}{\gcd(k,n)}$.

Proof: Order of powers

Let $d = \gcd(k, n)$. We show $\operatorname{ord}(a^k) = n/d$.

First, $(a^k)^{n/d} = a^{kn/d} = (a^n)^{k/d} = e$, so $\operatorname{ord}(a^k) \mid n/d$.

Conversely, let $m = \operatorname{ord}(a^k)$. Then $a^{km} = e$, so $n \mid km$. Write $k = d \cdot k'$ and $n = d \cdot n'$ where $\gcd(k', n') = 1$. Then $dn' \mid dk'm$, so $n' \mid k'm$. Since $\gcd(k', n') = 1$, we have $n' \mid m$, i.e., $n/d \mid m$. Thus $m = n/d$. □

Corollary 2.6.1 Generators of cyclic groups

In $\langle a \rangle$ of order n, the element a^k is a generator iff $\gcd(k, n) = 1$.

Example 2.6.1

In $\mathbb{Z}/12$, the generators are $1, 5, 7, 11$ (the four elements coprime to 12). The element 4 has order $12/\gcd(4, 12) = 12/4 = 3$.

Question 4

1. In a group G, if $a^{12} = e$ and $a^8 = e$, what are the possible orders of a?

2. Prove: if $\operatorname{ord}(a) = n$ and $\operatorname{ord}(b) = m$ with $\gcd(n, m) = 1$ and $ab = ba$, then $\operatorname{ord}(ab) = nm$.

3. Find the order of $(2, 3)$ in $\mathbb{Z}/6 \times \mathbb{Z}/8$.

Solution

(1) $\operatorname{ord}(a)$ divides both 12 and 8, so $\operatorname{ord}(a) \mid \gcd(12, 8) = 4$. Possible orders: 1, 2, 4.
(2) Let $k = \operatorname{ord}(ab)$. Then $e = (ab)^{nm} = a^{nm}b^{nm} = (a^n)^m(b^m)^n = e$ (using commutativity), so $k \mid nm$.
Now $e = (ab)^{kn} = a^{kn}b^{kn} = b^{kn}$ (since $a^n = e$), so $m \mid kn$. Since $\gcd(m, n) = 1$, $m \mid k$. Similarly, $n \mid k$. Since $\gcd(m, n) = 1$, $mn \mid k$. Thus $k = mn$.
(3) $\operatorname{ord}(2, 3) = \operatorname{lcm}(\operatorname{ord}(2), \operatorname{ord}(3)) = \operatorname{lcm}(6/\gcd(2, 6), 8/\gcd(3, 8)) = \operatorname{lcm}(3, 8) = 24$.

Lecture 3.

Group Relations

We say that two sets have the same **cardinality** if $\exists f : X \hookrightarrow Y$, and we write $|X| \leqslant |Y|$. This allows us to inter-relate cardinality and function relations via a special theorem.

Theorem 3.0.1 Schröder-Berstein Theorem

If $\exists f : X \hookrightarrow Y$ and $g : Y \hookrightarrow X$ then $|X| = |Y|$.

The Schröder-Berstein theorem provides some immediate results for allowing explicit bijections.

Proposition 3.0.1 Cardinality of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ to any power

Using Schröder-Berstein we can create explicit bijections $f : \mathbb{Z} \leftrightarrow \mathbb{N}$ given by $n \mapsto \begin{cases} 2n & \text{if } n \geqslant 0; \\ -(2n+1) & \text{if } n < 0. \end{cases}$

Proposition 3.0.2 $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}| = \mathbb{N}^n$

We create the explicit bijection via $f : \mathbb{N} \times \mathbb{N} \leftrightarrow \mathbb{N}$, $(k, 1) \mapsto 2^k(2l + 1) - 1$. It follows then that $\mathbb{N}^a, \mathbb{Z}^b, \mathbb{Q}^c, \forall a, b, c \in \mathbb{N}$ permits a similar bijection.

Proof: Proof of Prop. 3.1.2

Notice that any natural number can be written as $2^k(m)$ where m is odd, i.e., $m = 2n + 1$. A bijection is automatically created here with $2^k(2l+1)$, and the additional -1 at the end of the statement is there to shift the entire bijection down one. So instead of $(0, 0) \mapsto 2^0(2(0)+1) = 1$ we get $(0, 0) \mapsto 2^0(2(0)+1)-1 = 0$. $\square$

Proposition 3.0.3 Countable vs uncountable infinites

$|\mathbb{R}| > |\mathbb{N}|$.

Proof: Proof by Cantor

Given by Cantor's diagonal argument, no map $f : \mathbb{N} \to \mathbb{R}$ be surjective, because:

$$\begin{cases} f(0) & = a_{00}a_{01}a_{02}a_{03}\dots \\ f(1) & = a_{10}a_{11}a_{12}a_{13}\dots \\ f(2) & = a_{20}a_{21}a_{22}a_{23}\dots \end{cases}$$

then let $y = b_0b_1b_2b_3$ where $b_j \neq a_{jj}$ for each j. Now look at the j^{th} digit, $y \neq f(j)$, $\forall j \in \mathbb{N}$, so f can't be surjective. $\square$

Proposition 3.0.4 Cardinality of mappings

$|\mathbb{R}^{\mathbb{N}}| = |\mathbb{R}|$

Proof: Proof by Cantor's diagonal used for mappings

We can use the fact that $|(0,1)| = |\mathbb{R}|$ in addition to Cantor's diagonal argument as a decimal expansion—$t = 0.t_1t_2t_3\ldots$—with t_i as an integer between 0 and 9. So now to show $\exists f \mid f : (0,1) \leftrightarrow (0,1)^{\mathbb{N}}$, we create some $s_j \in (0,1)^{\mathbb{N}}$ and construct the bijection via:

$$\begin{cases} s_1 &= 0.t_1t_3t_5\ldots \\ s_2 &= 0.t_2t_6t_{10}\ldots \\ s_3 &= 0.t_4t_{12}t_{20}\ldots \end{cases}$$

We say that s_1 is made up of all t_i that are odd; s_2 is made up of all i which are odd such that $i \cdot 2 \forall i \in s_2$; s_3 is made up of all i which are odd such that $i \cdot 2^2, \forall i \in s_3$, $i \cdot 2^3 \in s_3$ and so on (the general form being $i \cdot 2^n \mid n = j - 1$). Therefore there exists a bijection between $(0,1) \leftrightarrow (0,1)^{\mathbb{N}}$. $\square$

Proposition 3.0.5 There does not exist a largest cardinality

Let $\mathcal{P}(S)$ be the standard powerset. It follows then that $\mathcal{P}(S) = \{0,1\}^S$ and $|\mathcal{P}(S)| > |S|$ for any S.

Proof: Proof of non-existence of largest cardinality

Recall that $\{0,1\}^S = \{\text{all } f \text{ such that } f : S \to \{0,1\}\}$. There must exist a bijection since $|\mathcal{P}(S)| = 2^{|S|}$ and $|\{0,1\}^S| = 2^{|S|}$. Now to show $|\mathcal{P}(S)| > |S|$ we use proof by contradiction: suppose $\exists \alpha : S \leftrightarrow \mathcal{P}(S)$ and let $A \subset S$ defined by $A \coloneqq \{s \in S \mid s \neq \alpha(s)\}$, now for $A = \alpha(t)$ for some $t \in A$ we notice that $t \notin \alpha(t) = A$ means that $t \in A$, but to say $t \in \alpha(t) \implies t \notin A$, thus we have a contradiction. $\square$

Another option for this proof, though less formal, can be found by showing that $\mathcal{P}(S) = \{0,1\}^S$ and that subsets of S have f such that $f \mapsto f^{-1}$; let $A \subset S$ then

$$A \mapsto \left(\nVdash_A : x \mapsto \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases} \right)$$

Note that if S is finite then $|S| = n$ and $\mathcal{P}(S) = 2^n$. This naturally leads us into our next theorem regarding the infinitude of the cardinality of S in relation to $\mathcal{P}(S)$.

3.1 Quotient groups

Consider $\mathbb{Z}/n$. Under the equivalence relation congruence mod n we can take the sum of the various subsets that mod n partitions $\mathbb{Z}$ into as the law of composition given by $(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = a + b + n\mathbb{Z}$. Notice that we can let T be the set of equivalence classes under mod n, and thus we have a homomorphic map from $\mathbb{Z} \to T$. Finally, $T = \mathbb{Z}/n$.

Why stop at integers? What if we want to make congruence mod H where H is a subgroup $H \subset G$? Under what cases, given that we can make mod H equivalence classes (we will define this in a bit) is the quotient map $G \to T$ a homomorphism? This question is the question of quotient groups, and its answer—normality—is of note and importance to much of group theory (i.e. it is foundational material).

Example 3.1.1 (Subgroups of S_3)

The group S_3 has order 6 and is non-abelian (in particular, $S_3 \not\cong \mathbb{Z}/6$). Writing $S_3 = \{e, \rho, \rho^2, \mu, \rho\mu, \rho^2\mu\}$ where $\rho = (123)$ and $\mu = (12)$, the subgroups are:

- One subgroup of order 3: $\langle\rho\rangle = \{e, \rho, \rho^2\} \cong \mathbb{Z}/3$ (this is A_3, the unique normal subgroup of S_3).
- Three subgroups of order 2: $\langle\mu\rangle$, $\langle\rho\mu\rangle$, and $\langle\rho^2\mu\rangle$ (each generated by a reflection).

$$\begin{tikzcd}[column sep=small]
 & & & S_3 \arrow[dlll, dash] \arrow[dl, dash] \arrow[dr, dash] \arrow[drrr, dash] & & & \\
\langle\rho\rangle & & \langle\mu\rangle & & \langle\rho\mu\rangle & & \langle\rho^2\mu\rangle \\
 & & & \{e\} \arrow[ulll, dash] \arrow[ul, dash] \arrow[ur, dash] \arrow[urrr, dash] & & &
\end{tikzcd}$$

3.1.1 Congruence mod H and cosets

For any two elements $a, b \in G$ we define $a \sim b \iff ab^{-1} \in H$, this is to say that $a \in bH$. We thus have $a \sim a$ since $e \in H$, $a \sim b \implies ab^{-1} \in H$, and $ba^{-1} = (ab^{-1})^{-1} \in H$, thus $b \sim a$. Finally, $a \sim b, b \sim c$ meaning that ab^{-1} and bc^{-1} are both in H, so $ac^{-1} = (ab^{-1})(bc^{-1}) \in H$ since H is closed under the group operation, thus $a \sim c$ and transitivity holds. We have shown congruence mod H is indeed an equivalence relation.

Definition 3.1.1: Cosets

Given a subgroup $H \subset G$ and an element $a \in G$, the **left coset** of H containing a is the set $aH = \{ah \mid h \in H\}$. Similarly, the **right coset** is $Ha = \{ha \mid h \in H\}$. The equivalence classes under congruence mod H are precisely the left cosets of H.

Proposition 3.1.1 Cosets partition G

The left cosets of H form a partition of G. Every element $g \in G$ belongs to exactly one coset, namely gH, and two cosets aH, bH are either identical or disjoint.

Proof: Proof of coset partition

This follows immediately from the fact that cosets are equivalence classes: $g \in gH$ since $g = ge$ and $e \in H$. For disjointness, suppose $aH \cap bH \neq \emptyset$, so there exists $c \in aH \cap bH$. Then $c = ah_1 = bh_2$ for some $h_1, h_2 \in H$, which gives $a = bh_2h_1^{-1} \in bH$. For any $ah \in aH$, we have $ah = bh_2h_1^{-1}h \in bH$, so $aH \subseteq bH$. By symmetry $bH \subseteq aH$, thus $aH = bH$. $\square$

Here is the key observation that makes quotient theory work: all cosets have the same size!

Proposition 3.1.2 Cosets are equinumerous

For any $a \in G$, the map $\lambda_a : H \to aH$ given by $h \mapsto ah$ is a bijection. In particular, $|aH| = |H|$ for all $a \in G$.

Proof: Proof of coset bijection

Surjectivity is immediate from the definition of aH. For injectivity, suppose $ah_1 = ah_2$. Left-multiplying by a^{-1} gives $h_1 = h_2$. (This is just left-translation, which is always a bijection in a group—a fact we will revisit when discussing group actions.) $\square$

Theorem 3.1.1 Lagrange's Theorem

Let G be a finite group and $H \subset G$ a subgroup. Then $|H|$ divides $|G|$, and the quotient $|G|/|H| = [G : H]$ equals the number of distinct left cosets of H in G.

Proof: Proof of Lagrange's Theorem

The left cosets of H partition G into disjoint subsets, each of cardinality $|H|$. If there are k distinct cosets, then $|G| = k \cdot |H|$, so $|H| \mid |G|$ and $k = [G : H]$. $\square$

Definition 3.1.2: Index

The **index** of a subgroup H in G, denoted $[G : H]$, is the number of distinct left cosets of H in G. For finite groups, $[G : H] = |G|/|H|$.

Note:-

Lagrange's theorem has immediate corollaries: the order of any element divides $|G|$ (since $\langle g \rangle$ is a subgroup), and groups of prime order are cyclic (the only subgroups have order 1 or p). The converse of Lagrange is false in general: A_4 has order 12 but no subgroup of order 6. However, for abelian groups and more generally for solvable groups, partial converses exist (feel free to use the ol' google machine for more info).

A question naturally arises then: when is G/H a group? If we try to define $(aH)(bH) = abH$, we need this to be well-defined, meaning if $aH = a'H$ and $bH = b'H$, then $abH = a'b'H$. This does not always work!

Example 3.1.2

Consider $G = S_3$ and $H = \{e, (12)\}$. We have three left cosets: $H = \{e, (12)\}$, $(13)H = \{(13), (132)\}$, and $(23)H = \{(23), (123)\}$. Take representatives $a = (13)$ and $a' = (132)$ from the same coset; both satisfy $aH = a'H = (13)H$. Now take $b = (23)$. We compute:

$$\begin{aligned} ab &= (13)(23) = (132), \quad \text{so } abH = (132)H = (13)H \\ a'b &= (132)(23) = (13), \quad \text{so } a'bH = (13)H \end{aligned}$$

Okay, this worked (yay)! But now try $b = (13)$ and $b' = (132)$ (same coset), with $a = (23)$:

$$\begin{aligned} ab &= (23)(13) = (123), \quad \text{so } abH = (123)H = (23)H \\ ab' &= (23)(132) = (12), \quad \text{so } ab'H = (12)H = H \end{aligned}$$

Disaster! We got $(23)H \neq H$, so coset multiplication is not well-defined for this H.

What went wrong? The issue is that $H = \{e, (12)\}$ is not "symmetric" enough—left cosets and right cosets differ. For instance, $(13)H = \{(13), (132)\}$ but $H(13) = \{(13), (123)\}$.

3.1.2 Normal subgroups and quotient groups

Definition 3.1.3: Normal subgroup

A subgroup $N \subset G$ is **normal** (written $N \trianglelefteq G$) if any of the following equivalent conditions hold:

1. $gNg^{-1} = N$ for all $g \in G$ (conjugation preserves N);
2. $gN = Ng$ for all $g \in G$ (left cosets equal right cosets);
3. $gNg^{-1} \subseteq N$ for all $g \in G$ (one inclusion suffices);
4. N is the kernel of some homomorphism $\varphi : G \to H$.

Proof: Equivalence of normality conditions

(1) $\Rightarrow$ (2): If $gNg^{-1} = N$, then $gN = Ng$ by right-multiplying by g.

(2) $\Rightarrow$ (3): If $gN = Ng$, then for $n \in N$, we have $gn \in gN = Ng$, so $gn = n'g$ for some $n' \in N$, giving $gng^{-1} = n' \in N$.

(3) $\Rightarrow$ (1): We have $gNg^{-1} \subseteq N$ by assumption. For the reverse, take $n \in N$. We want $n \in gNg^{-1}$, i.e., $g^{-1}ng \in N$. But applying (3) with g^{-1}: $g^{-1}N(g^{-1})^{-1} = g^{-1}Ng \subseteq N$, so $g^{-1}ng \in N$.

(4) $\Rightarrow$ (1): If $N = \ker(\varphi)$, then for $n \in N$ and $g \in G$: $\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g)^{-1} = \varphi(g)e_H\varphi(g)^{-1} = e_H$, so $gng^{-1} \in \ker(\varphi) = N$.

(1) $\Rightarrow$ (4): This is the content of the next theorem—every normal subgroup is a kernel! □

Theorem 3.1.2 Quotient groups exist for normal subgroups

If $N \trianglelefteq G$, then G/N is a group under the operation $(aN)(bN) = abN$, called the **quotient group**. The map $\pi : G \to G/N$ given by $\pi(g) = gN$ is a surjective homomorphism (the **canonical projection**) with $\ker(\pi) = N$.

Proof: Proof of quotient group structure

Well-definedness: Suppose $aN = a'N$ and $bN = b'N$, so $a' = an_1$ and $b' = bn_2$ for some $n_1, n_2 \in N$. Then

$$a'b' = an_1bn_2 = a(n_1b)n_2.$$

Since N is normal, $n_1b = bn_3$ for some $n_3 \in N$. We have $n_1b \in Nb = bN$, so $n_1b = bn_3$ for some $n_3 \in N$. Thus $a'b' = abn_3n_2 \in abN$, so $a'b'N = abN$.

Group axioms: Associativity: $(aN \cdot bN) \cdot cN = abN \cdot cN = (ab)cN = a(bc)N = aN \cdot bcN = aN \cdot (bN \cdot cN)$. Identity: $eN = N$ satisfies $N \cdot aN = eaN = aN$. Inverses: $(aN)^{-1} = a^{-1}N$ since $aN \cdot a^{-1}N = aa^{-1}N = N$.

The projection: $\pi(ab) = abN = aN \cdot bN = \pi(a)\pi(b)$, so π is a homomorphism. Surjectivity is clear. Finally, $\ker(\pi) = \{g \in G : gN = N\} = \{g \in G : g \in N\} = N$. □

Question 1

Let G be a group and $Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}$ the center of G.

1. Prove that $Z(G) \trianglelefteq G$.
2. Prove that if $G/Z(G)$ is cyclic, then G is abelian.
3. Find $Z(D_4)$ and describe $D_4/Z(D_4)$.

Solution

(1) We verify normality via condition (1). For any $z \in Z(G)$ and $g \in G$, we have $gzg^{-1} = zgg^{-1} = z \in Z(G)$ (using that z commutes with everything). Thus $gZ(G)g^{-1} \subseteq Z(G)$ for all g, so $Z(G) \trianglelefteq G$.
(2) Suppose $G/Z(G) = \langle aZ(G) \rangle$ is cyclic. Then every element of G has the form a^kz for some $k \in \mathbb{Z}$ and $z \in Z(G)$. Take any two elements $g = a^iz_1$ and $h = a^jz_2$. Then:

$$gh = a^iz_1a^jz_2 = a^ia^jz_1z_2 = a^{i+j}z_1z_2$$

(since z_1 commutes with a^j and z_2), and similarly $hg = a^{j+i}z_2z_1 = a^{i+j}z_1z_2 = gh$ (since z_1, z_2 commute with everything and with each other). Thus G is abelian.
(3) Recall $D_4 = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\}$ with $r^4 = s^2 = e$ and $srs = r^{-1}$. For $z \in Z(D_4)$, we need $zr = rz$ and $zs = sz$.
For rotations: $r^k \in Z(D_4)$ requires $sr^k = r^ks$. We have $sr^k = r^{-k}s$ (by induction using $srs = r^{-1}$), so $r^{-k}s = r^ks$, giving $r^{2k} = e$, so $k \in \{0, 2\}$. Thus e, r^2 are central among rotations.
For reflections: $sr^j \in Z(D_4)$ requires $r(sr^j) = (sr^j)r$, i.e., $rsr^j = sr^{j+1}$. But $rsr^j = r \cdot r^{-j}s = r^{1-j}s$, so we need $r^{1-j}s = sr^{j+1}$, giving $r^{1-j} = r^{-(j+1)} = r^{-j-1}$, so $r^2 = e$, contradiction since r has order 4. No reflections are central.

Therefore $Z(D_4) = \{e, r^2\} \cong \mathbb{Z}/2$. The quotient $D_4/Z(D_4)$ has order $8/2 = 4$. The cosets are $\{e, r^2\}, \{r, r^3\}, \{s, sr^2\}, \{sr, sr^3\}$. One can verify this is $\cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (the Klein four-group), not $\mathbb{Z}/4$, since every non-identity element has order 2 in the quotient.

As a final remark before moving onto our next topic, I wish to introduce the concept of simple groups. Simply put, no pun intended, a simple group is a group that cannot be decomposed into further normal subgroups other than the trivial subgroup and itself. Formally:

Definition 3.1.4: Simple groups

We say a group is simple if it has no normal subgroups other than itself and the trivial subgroup.

3.2 Group actions and symmetry

The notion of a group "acting" on a set is one of the main "substances" which underlies much of our study of mathematics. When we study symmetries of geometric objects, solutions to polynomial equations, or even the structure of groups themselves, we are studying group actions. This section develops the theory systematically—pay close attention, as these ideas will reappear throughout when we discuss representation theory (where groups act on vector spaces via linear maps).

3.2.1 Group actions

Definition 3.2.1: Group action

A **(left) group action** of a group G on a set X is a map $G \times X \to X$, written $(g, x) \mapsto g \cdot x$, satisfying:

1. $e \cdot x = x$ for all $x \in X$ (identity acts trivially);
2. $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$ (compatibility).

We say G **acts on** X, or that X is a **G-set**.

Note:-

There is an equivalent, perhaps more fun, formulation: a group action is a homomorphism $\rho : G \to \mathrm{Sym}(X)$, where $\mathrm{Sym}(X)$ is the group of all bijections $X \to X$. Given an action, define $\rho(g)(x) = g \cdot x$; the axioms ensure each $\rho(g)$ is a bijection (with inverse $\rho(g^{-1})$) and that ρ is a homomorphism. Conversely, any such homomorphism defines an action. This perspective is crucial: group actions are the same as representations of G into symmetric groups.

Example 3.2.1

Some fundamental examples of group actions:

1. **Left multiplication:** G acts on itself by $g \cdot h = gh$. This action is always faithful (only e acts trivially) and transitive (any element can be sent to any other).
2. **Conjugation:** G acts on itself by $g \cdot h = ghg^{-1}$. The kernel is $Z(G)$, the center.
3. **Conjugation on subgroups:** G acts on the set of its subgroups by $g \cdot H = gHg^{-1}$. Normal subgroups are precisely the fixed points!
4. **Coset action:** If $H \subset G$, then G acts on G/H (left cosets) by $g \cdot (aH) = (ga)H$.
5. **Linear actions:** $\mathrm{GL}_n(\mathbb{R})$ acts on $\mathbb{R}^n$ by matrix multiplication. This is the prototype for representation theory.

Definition 3.2.2: Orbits and stabilizers

Let G act on X. For $x \in X$:

1. The **orbit** of x is $G \cdot x = \{g \cdot x : g \in G\} \subseteq X$.
2. The **stabilizer** (or **isotropy group**) of x is $G_x = \mathrm{Stab}_G(x) = \{g \in G : g \cdot x = x\} \subseteq G$.

Proposition 3.2.1 Stabilizers are subgroups

For any $x \in X$, the stabilizer G_x is a subgroup of G.

Proof: Proof that stabilizers are subgroups

Identity: $e \cdot x = x$, so $e \in G_x$. Closure: if $g, h \in G_x$, then $(gh) \cdot x = g \cdot (h \cdot x) = g \cdot x = x$, so $gh \in G_x$. Inverses: if $g \in G_x$, then $x = e \cdot x = (g^{-1}g) \cdot x = g^{-1} \cdot (g \cdot x) = g^{-1} \cdot x$, so $g^{-1} \in G_x$. □

Proposition 3.2.2 Orbits partition X

The orbits of a group action form a partition of X. Define $x \sim y$ iff $y = g \cdot x$ for some $g \in G$; this is an equivalence relation whose equivalence classes are exactly the orbits.

Proof: Proof that orbits partition

Reflexivity: $x = e \cdot x$, so $x \sim x$. Symmetry: if $y = g \cdot x$, then $x = g^{-1} \cdot y$, so $y \sim x$ implies $x \sim y$. Transitivity: if $y = g \cdot x$ and $z = h \cdot y$, then $z = h \cdot (g \cdot x) = (hg) \cdot x$, so $x \sim z$. □

The following theorem is one of the most useful results in all of group theory. It connects the size of an orbit to the index of a stabilizer.

Theorem 3.2.1 Orbit-Stabilizer Theorem

Let G act on X and let $x \in X$. There is a bijection between the orbit $G \cdot x$ and the set of left cosets G/G_x, given by $g \cdot x \mapsto gG_x$. In particular, if G is finite,

$$|G \cdot x| = [G : G_x] = \frac{|G|}{|G_x|}.$$

Proof: Proof of Orbit-Stabilizer

Define $\varphi : G/G_x \to G \cdot x$ by $\varphi(gG_x) = g \cdot x$.

Well-defined: If $gG_x = hG_x$, then $h = gk$ for some $k \in G_x$, so $h \cdot x = (gk) \cdot x = g \cdot (k \cdot x) = g \cdot x$.

Injective: If $g \cdot x = h \cdot x$, then $h^{-1}g \cdot x = x$, so $h^{-1}g \in G_x$, giving $gG_x = hG_x$.

Surjective: Every element of $G \cdot x$ has the form $g \cdot x = \varphi(gG_x)$. □

Example 3.2.2

Consider S_4 acting on $\{1, 2, 3, 4\}$ in the natural way. Take $x = 1$. The orbit is $S_4 \cdot 1 = \{1, 2, 3, 4\}$ (any element can be sent to any other). The stabilizer is $(S_4)_1 = \{\sigma \in S_4 : \sigma(1) = 1\}$, which consists of all permutations of $\{2, 3, 4\}$, so $(S_4)_1 \cong S_3$ and $|(S_4)_1| = 6$. Check: $|S_4 \cdot 1| = 4 = 24/6 = |S_4|/|(S_4)_1|$. ✓

3.2.2 Cayley's theorem and applications

Theorem 3.2.2 Cayley's Theorem

Every group G is isomorphic to a subgroup of $\mathrm{Sym}(G)$. If $|G| = n$, then G embeds into S_n.

Proof: Proof of Cayley's Theorem

Consider the left multiplication action of G on itself: $g \cdot h = gh$. This gives a homomorphism $\rho : G \to \mathrm{Sym}(G)$ where $\rho(g)$ is the bijection $h \mapsto gh$.

Injectivity: If $\rho(g) = \rho(g')$, then for all $h \in G$, $gh = g'h$. Taking $h = e$: $g = g'$. Thus $\ker(\rho) = \{e\}$, so ρ is injective.

By the First Isomorphism Theorem, $G \cong \mathrm{Im}(\rho) \subset \mathrm{Sym}(G)$. If $|G| = n$, then $\mathrm{Sym}(G) \cong S_n$. $\square$

Note:-

Cayley's theorem is simultaneously profound and useless. Profound: every abstract group is "really" a permutation group; the group axioms capture exactly the structure of bijection composition. Useless: the embedding is almost always far larger than necessary. S_n has order $n!$, so embedding a group of order n into S_n wastes enormous space. Better embeddings come from finding smaller faithful actions. 'Tis fun nonetheless.

Proposition 3.2.3 Improved Cayley embedding

Let $H \subset G$ with $[G : H] = n$. The action of G on G/H gives a homomorphism $\rho : G \to S_n$ with $\ker(\rho) = \bigcap_{g \in G} gHg^{-1}$, the largest normal subgroup of G contained in H.

Proof: Proof of improved embedding

The action $g \cdot (aH) = (ga)H$ is well-defined (check!). The kernel consists of all g fixing every coset: $gaH = aH$ for all a, i.e., $a^{-1}ga \in H$ for all a, i.e., $g \in aHa^{-1}$ for all a. Thus $\ker(\rho) = \bigcap_{a \in G} aHa^{-1}$. $\square$

Example 3.2.3

Let $G = S_4$ and $H = S_3$ (permutations fixing 4). Then $[G : H] = 4$, and the action on G/H identifies with the natural action on $\{1, 2, 3, 4\}$. The kernel is trivial (since the natural action is faithful), recovering the identity $S_4 \hookrightarrow S_4$. But: if we take $G = A_5$ and $H = A_4$ (stabilizer of a point), then $[G : H] = 5$ and we get $A_5 \hookrightarrow S_5$. Since $|A_5| = 60 < 120 = |S_5|$, this is a non-surjective embedding, showing A_5 as a proper subgroup of S_5.

Question 2

Let G be a group of order $2p$ where p is an odd prime.

1. Prove that G has a unique subgroup of order p, which is therefore normal.
2. Conclude that $G \cong \mathbb{Z}/2p$ or $G \cong D_p$ (the dihedral group of order $2p$).

Solution

(1) By Cauchy's theorem (or direct counting), G has an element a of order p, so $H = \langle a \rangle$ is a subgroup of order p. Since $[G : H] = 2$, we have $H \trianglelefteq G$ (index 2 subgroups are always normal: $G = H \sqcup gH = H \sqcup Hg$ for any $g \notin H$, so $gH = Hg$).
For uniqueness: suppose K is another subgroup of order p. Since $|H| = |K| = p$ is prime, $H \cap K$ is a subgroup of both with order dividing p, so $|H \cap K| \in \{1, p\}$. If $|H \cap K| = p$, then $H = K$. Otherwise, $H \cap K = \{e\}$, and since elements of H and K (excluding e) have order p, we have $|H \cup K| = 1 + (p-1) + (p-1) = 2p-1$ elements of orders 1 or p. But then G has only one element not in $H \cup K$, call

it b. We must have $b^2 \in H \cup K$ (since $b \notin H \cup K$), but b has order dividing $2p$... this gets complicated. Easier: if $H \neq K$, then HK is a subgroup (since $H \trianglelefteq G$) of order $|H||K|/|H \cap K| = p^2 > 2p = |G|$, contradiction.

(2) Let $H = \langle a \rangle \cong \mathbb{Z}/p$ be the unique subgroup of order p. Pick $b \in G \setminus H$; then b has order 2 (if $|b| = p$, then $b \in H$ by uniqueness; if $|b| = 2p$, then $G = \langle b \rangle$ is cyclic and $G \cong \mathbb{Z}/2p$).

Assume $|b| = 2$. Since $H \trianglelefteq G$, conjugation by b gives an automorphism of $H \cong \mathbb{Z}/p$. We have $bab^{-1} = a^k$ for some k. Since $b^2 = e$: $a = b^2ab^{-2} = b(bab^{-1})b^{-1} = ba^kb^{-1} = (bab^{-1})^k = a^{k^2}$, so $k^2 \equiv 1 \pmod p$, giving $k \equiv \pm 1 \pmod p$.

If $k \equiv 1$: then $bab^{-1} = a$, so $ab = ba$, and $G = \langle a, b \rangle$ is abelian of order $2p$ with $|a| = p, |b| = 2$, so $G \cong \mathbb{Z}/p \times \mathbb{Z}/2 \cong \mathbb{Z}/2p$.

If $k \equiv -1$: then $bab^{-1} = a^{-1}$, i.e., $ba = a^{-1}b$. This is exactly the dihedral relation! We have $G = \langle a, b : a^p = b^2 = e,\ bab^{-1} = a^{-1} \rangle \cong D_p$.

3.3 The isomorphism theorems

The isomorphism theorems we have already introduced somewhat. They describe how quotients, subgroups, and homomorphisms interact. These results will reappear essentially unchanged when we study rings, modules, and vector spaces—they are instances of a more general phenomenon that category theory makes precise.

3.3.1 Kernels, images, and the First Isomorphism Theorem

Definition 3.3.1: Kernel

The **kernel** of a group homomorphism $\varphi : G \to H$ is

$$\ker(\varphi) = \{g \in G : \varphi(g) = e_H\} = \varphi^{-1}(\{e_H\}).$$

Definition 3.3.2: Image

The **image** of a group homomorphism $\varphi : G \to H$ is

$$\mathrm{Im}(\varphi) = \varphi(G) = \{h \in H : \exists g \in G,\ \varphi(g) = h\}.$$

Proposition 3.3.1 Basic properties of kernels and images

Let $\varphi : G \to H$ be a homomorphism.

1. $\ker(\varphi)$ is a normal subgroup of G.
2. $\mathrm{Im}(\varphi)$ is a subgroup of H (not necessarily normal).
3. φ is injective $\iff \ker(\varphi) = \{e_G\}$.
4. $\varphi(g_1) = \varphi(g_2) \iff g_1^{-1}g_2 \in \ker(\varphi) \iff g_1 \ker(\varphi) = g_2 \ker(\varphi)$.

Proof: Proofs of kernel/image properties

(1) First, $\ker(\varphi)$ is a subgroup: $\varphi(e) = e$, so $e \in \ker(\varphi)$; if $\varphi(a) = \varphi(b) = e$, then $\varphi(ab) = ee = e$; if $\varphi(a) = e$, then $\varphi(a^{-1}) = \varphi(a)^{-1} = e^{-1} = e$. For normality: if $k \in \ker(\varphi)$, then $\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g)^{-1} = \varphi(g) \cdot e \cdot \varphi(g)^{-1} = e$.

(2) Subgroup: $e = \varphi(e) \in \mathrm{Im}(\varphi)$; if $\varphi(a), \varphi(b) \in \mathrm{Im}(\varphi)$, then $\varphi(a)\varphi(b) = \varphi(ab) \in \mathrm{Im}(\varphi)$; $\varphi(a)^{-1} = \varphi(a^{-1}) \in \mathrm{Im}(\varphi)$.

(3) If $\ker(\varphi) = \{e\}$ and $\varphi(g_1) = \varphi(g_2)$, then $\varphi(g_1^{-1}g_2) = e$, so $g_1^{-1}g_2 = e$, giving $g_1 = g_2$. Conversely, if φ is injective and $k \in \ker(\varphi)$, then $\varphi(k) = e = \varphi(e)$, so $k = e$.

(4) $\varphi(g_1) = \varphi(g_2) \iff \varphi(g_1)^{-1}\varphi(g_2) = e \iff \varphi(g_1^{-1}g_2) = e \iff g_1^{-1}g_2 \in \ker(\varphi)$. The last equivalence with cosets is the definition of coset equality. □

Theorem 3.3.1 First Isomorphism Theorem

Let $\varphi : G \to H$ be a homomorphism. Then:

1. $\ker(\varphi) \trianglelefteq G$;
2. The map $\bar{\varphi} : G/\ker(\varphi) \to H$ defined by $\bar{\varphi}(g \cdot \ker(\varphi)) = \varphi(g)$ is a well-defined injective homomorphism;
3. $\mathrm{Im}(\bar{\varphi}) = \mathrm{Im}(\varphi)$, so $G/\ker(\varphi) \cong \mathrm{Im}(\varphi)$.

In slogan form: the image of a homomorphism is isomorphic to the domain modulo the kernel.

Proof: Proof of First Isomorphism Theorem

Part (1) was proved above. For (2): $\bar{\varphi}$ is well-defined because $g_1\ker(\varphi) = g_2\ker(\varphi)$ implies $\varphi(g_1) = \varphi(g_2)$ by property (4). It's a homomorphism: $\bar{\varphi}(g_1K \cdot g_2K) = \bar{\varphi}(g_1g_2K) = \varphi(g_1g_2) = \varphi(g_1)\varphi(g_2) = \bar{\varphi}(g_1K)\bar{\varphi}(g_2K)$ (writing $K = \ker(\varphi)$). It's injective: $\bar{\varphi}(gK) = e \implies \varphi(g) = e \implies g \in K \implies gK = K$.

For (3): $\mathrm{Im}(\bar{\varphi}) = \{\bar{\varphi}(gK) : g \in G\} = \{\varphi(g) : g \in G\} = \mathrm{Im}(\varphi)$. Since $\bar{\varphi}$ is an injective homomorphism onto $\mathrm{Im}(\varphi)$, it's an isomorphism $G/\ker(\varphi) \xrightarrow{\sim} \mathrm{Im}(\varphi)$. □

Note:-

The First Isomorphism Theorem is sometimes called the "fundamental theorem of homomorphisms." It says that every homomorphism $\varphi : G \to H$ factors as $G \twoheadrightarrow G/\ker(\varphi) \xrightarrow{\sim} \mathrm{Im}(\varphi) \hookrightarrow H$: a surjection, followed by an isomorphism, followed by an inclusion. This factorization is unique and natural—in the categorical sense, which you'll see later.

3.3.2 The Second and Third Isomorphism Theorems

Theorem 3.3.2 Second Isomorphism Theorem (Diamond Theorem)

Let G be a group, $H \subset G$ a subgroup, and $N \trianglelefteq G$ a normal subgroup. Then:

1. $HN = \{hn : h \in H, n \in N\}$ is a subgroup of G;
2. $H \cap N \trianglelefteq H$;
3. $H/(H \cap N) \cong HN/N$.

Proof: Proof of Second Isomorphism Theorem

(1) HN is a subgroup: $e = e \cdot e \in HN$. For closure: $(h_1n_1)(h_2n_2) = h_1(n_1h_2)n_2$. Since N is normal, $n_1h_2 = h_2n_3$ for some $n_3 \in N$, so $(h_1n_1)(h_2n_2) = h_1h_2n_3n_2 \in HN$. For inverses: $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}(hn^{-1}h^{-1}) \in HN$ since $hn^{-1}h^{-1} \in N$ by normality.

(2) $H \cap N$ is a subgroup of H (intersection of subgroups). For normality in H: if $x \in H \cap N$ and $h \in H$, then $hxh^{-1} \in H$ (since H is a subgroup) and $hxh^{-1} \in N$ (since $N \trianglelefteq G$), so $hxh^{-1} \in H \cap N$.

(3) Define $\varphi : H \to HN/N$ by $\varphi(h) = hN$. This is a homomorphism: $\varphi(h_1h_2) = h_1h_2N = (h_1N)(h_2N) = \varphi(h_1)\varphi(h_2)$.

Surjectivity: Any element of HN/N has the form $hnN = hN$ (since $n \in N$), which equals $\varphi(h)$.

Kernel: $\ker(\varphi) = \{h \in H : hN = N\} = \{h \in H : h \in N\} = H \cap N$.

By the First Isomorphism Theorem, $H/(H \cap N) = H/\ker(\varphi) \cong \mathrm{Im}(\varphi) = HN/N$. □

Note:-

The name "Diamond Theorem" comes from the lattice diagram: draw G at top, HN below it, H and N below HN on left and right, and $H \cap N$ at the bottom. The isomorphism $H/(H \cap N) \cong HN/N$ says the "left edge" quotient equals the "right edge" quotient. This is a manifestation of a general principle called the modular law in lattice theory.

Theorem 3.3.3 Third Isomorphism Theorem

Let G be a group and $N \subset M$ be normal subgroups of G (so $N \trianglelefteq G$ and $M \trianglelefteq G$ with $N \subset M$). Then:

1. $M/N \trianglelefteq G/N$;
2. $(G/N)/(M/N) \cong G/M$.

Proof: Proof of Third Isomorphism Theorem

(1) Elements of G/N are cosets gN; elements of M/N are cosets mN for $m \in M$. We verify $M/N \trianglelefteq G/N$: for $gN \in G/N$ and $mN \in M/N$, we have $(gN)(mN)(gN)^{-1} = gmg^{-1}N$. Since $M \trianglelefteq G$, $gmg^{-1} \in M$, so $gmg^{-1}N \in M/N$.

(2) Define $\pi : G/N \to G/M$ by $\pi(gN) = gM$.

Well-defined: If $g_1N = g_2N$, then $g_1^{-1}g_2 \in N \subset M$, so $g_1M = g_2M$.

Homomorphism: $\pi(g_1N \cdot g_2N) = \pi(g_1g_2N) = g_1g_2M = (g_1M)(g_2M) = \pi(g_1N)\pi(g_2N)$.

Surjective: For any $gM \in G/M$, we have $\pi(gN) = gM$.

Kernel: $\ker(\pi) = \{gN : gM = M\} = \{gN : g \in M\} = M/N$.

By the First Isomorphism Theorem, $(G/N)/\ker(\pi) = (G/N)/(M/N) \cong \mathrm{Im}(\pi) = G/M$. □

Note:-

The Third Isomorphism Theorem says "quotients of quotients are quotients": $(G/N)/(M/N) \cong G/M$. The N's "cancel." This is exactly what you'd hope for, and it's a sign that quotient groups are a natural and well-behaved construction.

3.3.3 The Correspondence Theorem

The final structural theorem describes the relationship between subgroups of a quotient and subgroups of the original group.

Theorem 3.3.4 Correspondence Theorem (Fourth Isomorphism Theorem)

Let $N \trianglelefteq G$ and $\pi : G \to G/N$ be the canonical projection. There is a bijection

$$\{\text{subgroups of } G \text{ containing } N\} \longleftrightarrow \{\text{subgroups of } G/N\}$$

given by $H \mapsto H/N = \pi(H)$ with inverse $\bar{H} \mapsto \pi^{-1}(\bar{H})$. Moreover:

1. $H_1 \subset H_2 \iff H_1/N \subset H_2/N$, and $[H_2 : H_1] = [H_2/N : H_1/N]$;
2. $H \trianglelefteq G \iff H/N \trianglelefteq G/N$.

Proof: Proof of Correspondence Theorem

Given $H \supset N$, we have $H/N = \{hN : h \in H\}$, which is a subgroup of G/N. Given a subgroup $\bar{H} \subset G/N$, define $H = \pi^{-1}(\bar{H}) = \{g \in G : gN \in \bar{H}\}$. This contains N (since $eN = N \in \bar{H}$) and is a subgroup.

These operations are inverses: starting with $H \supset N$, we get $\pi^{-1}(H/N) = \{g : gN \in H/N\} = \{g : gN = hN \text{ for some } h \in H\} = \{g : g \in hN \text{ for some } h \in H\} = HN = H$ (since $N \subset H$).

Starting with $\bar{H} \subset G/N$: $\pi(\pi^{-1}(\bar{H})) = \{gN : gN \in \bar{H}\} = \bar{H}$.

(1) $H_1 \subset H_2 \implies H_1/N \subset H_2/N$ is clear. Conversely, if $H_1/N \subset H_2/N$ and $h \in H_1$, then $hN \in H_1/N \subset H_2/N$, so $hN = h'N$ for some $h' \in H_2$, giving $h \in h'N \subset H_2$. The index statement follows since the correspondence preserves cosets.

(2) If $H \trianglelefteq G$, then for $gN \in G/N$ and $hN \in H/N$: $(gN)(hN)(gN)^{-1} = ghg^{-1}N \in H/N$ since $ghg^{-1} \in H$. Conversely, if $H/N \trianglelefteq G/N$, then for $g \in G, h \in H$: $ghg^{-1}N = (gN)(hN)(gN)^{-1} \in H/N$, so $ghg^{-1} \in H$. $\square$

Question 3

Let $G = \mathbb{Z}$ and $N = 12\mathbb{Z}$.

1. List all subgroups of $G/N \cong \mathbb{Z}/12$.
2. For each, find the corresponding subgroup of $\mathbb{Z}$ containing $12\mathbb{Z}$.
3. Which of these are normal in $\mathbb{Z}$? (Trick question!)
4. Verify the Third Isomorphism Theorem: check that $(\mathbb{Z}/12\mathbb{Z})/(4\mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$.

Solution

(1) $\mathbb{Z}/12$ is cyclic of order 12, so its subgroups correspond to divisors of 12: they are $\langle\bar{0}\rangle, \langle\bar{6}\rangle, \langle\bar{4}\rangle, \langle\bar{3}\rangle, \langle\bar{2}\rangle, \langle\bar{1}\rangle$ of orders 1, 2, 3, 4, 6, 12 respectively.
(2) The corresponding subgroups of $\mathbb{Z}$ are those containing $12\mathbb{Z}$, i.e., of the form $d\mathbb{Z}$ where $d \mid 12$:

$$\langle\bar{0}\rangle \leftrightarrow 12\mathbb{Z}, \quad \langle\bar{6}\rangle \leftrightarrow 6\mathbb{Z}, \quad \langle\bar{4}\rangle \leftrightarrow 4\mathbb{Z},$$
$$\langle\bar{3}\rangle \leftrightarrow 3\mathbb{Z}, \quad \langle\bar{2}\rangle \leftrightarrow 2\mathbb{Z}, \quad \langle\bar{1}\rangle \leftrightarrow \mathbb{Z}.$$

(3) All subgroups of $\mathbb{Z}$ are normal because $\mathbb{Z}$ is abelian!
(4) We have $4\mathbb{Z}/12\mathbb{Z} = \{\bar{0}, \bar{4}, \bar{8}\} \cong \mathbb{Z}/3$. The quotient $(\mathbb{Z}/12)/(4\mathbb{Z}/12\mathbb{Z})$ has order $12/3 = 4$. Explicitly, the cosets are $\bar{0} + 4\mathbb{Z}/12\mathbb{Z} = \{\bar{0}, \bar{4}, \bar{8}\}$, $\bar{1} + \cdots = \{\bar{1}, \bar{5}, \bar{9}\}$, $\bar{2} + \cdots = \{\bar{2}, \bar{6}, \overline{10}\}$, $\bar{3} + \cdots = \{\bar{3}, \bar{7}, \overline{11}\}$. This is cyclic of order 4, generated by $\bar{1} + 4\mathbb{Z}/12\mathbb{Z}$, hence $\cong \mathbb{Z}/4$.
The Third Isomorphism Theorem says $(\mathbb{Z}/12\mathbb{Z})/(4\mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$, which checks out!

3.4 The symmetric group

The symmetric group S_n is arguably the most important finite group. By Cayley's theorem, every finite group embeds in some S_n. But S_n is also interesting in its own right: its structure illuminates concepts like conjugacy classes, normal subgroups, and simplicity. Moreover, symmetric groups connect to linear algebra through the sign homomorphism and determinants, and to Galois theory through their role in permuting roots of polynomials.

3.4.1 Cycle structure and decomposition

Definition 3.4.1: Symmetric group

The **symmetric group** S_n is the group of all bijections $\sigma : \{1, \dots, n\} \to \{1, \dots, n\}$ under composition. It has order $n!$.

Definition 3.4.2: Cycle notation

A **k-cycle** (or cycle of length k) is a permutation $(a_1\, a_2\, \cdots\, a_k)$ that acts by $a_1 \mapsto a_2 \mapsto \cdots \mapsto a_k \mapsto a_1$ and fixes all other elements. A 2-cycle is called a **transposition**. Two cycles are **disjoint** if they have no elements in common; disjoint cycles commute.

Theorem 3.4.1 Cycle Decomposition Theorem

Every permutation $\sigma \in S_n$ can be written as a product of disjoint cycles, uniquely up to the order of the factors. The decomposition is obtained by writing out the orbits of σ acting on $\{1, \ldots, n\}$.

Proof: Existence and uniqueness of cycle decomposition

Existence: Consider the action of $\langle\sigma\rangle$ on $\{1, \ldots, n\}$. The orbits partition $\{1, \ldots, n\}$. For each orbit $\{a_1, a_2, \ldots, a_k\}$ where $\sigma(a_i) = a_{i+1}$ (indices mod k), define the cycle $c = (a_1\, a_2 \cdots a_k)$. Fixed points give 1-cycles (usually omitted). Since orbits are disjoint, these cycles are disjoint.

To verify σ equals the product of these cycles: for any i, if i is in the orbit giving cycle c, then $c(i) = \sigma(i)$ and all other cycles fix i, so the product sends $i \mapsto \sigma(i)$.

Uniqueness: Suppose $\sigma = c_1 c_2 \cdots c_r = d_1 d_2 \cdots d_s$ are two decompositions into disjoint cycles. For any i, exactly one c_j moves i (the others fix it), and that c_j must generate the orbit of i under σ. Similarly for the d's. Thus c_j and the d_k containing i describe the same orbit, hence are the same cycle (up to cyclic rotation of the notation). □

Definition 3.4.3: Cycle type

The **cycle type** of $\sigma \in S_n$ is the partition of n given by the lengths of the cycles in its disjoint cycle decomposition (including 1-cycles for fixed points). We write it as $(1^{m_1} 2^{m_2} \cdots n^{m_n})$ where m_k is the number of k-cycles.

Example 3.4.1

In S_6, the permutation $\sigma = (1\,3\,5)(2\,4)$ has cycle type $(1^1 2^1 3^1)$ since there's one fixed point (6), one 2-cycle, and one 3-cycle. The identity has cycle type (1^6). A single 6-cycle has type (6^1).

3.4.2 Conjugacy in S_n

Conjugacy classes are fundamental to representation theory. In S_n, they have an elegant characterization.

Theorem 3.4.2 Conjugacy in S_n

Two permutations in S_n are conjugate if and only if they have the same cycle type.

Proof: Proof of conjugacy classification

Key computation: Let $\tau \in S_n$ and $(a_1\, a_2 \cdots a_k)$ be a cycle. Then

$$\tau(a_1\, a_2 \cdots a_k)\tau^{-1} = (\tau(a_1)\, \tau(a_2) \cdots \tau(a_k)).$$

Proof: For any i, if $i = \tau(a_j)$, then $\tau(a_1 \cdots a_k)\tau^{-1}(i) = \tau(a_{j+1}) = \tau(a_{j+1})$. If $i \notin \{\tau(a_1), \ldots, \tau(a_k)\}$, then $\tau^{-1}(i) \notin \{a_1, \ldots, a_k\}$, so the cycle fixes $\tau^{-1}(i)$, and conjugation gives $\tau(\tau^{-1}(i)) = i$.

Same type ⇒ conjugate: Let $\sigma = (a_1 \cdots a_{k_1})(b_1 \cdots b_{k_2}) \cdots$ and $\rho = (c_1 \cdots c_{k_1})(d_1 \cdots d_{k_2}) \cdots$ have the same cycle type. Define τ by $\tau(a_i) = c_i, \tau(b_j) = d_j$, etc. Then by the key computation, $\tau\sigma\tau^{-1} = \rho$.

Conjugate ⇒ same type: If $\rho = \tau\sigma\tau^{-1}$, write σ as a product of disjoint cycles. By the key computation, ρ is the product of the conjugated cycles, which have the same lengths. Disjointness is preserved since τ is a bijection. □

Proposition 3.4.1 Number of conjugacy classes in S_n

The number of conjugacy classes in S_n equals $p(n)$, the number of partitions of n. For example, $p(1) = 1, p(2) = 2, p(3) = 3, p(4) = 5, p(5) = 7, p(6) = 11$.

Note:-

This fact that conjugacy classes correspond to partitions is one reason why the representation theory of S_n is so whimsical. The irreducible representations of S_n are also indexed by partitions of n, and the character table can be computed combinatorially. This is a whole subject in itself.

3.4.3 The sign homomorphism and alternating group

Every permutation can be written as a product of transpositions. The number of transpositions is not unique, but its parity is!

Proposition 3.4.2 Transposition generation

S_n is generated by transpositions. More precisely, every k-cycle can be written as a product of $k-1$ transpositions:

$$(a_1\, a_2\, \cdots\, a_k) = (a_1\, a_k)(a_1\, a_{k-1}) \cdots (a_1\, a_2).$$

Proof: Verification of transposition decomposition

Apply the right side to a_1: $(a_1\, a_2)$ sends $a_1 \mapsto a_2$, then subsequent transpositions fix a_2. So $a_1 \mapsto a_2$. ✓

Apply to a_j for $2 \leqslant j \leqslant k-1$: transpositions $(a_1\, a_2), \ldots, (a_1\, a_{j-1})$ fix a_j. Then $(a_1\, a_j)$ sends $a_j \mapsto a_1$. Then $(a_1\, a_{j+1})$ sends $a_1 \mapsto a_{j+1}$. Subsequent transpositions fix a_{j+1}. So $a_j \mapsto a_{j+1}$. ✓

Apply to a_k: all transpositions except $(a_1\, a_k)$ fix a_k, which sends $a_k \mapsto a_1$. ✓ □

Theorem 3.4.3 Well-definedness of parity

If $\sigma = \tau_1 \cdots \tau_r = \tau_1' \cdots \tau_s'$ are two expressions of σ as products of transpositions, then $r \equiv s \pmod 2$.

Proof: Proof of parity well-definedness

Consider the polynomial $P(x_1, \ldots, x_n) = \prod_{1 \leqslant i < j \leqslant n}(x_i - x_j)$. For $\sigma \in S_n$, define $\sigma \cdot P = \prod_{i<j}(x_{\sigma(i)} - x_{\sigma(j)})$. Each factor $(x_{\sigma(i)} - x_{\sigma(j)})$ equals $\pm(x_k - x_l)$ for some $k < l$, so $\sigma \cdot P = \pm P$.

For a transposition $\tau = (a\, b)$ with $a < b$: the factor $(x_a - x_b)$ becomes $(x_b - x_a) = -(x_a - x_b)$. For $i < a < b < j$, the factor $(x_i - x_j)$ is unchanged. For $i < a$, the factors $(x_i - x_a)$ and $(x_i - x_b)$ are swapped, contributing no sign change. Similarly for other cases. Careful counting shows $\tau \cdot P = -P$ (the key is that one factor picks up a sign, and swaps come in pairs).

Thus if $\sigma = \tau_1 \cdots \tau_r$, then $\sigma \cdot P = (-1)^r P$. If also $\sigma = \tau_1' \cdots \tau_s'$, then $(-1)^r = (-1)^s$, so $r \equiv s \pmod 2$. □

Definition 3.4.4: Sign of a permutation

The **sign** (or **signature**) of $\sigma \in S_n$ is $\operatorname{sgn}(\sigma) = (-1)^r$ where $\sigma = \tau_1 \cdots \tau_r$ is any expression as a product of transpositions. Equivalently, $\operatorname{sgn}(\sigma) = (-1)^{n-c(\sigma)}$ where $c(\sigma)$ is the number of cycles (including 1-cycles). A permutation is **even** if $\operatorname{sgn}(\sigma) = 1$ and **odd** if $\operatorname{sgn}(\sigma) = -1$.

Theorem 3.4.4 Sign is a homomorphism

The map $\operatorname{sgn} : S_n \to \{1, -1\}$ is a surjective group homomorphism.

Proof: Proof that sign is a homomorphism

If $\sigma = \tau_1 \cdots \tau_r$ and $\rho = \tau_1' \cdots \tau_s'$, then $\sigma\rho = \tau_1 \cdots \tau_r \tau_1' \cdots \tau_s'$, a product of $r + s$ transpositions. Thus $\operatorname{sgn}(\sigma\rho) = (-1)^{r+s} = (-1)^r(-1)^s = \operatorname{sgn}(\sigma)\operatorname{sgn}(\rho)$. Surjectivity: any transposition has sign -1. □

Definition 3.4.5: Alternating group

The **alternating group** $A_n = \ker(\mathrm{sgn}) = \{\sigma \in S_n : \mathrm{sgn}(\sigma) = 1\}$ is the group of even permutations.

Proposition 3.4.3 Properties of A_n

We have:

1. $A_n \trianglelefteq S_n$ with index $[S_n : A_n] = 2$, so $|A_n| = n!/2$.
2. $S_n/A_n \cong \mathbb{Z}/2$.
3. A_n is generated by 3-cycles. In fact, $A_n = \langle (1\,2\,3), (1\,2\,4), \ldots, (1\,2\,n) \rangle$.
4. For $n \geqslant 5$, A_n is **simple** (has no proper nontrivial normal subgroups).

Proof: Proof of A_n properties (1)-(3)

(1)-(2) follow from the First Isomorphism Theorem: $S_n/\ker(\mathrm{sgn}) \cong \mathrm{Im}(\mathrm{sgn}) = \{1, -1\} \cong \mathbb{Z}/2$.

(3) Every even permutation is a product of an even number of transpositions. We can pair them: $(a\,b)(c\,d)$. If $\{a, b\} \cap \{c, d\} = \emptyset$, then $(a\,b)(c\,d) = (a\,c\,b)(a\,c\,d)$ (verify!). If $\{a, b\} \cap \{c, d\} = \{b\} = \{c\}$, then $(a\,b)(b\,d) = (a\,b\,d)$. Thus A_n is generated by 3-cycles. For the specific generating set:

$$(a\,b\,c) = (1\,2\,a)(1\,2\,b)(1\,2\,c)(1\,2\,a)^{-1}(1\,2\,b)^{-1}$$

(if a, b, c all differ from 1,2; similar manipulations handle other cases). □

Note:-

The simplicity of A_n for $n \geqslant 5$ is a historically revolutionary result (it regards the non-existence of the quintic formula). It implies that A_n has no quotients except $\{e\}$ and itself, which means A_n is "irreducible" in a sense. Via Galois theory, this simplicity is precisely why the general quintic polynomial has no solution in radicals: the Galois group of the "generic" degree-n polynomial is S_n, and solvability by radicals requires a composition series with abelian quotients. For $n \geqslant 5$, A_n blocks this because it's simple and non-abelian.

Note:-

The sign homomorphism is intimately related to the determinant: for a matrix $A \in GL_n(\mathbb{R})$, we have $\det(A) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}$. The sign ensures the determinant is alternating in the rows (swapping rows negates the determinant). This is no coincidence: the determinant is the unique alternating multilinear function on n vectors with $\det(I) = 1$, and "alternating" is encoded by the sign.

Question 4

1. Compute the cycle type and sign of $\sigma = (1\,2\,3)(4\,5)(1\,4)(2\,5\,3)$ in S_5. (First simplify!)
2. How many elements of S_7 have cycle type $(1^1 2^1 4^1)$?
3. Prove that the only normal subgroups of S_3 are $\{e\}, A_3, S_3$.
4. Let $H = \{e, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\} \subset S_4$. Prove $H \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ and $H \trianglelefteq S_4$. What is S_4/H?

Solution

(1) First, multiply the cycles right-to-left. Trace each element:

$$1 \mapsto 4 \mapsto 5 \mapsto 5 \mapsto 4 \mapsto 1 : \text{let's be systematic:}$$

Write $\sigma = (1\,2\,3)(4\,5)(1\,4)(2\,5\,3)$. Reading right-to-left: start with $(2\,5\,3)$, then $(1\,4)$, then $(4\,5)$, then $(1\,2\,3)$.

Trace 1: $1 \xrightarrow{(2\,5\,3)} 1 \xrightarrow{(1\,4)} 4 \xrightarrow{(4\,5)} 5 \xrightarrow{(1\,2\,3)} 5$. So $1 \mapsto 5$.
Trace 2: $2 \xrightarrow{(2\,5\,3)} 5 \xrightarrow{(1\,4)} 5 \xrightarrow{(4\,5)} 4 \xrightarrow{(1\,2\,3)} 4$. So $2 \mapsto 4$.
Trace 3: $3 \xrightarrow{(2\,5\,3)} 2 \xrightarrow{(1\,4)} 2 \xrightarrow{(4\,5)} 2 \xrightarrow{(1\,2\,3)} 3$. So $3 \mapsto 3$ (fixed).
Trace 4: $4 \xrightarrow{(2\,5\,3)} 4 \xrightarrow{(1\,4)} 1 \xrightarrow{(4\,5)} 1 \xrightarrow{(1\,2\,3)} 2$. So $4 \mapsto 2$.
Trace 5: $5 \xrightarrow{(2\,5\,3)} 3 \xrightarrow{(1\,4)} 3 \xrightarrow{(4\,5)} 3 \xrightarrow{(1\,2\,3)} 1$. So $5 \mapsto 1$.
Thus $\sigma = (1\,5)(2\,4)$, cycle type $(1^1 2^2)$. Sign: two 2-cycles = two transpositions, so $\operatorname{sgn}(\sigma) = (-1)^2 = 1$ (even).
(2) Cycle type $(1^1 2^1 4^1)$ means one fixed point, one 2-cycle, one 4-cycle. Choose the fixed point: 7 ways. Choose 4 elements for the 4-cycle from the remaining 6: $\binom{6}{4} = 15$ ways. Arrange them in a 4-cycle: $(4-1)! = 6$ ways. The remaining 2 elements form the 2-cycle: 1 way. Total: $7 \cdot 15 \cdot 6 \cdot 1 = 630$.
(3) S_3 has order 6. By Lagrange, subgroups have order 1, 2, 3, or 6. Normal subgroups must be unions of conjugacy classes. Conjugacy classes in S_3: $\{e\}$ (size 1), $\{(12), (13), (23)\}$ (size 3), $\{(123), (132)\}$ (size 2).
A normal subgroup of order 2 would need size-1 and size-1 classes, but $\{e\}$ has size 1 and we'd need another element with a size-1 class—none exist. A normal subgroup of order 3 must be $\{e\} \cup \{(123), (132)\} = A_3$. ✓
(4) Check H is a subgroup: $(1\,2)(3\,4) \cdot (1\,3)(2\,4) = (1\,4)(2\,3)$, etc.—closure holds (verify all products give elements of H). Each non-identity element has order 2. So $H \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (Klein four-group).
Normality: H consists of e plus all products of two disjoint transpositions. These form a conjugacy class in S_4 (cycle type (2^2)), so H is a union of conjugacy classes, hence normal.
S_4/H has order $24/4 = 6$. The quotient is non-abelian (since S_4 is non-abelian and H is not all of S_4), so $S_4/H \cong S_3$. Explicitly: S_4 acts on the three partitions $\{\{1,2\},\{3,4\}\}, \{\{1,3\},\{2,4\}\}, \{\{1,4\},\{2,3\}\}$, and H is the kernel.

3.5 Exact sequences

The isomorphism theorems reveal a pattern: kernels and images of homomorphisms are intimately related. Exact sequences provide elegant jargon to express these relationships, and form the foundation of homological algebra—a subject that permeates modern mathematics from algebraic topology to representation theory to algebraic geometry. We introduce the basic concepts here.

3.5.1 Exact sequences and their meaning

Definition 3.5.1: Exact sequence

A sequence of groups and homomorphisms

$$\cdots \to G_{i-1} \xrightarrow{\varphi_{i-1}} G_i \xrightarrow{\varphi_i} G_{i+1} \to \cdots$$

is **exact at G_i** if $\operatorname{Im}(\varphi_{i-1}) = \ker(\varphi_i)$. The sequence is **exact** if it is exact at every group (except possibly the endpoints).

The condition $\operatorname{Im}(\varphi_{i-1}) = \ker(\varphi_i)$ says: "everything that comes from the left dies when we go right." This is stronger than $\operatorname{Im}(\varphi_{i-1}) \subseteq \ker(\varphi_i)$ (which just says $\varphi_i \circ \varphi_{i-1} = 0$, i.e., the composition is the zero map).

Example 3.5.1 (Exactness encodes injectivity and surjectivity)
Consider short sequences with trivial groups $\{e\}$:

1. $\{e\} \to A \xrightarrow{\varphi} B$ is exact at A iff $\ker(\varphi) = \operatorname{Im}(\{e\} \to A) = \{e_A\}$, i.e., φ is **injective**.
2. $A \xrightarrow{\varphi} B \to \{e\}$ is exact at B iff $\operatorname{Im}(\varphi) = \ker(B \to \{e\}) = B$, i.e., φ is **surjective**.

3. $\{e\} \to A \xrightarrow{\varphi} B \to \{e\}$ is exact iff φ is an **isomorphism**.

3.5.2 Short exact sequences

The most important type of exact sequence is the short exact sequence, which captures the structure of quotient groups.

Definition 3.5.2: Short exact sequence

A **short exact sequence** (SES) is an exact sequence of the form

$$\{e\} \to N \xrightarrow{\iota} G \xrightarrow{\pi} Q \to \{e\}$$

where exactness means: ι is injective, π is surjective, and $\mathrm{Im}(\iota) = \ker(\pi)$.

Note:-

We often write a short exact sequence as $1 \to N \to G \to Q \to 1$ (using 1 for the trivial group in multiplicative notation) or $0 \to N \to G \to Q \to 0$ (in additive notation for abelian groups).

Proposition 3.5.1 Short exact sequences and quotients

A sequence $1 \to N \xrightarrow{\iota} G \xrightarrow{\pi} Q \to 1$ is exact if and only if:

1. ι identifies N with a normal subgroup of G (namely $\iota(N) = \ker(\pi) \trianglelefteq G$);
2. π induces an isomorphism $G/\iota(N) \cong Q$.

In other words, short exact sequences encode the statement "$G/N \cong Q$" or "G is an extension of Q by N."

Proof: Proof of SES characterization

($\Rightarrow$) Suppose the sequence is exact. Then ι is injective, so $N \cong \iota(N) \subseteq G$. Exactness at G gives $\iota(N) = \ker(\pi)$, which is normal in G. By the First Isomorphism Theorem, $G/\ker(\pi) \cong \mathrm{Im}(\pi) = Q$ (using surjectivity of π).

($\Leftarrow$) If $\iota(N) \trianglelefteq G$ and $G/\iota(N) \cong Q$, take $\pi : G \to G/\iota(N) \cong Q$ to be the quotient map (composed with the isomorphism). Then $\ker(\pi) = \iota(N) = \mathrm{Im}(\iota)$, giving exactness. $\square$

Example 3.5.2 (Fundamental examples of short exact sequences)

1. **Integers mod n:** $0 \to n\mathbb{Z} \xrightarrow{\iota} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0$, where ι is inclusion and π is reduction mod n.
2. **Determinant:** $1 \to SL_n(\mathbb{R}) \xrightarrow{\iota} GL_n(\mathbb{R}) \xrightarrow{\det} \mathbb{R}^* \to 1$. The kernel of det is exactly SL_n.
3. **Sign:** $1 \to A_n \xrightarrow{\iota} S_n \xrightarrow{\mathrm{sgn}} \{\pm 1\} \to 1$. The alternating group is the kernel of the sign homomorphism.
4. **Center:** For any group G, we have $1 \to Z(G) \xrightarrow{\iota} G \xrightarrow{\pi} G/Z(G) \to 1$. The quotient $G/Z(G)$ is called the inner automorphism group (it acts on G by conjugation).
5. **Product decomposition:** $1 \to N \xrightarrow{\iota} N \times Q \xrightarrow{\pi} Q \to 1$, where $\iota(n) = (n, e)$ and $\pi(n, q) = q$. This is a "trivial" extension.

3.5.3 Splitting and semidirect products

When does a short exact sequence "come from" a direct product? This leads to the important notion of splitting.

Definition 3.5.3: Split exact sequence

A short exact sequence $1 \to N \xrightarrow{\iota} G \xrightarrow{\pi} Q \to 1$ is **split** (or **splits**) if any of the following equivalent conditions holds:

1. There exists a homomorphism $s : Q \to G$ with $\pi \circ s = \mathrm{id}_Q$ (a **section** or **right inverse**);
2. There exists a homomorphism $r : G \to N$ with $r \circ \iota = \mathrm{id}_N$ (a **retraction** or **left inverse**);
3. $G \cong N \rtimes Q$ for some semidirect product structure.

Note:-

For abelian groups, split exact sequences correspond exactly to direct products: if $1 \to A \to B \to C \to 1$ splits with A, B, C abelian, then $B \cong A \times C$. For non-abelian groups, the situation is more subtle—we get semidirect products, which we'll study in detail later.

Example 3.5.3 (Split vs non-split)

1. **Splits:** $1 \to A_3 \to S_3 \to \mathbb{Z}/2 \to 1$. A section $s : \mathbb{Z}/2 \to S_3$ sends the generator to any transposition, e.g., $s(1) = (12)$. Thus $S_3 \cong A_3 \rtimes \mathbb{Z}/2$.
2. **Does not split:** $0 \to \mathbb{Z}/2 \xrightarrow{\times 2} \mathbb{Z}/4 \xrightarrow{\text{mod } 2} \mathbb{Z}/2 \to 0$. If it split, we'd have $\mathbb{Z}/4 \cong \mathbb{Z}/2 \times \mathbb{Z}/2$, but $\mathbb{Z}/4$ has an element of order 4 while $\mathbb{Z}/2 \times \mathbb{Z}/2$ doesn't. The sequence is exact but not split.

Proposition 3.5.2 Splitting lemma (partial)

If $1 \to N \xrightarrow{\iota} G \xrightarrow{\pi} Q \to 1$ has a section $s : Q \to G$, then every element $g \in G$ can be written uniquely as $g = \iota(n) \cdot s(q)$ for some $n \in N$ and $q \in Q$.

Proof: Proof of splitting decomposition

Existence: Given $g \in G$, let $q = \pi(g) \in Q$. Then $\pi(g \cdot s(q)^{-1}) = \pi(g) \cdot \pi(s(q))^{-1} = q \cdot q^{-1} = e_Q$, so $g \cdot s(q)^{-1} \in \ker(\pi) = \iota(N)$. Thus $g \cdot s(q)^{-1} = \iota(n)$ for some $n \in N$, giving $g = \iota(n) \cdot s(q)$.

Uniqueness: Suppose $\iota(n_1) \cdot s(q_1) = \iota(n_2) \cdot s(q_2)$. Apply π: $e \cdot q_1 = e \cdot q_2$, so $q_1 = q_2$. Then $\iota(n_1) = \iota(n_2)$, and since ι is injective, $n_1 = n_2$. □

3.5.4 The extension problem

The short exact sequence perspective leads to a fundamental question in group theory.

Definition 3.5.4: Group extension

Given groups N and Q, an **extension of Q by N** is a group G fitting into a short exact sequence $1 \to N \to G \to Q \to 1$.

The **extension problem** asks: given N and Q, classify all groups G that are extensions of Q by N. This is a deep problem that leads to:

1. Semidirect products (when the sequence splits);
2. Group cohomology $H^2(Q, N)$ (classifying non-split extensions);
3. The structure theory of finite groups (building groups from simple groups).

Example 3.5.4 (Extensions of $\mathbb{Z}/2$ by $\mathbb{Z}/2$)

What groups G fit into $1 \to \mathbb{Z}/2 \to G \to \mathbb{Z}/2 \to 1$? Such G has order 4. We know there are exactly two groups of order 4: $\mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$. Both work:

- $\mathbb{Z}/4$: the sequence $0 \to \mathbb{Z}/2 \xrightarrow{\times 2} \mathbb{Z}/4 \to \mathbb{Z}/2 \to 0$ (non-split).
- $\mathbb{Z}/2 \times \mathbb{Z}/2$: the sequence $0 \to \mathbb{Z}/2 \to \mathbb{Z}/2 \times \mathbb{Z}/2 \to \mathbb{Z}/2 \to 0$ (split).

So there are exactly two extensions, corresponding to the two groups of order 4.

Question 5

1. Verify that $1 \to \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \xrightarrow{\text{mod } n} \mathbb{Z}/n \to 1$ is exact.
2. Show that the sequence $1 \to \langle r \rangle \to D_n \to \mathbb{Z}/2 \to 1$ is exact, where $\langle r \rangle \cong \mathbb{Z}/n$ is the rotation subgroup. Does it split?
3. Let G be a group with normal subgroup N. Prove that the sequence $1 \to N \to G \to G/N \to 1$ is exact (with the obvious maps).
4. Suppose $1 \to A \xrightarrow{f} B \xrightarrow{g} C \to 1$ and $1 \to C \xrightarrow{h} D \xrightarrow{k} E \to 1$ are exact. Prove that $1 \to A \xrightarrow{f} B \xrightarrow{h \circ g} D \xrightarrow{k} E \to 1$ is exact.

Solution

(1) Let $\iota : \mathbb{Z} \to \mathbb{Z}$ be $\iota(k) = nk$ and $\pi : \mathbb{Z} \to \mathbb{Z}/n$ be reduction mod n.
Exactness at first $\mathbb{Z}$: $\ker(\iota) = \{k : nk = 0\} = \{0\}$, and $\operatorname{Im}(1 \to \mathbb{Z}) = \{0\}$. ✓
Exactness at second $\mathbb{Z}$: $\operatorname{Im}(\iota) = n\mathbb{Z}$ and $\ker(\pi) = n\mathbb{Z}$. ✓
Exactness at $\mathbb{Z}/n$: $\operatorname{Im}(\pi) = \mathbb{Z}/n$ and $\ker(\mathbb{Z}/n \to 1) = \mathbb{Z}/n$. ✓
(2) The rotation subgroup $\langle r \rangle = \{e, r, r^2, \ldots, r^{n-1}\}$ is normal in D_n (we showed this earlier). The quotient $D_n/\langle r \rangle$ has order $2n/n = 2$, so $D_n/\langle r \rangle \cong \mathbb{Z}/2$. The sequence $1 \to \langle r \rangle \xrightarrow{\iota} D_n \xrightarrow{\pi} \mathbb{Z}/2 \to 1$ is exact.
Does it split? Yes! Define $s : \mathbb{Z}/2 \to D_n$ by $s(0) = e$ and $s(1) = s$ (a reflection). Then $\pi(s(1)) = \pi(s) = 1$ (since $s \notin \langle r \rangle$), so $\pi \circ s = \mathrm{id}$. Thus $D_n \cong \mathbb{Z}/n \rtimes \mathbb{Z}/2$.
(3) Let $\iota : N \hookrightarrow G$ be inclusion and $\pi : G \to G/N$ be the quotient map.
Exactness at N: ι is injective, so $\ker(\iota) = \{e\} = \operatorname{Im}(1 \to N)$. ✓
Exactness at G: $\operatorname{Im}(\iota) = N$ and $\ker(\pi) = N$. ✓
Exactness at G/N: π is surjective, so $\operatorname{Im}(\pi) = G/N = \ker(G/N \to 1)$. ✓
(4) We check exactness at each position.
At A: Same as the original sequence—f is injective.
At B: We need $\operatorname{Im}(f) = \ker(h \circ g)$. We have $\operatorname{Im}(f) = \ker(g)$ (exactness of first sequence). If $b \in \ker(g)$, then $g(b) = e_C$, so $(h \circ g)(b) = h(e_C) = e_D$, giving $b \in \ker(h \circ g)$. Conversely, if $(h \circ g)(b) = e_D$, then $h(g(b)) = e_D$, so $g(b) \in \ker(h) = \{e_C\}$ (since h is injective). Thus $g(b) = e_C$, so $b \in \ker(g) = \operatorname{Im}(f)$.
At D: We need $\operatorname{Im}(h \circ g) = \ker(k)$. From exactness of second sequence, $\operatorname{Im}(h) = \ker(k)$. Since g is surjective, $\operatorname{Im}(h \circ g) = h(g(B)) = h(C) = \operatorname{Im}(h) = \ker(k)$. ✓
At E: Same as the original sequence—k is surjective.

Note:-

Exact sequences are the beginning of homological algebra. The ideas extend to longer exact sequences, to chain complexes (sequences where consecutive compositions are zero, not necessarily exact), and to derived functors that measure the failure of exactness. In representation theory, exact sequences of representations encode how representations decompose. In topology, the long exact sequence of a fibration is a powerful tool. The language you've learned here will reappear throughout mathematics.

3.6 Free groups and presentations

Every group we've seen so far has been defined concretely: permutations, matrices, integers mod n. But there's a fundamentally different approach: describe a group by generators and relations. This leads to group presentations, which are both computationally useful and theoretically profound. The key player is the free group—the "most general" group on a given set of generators.

3.6.1 Words and reduced words

Definition 3.6.1: Alphabet and words

Let S be a set (the **alphabet** or **generating set**). Define $S^{-1} = \{s^{-1} : s \in S\}$, a formal set of "inverse symbols" disjoint from S. A **word** in S is a finite sequence

$$w = a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n}$$

where each $a_i \in S$ and $\epsilon_i \in \{1, -1\}$. The **empty word**, denoted e or ε, is the word with no letters.

Example 3.6.1

If $S = \{a, b\}$, then $ab^{-1}a^{-1}ba$, $aaab$, $b^{-1}b^{-1}a$, and e are all words. We write a^3 for aaa and a^{-2} for $a^{-1}a^{-1}$.

Definition 3.6.2: Concatenation

The **concatenation** of words $w_1 = a_1 \cdots a_m$ and $w_2 = b_1 \cdots b_n$ is $w_1 w_2 = a_1 \cdots a_m b_1 \cdots b_n$. This operation is associative with identity e.

Now, we want words to form a group. The issue: aa^{-1} should equal e, but as sequences they're different. We need to identify words that "should be equal."

Definition 3.6.3: Elementary reduction

An **elementary reduction** of a word w is the operation of removing a subword of the form ss^{-1} or $s^{-1}s$ (for $s \in S$). A word is **reduced** if it contains no such subwords—i.e., no letter is immediately followed by its inverse.

Example 3.6.2

The word $ab^{-1}ba^{-1}a$ reduces as follows:

$$ab^{-1}ba^{-1}a \to ab^{-1}b \cdot e \cdot a^{-1}a \to ab^{-1}b \to a \cdot e = a.$$

Theorem 3.6.1 Unique reduced form

Every word can be reduced to a unique reduced word by a sequence of elementary reductions. The order of reductions doesn't matter.

Proof: Sketch of unique reduction

Termination: Each reduction decreases the length by 2, so the process terminates.

Uniqueness (sketch): This is the key point. Suppose w reduces to both u and v. We use the diamond lemma: if we can perform two different reductions $w \to w_1$ and $w \to w_2$, then there exists w' reachable from both w_1 and w_2.

The only interesting case is when the two reductions overlap, e.g., $w = \cdots ss^{-1}s \cdots$. Reducing ss^{-1} gives $\cdots s \cdots$; reducing $s^{-1}s$ also gives $\cdots s \cdots$. They agree! So all reduction paths lead to the same result. ☐

3.6.2 The free group

Definition 3.6.4: Free group

The <u>**free group on** S</u>, denoted $F(S)$ or F_n (where $n = |S|$), is the set of reduced words in S with the group operation:

$$w_1 \cdot w_2 = \text{reduce}(w_1 w_2).$$

The identity is the empty word e, and the inverse of $a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}$ is $a_n^{-\epsilon_n} \cdots a_1^{-\epsilon_1}$.

Proposition 3.6.1 Free group is a group

The free group $F(S)$ satisfies the group axioms.

Proof: Verification of group axioms for $F(S)$

Closure: The product of two reduced words, after reduction, is a reduced word.

Associativity: This is subtle! We need $(w_1 \cdot w_2) \cdot w_3 = w_1 \cdot (w_2 \cdot w_3)$. Both equal the reduction of $w_1 w_2 w_3$—but the reductions might happen in different orders. By uniqueness of reduced form, they give the same result.

Identity: $e \cdot w = \text{reduce}(ew) = \text{reduce}(w) = w$ (since w is already reduced). Similarly $w \cdot e = w$.

Inverses: Let $w = a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}$ and $w^{-1} = a_n^{-\epsilon_n} \cdots a_1^{-\epsilon_1}$. Then $ww^{-1} = a_1^{\epsilon_1} \cdots a_n^{\epsilon_n} a_n^{-\epsilon_n} \cdots a_1^{-\epsilon_1}$. Reducing: $a_n^{\epsilon_n} a_n^{-\epsilon_n}$ cancels, then $a_{n-1}^{\epsilon_{n-1}} a_{n-1}^{-\epsilon_{n-1}}$ cancels, etc., until we get e. □

Example 3.6.3 (Free group on one generator)

$F(\{a\}) = F_1$ consists of reduced words in a, a^{-1}: these are $\ldots, a^{-2}, a^{-1}, e, a, a^2, \ldots$. This is just $\{a^n : n \in \mathbb{Z}\}$ with $a^m \cdot a^n = a^{m+n}$. So $F_1 \cong \mathbb{Z}$.

Example 3.6.4 (Free group on two generators)

$F_2 = F(\{a, b\})$ is much more interesting. Elements include $e, a, b, a^{-1}, b^{-1}, ab, ba, a^2, ab^{-1}, aba, bab^{-1}a^{-1}, \ldots$—infinitely many distinct reduced words. Notably, $ab \neq ba$ (both are reduced but different), so F_2 is non-abelian. In fact, F_2 contains copies of every countable free group!

3.6.3 The universal property

The free group is "free" because it has no relations other than the group axioms. This is captured by a <u>universal property</u>.

Theorem 3.6.2 Universal property of free groups

Let $F(S)$ be the free group on S, and let $\iota : S \to F(S)$ be the inclusion $s \mapsto s$ (the one-letter word). For any group G and any function $f : S \to G$, there exists a unique homomorphism $\tilde{f} : F(S) \to G$ such that $\tilde{f} \circ \iota = f$.

$$\begin{tikzcd} S \arrow[r, "\iota"] \arrow[dr, "f"'] & F(S) \arrow[d, dashed, "\exists! \tilde{f}"] \\ & G \end{tikzcd}$$

Proof: Proof of universal property

Existence: Define $\tilde{f}(a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}) = f(a_1)^{\epsilon_1} \cdots f(a_n)^{\epsilon_n}$. This is well-defined on reduced words and is a homomorphism: $\tilde{f}(w_1 w_2) = \tilde{f}(w_1)\tilde{f}(w_2)$ because the reductions in $F(S)$ correspond to the group relations $f(s)f(s)^{-1} = e_G$ in G.

Uniqueness: Any homomorphism $\varphi : F(S) \to G$ with $\varphi(s) = f(s)$ for all $s \in S$ must satisfy $\varphi(a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}) = \varphi(a_1)^{\epsilon_1} \cdots \varphi(a_n)^{\epsilon_n} = f(a_1)^{\epsilon_1} \cdots f(a_n)^{\epsilon_n} = \tilde{f}(w)$. □

Note:-

The universal property says: to define a homomorphism out of $F(S)$, you just need to say where the generators go—there are no constraints! Any choice of images for S in any group G extends uniquely to a homomorphism. This is why $F(S)$ is "free"—you have complete freedom in defining maps from it. Categorically, the free group functor is left adjoint to the forgetful functor from groups to sets.

3.6.4 Group presentations

The universal property immediately gives us a way to describe groups by generators and relations.

Definition 3.6.5: Group presentation

A **presentation** of a group G is an expression

$$G = \langle S \mid R \rangle$$

where S is a set of **generators** and $R \subseteq F(S)$ is a set of **relators**. This means $G \cong F(S)/N$, where $N = \langle\langle R \rangle\rangle$ is the **normal closure** of R—the smallest normal subgroup of $F(S)$ containing R. Equivalently, if $R = \{r_1, r_2, \ldots\}$, we write $G = \langle S \mid r_1 = e, r_2 = e, \ldots \rangle$ or $G = \langle S \mid r_1, r_2, \ldots \rangle$.

Note:-

The normal closure $\langle\langle R \rangle\rangle$ consists of all products of conjugates of elements of R and their inverses: $\langle\langle R \rangle\rangle = \{g_1 r_1^{\pm 1} g_1^{-1} \cdots g_k r_k^{\pm 1} g_k^{-1} : g_i \in F(S), r_i \in R\}$. We need normal closure (not just the subgroup generated by R) because we're quotienting, and quotients require normal subgroups.

Example 3.6.5 (Presentations of familiar groups)

1. **Cyclic group:** $\mathbb{Z}/n = \langle a \mid a^n \rangle$. One generator, one relation.
2. **Infinite cyclic:** $\mathbb{Z} = \langle a \mid \rangle$ (no relations). The free group on one generator.
3. **Klein four-group:** $\mathbb{Z}/2 \times \mathbb{Z}/2 = \langle a, b \mid a^2, b^2, aba^{-1}b^{-1} \rangle$. The last relation says $ab = ba$ (commutativity).
4. **Dihedral group:** $D_n = \langle r, s \mid r^n, s^2, srs^{-1}r \rangle$. The last relation says $srs^{-1} = r^{-1}$, i.e., $sr = r^{-1}s$.
5. **Symmetric group:** $S_n = \langle s_1, \ldots, s_{n-1} \mid s_i^2, (s_i s_{i+1})^3, (s_i s_j)^2 \text{ for } |i - j| \geqslant 2 \rangle$, where $s_i = (i\ i+1)$. These are the Coxeter relations.
6. **Free abelian group:** $\mathbb{Z}^n = \langle a_1, \ldots, a_n \mid [a_i, a_j] \text{ for all } i, j \rangle$, where $[a, b] = aba^{-1}b^{-1}$ is the commutator.

Proposition 3.6.2 Every group has a presentation

For any group G, there exist S and R such that $G \cong \langle S \mid R \rangle$.

Proof: Existence of presentation

Take $S = G$ (yes, all of G!) and define $f : S \to G$ by $f(g) = g$. By the universal property, this extends to a surjective homomorphism $\tilde{f} : F(G) \to G$. Let $R = \ker(\tilde{f})$. Then $G \cong F(G)/R = \langle G \mid R \rangle$.

This is a "stupid" presentation (huge!), but it exists. Finding nice presentations is an art. □

3.6.5 The word problem

Presentations raise a fundamental computational question.

Definition 3.6.6: Word problem

Given a presentation $G = \langle S \mid R\rangle$ and two words w_1, w_2 in S, the **word problem** asks: do w_1 and w_2 represent the same element of G? Equivalently: is $w_1 w_2^{-1}$ in the normal closure of R?

Theorem 3.6.3 Undecidability of the word problem (Novikov-Boone)

There exists a finitely presented group $G = \langle S \mid R\rangle$ (with S, R finite) for which the word problem is **undecidable**—no algorithm can determine, for all pairs of words, whether they're equal in G.

Note:-

This is a somewhat saddening (or perhaps, better said sobering) result. Group presentations look like a nice computational tool, but they're actually extremely powerful—so powerful, in facct, that they can encode arbitrary computation. The word problem for G can simulate a Turing machine (I needn't expound on the insanity and awesomeness of this)! This doesn't mean presentations are useless; for specific groups (like S_n, D_n, free groups), the word problem is decidable. But there's no universal algorithm.

Question 6

1. Prove that F_2 contains a subgroup isomorphic to F_3.
2. Show that $\langle a, b \mid aba^{-1}b^{-1}\rangle \cong \mathbb{Z}^2$.
3. Find a presentation for the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.
4. Prove that $\langle a, b \mid a^2, b^3, (ab)^2\rangle \cong S_3$.

Solution

(1) In $F_2 = F(\{a, b\})$, consider the subgroup $H = \langle a^2, ab, b^2\rangle$. We claim $H \cong F_3$.
To see this, define $f : \{x, y, z\} \to F_2$ by $f(x) = a^2, f(y) = ab, f(z) = b^2$. By the universal property, this extends to a homomorphism $\tilde{f} : F_3 \to F_2$ with image H. To show $\tilde{f}$ is injective, we must show no nontrivial reduced word in x, y, z maps to e.
Suppose $w(x, y, z) \mapsto e$ in F_2. Substituting: $w(a^2, ab, b^2) = e$ as a reduced word in a, b. The key observation: a^2, ab, b^2 have the property that no product of them (with exponents ± 1) reduces to e unless the product was empty. (This can be shown by analyzing how cancellation works—the words have distinct "shapes.") This is the <u>Schreier basis</u> approach. Thus $\tilde{f}$ is injective, and $H \cong F_3$.
(2) The presentation $G = \langle a, b \mid aba^{-1}b^{-1}\rangle$ has one relation: $aba^{-1}b^{-1} = e$, i.e., $ab = ba$. So G is the free group on a, b modulo the relation that a and b commute. This is exactly the <u>free abelian group</u> on two generators, which is $\mathbb{Z}^2$.
Explicitly: define $\varphi : G \to \mathbb{Z}^2$ by $\varphi(a) = (1, 0)$ and $\varphi(b) = (0, 1)$. This is well-defined (the relation $ab = ba$ holds in $\mathbb{Z}^2$). It's surjective since $(1, 0), (0, 1)$ generate $\mathbb{Z}^2$. For injectivity: any element of G can be written as $a^m b^n$ (using commutativity to collect), and $\varphi(a^m b^n) = (m, n)$; if this equals $(0, 0)$, then $m = n = 0$.
(3) The quaternion group Q_8 has elements $\{1, -1, i, -i, j, -j, k, -k\}$ with relations $i^2 = j^2 = k^2 = ijk = -1$.
A presentation: $Q_8 = \langle i, j \mid i^4, i^2 j^{-2}, iji^{-1}j\rangle$.
Let's verify: $i^4 = 1$ says i has order dividing 4 (actually 4). $i^2 = j^2$ says both square to the same element (which will be -1). $iji^{-1}j = e$ means $ij = j^{-1}i^{-1} = j^{-1}i^3 = j^{-1}(i^{-1}) = (ji)^{-1}$, which combined with $i^2 = j^2$ gives $ij = k$ where $k = i^{-1}j^{-1}$, and one can derive all the quaternion relations.

Alternative (cleaner): $Q_8 = \langle x, y \mid x^4, x^2y^{-2}, yxy^{-1}x \rangle$.
(4) Let $G = \langle a, b \mid a^2, b^3, (ab)^2 \rangle$. We have $|a| \mid 2$, $|b| \mid 3$, and $(ab)^2 = e$ means $abab = e$, so $ab = b^{-1}a^{-1} = b^{-1}a = b^2a$ (using $a^2 = e$).
So $ab = b^2a$, i.e., $aba^{-1} = b^2 = b^{-1}$. This is exactly the dihedral relation! Compare with $D_3 = \langle r, s \mid r^3, s^2, srs^{-1} = r^{-1} \rangle$. Setting $r = b, s = a$, we get the same relations. Thus $G \cong D_3 \cong S_3$.

Lecture 4.

Vector Spaces and Linear Algebra

This lecture marks our transition from the purely algebraic study of groups to the more geometric realm of linear algebra. Vector spaces are arguably the most tractable algebraic structures—unlike groups, where classification is hopelessly complex, finite-dimensional vector spaces over a fixed field are completely classified by a single integer: the dimension. We begin by establishing the algebraic prerequisites (ring and field homomorphisms, polynomial rings) before developing the theory of vector spaces systematically.

4.1 Ring and Field Homomorphisms

Just as group homomorphisms preserve the group structure, ring and field homomorphisms preserve both operations simultaneously. These maps are fundamental for understanding how algebraic structures relate to one another.

Definition 4.1.1: Ring homomorphism

A **ring homomorphism** is a map $\varphi : R \to S$ between rings that respects both operations:

1. $\varphi(a + b) = \varphi(a) + \varphi(b)$ for all $a, b \in R$;
2. $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in R$;
3. $\varphi(1_R) = 1_S$.

A **field homomorphism** is a ring homomorphism between fields.

Note:-

Condition (1) implies that φ is a group homomorphism $(R, +) \to (S, +)$, so we automatically get $\varphi(0) = 0$ and $\varphi(-a) = -\varphi(a)$. However, condition (3) does not follow from condition (2), even for fields! Consider $\varphi = 0$: this satisfies $\varphi(ab) = 0 = 0 \cdot 0 = \varphi(a)\varphi(b)$, but $\varphi(1) = 0 \neq 1$. We must include (3) as a separate axiom.

Proposition 4.1.1 Field homomorphisms are injective

If $\varphi : k \to K$ is a field homomorphism, then φ is injective.

Proof: Let $a \in k$ with $a \neq 0$. Since k is a field, there exists $b = a^{-1}$ such that $ab = 1_k$. Applying φ:

$$\varphi(a)\varphi(b) = \varphi(ab) = \varphi(1_k) = 1_K \neq 0_K.$$

This implies $\varphi(a) \neq 0_K$ (if $\varphi(a) = 0$, then $\varphi(a)\varphi(b) = 0 \neq 1_K$).

Thus $\ker(\varphi) = \{a \in k : \varphi(a) = 0_K\} = \{0_k\}$, so φ is injective. □

Note:-

This means every field homomorphism is an embedding. There are no "collapsing" maps between fields. This stands in sharp contrast to group and ring homomorphisms, which can have large kernels. The reason is that fields have no nontrivial ideals (a consequence of having multiplicative inverses), so the kernel of any field homomorphism must be either $\{0\}$ or the entire field—but the latter is ruled out by $\varphi(1) = 1 \neq 0$.

4.2 Polynomial Rings and Field Extensions

Polynomials are central objects connecting algebra to analysis, number theory, and geometry. Given a field k, we construct the ring of polynomials $k[x]$ and explore how to enlarge k to contain roots of polynomials.

4.2.1 The polynomial ring $k[x]$

Definition 4.2.1: Polynomial ring

Given a field k, the **polynomial ring in one variable** is

$$k[x] := \left\{ a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \;\middle|\; a_i \in k,\, n \in \mathbb{N} \right\}.$$

Here x is a **formal variable**—it is not an element of k or any particular set. A polynomial is simply a finite sequence of coefficients, and we add and multiply term by term using the familiar rules.

Note:-

The adjective "formal" is important. When we write $x^2 + 1 \in \mathbb{R}^x$, we are not thinking of x as a real number—we are thinking of $x^2 + 1$ as a symbolic expression. However, we can evaluate a polynomial at any element of k (or any field extension of k): given $f(x) = a_0 + a_1 x + \cdots + a_n x^n$ and $r \in k$, we define $f(r) = a_0 + a_1 r + \cdots + a_n r^n \in k$.

The polynomial ring $k[x]$ is indeed a ring: it is closed under addition and multiplication, has identity 1, and satisfies all the ring axioms. However, it is not a field.

Proposition 4.2.1 $k[x]$ is not a field

The polynomial ring $k[x]$ has no multiplicative inverses for non-constant polynomials. For instance, x has no inverse in $k[x]$.

Proof: Suppose $f(x) \in k[x]$ satisfies $x \cdot f(x) = 1$. Then $\deg(x \cdot f(x)) = 1 + \deg(f(x)) \geqslant 1$, but $\deg(1) = 0$. Contradiction. □

4.2.2 The field of rational functions

To make $k[x]$ into a field, we must allow "fractions" of polynomials, just as we pass from $\mathbb{Z}$ to $\mathbb{Q}$.

Definition 4.2.2: Field of rational functions

The **field of rational functions** over k is

$$k(x) := \left\{ \frac{p}{q} \;\middle|\; p, q \in k[x],\, q \neq 0 \right\} \Big/ \sim$$

where $\frac{p}{q} \sim \frac{p'}{q'}$ iff $pq' = p'q$. Addition and multiplication are defined as for ordinary fractions.

This construction generalizes naturally to polynomials in any number of variables, giving $k(x_1, \ldots, x_n)$, the field of rational functions in n variables.

4.2.3 Formal power series

Another way to extend $k[x]$ is to allow infinite sums—formal power series.

Definition 4.2.3: Formal power series

The **ring of formal power series** over k is

$$k[\![x]\!] := \left\{ \sum_{i=0}^{\infty} a_i x^i \;\middle|\; a_i \in k \right\}.$$

Addition and multiplication are defined term by term: given $f = \sum a_i x^i$ and $g = \sum b_i x^i$, we have $f + g = \sum (a_i + b_i) x^i$ and $fg = \sum c_k x^k$ where $c_k = \sum_{i+j=k} a_i b_j$.

Note:-

The product formula makes sense because each coefficient c_k is a finite sum—only finitely many pairs (i, j) satisfy $i + j = k$ with $i, j \geqslant 0$. This is why we call these "formal" power series: we never worry about convergence. The ring $k[\![x]\!]$ contains $k[x]$ as a subring (polynomials are power series with only finitely many nonzero coefficients).

Unlike $k[x]$, the ring $k[\![x]\!]$ has many more invertible elements.

Lemma 4.2.1 Invertibility in $k[\![x]\!]$

A power series $f = \sum_{i=0}^{\infty} a_i x^i \in k[\![x]\!]$ has a multiplicative inverse if and only if $a_0 \neq 0$.

Proof: ($\Rightarrow$) If $fg = 1$ for some $g = \sum b_i x^i$, then comparing constant terms gives $a_0 b_0 = 1$, so $a_0 \neq 0$.
($\Leftarrow$) Assume $a_0 \neq 0$. We construct $g = \sum b_i x^i$ satisfying $fg = 1$ inductively.
The constant term equation is $a_0 b_0 = 1$, so $b_0 = a_0^{-1}$ (which exists since k is a field).
The coefficient of x^1 equation is $a_0 b_1 + a_1 b_0 = 0$, so $b_1 = -a_0^{-1}(a_1 b_0) = -a_0^{-1} a_1 a_0^{-1}$.
In general, the coefficient of x^n gives $\sum_{i=0}^{n} a_i b_{n-i} = 0$ for $n \geqslant 1$, i.e.,

$$a_0 b_n + \sum_{i=1}^{n} a_i b_{n-i} = 0 \implies b_n = -a_0^{-1} \sum_{i=1}^{n} a_i b_{n-i}.$$

Each b_n depends only on $b_0, \ldots, b_{n-1}$, so we can solve inductively. □

Example 4.2.1

The geometric series: in $k[\![x]\!]$, we have $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \cdots = \sum_{i=0}^{\infty} x^i$. This is verified directly: $(1 - x)(1 + x + x^2 + \cdots) = 1$ since all higher terms cancel.

4.2.4 Laurent series and field of fractions

Every nonzero element of $k[\![x]\!]$ can be written as $x^m(a_m + a_{m+1}x + \cdots)$ where $a_m \neq 0$. To get a field, we need to invert x^m—i.e., allow negative powers.

Definition 4.2.4: Field of Laurent series

The **field of Laurent series** over k is

$$k(\!(x)\!) := \left\{ \sum_{i=m}^{\infty} a_i x^i \;\middle|\; m \in \mathbb{Z},\ a_i \in k \right\}.$$

Elements are "power series starting at some (possibly negative) power of x."

Proposition 4.2.2 $k((x))$ is a field

Every nonzero Laurent series has a multiplicative inverse.

Proof: Let $f = \sum_{i=m}^{\infty} a_i x^i$ with $a_m \neq 0$. Then $f = x^m g$ where $g = a_m + a_{m+1}x + \cdots \in k[\![x]\!]$ has nonzero constant term. By the lemma, g^{-1} exists in $k[\![x]\!]$. Thus $f^{-1} = x^{-m}g^{-1} \in k((x))$. □

4.2.5 Roots of polynomials and field extensions

Given a polynomial $f \in k[x]$ with $\deg(f) \geqslant 1$, we can ask: does f have a root in k? If not, can we extend k to a larger field K containing a root?

Definition 4.2.5: Root and field extension

Let $f \in k[x]$ and K be a field containing k. An element $r \in K$ is a **root** of f if $f(r) = 0$. We call K a **field extension** of k, written K/k or $k \subset K$.

Example 4.2.2

The polynomial $x^2 - 2 \in \mathbb{Q}^x$ has no roots in $\mathbb{Q}$, but we can form the extension

$$\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\},$$

which is a field containing $\mathbb{Q}$ and the root $\sqrt{2}$. To verify this is a field, we need to check multiplicative inverses: for $a + b\sqrt{2} \neq 0$,

$$\frac{1}{a + b\sqrt{2}} = \frac{a - b\sqrt{2}}{a^2 - 2b^2} = \frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2}\sqrt{2} \in \mathbb{Q}(\sqrt{2}).$$

Note that $a^2 - 2b^2 \neq 0$ when $(a, b) \neq (0, 0)$, since $\sqrt{2} \notin \mathbb{Q}$.

Example 4.2.3

Similarly, $x^2 + 1 \in \mathbb{R}^x$ has no real roots. The extension

$$\mathbb{R}(i) = \mathbb{R}(\sqrt{-1}) = \{a + bi : a, b \in \mathbb{R}\} = \mathbb{C}$$

is the field of complex numbers, which contains roots of $x^2 + 1$ (namely $\pm i$).

Definition 4.2.6: Algebraically closed field

A field k is **algebraically closed** if every non-constant polynomial $f \in k[x]$ has a root in k. Equivalently, every polynomial factors completely into linear factors over k.

Note:-

The Fundamental Theorem of Algebra states that $\mathbb{C}$ is algebraically closed. Over an algebraically closed field, every non-constant polynomial has a root, and there are no further algebraic extensions. This is why $\mathbb{C}$ is often called the “algebraic closure” of $\mathbb{R}$. In contrast, $\mathbb{Q}$ is very far from algebraically closed—there are infinitely many distinct algebraic extensions of $\mathbb{Q}$.

4.3 Characteristic of a Field

Every field k has a distinguished ring homomorphism from the integers, whose kernel determines an important invariant.

Proposition 4.3.1 Universal map from $\mathbb{Z}$

For any field k, there is a unique ring homomorphism $\varphi : \mathbb{Z} \to k$ defined by $\varphi(1) = 1_k$. Explicitly, $\varphi(n) = n \cdot 1_k$ (add 1_k to itself n times for $n > 0$; negate for $n < 0$).

Proof: Uniqueness: Any ring homomorphism must satisfy $\varphi(1) = 1_k$, and since $\mathbb{Z} = \langle 1 \rangle$ as an additive group, φ is determined by its value on 1.

Existence: Define $\varphi(n) = n \cdot 1_k$. Then $\varphi(m+n) = (m+n) \cdot 1_k = m \cdot 1_k + n \cdot 1_k = \varphi(m) + \varphi(n)$ and $\varphi(mn) = (mn) \cdot 1_k = (m \cdot 1_k)(n \cdot 1_k) = \varphi(m)\varphi(n)$ (verify by induction). Also $\varphi(1) = 1_k$. $\square$

The kernel of φ is a subgroup of $\mathbb{Z}$, hence of the form $n\mathbb{Z}$ for some $n \geqslant 0$.

Proposition 4.3.2 Kernel is zero or prime

The kernel $\ker(\varphi)$ equals $n\mathbb{Z}$ where either $n = 0$ or $n = p$ is prime.

Proof: Suppose $n > 0$ and n is not prime, so $n = ab$ with $1 < a, b < n$. Then

$$\varphi(n) = \varphi(ab) = \varphi(a)\varphi(b) = 0_k.$$

In a field, $xy = 0$ implies $x = 0$ or $y = 0$. So $\varphi(a) = 0$ or $\varphi(b) = 0$, meaning $a \in \ker(\varphi)$ or $b \in \ker(\varphi)$. But $\ker(\varphi) = n\mathbb{Z}$ and $a, b < n$, so a or b is in $n\mathbb{Z}$ only if $a = 0$ or $b = 0$—contradicting $a, b > 1$.

Thus if $n > 0$, then n must be prime. $\square$

Definition 4.3.1: Characteristic

The **characteristic** of a field k, denoted $\mathrm{char}(k)$, is:

- 0 if $\ker(\varphi) = \{0\}$, i.e., $\varphi : \mathbb{Z} \hookrightarrow k$ is injective;
- p if $\ker(\varphi) = p\mathbb{Z}$ for a prime p.

Equivalently, $\mathrm{char}(k) = p$ means $\underbrace{1_k + 1_k + \cdots + 1_k}_{p \text{ times}} = 0_k$, and p is the smallest such positive integer.

Example 4.3.1

- $\mathrm{char}(\mathbb{Q}) = \mathrm{char}(\mathbb{R}) = \mathrm{char}(\mathbb{C}) = 0$. The map $\varphi : \mathbb{Z} \to \mathbb{Q}$ is injective.
- $\mathrm{char}(\mathbb{Z}/p) = p$ for prime p. Indeed, $p \cdot 1 = 0$ in $\mathbb{Z}/p$.
- $\mathrm{char}(k(x)) = \mathrm{char}(k[\![x]\!]) = \mathrm{char}(k(\!(x)\!)) = \mathrm{char}(k)$.

Note:-

The field $\mathbb{Z}/p$ is often denoted $\mathbb{F}_p$ and called the **prime field** of characteristic p. Every field of characteristic p contains a copy of $\mathbb{F}_p$; every field of characteristic 0 contains a copy of $\mathbb{Q}$. Thus $\mathbb{F}_p$ and $\mathbb{Q}$ are the "minimal" fields in their respective characteristics.

Theorem 4.3.1 Existence and uniqueness of finite fields

For every prime p and $n \geqslant 1$, there exists a unique (up to isomorphism) field with exactly p^n elements, denoted $\mathbb{F}_{p^n}$ or $GF(p^n)$. These are all the finite fields.

Note:-

We will not prove this theorem here. The finite field $\mathbb{F}_{p^n}$ can be constructed as a splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$. There are also infinite fields of characteristic p—for example, $\mathbb{F}_p(x)$, the field of rational functions over $\mathbb{F}_p$.

4.4 Vector Spaces

We now arrive at the central structure of linear algebra. A vector space is a set with two operations—addition and scalar multiplication—satisfying axioms that generalize familiar properties of $\mathbb{R}^n$.

Definition 4.4.1: Vector space

Fix a field k. A **vector space over k** (or **k-vector space**) is a set V equipped with two operations:

1. **Vector addition:** $+ : V \times V \to V$;
2. **Scalar multiplication:** $\cdot : k \times V \to V$;

satisfying the following axioms:

1. $(V, +)$ is an abelian group with identity 0 (the **zero vector**);
2. $1 \cdot v = v$ for all $v \in V$ (identity for scalar multiplication);
3. $(ab) \cdot v = a \cdot (b \cdot v)$ for all $a, b \in k$ and $v \in V$ (associativity);
4. $(a + b) \cdot v = a \cdot v + b \cdot v$ for all $a, b \in k$ and $v \in V$ (distributivity over scalar addition);
5. $a \cdot (v + w) = a \cdot v + a \cdot w$ for all $a \in k$ and $v, w \in V$ (distributivity over vector addition).

Elements of V are called **vectors**; elements of k are called **scalars**.

Note:-

The distributive laws (4) and (5) connect the two operations. Note that axiom (1) implies $0_V + v = v$ and $v + (-v) = 0_V$, while axiom (2)-(5) give us things like $0_k \cdot v = 0_V$ (proof: $0_k \cdot v = (0_k + 0_k) \cdot v = 0_k \cdot v + 0_k \cdot v$, so $0_k \cdot v = 0_V$ by cancellation). We also get $(-1) \cdot v = -v$.

4.4.1 Fundamental examples

Example 4.4.1

The standard vector space k^n: For any $n \geqslant 1$, the set

$$k^n = \{(a_1, a_2, \dots, a_n) : a_i \in k\}$$

is a vector space with componentwise operations:

$$\begin{aligned}(a_1, \dots, a_n) + (b_1, \dots, b_n) &= (a_1 + b_1, \dots, a_n + b_n),\\ c \cdot (a_1, \dots, a_n) &= (ca_1, \dots, ca_n).\end{aligned}$$

This is the prototypical n-dimensional vector space.

Example 4.4.2

Infinite sequences k^∞: The set of all sequences in k,

$$k^\infty = \{(a_i)_{i \in \mathbb{N}} : a_i \in k\} = \{(a_1, a_2, a_3, \dots) : a_i \in k\},$$

is a vector space with componentwise operations. It contains the subspace of **eventually zero sequences**—those with $a_i = 0$ for all sufficiently large i.

Example 4.4.3

Function spaces: For any set S, the set

$$k^S = \{f : S \to k\}$$

of all functions from S to k is a vector space with pointwise operations: $(f+g)(s) = f(s) + g(s)$ and $(c \cdot f)(s) = c \cdot f(s)$.
When $S = \mathbb{N}$, we recover k^∞. When $S = \{1, \dots, n\}$, we recover k^n.

Example 4.4.4

Polynomial and power series spaces:

- $k[x]$ is a k-vector space (addition is polynomial addition; scalar multiplication is multiplying each coefficient by the scalar).
- $k[\![x]\!]$ is a k-vector space.
- For each $d \geqslant 0$, the set $k[x]_{\leqslant d} = \{f \in k[x] : \deg(f) \leqslant d\}$ of polynomials of degree at most d is a k-vector space.

Example 4.4.5

Spaces of maps $\mathbb{R} \to \mathbb{R}$: Consider $V = \{f : \mathbb{R} \to \mathbb{R}\}$, all functions from $\mathbb{R}$ to $\mathbb{R}$. This contains many interesting subspaces:

- Continuous functions $C(\mathbb{R})$;
- Differentiable functions $C^1(\mathbb{R})$;
- Smooth (infinitely differentiable) functions $C^\infty(\mathbb{R})$;
- Polynomials $\mathbb{R}^x$.

Each is closed under addition and scalar multiplication, forming a vector space.

4.4.2 Subspaces

Definition 4.4.2: Subspace

A **subspace** of a vector space V is a nonempty subset $W \subseteq V$ that is closed under addition and scalar multiplication:

1. $w_1, w_2 \in W \implies w_1 + w_2 \in W$;
2. $w \in W$, $a \in k \implies a \cdot w \in W$.

Equivalently, W is a subspace if it is a vector space under the inherited operations.

Proposition 4.4.1 Subspace criterion

A nonempty subset $W \subseteq V$ is a subspace if and only if $aw_1 + bw_2 \in W$ for all $a, b \in k$ and $w_1, w_2 \in W$.

Proof: ($\Rightarrow$) If W is a subspace, then $aw_1 \in W$ and $bw_2 \in W$ by scalar multiplication closure, and $aw_1 + bw_2 \in W$ by addition closure.

($\Leftarrow$) Taking $a = b = 1$ gives closure under addition. Taking $b = 0$ gives closure under scalar multiplication. $\square$

Note:-

The condition "nonempty" is required. Alternatively, we can replace it with "$0 \in W$," which follows from taking $a = 0$ in the scalar multiplication closure. So a subspace is precisely a nonempty subset closed under linear combinations.

Example 4.4.6

- $\{0\}$ and V itself are always subspaces (the **trivial subspaces**).
- In $\mathbb{R}^3$, subspaces are: $\{0\}$, lines through the origin, planes through the origin, and $\mathbb{R}^3$.
- In $k[x]$, the polynomials of degree $\leqslant n$ form a subspace. The polynomials of degree exactly n do not form a subspace (not closed under addition: $x^n + (-x^n) = 0$).

4.4.3 Direct Sums of Subspaces

When can we decompose a vector space into "independent" pieces? The notion of direct sum captures exactly this.

Definition 4.4.3: Sum of subspaces

Given subspaces $U, W \subseteq V$, their **sum** is

$$U + W = \{u + w : u \in U,\ w \in W\}.$$

This is the smallest subspace of V containing both U and W.

Proposition 4.4.2 Sum is a subspace

For subspaces $U, W \subseteq V$, the sum $U + W$ is a subspace of V.

Proof: Let $v_1 = u_1 + w_1$ and $v_2 = u_2 + w_2$ be elements of $U + W$, and let $\lambda \in k$. Then

$$v_1 + v_2 = (u_1 + u_2) + (w_1 + w_2) \in U + W$$

since $u_1 + u_2 \in U$ and $w_1 + w_2 \in W$. Similarly, $\lambda v_1 = \lambda u_1 + \lambda w_1 \in U + W$. The zero vector $0 = 0 + 0 \in U + W$. $\square$

Definition 4.4.4: Direct sum of subspaces

The sum $U + W$ is called a **direct sum**, written $U \oplus W$, if $U \cap W = \{0\}$. Equivalently, every element of $U + W$ can be written uniquely as $u + w$ with $u \in U$ and $w \in W$.

Proposition 4.4.3 Characterizations of direct sum

For subspaces $U, W \subseteq V$, the following are equivalent:

1. $U + W$ is a direct sum (i.e., $U \cap W = \{0\}$).
2. Every $v \in U + W$ has a unique representation $v = u + w$ with $u \in U$, $w \in W$.
3. $u + w = 0$ with $u \in U$, $w \in W$ implies $u = w = 0$.

Proof: (1) $\Rightarrow$ (2): Suppose $v = u_1 + w_1 = u_2 + w_2$. Then $u_1 - u_2 = w_2 - w_1$. The left side is in U, the right side is in W, so both are in $U \cap W = \{0\}$. Thus $u_1 = u_2$ and $w_1 = w_2$.

(2) $\Rightarrow$ (3): If $u + w = 0$, then $u + w = 0 + 0$. By uniqueness, $u = 0$ and $w = 0$.

(3) $\Rightarrow$ (1): If $v \in U \cap W$, then $v + (-v) = 0$ with $v \in U$ and $-v \in W$. By (3), $v = 0$. $\square$

Theorem 4.4.1 Dimension formula for sums

For finite-dimensional subspaces $U, W \subseteq V$:

$$\dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W).$$

In particular, if $U + W = U \oplus W$ is a direct sum, then $\dim(U \oplus W) = \dim(U) + \dim(W)$.

Proof: Let $\{v_1, \dots, v_k\}$ be a basis of $U \cap W$. Extend to a basis $\{v_1, \dots, v_k, u_1, \dots, u_m\}$ of U and to a basis $\{v_1, \dots, v_k, w_1, \dots, w_n\}$ of W.

Claim: $\{v_1, \dots, v_k, u_1, \dots, u_m, w_1, \dots, w_n\}$ is a basis of $U + W$.

Spanning: Any $x \in U + W$ is $x = u + w$ with $u \in U$, $w \in W$. Express u and w in terms of the bases above; the result is a linear combination of all the listed vectors.

Independence: Suppose $\sum a_i v_i + \sum b_j u_j + \sum c_\ell w_\ell = 0$. Then

$$\sum c_\ell w_\ell = -\sum a_i v_i - \sum b_j u_j \in U.$$

But $\sum c_\ell w_\ell \in W$, so $\sum c_\ell w_\ell \in U \cap W = \operatorname{Span}\{v_1, \dots, v_k\}$. Since $\{v_1, \dots, v_k, w_1, \dots, w_n\}$ is independent, all $c_\ell = 0$. Similarly, all $b_j = 0$, and then all $a_i = 0$.

Thus $\dim(U+W) = k + m + n = (k+m) + (k+n) - k = \dim(U) + \dim(W) - \dim(U \cap W)$. □

Definition 4.4.5: Internal direct sum decomposition

We say V is the **(internal) direct sum** of subspaces U and W, written $V = U \oplus W$, if:

1. $V = U + W$ (every vector is a sum of elements from U and W), and
2. $U \cap W = \{0\}$ (the decomposition is unique).

More generally, $V = U_1 \oplus U_2 \oplus \cdots \oplus U_n$ if every $v \in V$ can be written uniquely as $v = u_1 + u_2 + \cdots + u_n$ with $u_i \in U_i$.

Example 4.4.7

- In $\mathbb{R}^3$, let $U = \{(x, 0, 0) : x \in \mathbb{R}\}$ (the x-axis) and $W = \{(0, y, z) : y, z \in \mathbb{R}\}$ (the yz-plane). Then $\mathbb{R}^3 = U \oplus W$.
- Let $V = k^n$ and $U = \operatorname{Span}\{e_1, \dots, e_m\}$, $W = \operatorname{Span}\{e_{m+1}, \dots, e_n\}$. Then $k^n = U \oplus W$.
- If V is the space of $n \times n$ matrices, let S = symmetric matrices and A = antisymmetric matrices. Then $V = S \oplus A$ (any matrix $M = \frac{1}{2}(M + M^T) + \frac{1}{2}(M - M^T)$).

Note:-

When we diagonalize a linear operator, we decompose the vector space as a direct sum of eigenspaces. When we study quotient spaces, we often use the fact that $V = U \oplus W$ implies $V/U \cong W$; if that doesn't make sense right now, don't worry, we'll get to it. The direct sum structure makes computations tractable by breaking a problem into independent pieces.

4.4.4 External Direct Sums and Products

We can also construct new vector spaces from old ones by taking products or direct sums "externally."

Definition 4.4.6: Direct product (external direct sum)

Given vector spaces $V_1, \dots, V_n$ over the same field k, their **direct product** (or **external direct sum**) is

$$V_1 \times V_2 \times \cdots \times V_n = \{(v_1, v_2, \dots, v_n) : v_i \in V_i\}$$

with componentwise operations:

$$(v_1, \dots, v_n) + (w_1, \dots, w_n) = (v_1 + w_1, \dots, v_n + w_n),$$
$$\lambda(v_1, \dots, v_n) = (\lambda v_1, \dots, \lambda v_n).$$

This is also denoted $V_1 \oplus V_2 \oplus \cdots \oplus V_n$ or $\bigoplus_{i=1}^n V_i$.

Proposition 4.4.4 Dimension of direct product

If each V_i is finite-dimensional, then

$$\dim(V_1 \times \cdots \times V_n) = \dim(V_1) + \cdots + \dim(V_n).$$

Proof: If $\mathcal{B}_i = \{v_{i,1}, \dots, v_{i,d_i}\}$ is a basis of V_i, then

$$\{(v_{1,1}, 0, \dots, 0), \dots, (v_{1,d_1}, 0, \dots, 0), (0, v_{2,1}, 0, \dots, 0), \dots, (0, \dots, 0, v_{n,d_n})\}$$

is a basis of $V_1 \times \cdots \times V_n$. This has $d_1 + \cdots + d_n$ elements. □

Note:-

The external direct sum $V_1 \times V_2$ and the internal direct sum $U \oplus W \subseteq V$ are related as follows: if $V = U \oplus W$, then the map

$$U \times W \to V, \quad (u, w) \mapsto u + w$$

is an isomorphism. Conversely, $V_1 \times V_2$ contains copies of V_1 and V_2 as subspaces (embedded as $V_1 \times \{0\}$ and $\{0\} \times V_2$), and $V_1 \times V_2$ is their internal direct sum.

Example 4.4.8

The space $\mathbb{R}^{m+n}$ is isomorphic to $\mathbb{R}^m \times \mathbb{R}^n$ via $(x_1, \dots, x_m, y_1, \dots, y_n) \mapsto ((x_1, \dots, x_m), (y_1, \dots, y_n))$. This is why we freely identify $\mathbb{R}^m \oplus \mathbb{R}^n$ with $\mathbb{R}^{m+n}$.

4.5 Span, Linear Independence, and Basis

These three concepts—span, linear independence, and basis—are the building blocks of linear algebra. They allow us to describe vector spaces in terms of generating sets and coordinates.

4.5.1 Linear combinations and span

Definition 4.5.1: Linear combination

A **linear combination** of vectors $v_1, \dots, v_n \in V$ is any vector of the form

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

where $a_1, \dots, a_n \in k$ are scalars.

Definition 4.5.2: Span

Given vectors $v_1, \dots, v_n \in V$, the **span** of $\{v_1, \dots, v_n\}$ is the set of all linear combinations:

$$\mathrm{Span}(v_1, \dots, v_n) = \{a_1 v_1 + \cdots + a_n v_n : a_i \in k\}.$$

This is the **smallest subspace** of V containing $v_1, \dots, v_n$.
We say $v_1, \dots, v_n$ **span** V (or are a **spanning set** for V) if $\mathrm{Span}(v_1, \dots, v_n) = V$.

Proposition 4.5.1 Span is a subspace

For any $v_1, \dots, v_n \in V$, the span $\mathrm{Span}(v_1, \dots, v_n)$ is a subspace of V.

Proof: Let $W = \mathrm{Span}(v_1, \dots, v_n)$. Clearly $0 = 0 \cdot v_1 + \cdots + 0 \cdot v_n \in W$, so $W \neq \emptyset$. For closure: if $w = \sum a_i v_i$ and $w' = \sum a_i' v_i$ are in W, and $c, c' \in k$, then

$$cw + c'w' = \sum (ca_i + c'a_i')v_i \in W.$$

$\square$

The span can also be defined for infinite (or even arbitrary) sets of vectors.

Definition 4.5.3: Span of an arbitrary set

For $S \subseteq V$, the **span** of S is the set of all finite linear combinations of elements of S:

$$\mathrm{Span}(S) = \left\{ \sum_{i=1}^{n} a_i v_i : n \geqslant 0,\ a_i \in k,\ v_i \in S \right\}.$$

When $n = 0$, the empty sum is 0, so $0 \in \mathrm{Span}(S)$ always.

4.5.2 Linear independence

Definition 4.5.4: Linear independence

Vectors $v_1, \dots, v_n \in V$ are **linearly independent** if the only way to write 0 as a linear combination is the trivial way:

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n = 0 \implies a_1 = a_2 = \cdots = a_n = 0.$$

Otherwise, they are **linearly dependent**—there exist scalars, not all zero, such that $\sum a_i v_i = 0$.

Note:-

Linear dependence means some vector can be expressed as a linear combination of the others. If $a_1 v_1 + \cdots + a_n v_n = 0$ with (say) $a_1 \neq 0$, then

$$v_1 = -\frac{a_2}{a_1} v_2 - \cdots - \frac{a_n}{a_1} v_n.$$

So v_1 is "redundant"—it's already in the span of the others.

Proposition 4.5.2 Equivalent characterizations of linear independence

The following are equivalent for $v_1, \dots, v_n \in V$:

1. $v_1, \dots, v_n$ are linearly independent.

2. No v_i is a linear combination of the others.

3. The linear map $\phi : k^n \to V$ defined by $\phi(a_1, \dots, a_n) = \sum a_i v_i$ is injective.

Proof: (1) $\Leftrightarrow$ (3): $\ker(\phi) = \{(a_1, \dots, a_n) : \sum a_i v_i = 0\}$. Linear independence says $\ker(\phi) = \{0\}$, which is equivalent to ϕ being injective.

(1) $\Rightarrow$ (2): Suppose $v_j = \sum_{i \neq j} b_i v_i$. Then $\sum_{i \neq j} b_i v_i + (-1)v_j = 0$ is a nontrivial linear combination (coefficient of v_j is $-1 \neq 0$), contradicting independence.

(2) $\Rightarrow$ (1): Suppose $\sum a_i v_i = 0$ with some $a_j \neq 0$. Then $v_j = -a_j^{-1} \sum_{i \neq j} a_i v_i$, contradicting (2). □

Similarly, we can characterize spanning:

Proposition 4.5.3 Characterization of spanning

The following are equivalent:

1. $v_1, \dots, v_n$ span V.

2. Every $v \in V$ can be written as $\sum a_i v_i$ for some $a_i \in k$.

3. The linear map $\phi : k^n \to V$ defined by $\phi(a_1, \dots, a_n) = \sum a_i v_i$ is surjective.

4.5.3 Basis

Definition 4.5.5: Basis

A **basis** for V is a set $\{v_1, \dots, v_n\}$ of vectors that are both linearly independent and span V. Equivalently, the linear map $\phi : k^n \to V$ is bijective.

Proposition 4.5.4 Unique representation

If $\{v_1, \dots, v_n\}$ is a basis for V, then every vector $v \in V$ can be written uniquely as

$$v = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

for some $a_i \in k$. The scalars $(a_1, \dots, a_n)$ are called the **coordinates** of v with respect to this basis.

Proof: Existence: Since $\{v_i\}$ spans V, every v is some linear combination.

Uniqueness: If $v = \sum a_i v_i = \sum a_i' v_i$, then $\sum (a_i - a_i') v_i = 0$. By linear independence, $a_i - a_i' = 0$ for all i, so $a_i = a_i'$. □

Example 4.5.1

Standard basis of k^n: The vectors

$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, 0, \dots, 0), \quad \dots, \quad e_n = (0, \dots, 0, 1)$$

form the **standard basis** of k^n. Any $(a_1, \dots, a_n) = a_1 e_1 + \cdots + a_n e_n$.

Example 4.5.2

Non-standard basis of k^2: In k^2, the vectors $(1, 0)$ and $(0, 1)$ form the standard basis. But so do $(1, 1)$ and $(1, -1)$ (for most k—we need $\mathrm{char}(k) \neq 2$ for these to be independent).

Example 4.5.3

Basis of $k[x]_{\leqslant n}$: The polynomials $\{1, x, x^2, \ldots, x^n\}$ form a basis for $k[x]_{\leqslant n}$, the space of polynomials of degree $\leqslant n$.

Question 1

1. Does $k[x]$ have a finite basis? What is a basis for $k[x]$?
2. Does $k[\![x]\!]$ have a basis?

Solution

(1) The polynomial ring $k[x]$ does not have a finite basis, because any finite set $\{p_1, \ldots, p_n\}$ of polynomials has bounded degree, so cannot span polynomials of higher degree.
However, $k[x]$ has the infinite basis $\{1, x, x^2, x^3, \ldots\}$. Every polynomial is a finite linear combination of these monomials, and they are linearly independent (a nontrivial linear combination would give a nonzero polynomial equaling 0).
(2) By the axiom of choice, $k[\![x]\!]$ has a basis (every vector space does). But this basis cannot be explicitly described—it is uncountably infinite and highly non-constructive. Unlike $k[x]$, the monomials $\{1, x, x^2, \ldots\}$ do not form a basis of $k[\![x]\!]$: while they are linearly independent, they do not span (a power series like $\sum_{i=0}^{\infty} x^i$ is not a finite linear combination of monomials).

4.6 Finite-Dimensional Vector Spaces and Dimension

The notion of dimension will appear constantly throughout linear algebra and beyond. Unlike groups, where classification is intractable, finite-dimensional vector spaces are completely determined by a single integer—their dimension. We establish this remarkable fact through a sequence of lemmas about the interplay between spanning sets and linearly independent sets.

Definition 4.6.1: Finite-dimensional

A vector space V is **finite-dimensional** if there exists a finite spanning set for V, i.e., vectors $v_1, \ldots, v_n$ such that $\mathrm{Span}(v_1, \ldots, v_n) = V$. Otherwise, V is **infinite-dimensional**.

4.6.1 Extracting bases from spanning sets

Our first key result shows that every spanning set contains a basis.

Lemma 4.6.1 Spanning sets contain bases

If $\{v_1, \ldots, v_n\}$ spans V, then some subset of $\{v_1, \ldots, v_n\}$ is a basis for V.

Proof: We construct a basis by iteratively removing redundant vectors.

If $\{v_1, \ldots, v_n\}$ is linearly independent, we are done—it is already a basis.

Otherwise, there exists a nontrivial relation $\sum_{i=1}^{n} a_i v_i = 0$ with some $a_j \neq 0$. Then

$$v_j = -\frac{1}{a_j} \sum_{i \neq j} a_i v_i \in \mathrm{Span}(\{v_i : i \neq j\}).$$

So $\{v_i : i \neq j\}$ still spans V (any linear combination involving v_j can be rewritten using the other vectors).

Remove v_j and repeat. Since we start with finitely many vectors, this process terminates. When it does, the remaining vectors are linearly independent (else we could remove another) and still span V, hence form a basis. □

Corollary 4.6.1 Finite-dimensional spaces have bases

Every finite-dimensional vector space has a basis.

4.6.2 Extending linearly independent sets to bases

The dual result shows that any linearly independent set can be enlarged to a basis.

Lemma 4.6.2 Linearly independent sets extend to bases

If V is finite-dimensional and $\{v_1, \ldots, v_m\}$ is linearly independent, then there exists a basis of V containing $\{v_1, \ldots, v_m\}$.

Proof: Let $\{w_1, \ldots, w_r\}$ be a finite spanning set for V (exists by finite-dimensionality). We enlarge $\{v_1, \ldots, v_m\}$ to a basis using vectors from $\{w_j\}$.

Define $W_0 = \text{Span}(v_1, \ldots, v_m)$, with basis $\{v_1, \ldots, v_m\}$ (our independent set).

Induction step: Suppose we have constructed a basis $\mathcal{B}_{j-1}$ for $W_{j-1} = \text{Span}(v_1, \ldots, v_m, w_{i_1}, \ldots, w_{i_{j-1}})$.

Consider w_j:

- If $w_j \in W_{j-1}$, set $W_j = W_{j-1}$ and $\mathcal{B}_j = \mathcal{B}_{j-1}$.
- If $w_j \notin W_{j-1}$, then $\mathcal{B}_{j-1} \cup \{w_j\}$ is linearly independent (otherwise w_j would be a linear combination of $\mathcal{B}_{j-1}$, contradicting $w_j \notin W_{j-1}$). Set $W_j = \text{Span}(\mathcal{B}_{j-1} \cup \{w_j\})$ and $\mathcal{B}_j = \mathcal{B}_{j-1} \cup \{w_j\}$.

After processing all w_j, we have $W_r = \text{Span}(v_1, \ldots, v_m, w_1, \ldots, w_r) \supseteq \text{Span}(w_1, \ldots, w_r) = V$. So $\mathcal{B}_r$ is a linearly independent set spanning V, i.e., a basis containing $\{v_1, \ldots, v_m\}$. □

4.6.3 Invariance of basis size

The following theorem shows that the size of a basis is intrinsic to the vector space, not dependent on the choice of basis.

Theorem 4.6.1 All bases have the same size

If V has a basis with n elements and another basis with m elements, then $m = n$.

Proof: Let $\{v_1, \ldots, v_n\}$ and $\{w_1, \ldots, w_m\}$ be bases for V. We show $m \leqslant n$ and $n \leqslant m$, hence $m = n$.

Claim: $\{v_1, \ldots, v_{m-1}, w_j\}$ is a basis for some choice of reordering, for each j we process.

We proceed by a "replacement" argument. Start with the basis $\{v_1, \ldots, v_n\}$.

Since $w_1 \in V = \text{Span}(v_1, \ldots, v_n)$, we can write $w_1 = \sum_{i=1}^n a_i v_i$. Not all $a_i = 0$ (else $w_1 = 0$, contradicting independence of $\{w_j\}$). By reordering, assume $a_n \neq 0$.

Then

$$v_n = a_n^{-1}(w_1 - \sum_{i=1}^{n-1} a_i v_i) \in \text{Span}(v_1, \ldots, v_{n-1}, w_1).$$

So

$$\text{Span}(v_1, \ldots, v_{n-1}, w_1) \supseteq \text{Span}(v_1, \ldots, v_n) = V,$$

and thus $\{v_1, \ldots, v_{n-1}, w_1\}$ spans V. Since it has n elements and spans, it contains a basis—but removing any element would give $n-1$ vectors. We verify it's actually linearly independent: if

$$\sum_{i=1}^{n-1} b_i v_i + c w_1 = 0$$

with $c \neq 0$, then $w_1 \in \text{Span}(v_1, \ldots, v_{n-1})$, so $v_n \in \text{Span}(v_1, \ldots, v_{n-1})$, contradicting independence of $\{v_i\}$. Thus $c = 0$, and then all $b_i = 0$. So $\{v_1, \ldots, v_{n-1}, w_1\}$ is a basis.

Repeat: $w_2 = \sum_{i=1}^{n-1} a_i v_i + c w_1$. Some $a_i \neq 0$ (else $w_2 \in \text{Span}(w_1)$, contradicting independence). Replace that v_i with w_2.

Continue this process, replacing one v with one w each time. After m steps, we have a basis

$$\{v_{i_1}, \ldots, v_{i_{n-m}}, w_1, \ldots, w_m\}$$

(assuming $m \leqslant n$).

If $m > n$, the process would terminate before incorporating all w_j, but then some w_j would be in the span of earlier w's, contradicting independence. Hence $m \leqslant n$.

By symmetry, $n \leqslant m$. Thus $m = n$. □

Definition 4.6.2: Dimension

The **dimension** of a finite-dimensional vector space V, denoted $\dim(V)$ or $\dim_k(V)$, is the number of elements in any basis. The theorem above shows this is well-defined.
By convention, $\dim(\{0\}) = 0$ (the empty set is a basis for the zero space).

Corollary 4.6.2 Isomorphism with k^n

Every finite-dimensional vector space V with $\dim(V) = n$ is isomorphic to k^n.

Proof: Let $\{v_1, \ldots, v_n\}$ be a basis for V. The map

$$\phi : k^n \to V, \quad (a_1, \ldots, a_n) \mapsto \sum_{i=1}^{n} a_i v_i$$

is linear (easily verified), injective (by linear independence), and surjective (by spanning). Hence ϕ is an isomorphism. □

Note:-

The isomorphism $\phi : k^n \xrightarrow{\sim} V$ depends on the choice of basis. Different bases give different isomorphisms. This is the key insight of linear algebra: choosing a basis amounts to choosing coordinates, and we represent elements of V by **column vectors** $\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in k^n$.

Example 4.6.1

Dimensions:

- $\dim(k^n) = n$. The standard basis has n elements.
- $\dim(k[x]_{\leqslant d}) = d + 1$. The monomials $\{1, x, \ldots, x^d\}$ form a basis.
- $k[x]$ is infinite-dimensional: any finite set of polynomials has bounded degree and cannot span.
- $k[\![x]\!]$ is infinite-dimensional (and in a "larger" sense—its bases are uncountable).

Lecture 5.

Linear Maps

Having established the structure of vector spaces, we now study the morphisms between them. Just as group homomorphisms preserve the group operation, linear maps preserve both vector space operations.

Definition 5.0.1: Linear map

Let V and W be vector spaces over k. A **linear map** (or **k-linear map**, or **homomorphism of vector spaces**) is a function $\varphi : V \to W$ satisfying:

1. $\varphi(u + v) = \varphi(u) + \varphi(v)$ for all $u, v \in V$ (additivity);
2. $\varphi(\lambda v) = \lambda\varphi(v)$ for all $\lambda \in k$ and $v \in V$ (homogeneity).

Equivalently, φ preserves all linear combinations:

$$\varphi\left(\sum_{i=1}^{n} a_i v_i\right) = \sum_{i=1}^{n} a_i \varphi(v_i).$$

Note:-

A linear map is completely determined by its values on a basis. If $\{v_1, \dots, v_n\}$ is a basis for V, and we specify $\varphi(v_i) = w_i \in W$ for each i, then for any $v = \sum a_i v_i$:

$$\varphi(v) = \varphi\left(\sum a_i v_i\right) = \sum a_i \varphi(v_i) = \sum a_i w_i.$$

This is the **universal property** of bases: specifying a linear map from V is equivalent to choosing where to send the basis vectors.

Definition 5.0.2: Space of linear maps

The set of all linear maps from V to W is denoted

$$\mathrm{Hom}_k(V, W) \quad \text{or simply} \quad \mathrm{Hom}(V, W).$$

When $V = W$, we write $\mathrm{End}(V) = \mathrm{Hom}(V, V)$ and call elements **endomorphisms**.

Proposition 5.0.1 $\mathrm{Hom}(V, W)$ is a vector space

The set $\mathrm{Hom}(V, W)$ is itself a vector space over k, with operations:

- **Addition:** $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$ for all $v \in V$;
- **Scalar multiplication:** $(\lambda\varphi)(v) = \lambda \cdot \varphi(v)$ for all $v \in V$.

Proof: We must verify that $\varphi + \psi$ and $\lambda\varphi$ are linear maps when φ, ψ are.

For $\varphi + \psi$: Let $u, v \in V$ and $\mu \in k$.

$$\begin{aligned}(\varphi + \psi)(u + v) &= \varphi(u + v) + \psi(u + v) = \varphi(u) + \varphi(v) + \psi(u) + \psi(v)\\ &= (\varphi(u) + \psi(u)) + (\varphi(v) + \psi(v)) = (\varphi + \psi)(u) + (\varphi + \psi)(v).\\ (\varphi + \psi)(\mu v) &= \varphi(\mu v) + \psi(\mu v) = \mu\varphi(v) + \mu\psi(v) = \mu(\varphi(v) + \psi(v)) = \mu(\varphi + \psi)(v).\end{aligned}$$

For $\lambda\varphi$: Similarly,

$$(\lambda\varphi)(u+v) = \lambda(\varphi(u+v)) = \lambda(\varphi(u)+\varphi(v)) = \lambda\varphi(u)+\lambda\varphi(v) = (\lambda\varphi)(u)+(\lambda\varphi)(v).$$
$$(\lambda\varphi)(\mu v) = \lambda(\varphi(\mu v)) = \lambda(\mu\varphi(v)) = \mu(\lambda\varphi(v)) = \mu(\lambda\varphi)(v).$$

The vector space axioms for $\mathrm{Hom}(V,W)$ follow from those of W (operations are defined pointwise). The zero vector is the zero map $v \mapsto 0_W$, and additive inverses are $(-\varphi)(v) = -\varphi(v)$. □

Note:-

The verification that $\varphi+\psi$ and $\lambda\varphi$ are linear is "rather boring, but worth checking if you're not sure." The operations on $\mathrm{Hom}(V,W)$ are inherited pointwise from W, so the axioms follow routinely.

Theorem 5.0.1 Dimension of $\mathrm{Hom}(V,W)$

If $\dim(V) = n$ and $\dim(W) = m$, then

$$\dim(\mathrm{Hom}(V,W)) = mn.$$

The proof of this theorem will emerge naturally when we establish the correspondence between linear maps and matrices in the next section.

Note:-

We will soon see: if $\dim(V) = n$ and $\dim(W) = m$, then $\dim(\mathrm{Hom}(V,W)) = mn$. In terms of bases for V and W, linear maps become $m \times n$ matrices!

5.0.1 Dependence on the scalar field

The notions of basis, dimension, and linearity depend crucially on the choice of scalar field k. The same set can be a vector space over different fields, with different dimensions.

Example 5.0.1

$\mathbb{C}$ as a vector space over different fields: Consider $\mathbb{C}$ as a vector space:

- Over $\mathbb{C}$: $\dim_{\mathbb{C}}(\mathbb{C}) = 1$, with basis $\{1\}$. Every complex number z is $z \cdot 1$.
- Over $\mathbb{R}$: $\dim_{\mathbb{R}}(\mathbb{C}) = 2$, with basis $\{1, i\}$. Every complex number $a+bi$ is the $\mathbb{R}$-linear combination $a \cdot 1 + b \cdot i$.

Note:-

More generally, if $k' \subset k$ is a subfield (e.g., $\mathbb{R} \subset \mathbb{C}$ or $\mathbb{Q} \subset \mathbb{R}$), then any k-vector space V is automatically a k'-vector space by "restriction of scalars"—we simply forget that we can multiply by elements of $k \setminus k'$. The k'-dimension is always at least as large as the k-dimension.

Proposition 5.0.2 Linearity over subfields

If V and W are k-vector spaces and $k' \subset k$ is a subfield, then:

1. V is a k'-vector space (by restriction of scalars).
2. Any k-linear map $\varphi : V \to W$ is also k'-linear.
3. The converse fails: not every k'-linear map is k-linear.

Thus $\mathrm{Hom}_k(V,W) \subsetneq \mathrm{Hom}_{k'}(V,W)$ in general.

Example 5.0.2

Complex conjugation is $\mathbb{R}$-linear but not $\mathbb{C}$-linear: Consider the map

$$\bar{\ }:\mathbb{C}\to\mathbb{C},\quad z=a+bi\mapsto\bar{z}=a-bi.$$

$\mathbb{R}$-linearity: For $z_1,z_2\in\mathbb{C}$ and $r\in\mathbb{R}$:

- $\overline{z_1+z_2}=\bar{z}_1+\bar{z}_2$ ✓
- $\overline{r\cdot z}=r\cdot\bar{z}$ (since $\bar{r}=r$ for $r\in\mathbb{R}$) ✓

Not $\mathbb{C}$-linear: For $\lambda=i$ and $z=1$:

$$\overline{i\cdot 1}=\bar{i}=-i,\quad\text{but}\quad i\cdot\bar{1}=i\cdot 1=i.$$

Since $-i\neq i$, complex conjugation is not $\mathbb{C}$-linear.

This example illustrates a profound point: the "natural" maps between spaces can depend on what scalars we consider. Complex conjugation is an $\mathbb{R}$-linear automorphism of $\mathbb{C}$ that is <u>not</u> $\mathbb{C}$-linear—it does not commute with multiplication by i.

5.1 Matrix Representation of Linear Maps

The correspondence between linear maps and matrices is one of the most powerful ideas in linear algebra. It reduces abstract questions about linear maps to concrete computations with arrays of numbers.

5.1.1 The matrix of a linear map

Fix bases $\mathcal{B}=(v_1,\dots,v_n)$ for V and $\mathcal{C}=(w_1,\dots,w_m)$ for W. Given a linear map $\varphi:V\to W$, we can express each image $\varphi(v_j)$ as a linear combination of the basis vectors of W:

$$\varphi(v_j)=\sum_{i=1}^{m}a_{ij}w_i=a_{1j}w_1+a_{2j}w_2+\cdots+a_{mj}w_m.$$

Definition 5.1.1: Matrix of a linear map

The <u>**matrix of φ**</u> with respect to bases $\mathcal{B}$ and $\mathcal{C}$ is the $m\times n$ matrix

$$A=(a_{ij})=\begin{pmatrix}a_{11}&a_{12}&\cdots&a_{1n}\\a_{21}&a_{22}&\cdots&a_{2n}\\\vdots&\vdots&\ddots&\vdots\\a_{m1}&a_{m2}&\cdots&a_{mn}\end{pmatrix}\in\mathcal{M}_{m\times n}(k).$$

The j-th <u>**column**</u> of A is the coordinate vector of $\varphi(v_j)$ in the basis $\mathcal{C}$.

Note:-

The key insight: **columns of A are the images of basis vectors, expressed in coordinates.** This is worth memorizing: the j-th column of the matrix tells you where the j-th basis vector goes.

5.1.2 Matrix-vector multiplication

The power of the matrix representation is that applying φ corresponds to matrix multiplication.

Represent a vector $x \in V$ by its coordinate column vector: if $x = \sum_{j=1}^{n} x_j v_j$, we write

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in k^n.$$

Similarly, $y = \varphi(x) \in W$ has coordinates $Y = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$ where $y = \sum_{i=1}^{m} y_i w_i$.

Proposition 5.1.1 Linear maps as matrix multiplication

With the notation above, $Y = AX$, i.e.,

$$y_i = \sum_{j=1}^{n} a_{ij} x_j.$$

Proof: Compute:

$$\begin{aligned} \varphi(x) = \varphi\left(\sum_{j=1}^{n} x_j v_j\right) &= \sum_{j=1}^{n} x_j \varphi(v_j) = \sum_{j=1}^{n} x_j \left(\sum_{i=1}^{m} a_{ij} w_i\right) \\ &= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j\right) w_i. \end{aligned}$$

Reading off the coefficient of w_i, we get $y_i = \sum_{j=1}^{n} a_{ij} x_j$, which is exactly the i-th component of AX. $\square$

Note:-

As a memory aid: the isomorphism $k^n \xrightarrow{\sim} V$ given by a basis can be written symbolically as

$$(v_1\ v_2\ \cdots\ v_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum_{j=1}^{n} x_j v_j.$$

Similarly, φ acts as

$$\varphi((v_1\ \cdots\ v_n)X) = (w_1\ \cdots\ w_m)AX.$$

This notation makes the matrix formula $Y = AX$ visually clear.

5.1.3 The isomorphism $\mathrm{Hom}(V, W) \cong \mathcal{M}_{m\times n}(k)$

Theorem 5.1.1 Linear maps correspond to matrices

Fix bases $\mathcal{B}$ for V and $\mathcal{C}$ for W. The map

$$\Phi : \mathrm{Hom}(V, W) \to \mathcal{M}_{m\times n}(k), \quad \varphi \mapsto [\varphi]_{\mathcal{B}}^{\mathcal{C}}$$

sending a linear map to its matrix is an **isomorphism of vector spaces**.

Proof: We verify Φ is linear, injective, and surjective.

Linearity: Let $\varphi, \psi \in \mathrm{Hom}(V, W)$ and $\lambda \in k$. The j-th column of $[\varphi + \psi]$ is the coordinate vector of $(\varphi+\psi)(v_j) = \varphi(v_j) + \psi(v_j)$, which is the sum of the j-th columns of $[\varphi]$ and $[\psi]$. Hence $[\varphi+\psi] = [\varphi] + [\psi]$. Similarly $[\lambda\varphi] = \lambda[\varphi]$.

Injectivity: If $[\varphi] = 0$ (the zero matrix), then every column is zero, meaning $\varphi(v_j) = 0$ for all basis vectors v_j. Since φ is linear and determined by its values on a basis, $\varphi = 0$.

Surjectivity: Given any matrix $A \in \mathcal{M}_{m\times n}(k)$, define $\varphi : V \to W$ by specifying $\varphi(v_j) = \sum_i a_{ij} w_i$ (the j-th column of A gives the coordinates of $\varphi(v_j)$). Extend linearly. Then $[\varphi] = A$. □

Corollary 5.1.1 Dimension formula

$$\dim(\mathrm{Hom}(V, W)) = \dim(\mathcal{M}_{m\times n}(k)) = mn.$$

Proof: The space $\mathcal{M}_{m\times n}(k)$ has the standard basis $\{E_{ij}\}$, where E_{ij} is the matrix with 1 in position (i, j) and 0 elsewhere. There are mn such matrices. □

Note:-

This result says: **linear maps are matrices, once we choose coordinates**. The abstract theory of linear maps reduces to the concrete theory of matrix computations. However, the matrix depends on the choice of bases—changing bases changes the matrix (we will study this "change of basis" in the near future).

The interplay between the abstract (linear maps) and the concrete (matrices) is what makes linear algebra, well linear algebra (duh). Our goal is to understand how much of a linear map's structure is intrinsic (independent of basis choice) versus coordinate-dependent.

5.2 Change of basis

What if we want to choose a different basis for V and/ or W? That is to say, what if we want to take vectors expressable in one basis to the same vectors expressed in another basis? Well, say that we are given two bases: (v_i) and (v_i') in V. If given an expression of (v_i) we want to plug in and get an expression in terms of v_i', we write this as

$$v_j' = \sum_{n}^{i=1} p_{ij} v_i.$$

So we get an $n \times m$ matrix P with entries (p_{ij}), whose j^{th} column is the expression of v_j' in the basis (v_i) as a column vector. So symbolically, we can think of this as $(v_i') = (v_i)P$. So $(v_i')X' = (v_i)PX'$, i.e. the element of V represented by a column vector X' in new basis (v_i') is represented by $X = PX'$ in the old basis (v_i) . We can think of this more conceptually as P being a linear map—i.e. a matrix transformation—by $P = \mathcal{M}(\mathrm{id}_V, (v_i'), (v_i))$. Breaking this down, we have the identity mapping taking the vector to itself (from one basis to another, but the same vector(s)); thus we are taking things from (v_i'), here, to the targets of (v_i). We can additionally do the same for W in reverse! Let $Q = \mathcal{M}(\mathrm{id}_W, (w_i)(w_i'))$, i.e. $(w_i) = (w_i')Q$. Hence:

$$\varphi((v_i'))X' = \varphi((v_i)PX') = (w_i)APX' = (w_i')QAPX'.$$

That is to say, $\mathcal{M}(\varphi, (v'), (w')) = QAP$.

Note:-

Matrix of identity from (v_i) to (v_i') would be P^{-1} which is the matrix such that pp^{-1} is the identity matrix.

If $V = W$ and we change basis using a single new basis (v_i') for both domain and codomain, then a remarkable simplification occurs. Let $\varphi \in \mathrm{End}(V) = \mathrm{Hom}(V, V)$ be an **endomorphism** (a linear map from a space to itself). With $P = \mathcal{M}(\mathrm{id}_V, (v_i'), (v_i))$ as above, and using the same basis change for both domain and codomain, we have $Q = P$ (since we're using (v_i) and (v_i') for W as well). Thus the formula $\mathcal{M}(\varphi, (v'), (w')) = QAP$ becomes:

Theorem 5.2.1 Change of basis for endomorphisms

Let $\varphi \in \mathrm{End}(V)$, let $\mathcal{B} = (v_1, \ldots, v_n)$ and $\mathcal{B}' = (v'_1, \ldots, v'_n)$ be two bases of V, and let $P = \mathcal{M}(\mathrm{id}_V, \mathcal{B}', \mathcal{B})$ be the change-of-basis matrix. If $A = \mathcal{M}(\varphi, \mathcal{B})$ and $A' = \mathcal{M}(\varphi, \mathcal{B}')$, then

$$A' = P^{-1}AP.$$

Proof: We have established that $\mathcal{M}(\varphi, \mathcal{B}', \mathcal{B}') = QAP$ where $Q = \mathcal{M}(\mathrm{id}_V, (v_i), (v'_i))$. But $Q = P^{-1}$ since applying P then Q (or vice versa) gives the identity transformation on coordinate vectors. Explicitly: if $X' = P^{-1}X$ converts old coordinates to new, and $X = PX'$ converts new to old, then composing identity maps gives $\mathcal{M}(\mathrm{id}_V, (v_i), (v'_i)) = P^{-1}$.

Thus $A' = P^{-1}AP$. □

This formula tells us that the matrices representing the same linear operator in different bases are related by **conjugation** by an invertible matrix.

5.2.1 Similar matrices

Definition 5.2.1: Similar matrices

Two $n \times n$ matrices A and B over a field k are **similar** if there exists an invertible matrix $P \in \mathrm{GL}_n(k)$ such that

$$B = P^{-1}AP.$$

We write $A \sim B$.

Proposition 5.2.1 Similarity is an equivalence relation

The similarity relation on $\mathcal{M}_{n\times n}(k)$ is an equivalence relation:

1. **Reflexive:** $A \sim A$ (take $P = I$).
2. **Symmetric:** If $A \sim B$, then $B \sim A$ (if $B = P^{-1}AP$, then $A = PBP^{-1} = (P^{-1})^{-1}B(P^{-1})$).
3. **Transitive:** If $A \sim B$ and $B \sim C$, then $A \sim C$ (if $B = P^{-1}AP$ and $C = Q^{-1}BQ$, then $C = Q^{-1}P^{-1}APQ = (PQ)^{-1}A(PQ)$).

As we can see, similarity classes of matrices correspond exactly to linear operators on k^n: two matrices are similar if and only if they represent the same linear operator in different bases.

Note:-

From the categorical perspective, similar matrices represent the same morphism in $\mathbf{Vect}_k$—they are simply different coordinate descriptions of one intrinsic object. This is why similarity-invariant properties (trace, determinant, eigenvalues, rank) are so important: they are properties of the operator itself, not of any particular matrix representation.

5.2.2 Invariants under similarity

What properties of a matrix are preserved under similarity? These are called **similarity invariants**—they depend only on the underlying linear operator, not on the choice of basis.

Definition 5.2.2: Trace of a matrix

The **trace** of an $n \times n$ matrix $A = (a_{ij})$ is the sum of its diagonal entries:

$$\operatorname{Tr}(A) = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}.$$

Lemma 5.2.1 Cyclic property of trace

For any matrices A and B such that both AB and BA are defined and square, we have $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$.

Proof: Let A be $m \times n$ and B be $n \times m$. Then

$$\operatorname{Tr}(AB) = \sum_{i=1}^{m} (AB)_{ii} = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} b_{ji}.$$

Similarly,

$$\operatorname{Tr}(BA) = \sum_{j=1}^{n} (BA)_{jj} = \sum_{j=1}^{n} \sum_{i=1}^{m} b_{ji} a_{ij}.$$

These double sums are identical (just reorder the summation), so $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$. $\square$

Proposition 5.2.2 Trace is a similarity invariant

If $A \sim B$, then $\operatorname{Tr}(A) = \operatorname{Tr}(B)$. In particular, the trace of a linear operator $\varphi \in \operatorname{End}(V)$ is well-defined: $\operatorname{Tr}(\varphi) := \operatorname{Tr}(\mathcal{M}(\varphi, \mathcal{B}))$ for any basis $\mathcal{B}$.

Proof: If $B = P^{-1}AP$, then using the cyclic property:

$$\operatorname{Tr}(B) = \operatorname{Tr}(P^{-1}AP) = \operatorname{Tr}((P^{-1}A)P) = \operatorname{Tr}(P(P^{-1}A)) = \operatorname{Tr}((PP^{-1})A) = \operatorname{Tr}(A).$$

$\square$

Proposition 5.2.3 Determinant is a similarity invariant

If $A \sim B$, then $\det(A) = \det(B)$. Hence the determinant of a linear operator is well-defined.

Proof: If $B = P^{-1}AP$, then by the multiplicativity of determinants:

$$\det(B) = \det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P) = \frac{1}{\det(P)} \cdot \det(A) \cdot \det(P) = \det(A).$$

$\square$

Proposition 5.2.4 Eigenvalues are similarity invariants

If $A \sim B$, then A and B have the same eigenvalues (with the same algebraic multiplicities). In particular, the eigenvalues of a linear operator are intrinsic to the operator, not dependent on the choice of basis.

Proof: A scalar λ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$. If $B = P^{-1}AP$, then

$$B - \lambda I = P^{-1}AP - \lambda P^{-1}IP = P^{-1}(A - \lambda I)P.$$

Thus $B - \lambda I \sim A - \lambda I$, so $\det(B - \lambda I) = \det(A - \lambda I)$ by the previous result. Hence A and B have identical characteristic polynomials, and therefore the same eigenvalues with the same multiplicities. $\square$

Note:-

Rank is also a similarity invariant: $\text{rank}(P^{-1}AP) = \text{rank}(A)$ since multiplying by invertible matrices doesn't change rank. More sophisticated invariants include the **minimal polynomial**, the **Jordan normal form**, and the **rational canonical form**—these completely classify operators up to similarity over algebraically closed fields. We will encounter these when we study eigenvalues in depth.

Example 5.2.1

Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ over $\mathbb{R}$. Both have $\text{Tr} = 4$, $\det = 4$, and eigenvalue $\lambda = 2$ with multiplicity 2. Yet $A \not\sim B$! The reason: $B = 2I$, so $P^{-1}BP = P^{-1}(2I)P = 2I = B$ for any P. Since $A \neq B$, they cannot be similar.

This shows that trace, determinant, and eigenvalues are necessary but not sufficient conditions for similarity. The difference lies in the **geometric multiplicity** of the eigenvalue 2: for B, the eigenspace $E_2 = \ker(B - 2I) = \mathbb{R}^2$ has dimension 2; for A, $E_2 = \ker(A - 2I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \text{Span}\{(1, 0)\}$ has dimension 1. Geometric multiplicity is also a similarity invariant!

Question 1

1. Let $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map with matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ with respect to the standard basis. Find the matrix of φ with respect to the basis $\mathcal{B}' = \{(1, 1), (1, -1)\}$.
2. Prove that an $n \times n$ matrix A is similar to a diagonal matrix if and only if there exists a basis of k^n consisting of eigenvectors of A.
3. Show that $\text{Tr}(AB) = \text{Tr}(BA)$ but in general $\det(A + B) \neq \det(A) + \det(B)$. Give a counterexample.

Solution

(1) First, find the change-of-basis matrix P from $\mathcal{B}'$ to the standard basis $\mathcal{E}$. The columns of P are the vectors of $\mathcal{B}'$ expressed in $\mathcal{E}$:

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then $P^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$.

The matrix of φ with respect to $\mathcal{B}'$ is:

$$A' = P^{-1}AP = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Computing AP first: $AP = \begin{pmatrix} 3 & -1 \\ 7 & -1 \end{pmatrix}$.

Then $P^{-1}(AP) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 7 & -1 \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ -2 & 0 \end{pmatrix}$.

So $A' = \begin{pmatrix} 5 & -1 \\ -2 & 0 \end{pmatrix}$. We can verify: $\text{Tr}(A') = 5 = \text{Tr}(A)$ and $\det(A') = 0 - 2 = -2 = \det(A)$. ✓

(2) ($\Rightarrow$) Suppose A is similar to a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, so $D = P^{-1}AP$ for some invertible P. Let v_i be the i-th column of P. Then $AP = PD$, which gives $Av_i = \lambda_i v_i$. So each column of P is an eigenvector of A. Since P is invertible, its columns form a basis of k^n.

($\Leftarrow$) Suppose $\{v_1, \dots, v_n\}$ is a basis of eigenvectors with $Av_i = \lambda_i v_i$. Let P be the matrix with columns v_i. Then $AP = PD$ where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, so $D = P^{-1}AP$.

(3) We proved $\mathsf{Tr}(\mathsf{AB}) = \mathsf{Tr}(\mathsf{BA})$ above. For the determinant counterexample:

$$\mathsf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathsf{B} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then $\det(\mathsf{A}) = 0$, $\det(\mathsf{B}) = 0$, but $\mathsf{A} + \mathsf{B} = \mathrm{I}$, so $\det(\mathsf{A} + \mathsf{B}) = 1 \neq 0 = \det(\mathsf{A}) + \det(\mathsf{B})$.

Lecture 6.

Kernals, images, and more

6.1 Kernel and Image

Every linear map carries two intrinsic subspaces that reveal its fundamental structure: the kernel (what gets annihilated) and the image (what gets hit). These are the linear-algebraic analogues of concepts we've seen before: the kernel of a group homomorphism and the image of any function. The intermingling betwixt (what a fun word) kernel and image is governed by the **Rank-Nullity Theorem**.

Note:-

From the categorical viewpoint, kernel and image are two ways to "measure" a morphism. The kernel measures the failure of injectivity—how much information the map destroys. The image measures the failure of surjectivity—how much of the codomain the map misses. The Rank-Nullity Theorem says these two failures are complementary: the more a linear map kills, the less it can produce, and vice versa. A linear map $\varphi : V \to W$ is completely determined by: (1) which subspace it kills (the kernel), and (2) which subspace it hits (the image). Once you know these, there's essentially only one way to fill in the rest—the map restricted to any complement of the kernel is an isomorphism onto the image. This is different from nonlinear maps, where knowing the zero set tells you almost nothing about the map's behavior elsewhere.

6.1.1 The Kernel

Definition 6.1.1: Kernel of a linear map

Let $\varphi : V \to W$ be a linear map. The **kernel** of φ is

$$\ker(\varphi) = \{v \in V : \varphi(v) = 0_W\}.$$

This is the set of all vectors that φ maps to zero.

Note:-

The terminology "kernel" comes from algebra (group homomorphisms), while "null space" is more common in applied linear algebra and matrix theory to refer to the generalized kernal (we will get to this later). In categorical language, the kernel is the equalizer of φ and the zero map.

Example 6.1.1 (Examples of kernels)

1. The zero map $0 : V \to W$ has $\ker(0) = V$.
2. The identity map $\mathrm{id}_V : V \to V$ has $\ker(\mathrm{id}_V) = \{0\}$.
3. Let $\varphi : \mathbb{R}^3 \to \mathbb{R}$ be defined by $\varphi(x, y, z) = x + 2y + 3z$. Then $\ker(\varphi) = \{(x, y, z) \in \mathbb{R}^3 : x + 2y + 3z = 0\}$, a plane through the origin.
4. Let $D : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ be differentiation, $Dp = p'$. Then $\ker(D)$ is the set of constant polynomials.
5. Let $T : \mathcal{M}_{n\times n}(k) \to k$ be the trace map, $T(A) = \mathrm{Tr}(A)$. Then $\ker(T)$ is the space of traceless matrices.

Proposition 6.1.1 The kernel is a subspace

Let $\varphi : V \to W$ be a linear map. Then $\ker(\varphi)$ is a subspace of V.

Proof: We verify the subspace criteria:

1. **Contains zero:** Since $\varphi(0_V) = 0_W$ (every linear map preserves zero), we have $0_V \in \ker(\varphi)$.
2. **Closed under addition:** If $u, v \in \ker(\varphi)$, then $\varphi(u+v) = \varphi(u)+\varphi(v) = 0+0 = 0$, so $u+v \in \ker(\varphi)$.
3. **Closed under scalar multiplication:** If $v \in \ker(\varphi)$ and $\lambda \in k$, then $\varphi(\lambda v) = \lambda\varphi(v) = \lambda \cdot 0 = 0$, so $\lambda v \in \ker(\varphi)$.

□

Note:-

Compare this with group theory: the kernel of a group homomorphism $f : G \to H$ is a normal subgroup of G. For linear maps, every subspace is "normal" in the appropriate sense—vector spaces are abelian, so there's no distinction between left and right cosets.

The kernel provides a clean characterization of injectivity:

Theorem 6.1.1 Injectivity via the kernel

Let $\varphi : V \to W$ be a linear map. Then

$$\varphi \text{ is injective} \iff \ker(\varphi) = \{0\}.$$

Proof: $(\Rightarrow)$ Suppose φ is injective. Let $v \in \ker(\varphi)$. Then $\varphi(v) = 0 = \varphi(0)$. Since φ is injective, $v = 0$. Thus $\ker(\varphi) \subseteq \{0\}$, and since $0 \in \ker(\varphi)$ always, we have $\ker(\varphi) = \{0\}$.

$(\Leftarrow)$ Suppose $\ker(\varphi) = \{0\}$. Let $u, v \in V$ with $\varphi(u) = \varphi(v)$. Then

$$\varphi(u - v) = \varphi(u) - \varphi(v) = 0,$$

so $u - v \in \ker(\varphi) = \{0\}$. Thus $u - v = 0$, i.e., $u = v$. Hence φ is injective. □

Corollary 6.1.1

An injective linear map from V to W embeds V as a subspace of W. More precisely, if $\varphi : V \to W$ is injective, then $V \cong \mathrm{Im}(\varphi) \subseteq W$.

Note:-

Here the kernal has a role more that merely verifying injectivity—it measures the failure of injectivity in a precise sense. Two vectors $u, v \in V$ have the same image under φ if and only if $u - v \in \ker(\varphi)$. Thus the fibers of φ (preimages of points) are exactly the cosets $v + \ker(\varphi)$. The map φ is injective precisely when these cosets are singletons, i.e., when $\ker(\varphi) = \{0\}$. This is why we can "mod out by the kernel" to obtain an injective map—the quotient $V/\ker(\varphi)$ collapses each fiber to a point.

6.1.2 The Image (Range)

Definition 6.1.2: Image of a linear map

Let $\varphi : V \to W$ be a linear map. The **image** (or **range**) of φ is

$$\mathrm{Im}(\varphi) = \{\varphi(v) : v \in V\} = \{w \in W : w = \varphi(v) \text{ for some } v \in V\}.$$

This is the set of all vectors that φ "hits."

Example 6.1.2 (Examples of images)

1. The zero map $0 : V \to W$ has $\mathrm{Im}(0) = \{0_W\}$.
2. The identity map $\mathrm{id}_V : V \to V$ has $\mathrm{Im}(\mathrm{id}_V) = V$.
3. Let $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ be projection $\pi(x, y, z) = (x, y)$. Then $\mathrm{Im}(\pi) = \mathbb{R}^2$.
4. Let $D : \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_{n-1}(\mathbb{R})$ be differentiation. Then $\mathrm{Im}(D) = \mathcal{P}_{n-1}(\mathbb{R})$ (every polynomial of degree $\leqslant n - 1$ is the derivative of some polynomial of degree $\leqslant n$).
5. Let $A \in \mathcal{M}_{m\times n}(k)$ define $L_A : k^n \to k^m$. Then $\mathrm{Im}(L_A) = \mathrm{ColSpace}(A)$, the column space of A.

Proposition 6.1.2 The image is a subspace

Let $\varphi : V \to W$ be a linear map. Then $\mathrm{Im}(\varphi)$ is a subspace of W.

Proof: We verify the subspace criteria:

1. **Contains zero:** $\varphi(0_V) = 0_W$, so $0_W \in \mathrm{Im}(\varphi)$.
2. **Closed under addition:** If $w_1, w_2 \in \mathrm{Im}(\varphi)$, then $w_1 = \varphi(v_1)$ and $w_2 = \varphi(v_2)$ for some $v_1, v_2 \in V$. Thus $w_1 + w_2 = \varphi(v_1) + \varphi(v_2) = \varphi(v_1 + v_2) \in \mathrm{Im}(\varphi)$.
3. **Closed under scalar multiplication:** If $w \in \mathrm{Im}(\varphi)$ and $\lambda \in k$, then $w = \varphi(v)$ for some $v \in V$, so $\lambda w = \lambda\varphi(v) = \varphi(\lambda v) \in \mathrm{Im}(\varphi)$.

$\square$

Definition 6.1.3: Surjectivity

A linear map $\varphi : V \to W$ is **surjective** (or **onto**) if $\mathrm{Im}(\varphi) = W$, i.e., every element of W is hit by some element of V.

Proposition 6.1.3 Characterization of surjectivity

Let $\varphi : V \to W$ be a linear map. The following are equivalent:

1. φ is surjective.
2. $\mathrm{Im}(\varphi) = W$.
3. If $\mathcal{B}$ spans V, then $\varphi(\mathcal{B}) = \{\varphi(v) : v \in \mathcal{B}\}$ spans W.

6.1.3 The Rank-Nullity Theorem

We now come to the central result connecting kernel and image. This theorem quantifies the fundamental trade-off in linear algebra: the more a map "kills," the less it can "produce."

Definition 6.1.4: Rank and nullity

Let $\varphi : V \to W$ be a linear map between finite-dimensional vector spaces. The **rank** of φ is

$$\operatorname{rank}(\varphi) = \dim(\operatorname{Im}(\varphi)),$$

and the **nullity** of φ is

$$\operatorname{null}(\varphi) = \dim(\ker(\varphi)).$$

For a matrix A, we define $\operatorname{rank}(A)$ and $\operatorname{null}(A)$ to be the rank and nullity of the corresponding linear map.

Theorem 6.1.2 The Rank-Nullity Theorem (Fundamental Theorem of Linear Maps)

Let $\varphi : V \to W$ be a linear map with V finite-dimensional. Then $\operatorname{Im}(\varphi)$ is finite-dimensional and

$$\dim(V) = \dim(\ker(\varphi)) + \dim(\operatorname{Im}(\varphi)).$$

Equivalently: $\dim(V) = \operatorname{null}(\varphi) + \operatorname{rank}(\varphi)$.

Proof: Let $\{u_1, \dots, u_m\}$ be a basis for $\ker(\varphi)$. By the basis extension theorem, we can extend this to a basis

$$\{u_1, \dots, u_m, v_1, \dots, v_r\}$$

of V, where $\dim(V) = m + r$.

Claim: $\{\varphi(v_1), \dots, \varphi(v_r)\}$ is a basis for $\operatorname{Im}(\varphi)$.

Spanning: Let $w \in \operatorname{Im}(\varphi)$, say $w = \varphi(x)$ for some $x \in V$. Write

$$x = a_1u_1 + \cdots + a_mu_m + b_1v_1 + \cdots + b_rv_r.$$

Applying φ:

$$w = \varphi(x) = a_1\varphi(u_1) + \cdots + a_m\varphi(u_m) + b_1\varphi(v_1) + \cdots + b_r\varphi(v_r).$$

Since each $u_i \in \ker(\varphi)$, we have $\varphi(u_i) = 0$, so

$$w = b_1\varphi(v_1) + \cdots + b_r\varphi(v_r).$$

Thus $\{\varphi(v_1), \dots, \varphi(v_r)\}$ spans $\operatorname{Im}(\varphi)$.

Linear independence: Suppose

$$c_1\varphi(v_1) + \cdots + c_r\varphi(v_r) = 0.$$

Then $\varphi(c_1v_1 + \cdots + c_rv_r) = 0$, so $c_1v_1 + \cdots + c_rv_r \in \ker(\varphi)$. Since $\{u_1, \dots, u_m\}$ is a basis for $\ker(\varphi)$, we can write

$$c_1v_1 + \cdots + c_rv_r = d_1u_1 + \cdots + d_mu_m$$

for some scalars d_j. Rearranging:

$$d_1u_1 + \cdots + d_mu_m - c_1v_1 - \cdots - c_rv_r = 0.$$

Since $\{u_1, \dots, u_m, v_1, \dots, v_r\}$ is linearly independent (it's a basis of V), all coefficients are zero. In particular, $c_1 = \cdots = c_r = 0$.

Thus $\{\varphi(v_1), \dots, \varphi(v_r)\}$ is a basis for $\operatorname{Im}(\varphi)$, so $\dim(\operatorname{Im}(\varphi)) = r$. We conclude:

$$\dim(V) = m + r = \dim(\ker(\varphi)) + \dim(\operatorname{Im}(\varphi)).$$

$\square$

Note:-

The Rank-Nullity Theorem is the linear algebra version of the **First Isomorphism Theorem** from group theory. Recall that for a group homomorphism $f : G \to H$, we have $G/\ker(f) \cong \mathrm{Im}(f)$. The linear version states $V/\ker(\varphi) \cong \mathrm{Im}(\varphi)$, and "counting dimensions" gives $\dim(V) - \dim(\ker(\varphi)) = \dim(\mathrm{Im}(\varphi))$.

6.1.4 Consequences of the Rank-Nullity Theorem

The Rank-Nullity Theorem has powerful implications for understanding linear maps.

Corollary 6.1.2 Dimension constraints on injectivity

Let $\varphi : V \to W$ be a linear map with $\dim(V) = n$ and $\dim(W) = m$.

1. If $n > m$, then φ cannot be injective.
2. If $n < m$, then φ cannot be surjective.

Proof: (1) If φ were injective, then $\dim(\ker(\varphi)) = 0$, so by Rank-Nullity:

$$\dim(\mathrm{Im}(\varphi)) = \dim(V) = n > m = \dim(W).$$

But $\mathrm{Im}(\varphi) \subseteq W$, so $\dim(\mathrm{Im}(\varphi)) \leqslant \dim(W) = m$. Contradiction.
(2) If φ were surjective, then $\dim(\mathrm{Im}(\varphi)) = \dim(W) = m$, so by Rank-Nullity:

$$\dim(\ker(\varphi)) = \dim(V) - \dim(\mathrm{Im}(\varphi)) = n - m < 0.$$

But dimensions are non-negative. Contradiction. □

Corollary 6.1.3 Equal dimensions: bijectivity criteria

Let $\varphi : V \to W$ be a linear map with $\dim(V) = \dim(W) = n$. The following are equivalent:

1. φ is injective.
2. φ is surjective.
3. φ is bijective (an isomorphism).
4. $\ker(\varphi) = \{0\}$.
5. $\mathrm{Im}(\varphi) = W$.
6. $\mathrm{rank}(\varphi) = n$.

Proof: By Rank-Nullity, $\mathrm{null}(\varphi) + \mathrm{rank}(\varphi) = n$. Since $\mathrm{rank}(\varphi) \leqslant \dim(W) = n$:

- φ injective $\iff \mathrm{null}(\varphi) = 0 \iff \mathrm{rank}(\varphi) = n$.
- φ surjective $\iff \mathrm{rank}(\varphi) = n \iff \mathrm{null}(\varphi) = 0$.

Thus injectivity, surjectivity, and bijectivity are equivalent when dimensions match. □

Note:-

For linear maps between equal-dimensional spaces, "one-to-one" and "onto" are the same condition. This fails for general functions and even for linear maps between infinite-dimensional spaces. Consider the **right shift** $R : k^\infty \to k^\infty$ defined by $R(x_1, x_2, x_3, \dots) = (0, x_1, x_2, \dots)$. This is injective (kernel is trivial) but not surjective (the sequence $(1, 0, 0, \dots)$ is not in the image). Conversely, the **left shift** $L(x_1, x_2, x_3, \dots) = (x_2, x_3, \dots)$ is surjective but not injective. In infinite dimensions, there's "room" for one without the other. The finite-dimensional equivalence is a manifestation of the pigeonhole principle applied to basis vectors.

Corollary 6.1.4 Existence of right/left inverses

Let $\varphi : V \to W$ be a linear map between finite-dimensional spaces.

1. φ has a left inverse if and only if φ is injective.
2. φ has a right inverse if and only if φ is surjective.

6.1.5 Matrix Interpretation

For a matrix $A \in \mathcal{M}_{m\times n}(k)$, the associated linear map is $L_A : k^n \to k^m$ given by $L_A(x) = Ax$. The Rank-Nullity Theorem becomes:

Proposition 6.1.4 Rank-Nullity for matrices

For $A \in \mathcal{M}_{m\times n}(k)$:

1. $\ker(L_A) = \{x \in k^n : Ax = 0\}$ is the null space of A.
2. $\operatorname{Im}(L_A) = \{Ax : x \in k^n\}$ is the column space of A (the span of the columns).
3. $\operatorname{null}(A) + \operatorname{rank}(A) = n$ (the number of columns).

Note:-

The rank of a matrix equals the number of linearly independent columns, which equals the number of linearly independent rows. The column space has dimension equal to the rank, and the null space has dimension equal to $n - \operatorname{rank}(A)$.
Looking ahead: when we study eigenvalues, the kernel will reappear in a crucial role. For a linear operator $T : V \to V$ and scalar λ, the **eigenspace** $E_\lambda = \ker(T - \lambda I)$ is precisely the kernel of the shifted operator. The dimension of this eigenspace (the geometric multiplicity of λ) measures "how degenerate" the eigenvalue is. The interplay between algebraic and geometric multiplicities—and the failure of these to match—is what makes Jordan normal form necessary.

Example 6.1.3 (Computing kernel and image)

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 3 & 5 \end{pmatrix}$ over $\mathbb{R}$.

Finding $\ker(A)$**:** Solve $Ax = 0$. Row reduce:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 3 & 5 \end{pmatrix} \xrightarrow{R_3 - R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Back-substituting: $x_2 = -2x_3$, $x_1 = -2x_2 - 3x_3 = 4x_3 - 3x_3 = x_3$. So

$$\ker(A) = \operatorname{Span}\{(1, -2, 1)^T\}, \quad \operatorname{null}(A) = 1.$$

Finding $\operatorname{Im}(A)$**:** The non-zero rows in RREF correspond to pivot columns. Columns 1 and 2 are pivot columns, so

$$\operatorname{Im}(A) = \operatorname{Span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \right\}, \quad \operatorname{rank}(A) = 2.$$

Verification: $\operatorname{null}(A) + \operatorname{rank}(A) = 1 + 2 = 3 = n$. ✓

Question 1

1. Let $D : \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be the differentiation operator. Find $\ker(D)$, $\operatorname{Im}(D)$, $\operatorname{null}(D)$, and $\operatorname{rank}(D)$. Verify the Rank-Nullity Theorem.
2. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be defined by $T(x_1, x_2, x_3, x_4) = (x_1 - x_2, x_2 - x_3, x_3 - x_4)$. Determine whether T is injective, surjective, or neither.
3. Prove that if $\varphi : V \to V$ is a linear operator on a finite-dimensional space with $\varphi^2 = \varphi$, then $V = \ker(\varphi) \oplus \operatorname{Im}(\varphi)$.
4. Let V and W be finite-dimensional vector spaces with $\dim(V) = \dim(W)$. Prove that a linear map $\varphi : V \to W$ is injective if and only if for every linearly independent list in V, its image under φ is linearly independent in W.

Solution

(1) The space $\mathcal{P}_3(\mathbb{R})$ has basis $\{1, x, x^2, x^3\}$, so $\dim(\mathcal{P}_3(\mathbb{R})) = 4$.
Kernel: $Dp = 0$ if and only if p is a constant polynomial. So $\ker(D) = \operatorname{Span}\{1\}$, which are the constant functions. Thus $\operatorname{null}(D) = 1$.
Image: Differentiating: $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, $D(x^3) = 3x^2$. The non-zero outputs span $\operatorname{Span}\{1, x, x^2\} = \mathcal{P}_2(\mathbb{R})$. Thus $\operatorname{Im}(D) = \mathcal{P}_2(\mathbb{R})$ and $\operatorname{rank}(D) = 3$.
Verification: $\operatorname{null}(D) + \operatorname{rank}(D) = 1 + 3 = 4 = \dim(\mathcal{P}_3(\mathbb{R}))$. ✓
(2) Injectivity: Find $\ker(T)$. Solve $T(x_1, x_2, x_3, x_4) = (0, 0, 0)$:

$$x_1 - x_2 = 0, \quad x_2 - x_3 = 0, \quad x_3 - x_4 = 0.$$

This gives $x_1 = x_2 = x_3 = x_4$. So $\ker(T) = \{(t, t, t, t) : t \in \mathbb{R}\} = \operatorname{Span}\{(1, 1, 1, 1)\}$.
Since $\ker(T) \neq \{0\}$, T is **not injective**.
Surjectivity: We have $\operatorname{null}(T) = 1$, so $\operatorname{rank}(T) = 4 - 1 = 3 = \dim(\mathbb{R}^3)$. Thus $\operatorname{Im}(T) = \mathbb{R}^3$, and T is **surjective**.
(3) We show $V = \ker(\varphi) + \operatorname{Im}(\varphi)$ and $\ker(\varphi) \cap \operatorname{Im}(\varphi) = \{0\}$.
Sum equals V: Let $v \in V$. Write $v = (v - \varphi(v)) + \varphi(v)$. Note that $\varphi(v) \in \operatorname{Im}(\varphi)$. Also:

$$\varphi(v - \varphi(v)) = \varphi(v) - \varphi^2(v) = \varphi(v) - \varphi(v) = 0,$$

so $v - \varphi(v) \in \ker(\varphi)$. Thus $v \in \ker(\varphi) + \operatorname{Im}(\varphi)$.
Intersection is trivial: Let $w \in \ker(\varphi) \cap \operatorname{Im}(\varphi)$. Then $\varphi(w) = 0$ and $w = \varphi(u)$ for some $u \in V$. So:

$$w = \varphi(u) = \varphi^2(u) = \varphi(\varphi(u)) = \varphi(w) = 0.$$

Thus $V = \ker(\varphi) \oplus \operatorname{Im}(\varphi)$.
(4) ($\Rightarrow$) Suppose φ is injective and $\{v_1, \ldots, v_m\}$ is linearly independent. Suppose $\sum c_i \varphi(v_i) = 0$. Then $\varphi(\sum c_i v_i) = 0$, so $\sum c_i v_i \in \ker(\varphi) = \{0\}$. Thus $\sum c_i v_i = 0$, and by linear independence of the v_i, all $c_i = 0$. Hence $\{\varphi(v_1), \ldots, \varphi(v_m)\}$ is linearly independent.
($\Leftarrow$) Suppose φ maps every linearly independent set to a linearly independent set. If $v \neq 0$, then $\{v\}$ is linearly independent, so $\{\varphi(v)\}$ is linearly independent, meaning $\varphi(v) \neq 0$. Thus $\ker(\varphi) = \{0\}$, so φ is injective.

6.2 Duality

The dual space of a vector space V consists of all linear maps from V to the base field k—these are called linear functionals. While this might seem like a minor variation on Hom-spaces, duality reveals useful structure: every basis has a "dual basis," every linear map has a "dual map," and the matrix of the dual map is the transpose of the original. In more advanced contexts (differential geometry, functional analysis, representation theory), duality becomes indispensable.

Note:-

The dual space construction is the first example of a **contravariant functor** you'll encounter. When we pass from V to V', linear maps $T : V \to W$ don't induce maps $V' \to W'$—instead, they induce maps $W' \to V'$, going the "wrong way." This reversal of arrows is the hallmark of contravariance. In category theory, the dual space functor $(-)' : \mathbf{Vect}_k^{\mathrm{op}} \to \mathbf{Vect}_k$ is representable: $V' = \mathrm{Hom}(V, k)$ is "the functor represented by k."

6.2.1 Linear Functionals and the Dual Space

Definition 6.2.1: Linear functional

A **linear functional** on a vector space V over a field k is a linear map $\ell : V \to k$. In other words, a linear functional is an element of $\mathrm{Hom}(V, k)$.

Example 6.2.1 (Examples of linear functionals)

1. **Coordinate projections:** On k^n, define $\pi_j(x_1, \dots, x_n) = x_j$. Each π_j is a linear functional.
2. **Weighted sums:** Fix $(c_1, \dots, c_n) \in k^n$. Define $\ell(x_1, \dots, x_n) = c_1x_1 + \cdots + c_nx_n$. This is a linear functional on k^n, and every linear functional on k^n has this form.
3. **Evaluation:** On $\mathcal{P}(\mathbb{R})$, define $\mathrm{ev}_a(p) = p(a)$ for fixed $a \in \mathbb{R}$. This is a linear functional.
4. **Integration:** On $\mathcal{P}(\mathbb{R})$, define $\ell(p) = \int_0^1 p(x)\,dx$. Linearity of the integral makes this a linear functional.
5. **Trace:** On $\mathcal{M}_{n\times n}(k)$, the trace $\mathrm{Tr}(A) = \sum_{i=1}^n a_{ii}$ is a linear functional.

Definition 6.2.2: Dual space

The **dual space** of V, denoted V^* or V', is the vector space of all linear functionals on V:

$$V^* = \mathrm{Hom}(V, k).$$

This is a vector space under pointwise addition and scalar multiplication: $(\ell_1 + \ell_2)(v) = \ell_1(v) + \ell_2(v)$ and $(\lambda\ell)(v) = \lambda \cdot \ell(v)$.

Note:-

We use V^* notation (common in differential geometry and physics) interchangeably with V' (common in algebra). Both mean the same thing. Some texts reserve V^* for the continuous dual in functional analysis, but for finite-dimensional spaces this distinction is irrelevant.

Proposition 6.2.1 Dimension of the dual space

If V is finite-dimensional with $\dim(V) = n$, then $\dim(V^*) = n$. Thus $V^* \cong V$ as vector spaces (non-canonically).

Proof: By our earlier result on Hom-spaces:

$$\dim(V^*) = \dim(\mathrm{Hom}(V, k)) = \dim(V) \cdot \dim(k) = n \cdot 1 = n.$$

□

Note:-

The isomorphism $V \cong V^*$ is non-canonical—it depends on a choice of basis. There's no "natural" way to identify vectors with linear functionals without extra structure. However, the double dual $V^{**} = (V^*)^*$

does have a canonical isomorphism with V, given by evaluation: $v \mapsto (\ell \mapsto \ell(v))$.

6.2.2 The Dual Basis

Given a basis of V, there's a natural associated basis of V^*.

Definition 6.2.3: Dual basis

Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis of V. The **dual basis** is the list $\mathcal{B}^* = \{\varphi_1, \dots, \varphi_n\}$ of linear functionals in V^* defined by

$$\varphi_j(v_i) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

In other words, φ_j "picks out" the j-th coordinate: if $v = c_1v_1 + \cdots + c_nv_n$, then $\varphi_j(v) = c_j$.

Proposition 6.2.2 Dual basis is a basis

The dual basis $\{\varphi_1, \dots, \varphi_n\}$ is indeed a basis of V^*.

Proof: **Linear independence:** Suppose $a_1\varphi_1 + \cdots + a_n\varphi_n = 0$ (the zero functional). Evaluating at v_k:

$$0 = (a_1\varphi_1 + \cdots + a_n\varphi_n)(v_k) = a_1\varphi_1(v_k) + \cdots + a_n\varphi_n(v_k) = a_k.$$

So all $a_k = 0$.
Spanning: Since we have n linearly independent vectors in V^* and $\dim(V^*) = n$, they form a basis. □

Proposition 6.2.3 Dual basis extracts coordinates

Let $\{v_1, \dots, v_n\}$ be a basis of V with dual basis $\{\varphi_1, \dots, \varphi_n\}$. For any $v \in V$:

$$v = \varphi_1(v)v_1 + \varphi_2(v)v_2 + \cdots + \varphi_n(v)v_n.$$

That is, $\varphi_j(v)$ is the j-th coordinate of v with respect to the basis $\{v_i\}$.

Proof: Write $v = c_1v_1 + \cdots + c_nv_n$. Applying φ_j:

$$\varphi_j(v) = c_1\varphi_j(v_1) + \cdots + c_n\varphi_j(v_n) = c_j.$$

Substituting back: $v = \varphi_1(v)v_1 + \cdots + \varphi_n(v)v_n$. □

Example 6.2.2 (Dual basis of the standard basis)

Let $\{e_1, \dots, e_n\}$ be the standard basis of k^n. The dual basis consists of the coordinate projections:

$$\varphi_j(x_1, \dots, x_n) = x_j.$$

These are the "row vectors" $\varphi_j = (0, \dots, 0, 1, 0, \dots, 0)$ (with 1 in position j).

6.2.3 The Dual Map

Just as a linear map $T : V \to W$ induces a map on Hom-spaces, it induces a map on dual spaces—but in the opposite direction.

Definition 6.2.4: Dual map (transpose)

Let $T : V \to W$ be a linear map. The **dual map** (or **transpose**) is the linear map $T^* : W^* \to V^*$ defined by

$$T^*(\psi) = \psi \circ T$$

for each $\psi \in W^*$. In other words, $T^*(\psi)$ is the linear functional on V given by $v \mapsto \psi(T(v))$.

Note:-

The dual map goes "backwards": $T : V \to W$ yields $T^* : W^* \to V^*$. This is because we're precomposing with T. If you think of linear functionals as "measurements," then $T^*(\psi)$ is the measurement you get by first applying T, then measuring with ψ. The reversal of direction is the essence of contravariance.

Proposition 6.2.4 The dual map is linear

For $T : V \to W$, the dual map $T^* : W^* \to V^*$ is indeed a linear map.

Proof: For $\psi_1, \psi_2 \in W^*$ and $\lambda \in k$:

$$\begin{aligned} T^*(\psi_1 + \psi_2) &= (\psi_1 + \psi_2) \circ T = \psi_1 \circ T + \psi_2 \circ T = T^*(\psi_1) + T^*(\psi_2), \\ T^*(\lambda\psi) &= (\lambda\psi) \circ T = \lambda(\psi \circ T) = \lambda T^*(\psi). \end{aligned}$$

□

Proposition 6.2.5 Algebraic properties of the dual map

Let $S : V \to W$ and $T : W \to U$ be linear maps.

1. $(S + T)^* = S^* + T^*$ (when both are defined).
2. $(\lambda T)^* = \lambda T^*$ for $\lambda \in k$.
3. $(T \circ S)^* = S^* \circ T^*$ (order reverses!).
4. $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$.

Proof: (3) is the key property. For $\psi \in U^*$:

$$(T \circ S)^*(\psi) = \psi \circ (T \circ S) = (\psi \circ T) \circ S = S^*(\psi \circ T) = S^*(T^*(\psi)) = (S^* \circ T^*)(\psi).$$

The other properties are similar. □

Note:-

Property (3) says the dual is a contravariant functor: it reverses the order of composition. Compare with the transpose of matrices: $(AB)^T = B^T A^T$. This is not a coincidence—the matrix of the dual map is the transpose!

6.2.4 Matrix of the Dual Map

The connection between dual maps and matrix transposes clarifies why the transpose operation is natural.

Theorem 6.2.1 Matrix of the dual map is the transpose

Let $T : V \to W$ be a linear map between finite-dimensional spaces. Fix bases $\mathcal{B} = \{v_1, \ldots, v_n\}$ of V and $\mathcal{C} = \{w_1, \ldots, w_m\}$ of W, with dual bases $\mathcal{B}^* = \{\varphi_1, \ldots, \varphi_n\}$ and $\mathcal{C}^* = \{\psi_1, \ldots, \psi_m\}$. If

$$A = \mathcal{M}(T; \mathcal{B}, \mathcal{C}) \in \mathcal{M}_{m \times n}(k)$$

is the matrix of T, then the matrix of T^* with respect to the dual bases is the transpose:

$$\mathcal{M}(T^*; \mathcal{C}^*, \mathcal{B}^*) = A^T \in \mathcal{M}_{n \times m}(k).$$

Proof: We need to show that $T^*(\psi_j) = \sum_{i=1}^{n} A_{ji}\varphi_i$ for each j.
By definition of the matrix A, we have $T(v_k) = \sum_{i=1}^{m} A_{ik}w_i$. Now compute:

$$T^*(\psi_j)(v_k) = \psi_j(T(v_k)) = \psi_j\left(\sum_{i=1}^{m} A_{ik}w_i\right) = \sum_{i=1}^{m} A_{ik}\psi_j(w_i) = A_{jk}.$$

On the other hand, if $T^*(\psi_j) = \sum_{i=1}^{n} c_i\varphi_i$, then

$$T^*(\psi_j)(v_k) = \sum_{i=1}^{n} c_i\varphi_i(v_k) = c_k.$$

Comparing: $c_k = A_{jk}$. Thus $T^*(\psi_j) = \sum_{k=1}^{n} A_{jk}\varphi_k$, which means the j-th column of $\mathcal{M}(T^*)$ is the j-th row of A. Hence $\mathcal{M}(T^*) = A^T$. □

Note:-

This theorem explains why the transpose is natural: it's not just a matrix operation, it's the coordinate representation of the dual map. The reversal of dimensions ($m \times n$ becomes $n \times m$) reflects the contravariance of the dual construction.

Corollary 6.2.1 Rank of transpose equals rank

For any matrix A, $\text{rank}(A^T) = \text{rank}(A)$.

Proof: The rank of A is $\dim(\text{Im}(T))$ where T is the associated linear map. By a result we'll see shortly, $\dim(\text{Im}(T^*)) = \dim(\text{Im}(T))$. Since the matrix of T^* is A^T, we get $\text{rank}(A^T) = \text{rank}(A)$. □

6.2.5 Kernel and Image of the Dual Map

The dual map interchanges the roles of kernel and image in a precise way.

Proposition 6.2.6 Relationships between T and T^*

Let $T : V \to W$ be a linear map between finite-dimensional spaces.

1. $\ker(T^*) = (\text{Im}(T))^\circ$, the annihilator of the image.
2. $\text{Im}(T^*) = (\ker(T))^\circ$, the annihilator of the kernel.
3. $\dim(\ker(T^*)) = \dim(W) - \text{rank}(T)$.
4. $\dim(\text{Im}(T^*)) = \text{rank}(T)$.

Here, for a subspace $U \subseteq V$, the **annihilator** is $U^\circ = \{\ell \in V^* : \ell(u) = 0 \text{ for all } u \in U\}$.

Note:-

The annihilator U° consists of all linear functionals that vanish on U. If $\dim(V) = n$ and $\dim(U) = k$, then $\dim(U^\circ) = n - k$. The annihilator provides a duality between subspaces of V and subspaces of V^*: bigger subspaces have smaller annihilators.

Question 2

1. Let $V = \mathbb{R}^3$ with standard basis $\{e_1, e_2, e_3\}$. Write down the dual basis explicitly as functions $\mathbb{R}^3 \to \mathbb{R}$.
2. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, y + z)$. Find the matrix of T with respect to the standard bases, then find the matrix of T^* with respect to the dual bases.
3. Prove that if $T : V \to W$ is surjective, then $T^* : W^* \to V^*$ is injective.
4. Prove that if V is finite-dimensional, the map $\Phi : V \to V^{**}$ defined by $\Phi(v)(\ell) = \ell(v)$ is an isomorphism.

Solution

(1) The dual basis consists of the coordinate projections:

$$\varphi_1(x, y, z) = x, \quad \varphi_2(x, y, z) = y, \quad \varphi_3(x, y, z) = z.$$

These are "row vectors" $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$.
(2) The matrix of T is:

$$A = \mathcal{M}(T) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

The matrix of T^* is the transpose:

$$A^T = \mathcal{M}(T^*) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Explicitly, if ψ_1, ψ_2 are the dual basis of $\mathbb{R}^2$ and $\varphi_1, \varphi_2, \varphi_3$ are the dual basis of $\mathbb{R}^3$:

$$T^*(\psi_1) = \varphi_1 + \varphi_2, \quad T^*(\psi_2) = \varphi_2 + \varphi_3.$$

(3) Suppose T is surjective and $T^*(\psi) = 0$, i.e., $\psi \circ T = 0$. For any $w \in W$, since T is surjective, $w = T(v)$ for some $v \in V$. Then:

$$\psi(w) = \psi(T(v)) = (\psi \circ T)(v) = 0.$$

Since this holds for all $w \in W$, we have $\psi = 0$. Thus $\ker(T^*) = \{0\}$, so T^* is injective.
(4) Well-defined and linear: For $v \in V$, $\Phi(v) : V^* \to k$ is defined by $\Phi(v)(\ell) = \ell(v)$. This is linear in ℓ (check!), so $\Phi(v) \in V^{**}$. Moreover, Φ is linear: $\Phi(v_1 + v_2)(\ell) = \ell(v_1 + v_2) = \ell(v_1) + \ell(v_2) = \Phi(v_1)(\ell) + \Phi(v_2)(\ell)$.
Injective: Suppose $\Phi(v) = 0$, meaning $\ell(v) = 0$ for all $\ell \in V^*$. If $v \neq 0$, extend $\{v\}$ to a basis $\{v, v_2, \ldots, v_n\}$. The dual basis element φ_1 satisfies $\varphi_1(v) = 1 \neq 0$, contradiction. So $v = 0$.
Surjective: Since $\dim(V) = \dim(V^*) = \dim(V^{**})$, injectivity implies surjectivity.

6.2.6 The Double Dual and Natural Isomorphisms

We've seen that $V \cong V^*$ for finite-dimensional V, but this isomorphism required a choice of basis. The double dual $V^{**} = (V^*)^*$ enjoys a more canonical relationship with V.

Definition 6.2.5: Evaluation map

The **evaluation map** is $\mathrm{ev} : V \to V^{**}$ defined by

$$\mathrm{ev}(v)(\ell) = \ell(v)$$

for $v \in V$ and $\ell \in V^*$. That is, $\mathrm{ev}(v)$ is the linear functional on V^* that "evaluates at v."

Proposition 6.2.7 Properties of the evaluation map

The evaluation map $\mathrm{ev} : V \to V^{**}$ satisfies:

1. ev is linear.
2. ev is injective.
3. If V is finite-dimensional, ev is an isomorphism.

Proof: **(1)** For $v, w \in V$, $\lambda \in k$, and $\ell \in V^*$:

$$\mathrm{ev}(v + \lambda w)(\ell) = \ell(v + \lambda w) = \ell(v) + \lambda\ell(w) = \mathrm{ev}(v)(\ell) + \lambda\,\mathrm{ev}(w)(\ell).$$

(2) Suppose $\mathrm{ev}(v) = 0$, meaning $\ell(v) = 0$ for all $\ell \in V^*$. If $v \neq 0$, extend $\{v\}$ to a basis and let ℓ be the dual basis element with $\ell(v) = 1$. Contradiction. So $v = 0$.

(3) For finite-dimensional V: $\dim(V^{**}) = \dim(V^*) = \dim(V)$. Since ev is injective, it's an isomorphism. ☐

Note:-

The crucial point: ev is natural—it doesn't depend on any choices. Given bases $\{e_1, \ldots, e_n\}$ of V, dual basis $\{e_1^*, \ldots, e_n^*\}$ of V^*, and double dual basis $\{e_1^{**}, \ldots, e_n^{**}\}$ of V^{**}, one can verify that $\mathrm{ev}(e_i) = e_i^{**}$. The evaluation map sends each basis element to its "double dual counterpart."

Note:-

For infinite-dimensional V, the situation is strikingly different. If V has a basis $\{e_i\}_{i\in I}$ indexed by an infinite set I, then every element of V is a finite linear combination $\sum_{i\in I} x_i e_i$ (with only finitely many $x_i \neq 0$). However, elements of V^* can be arbitrary: given any function $a : I \to k$, the map $\ell_a(\sum x_i e_i) = \sum x_i a_i$ is a well-defined linear functional. Thus $V^* \cong k^I = \prod_{i\in I} k$, the full product, which is much larger than the direct sum $V \cong \bigoplus_{i\in I} k$. The evaluation map $\mathrm{ev} : V \to V^{**}$ is still injective, but far from surjective.

Lecture 7.

More maps and spaces

7.0.1 Annihilators

The annihilator provides a precise duality between subspaces of V and subspaces of V^*.

Definition 7.0.1: Annihilator

For a subspace $U \subseteq V$, the **annihilator** of U is

$$\mathrm{Ann}(U) = U^\circ = \{\ell \in V^* : \ell(u) = 0 \text{ for all } u \in U\}.$$

Theorem 7.0.1 Dimension of the annihilator

If V is finite-dimensional and $U \subseteq V$ is a subspace, then

$$\dim(\mathrm{Ann}(U)) = \dim(V) - \dim(U).$$

Proof: The restriction map $\rho : V^* \to U^*$ defined by $\rho(\ell) = \ell|_U$ is a linear map with $\ker(\rho) = \mathrm{Ann}(U)$. We claim ρ is surjective: given $\psi \in U^*$, extend a basis of U to a basis of V, define ℓ to agree with ψ on U and be zero on the complementary basis vectors. Then $\rho(\ell) = \psi$.

By rank-nullity:

$$\dim(V^*) = \dim(\mathrm{Ann}(U)) + \dim(U^*).$$

Since $\dim(V^*) = \dim(V)$ and $\dim(U^*) = \dim(U)$, we get $\dim(\mathrm{Ann}(U)) = \dim(V) - \dim(U)$. □

Theorem 7.0.2 Annihilator isomorphism

For a subspace $U \subseteq V$, there is a natural isomorphism

$$\mathrm{Ann}(U) \cong (V/U)^*.$$

Proof: Define $\Phi : \mathrm{Ann}(U) \to (V/U)^*$ by $\Phi(\ell)(v + U) = \ell(v)$.

Well-defined: If $v + U = v' + U$, then $v - v' \in U$, so $\ell(v) - \ell(v') = \ell(v - v') = 0$ since $\ell \in \mathrm{Ann}(U)$.

Injective: If $\Phi(\ell) = 0$, then $\ell(v) = 0$ for all $v \in V$, so $\ell = 0$.

Surjective: Given $\psi \in (V/U)^*$, define $\ell = \psi \circ q$ where $q : V \to V/U$ is the quotient map. Then $\ell \in \mathrm{Ann}(U)$ and $\Phi(\ell) = \psi$. □

7.1 Quotient Spaces

Quotient spaces provide a systematic way to "collapse" a subspace to zero, creating a smaller vector space that captures the structure "perpendicular" to the subspace. This construction is fundamental in linear algebra and pervades abstract algebra.

7.1.1 Construction of the Quotient Space

Definition 7.1.1: Cosets and quotient space

Let V be a vector space over k and $U \subseteq V$ a subspace. For $v \in V$, the **coset** of v modulo U is

$$v + U = \{v + u : u \in U\}.$$

The **quotient space** is the set of all cosets:

$$V/U = \{v + U : v \in V\}.$$

Note:-

Two cosets $v + U$ and $w + U$ are equal if and only if $v - w \in U$. Equivalently, the cosets partition V into equivalence classes under the relation $v \sim w \Leftrightarrow v - w \in U$.

Proposition 7.1.1 Vector space structure on the quotient

The quotient V/U becomes a vector space over k under the operations:

1. **Addition:** $(v + U) + (w + U) = (v + w) + U$.
2. **Scalar multiplication:** $\lambda(v + U) = (\lambda v) + U$.

The zero element is $0 + U = U$, and $-(v + U) = (-v) + U$.

Proof: The key is well-definedness: if $v+U = v'+U$ and $w+U = w'+U$, then $v-v' \in U$ and $w-w' \in U$, so $(v + w) - (v' + w') = (v - v') + (w - w') \in U$, meaning $(v + w) + U = (v' + w') + U$.

Similarly for scalar multiplication: if $v + U = v' + U$, then $v - v' \in U$, so $\lambda(v - v') = \lambda v - \lambda v' \in U$, giving $\lambda v + U = \lambda v' + U$.

The vector space axioms follow from those in V. $\square$

7.1.2 The Quotient Map and Exact Sequences

Definition 7.1.2: Quotient map

The **quotient map** (or **canonical projection**) is $q : V \to V/U$ defined by $q(v) = v + U$.

Proposition 7.1.2 Properties of the quotient map

The quotient map $q : V \to V/U$ satisfies:

1. q is linear.
2. q is surjective.
3. $\ker(q) = U$.

Proof: (1) $q(v + \lambda w) = (v + \lambda w) + U = (v + U) + \lambda(w + U) = q(v) + \lambda q(w)$.

(2) Every coset $v + U$ is $q(v)$.

(3) $v \in \ker(q) \Leftrightarrow v + U = 0 + U \Leftrightarrow v \in U$. $\square$

Note:-

This gives a **short exact sequence**:

$$0 \longrightarrow U \xrightarrow{\iota} V \xrightarrow{q} V/U \longrightarrow 0$$

where $\iota : U \hookrightarrow V$ is the inclusion. "Exact" means: $\mathrm{Im}(\iota) = \ker(q)$, ι is injective, and q is surjective. Short exact sequences are a central organizing principle in algebra.

Theorem 7.1.1 Dimension of the quotient space

If V is finite-dimensional and $U \subseteq V$ is a subspace, then

$$\dim(V/U) = \dim(V) - \dim(U).$$

Proof: Apply rank-nullity to $q : V \to V/U$:

$$\dim(V) = \dim(\ker(q)) + \dim(\mathrm{Im}(q)) = \dim(U) + \dim(V/U).$$

$\square$

Example 7.1.1 (Concrete example)

Let $V = \mathbb{R}^3$ and $U = \mathrm{Span}\{(1, 0, 0)\}$, the x-axis. Then $V/U \cong \mathbb{R}^2$:
The coset $(a, b, c) + U$ consists of all points $\{(a + t, b, c) : t \in \mathbb{R}\}$—a horizontal line through (a, b, c).
Two points have the same coset iff they have the same y and z coordinates. The map $(a, b, c) + U \mapsto (b, c)$ is an isomorphism $V/U \xrightarrow{\sim} \mathbb{R}^2$.

7.1.3 Universal Property of the Quotient

The quotient space is characterized by a universal property that explains its importance.

Theorem 7.1.2 Universal property of the quotient

Let $\varphi : V \to W$ be a linear map with $U \subseteq \ker(\varphi)$. Then there exists a unique linear map $\bar{\varphi} : V/U \to W$ such that $\varphi = \bar{\varphi} \circ q$:

$$\begin{tikzcd} V \arrow[r, "\varphi"] \arrow[d, "q"'] & W \\ V/U \arrow[ur, dashed, "\exists!\bar{\varphi}"'] & \end{tikzcd}$$

Moreover, $\bar{\varphi}$ is defined by $\bar{\varphi}(v + U) = \varphi(v)$.

Proof: **Well-defined:** If $v + U = v' + U$, then $v - v' \in U \subseteq \ker(\varphi)$, so $\varphi(v) = \varphi(v')$.
Linearity: $\bar{\varphi}((v+U)+\lambda(w+U)) = \bar{\varphi}((v+\lambda w)+U) = \varphi(v+\lambda w) = \varphi(v)+\lambda\varphi(w) = \bar{\varphi}(v+U)+\lambda\bar{\varphi}(w+U)$.
Uniqueness: Any $\psi : V/U \to W$ with $\varphi = \psi \circ q$ must satisfy $\psi(v + U) = \psi(q(v)) = \varphi(v) = \bar{\varphi}(v + U)$. $\square$

Corollary 7.1.1 First Isomorphism Theorem

For any linear map $\varphi : V \to W$:

$$V/\ker(\varphi) \cong \mathrm{Im}(\varphi).$$

The isomorphism is $\bar{\varphi} : v + \ker(\varphi) \mapsto \varphi(v)$.

Proof: By the universal property, $\bar{\varphi} : V/\ker(\varphi) \to W$ is well-defined. Its image is exactly $\mathrm{Im}(\varphi)$. Injectivity: if $\bar{\varphi}(v + \ker(\varphi)) = 0$, then $\varphi(v) = 0$, so $v \in \ker(\varphi)$, meaning $v + \ker(\varphi) = 0 + \ker(\varphi)$. $\square$

Note:-

This theorem is remarkably useful. It says that every surjective linear map $\varphi : V \twoheadrightarrow W$ is "really" a quotient map: $V \to V/\ker(\varphi) \cong W$. Dually, every injective map $\iota : U \hookrightarrow V$ makes U isomorphic to a subspace of V.

7.1.4 Correspondence Between Subspaces

The quotient construction establishes a bijection between certain subspaces.

Theorem 7.1.3 Lattice correspondence

Let $U \subseteq V$ be a subspace. There is a bijection

$$\{\text{subspaces } W \subseteq V \text{ with } U \subseteq W\} \longleftrightarrow \{\text{subspaces of } V/U\}$$

given by:

- Forward: $W \mapsto W/U = \{w + U : w \in W\}$.
- Backward: $\bar{W} \mapsto q^{-1}(\bar{W}) = \{v \in V : v + U \in \bar{W}\}$.

Moreover, this bijection preserves inclusions: $W_1 \subseteq W_2 \Leftrightarrow W_1/U \subseteq W_2/U$.

Proof: We verify the maps are inverses.

Forward then backward: Starting with $W \supseteq U$, we get W/U, then $q^{-1}(W/U) = \{v \in V : v + U \in W/U\}$. But $v + U \in W/U$ iff $v + U = w + U$ for some $w \in W$, iff $v - w \in U \subseteq W$, iff $v \in W$. So $q^{-1}(W/U) = W$.

Backward then forward: Starting with $\bar{W} \subseteq V/U$, let $W = q^{-1}(\bar{W})$. Note $U = q^{-1}(0) \subseteq q^{-1}(\bar{W}) = W$. Then $W/U = q(W) = q(q^{-1}(\bar{W})) = \bar{W}$ (surjectivity of q ensures $q(q^{-1}(S)) = S$ for any $S \subseteq V/U$). $\square$

Note:-

Maximal proper subspaces of V containing U correspond to maximal proper subspaces of V/U—i.e., hyperplanes (codimension-1 subspaces). This perspective becomes useful in algebraic geometry and representation theory.

7.2 Linear Operators and Endomorphisms

When the domain and codomain of a linear map coincide, we gain additional algebraic structure: the ability to compose maps with themselves.

7.2.1 The Ring of Endomorphisms

Definition 7.2.1: Linear operator / Endomorphism

A **linear operator** or **endomorphism** of a vector space V is a linear map $\varphi : V \to V$. We write

$$\mathrm{End}(V) = \mathrm{Hom}(V, V)$$

for the set of all endomorphisms of V.

Proposition 7.2.1 Ring structure on $\mathrm{End}(V)$

$\mathrm{End}(V)$ is a **ring** (with identity) under:

- **Addition:** $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$.

- **Multiplication:** $(\varphi \circ \psi)(v) = \varphi(\psi(v))$ (composition).
- **Identity:** The identity map id_V.

Moreover, $\mathrm{End}(V)$ is a k-algebra: multiplication distributes over scalar multiplication.

Proof: Addition makes $\mathrm{End}(V)$ an abelian group (it's a subgroup of $\mathrm{Hom}(V,V)$ as a vector space). Composition is associative with identity id_V. Distributivity: $\varphi\circ(\psi+\rho) = \varphi\circ\psi+\varphi\circ\rho$ and $(\varphi+\psi)\circ\rho = \varphi\circ\rho+\psi\circ\rho$. These follow from linearity. □

Note:-

The ring $\mathrm{End}(V)$ is noncommutative unless $\dim(V) \leqslant 1$. For $V = k^n$, $\mathrm{End}(V) \cong M_n(k)$, the ring of $n \times n$ matrices. The noncommutativity of matrix multiplication reflects the noncommutativity of $\mathrm{End}(V)$.

7.2.2 Polynomials of Operators

Since endomorphisms can be composed, we can form "powers" and evaluate polynomials.

Definition 7.2.2: Powers and polynomials of operators

For $\varphi \in \mathrm{End}(V)$ and $n \in \mathbb{Z}_{\geqslant 0}$:

$$\varphi^0 = \mathrm{id}_V, \quad \varphi^n = \underbrace{\varphi \circ \varphi \circ \cdots \circ \varphi}_{n \text{ times}}.$$

For a polynomial $p(x) = a_0 + a_1x + \cdots + a_nx^n \in k[x]$, define

$$p(\varphi) = a_0\mathrm{id}_V + a_1\varphi + \cdots + a_n\varphi^n \in \mathrm{End}(V).$$

Proposition 7.2.2 Evaluation homomorphism

The map $\mathrm{ev}_\varphi : k[x] \to \mathrm{End}(V)$ defined by $p(x) \mapsto p(\varphi)$ is a ring homomorphism. That is:

1. $(p+q)(\varphi) = p(\varphi) + q(\varphi)$.
2. $(pq)(\varphi) = p(\varphi) \circ q(\varphi)$.
3. $1(\varphi) = \mathrm{id}_V$.

Proof: (1) and (3) are immediate. For (2), let $p(x) = \sum a_ix^i$ and $q(x) = \sum b_jx^j$. Then $(pq)(x) = \sum_k c_kx^k$ where $c_k = \sum_{i+j=k} a_ib_j$. So

$$(pq)(\varphi) = \sum_k c_k\varphi^k = \sum_k \left(\sum_{i+j=k} a_ib_j\right)\varphi^k = \left(\sum_i a_i\varphi^i\right) \circ \left(\sum_j b_j\varphi^j\right) = p(\varphi) \circ q(\varphi).$$

□

Note:-

This makes $\mathrm{End}(V)$ a $k[x]$-module: $p(x) \cdot v = p(\varphi)(v)$. The study of this module structure (particularly the kernel of ev_φ, called the minimal polynomial) is central to the theory of canonical forms.

7.3 Invariant Subspaces

Understanding an operator often begins with finding subspaces on which its behavior is simpler.

Definition 7.3.1: Invariant subspace

A subspace $W \subseteq V$ is **invariant** under $\varphi \in \mathrm{End}(V)$ (or **φ-invariant**) if

$$\varphi(W) \subseteq W,$$

i.e., $\varphi(w) \in W$ for all $w \in W$.

Example 7.3.1 (Basic examples of invariant subspaces)
For any $\varphi \in \mathrm{End}(V)$:

- $\{0\}$ and V are always φ-invariant.
- $\ker(\varphi)$ is φ-invariant: if $\varphi(v) = 0$, then $\varphi(\varphi(v)) = \varphi(0) = 0$.
- $\mathrm{Im}(\varphi)$ is φ-invariant: $\varphi(\mathrm{Im}(\varphi)) = \mathrm{Im}(\varphi^2) \subseteq \mathrm{Im}(\varphi)$.
- More generally, $\ker(\varphi^n)$ and $\mathrm{Im}(\varphi^n)$ are φ-invariant for all $n \geqslant 0$.

Proposition 7.3.1 Restriction to an invariant subspace
If $W \subseteq V$ is φ-invariant, the **restriction** $\varphi|_W : W \to W$ is a well-defined linear operator on W.

7.3.1 Block Diagonal Structure

Invariant subspaces lead to simpler matrix representations.

Theorem 7.3.1 Direct sum of invariant subspaces
Suppose $V = V_1 \oplus V_2$ where both V_1 and V_2 are φ-invariant. Choose bases $\mathcal{B}_1 = \{e_1, \dots, e_m\}$ of V_1 and $\mathcal{B}_2 = \{e_{m+1}, \dots, e_n\}$ of V_2, forming a basis $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$ of V. Then the matrix of φ with respect to $\mathcal{B}$ is **block diagonal**:

$$\mathcal{M}(\varphi; \mathcal{B}) = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

where $A_1 = \mathcal{M}(\varphi|_{V_1}; \mathcal{B}_1)$ and $A_2 = \mathcal{M}(\varphi|_{V_2}; \mathcal{B}_2)$.

Proof: For $i \leqslant m$: $\varphi(e_i) \in V_1$ (by φ-invariance), so $\varphi(e_i)$ is a linear combination of $e_1, \dots, e_m$ only. For $i > m$: $\varphi(e_i) \in V_2$, so $\varphi(e_i)$ is a linear combination of $e_{m+1}, \dots, e_n$ only. This gives the block structure. $\square$

Note:-
Conversely, if a matrix is block diagonal with blocks A_1 and A_2, then the spans of the corresponding basis vectors are invariant subspaces. The goal of much of spectral theory is to decompose V into a direct sum of invariant subspaces on which φ acts simply.

Corollary 7.3.1 Block triangular from one invariant subspace
If V_1 is φ-invariant (but we don't assume V_1 has an invariant complement), then extending a basis of V_1 to a basis of V gives a **block upper-triangular** matrix:

$$\mathcal{M}(\varphi) = \begin{pmatrix} A_1 & * \\ 0 & A_2 \end{pmatrix}.$$

The upper-right block $*$ may be nonzero; it measures the "interaction" between V_1 and V/V_1.

7.4 Eigenvalues and Eigenvectors

The simplest invariant subspaces are one-dimensional. These give rise to the fundamental notion of eigenvalues.

7.4.1 Definitions and Basic Properties

Definition 7.4.1: Eigenvector and eigenvalue

Let $\varphi \in \mathrm{End}(V)$. A nonzero vector $v \in V$ is an **eigenvector** of φ if

$$\varphi(v) = \lambda v$$

for some scalar $\lambda \in k$. The scalar λ is called the **eigenvalue** corresponding to v.

Note:-

The condition $\varphi(v) = \lambda v$ says that φ acts on the line spanned by v as scaling by λ. Equivalently, $\mathrm{Span}\{v\}$ is a one-dimensional φ-invariant subspace. Conversely, any one-dimensional invariant subspace $U = \mathrm{Span}\{v\}$ consists of eigenvectors (with the same eigenvalue) plus zero.

Proposition 7.4.1 Characterization of eigenvalues

For $\varphi \in \mathrm{End}(V)$ and $\lambda \in k$, the following are equivalent:

1. λ is an eigenvalue of φ.
2. $\varphi - \lambda \mathrm{id}_V$ is not injective.
3. $\ker(\varphi - \lambda \mathrm{id}_V) \neq \{0\}$.
4. (If V is finite-dimensional) $\varphi - \lambda \mathrm{id}_V$ is not surjective.
5. (If V is finite-dimensional) $\det(\varphi - \lambda \mathrm{id}_V) = 0$.

Proof: (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3): λ is an eigenvalue iff there exists nonzero v with $\varphi(v) = \lambda v$, iff $(\varphi - \lambda \mathrm{id})(v) = 0$ for some nonzero v, iff $\ker(\varphi - \lambda \mathrm{id}) \neq \{0\}$.

(3) $\Leftrightarrow$ (4) $\Leftrightarrow$ (5): In finite dimensions, a linear operator is injective iff surjective iff invertible iff has nonzero determinant. So $\ker(\varphi - \lambda \mathrm{id}) \neq \{0\}$ iff $\det(\varphi - \lambda \mathrm{id}) = 0$. □

Definition 7.4.2: Eigenspace

For an eigenvalue λ of φ, the **eigenspace** is

$$E(\lambda, \varphi) = \ker(\varphi - \lambda \mathrm{id}_V) = \{v \in V : \varphi(v) = \lambda v\}.$$

This is a subspace of V (it includes 0, though 0 is not an eigenvector).

Theorem 7.4.1 Eigenvectors with distinct eigenvalues are linearly independent

Let $v_1, \ldots, v_m$ be eigenvectors of φ with distinct eigenvalues $\lambda_1, \ldots, \lambda_m$. Then $v_1, \ldots, v_m$ are linearly independent.

Proof: Induction on m. Base case $m = 1$: a single eigenvector is nonzero, hence linearly independent.

Suppose the result holds for $m-1$ eigenvectors. Assume $\sum_{i=1}^{m} c_i v_i = 0$. Apply φ:

$$\sum_{i=1}^{m} c_i \lambda_i v_i = 0.$$

Multiply the original equation by λ_m and subtract:

$$\sum_{i=1}^{m-1} c_i(\lambda_i - \lambda_m) v_i = 0.$$

By the induction hypothesis (since $v_1, \ldots, v_{m-1}$ are eigenvectors with distinct eigenvalues $\lambda_1, \ldots, \lambda_{m-1}$), we have $c_i(\lambda_i - \lambda_m) = 0$ for $i < m$. Since $\lambda_i \neq \lambda_m$, we get $c_i = 0$. Substituting back: $c_m v_m = 0$, so $c_m = 0$ (since $v_m \neq 0$). ☐

Corollary 7.4.1 Bound on number of eigenvalues

A linear operator on an n-dimensional space has at most n distinct eigenvalues.

Proof: If $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues with eigenvectors $v_1, \ldots, v_m$, then $v_1, \ldots, v_m$ are linearly independent. In an n-dimensional space, any linearly independent set has at most n elements. ☐

Corollary 7.4.2 Sum of eigenspaces is direct

If $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of φ, then

$$E(\lambda_1, \varphi) + \cdots + E(\lambda_m, \varphi) = E(\lambda_1, \varphi) \oplus \cdots \oplus E(\lambda_m, \varphi).$$

is a direct sum. Furthermore, $\sum_i \dim E(\lambda_i, \varphi) \leqslant \dim V$.

7.4.2 Diagonalizable Operators

The best-case scenario: V has a basis of eigenvectors.

Definition 7.4.3: Diagonalizable operator

An operator $\varphi \in \mathrm{End}(V)$ is **diagonalizable** if there exists a basis of V consisting entirely of eigenvectors of φ. Equivalently, φ has a diagonal matrix with respect to some basis.

Note:-

If $\{v_1, \ldots, v_n\}$ is a basis of eigenvectors with $\varphi(v_i) = \lambda_i v_i$, then the matrix of φ is

$$\mathcal{M}(\varphi) = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

The eigenvalues appear on the diagonal (possibly with repetition).

Theorem 7.4.2 Characterizations of diagonalizability

For $\varphi \in \mathrm{End}(V)$ with $\dim(V) = n$ and distinct eigenvalues $\lambda_1, \ldots, \lambda_m$, the following are equivalent:

1. φ is diagonalizable.
2. V has a basis of eigenvectors of φ.
3. $V = E(\lambda_1, \varphi) \oplus \cdots \oplus E(\lambda_m, \varphi)$.
4. $\dim E(\lambda_1, \varphi) + \cdots + \dim E(\lambda_m, \varphi) = n$.

Proof: (1) $\Leftrightarrow$ (2): By definition.

(2) $\Rightarrow$ (3): If V has a basis of eigenvectors, partition it by eigenvalue. Each part is a basis of the corresponding eigenspace, and together they span V.

(3) $\Rightarrow$ (4): Take dimensions: $\dim V = \dim(E_1 \oplus \cdots \oplus E_m) = \sum \dim E_i$.

(4) $\Rightarrow$ (2): The sum $E_1 + \cdots + E_m$ is direct (earlier corollary), and has dimension $\sum \dim E_i = n = \dim V$. So $E_1 \oplus \cdots \oplus E_m = V$. Combining bases of each E_i gives a basis of V consisting of eigenvectors. $\square$

Corollary 7.4.3 Enough distinct eigenvalues implies diagonalizability

If $\varphi \in \mathrm{End}(V)$ has $\dim(V)$ distinct eigenvalues, then φ is diagonalizable.

Proof: Each eigenspace has dimension at least 1 (it contains an eigenvector). With $n = \dim V$ distinct eigenvalues, $\sum \dim E(\lambda_i) \geqslant n$. But this sum is also $\leqslant n$, so equality holds, and criterion (4) above is satisfied. $\square$

7.4.3 Examples: Diagonalizable and Non-Diagonalizable

Example 7.4.1 (Diagonal matrices)

The identity matrix I_n has eigenvalue 1 with eigenspace all of k^n. More generally, a diagonal matrix $\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ has the standard basis vectors as eigenvectors with eigenvalues $\lambda_1, \ldots, \lambda_n$. Such a matrix is (trivially) diagonalizable—it's already diagonal!

Example 7.4.2 (A non-diagonalizable matrix)

Consider $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ with matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The eigenvalues satisfy $\det(A - \lambda I) = (1-\lambda)^2 = 0$, so $\lambda = 1$ is the only eigenvalue.

The eigenspace: $E(1, \varphi) = \ker(A - I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \mathrm{Span}\{e_1\}$.

This has dimension 1, but $\dim(\mathbb{R}^2) = 2$. So φ is **not diagonalizable**.
Geometrically, φ is a "shear": it fixes the x-axis pointwise and slides other points parallel to the x-axis. The only eigenvector direction is along the x-axis.

Example 7.4.3 (No eigenvectors over $\mathbb{R}$)

Consider rotation by $90°$ in $\mathbb{R}^2$:

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The characteristic polynomial is $\det(R - \lambda I) = \lambda^2 + 1$, which has no real roots. So R has <u>no eigenvalues</u> over $\mathbb{R}$, hence no eigenvectors. Over $\mathbb{C}$, the eigenvalues are $\pm i$ with eigenvectors $(1, \mp i)^T$, and R is diagonalizable as a complex matrix. The eigenfairy doesn't visit $\mathbb{R}$ (sad).
This illustrates that diagonalizability depends on the scalar field!

Note:-

The rotation example shows that not every operator is diagonalizable. Even over $\mathbb{C}$, where every polynomial has roots, an operator may fail to be diagonalizable (like the shear above). Understanding non-diagonalizable operators requires the theory of Jordan normal form, which we'll develop later.

Question 1: Exercises on quotient spaces, operators, and eigenvalues

1. Let $V = \mathbb{R}^4$ and $U = \{(x, y, z, w) : x + y = 0, z = w\}$. Find $\dim(U)$, $\dim(V/U)$, and describe V/U explicitly.

2. Prove that if $\varphi, \psi \in \mathrm{End}(V)$ commute ($\varphi \circ \psi = \psi \circ \varphi$), then every eigenspace of φ is ψ-invariant.

3. Let $\varphi \in \mathrm{End}(V)$ be diagonalizable with eigenvalues $\lambda_1, \ldots, \lambda_m$. Prove that $p(\varphi) = 0$ where $p(x) = \prod_{i=1}^{m}(x - \lambda_i)$.

4. Show that the quotient map $q : V \to V/U$ induces a bijection between {linear maps $V/U \to W$} and {linear maps $V \to W$ vanishing on U}.

5. Let $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$. Find the eigenvalues and eigenspaces of A. Is A diagonalizable? If so, find a matrix P such that $P^{-1}AP$ is diagonal.

Solution

(1) The constraints $x + y = 0$ and $z = w$ give 2 independent linear equations on $\mathbb{R}^4$, so $\dim(U) = 4 - 2 = 2$. A basis: $\{(-1, 1, 0, 0), (0, 0, 1, 1)\}$.
By the dimension formula: $\dim(V/U) = \dim(V) - \dim(U) = 4 - 2 = 2$.
The map $\pi : \mathbb{R}^4 \to \mathbb{R}^2$ defined by $\pi(x, y, z, w) = (x + y, z - w)$ has $\ker(\pi) = U$. By the first isomorphism theorem, $V/U \cong \mathbb{R}^2$. Explicitly, the coset $(x, y, z, w)+U$ is determined by $(x+y, z-w)$.
(2) Let v be an eigenvector of φ with eigenvalue λ: $\varphi(v) = \lambda v$. We need $\psi(v) \in E(\lambda, \varphi)$, i.e., $\varphi(\psi(v)) = \lambda\psi(v)$.
Compute: $\varphi(\psi(v)) = \psi(\varphi(v)) = \psi(\lambda v) = \lambda\psi(v)$. ✓
(We used commutativity in the first equality.)
(3) Let $p(x) = \prod_{i=1}^{m}(x - \lambda_i)$. Since φ is diagonalizable, $V = \bigoplus_{i=1}^{m} E(\lambda_i, \varphi)$. For any eigenvector $v \in E(\lambda_j, \varphi)$:

$$(\varphi - \lambda_j \mathrm{id})(v) = 0.$$

So $p(\varphi)(v) = (\varphi - \lambda_1) \cdots (\varphi - \lambda_m)(v) = 0$ because one factor gives zero.
Since this holds for all eigenvectors and they span V, we have $p(\varphi) = 0$.
(4) Define $\Phi : \mathrm{Hom}(V/U, W) \to \{T \in \mathrm{Hom}(V, W) : U \subseteq \ker(T)\}$ by $\Phi(\bar{T}) = \bar{T} \circ q$.
Injective: If $\bar{T} \circ q = 0$, then $\bar{T}(v + U) = \bar{T}(q(v)) = 0$ for all v, so $\bar{T} = 0$.
Surjective: Given $T : V \to W$ with $U \subseteq \ker(T)$, define $\bar{T} : V/U \to W$ by $\bar{T}(v+U) = T(v)$. Well-defined because $v - v' \in U \Rightarrow T(v) = T(v')$. Then $\Phi(\bar{T}) = \bar{T} \circ q = T$.
(5) Characteristic polynomial: $\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2)$.
Eigenvalues: $\lambda_1 = 5$, $\lambda_2 = 2$.
Eigenspaces:

- $\lambda = 5$: $(A - 5I)v = 0$ gives $\begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$, so $x = 2y$. Eigenspace: $\mathrm{Span}\{(2, 1)^T\}$.
- $\lambda = 2$: $(A - 2I)v = 0$ gives $\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$, so $x = -y$. Eigenspace: $\mathrm{Span}\{(1, -1)^T\}$.

Since we have 2 linearly independent eigenvectors in $\mathbb{R}^2$, A is diagonalizable.
Let $P = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$ (columns are eigenvectors). Then $P^{-1}AP = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$.

7.5 Let's get general up in here!

7.5.1 Generalized Kernals (Null Spaces)

We want to look at general eigenvectors: things that are not *actually* eigenvectors but "kinda want to be" in a generalized sense. What in tarnation does that mean!? Simplest Eigenspace is just 0, so let's focus on that to get an understanding:

Definition 7.5.1: Generalized kernal

The generalized kernal of φ is $\text{gker}(\varphi) = \{v \in V \mid \exists m > 0 \text{ s.t. } \varphi^m(v) = 0\}$. These are all the vecrtors that are evnetually sent to 0 by repeatedly applying φ.

Note:-

We note that $\text{gker}(\varphi)$ is not standard notation. Auroux likes it though, so that is what we are going with! *If you meet a stranger on the street and bring up gker be sure to tell them what you mean, lest they be confused.*

The difference between the kernal and generalized kernal can be summed up by this: you are in the kernal if you go to 0 in one step; if you apply itterative steps—φ^m—and end up at 0 you are in the generalized kernal. If I look at the kernal of φ I notice that it is in the kernal of φ^2; then too this holds for more! They may or may not get larger but they are subsets:

$$\text{Ker}(\varphi) \subset \text{Ker}(\varphi^2) \subset \text{Ker}\,\varphi \subset \dots$$

Proposition 7.5.1

If $\text{Ker}(\varphi^m) = \text{Ker}(\varphi^{m+1})$ then there is no strictly larger kernal.

Proof: $\varphi^{m+2}(v) = \varphi^{m+1}(\varphi(V)) = 0 \iff \varphi(v) = \text{Ker}\,\varphi^m \iff \varphi^m(\varphi(v)) = \varphi^{m+1}(v) = 0$. So $\text{Ker}(\varphi^{m+1}) = \text{Ker}(\varphi^{m+2})$. Since the sequence stops increasing after at most $n = \dim V$ steps, $\text{gker}(\varphi) = \text{Ker}\,\varphi^n$. $\square$

Example 7.5.1

$\varphi : k^2 \to k^2$ with $e_1 \mapsto 0$, $e_2 \mapsto e_1$ and $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, then $\text{Ker}(\varphi) = k \cdot e_1$, but $\text{Ker}(\varphi^2) = \text{gker}(\varphi) = k^2$.

Note:-

The generalized kernal is often referred to as the null space (this is the terminology that Axler uses, for example). I may use this interchangeably.

Since it is not the case that $V = \text{gker}(\varphi) \oplus \text{Img}(\varphi)$ for every $\varphi \in \text{Hom}(V, V)$, we can use the following proposition as a substitute:

Proposition 7.5.2 Generalized kernel-image decomposition

Suppose $\varphi \in \text{End}(V)$ and let m be the smallest positive integer such that $\text{Ker}(\varphi^m) = \text{Ker}(\varphi^{m+1})$. Then

$$V = \text{Ker}(\varphi^m) \oplus \text{Im}(\varphi^m).$$

In particular, $V = \text{gker}(\varphi) \oplus \text{Im}(\varphi^{\dim V})$.

Proof: We verify the two conditions for a direct sum decomposition.

Intersection is trivial: Suppose $v \in \text{Ker}(\varphi^m) \cap \text{Im}(\varphi^m)$. Write $v = \varphi^m(u)$ for some $u \in V$. Then $\varphi^m(v) = \varphi^{2m}(u) = 0$, so $u \in \text{Ker}(\varphi^{2m})$.

We claim $\operatorname{Ker}(\varphi^{2m}) = \operatorname{Ker}(\varphi^m)$. By the stabilization property, once the kernel sequence stabilizes at step m, it stays constant: $\operatorname{Ker}(\varphi^m) = \operatorname{Ker}(\varphi^{m+1}) = \cdots = \operatorname{Ker}(\varphi^{2m})$. Thus $u \in \operatorname{Ker}(\varphi^m)$, which means $v = \varphi^m(u) = 0$.

Sum is all of V: By rank-nullity applied to φ^m:

$$\dim V = \dim \operatorname{Ker}(\varphi^m) + \dim \operatorname{Im}(\varphi^m).$$

Since we just showed $\operatorname{Ker}(\varphi^m) \cap \operatorname{Im}(\varphi^m) = \{0\}$, the sum $\operatorname{Ker}(\varphi^m) + \operatorname{Im}(\varphi^m)$ is direct, and its dimension equals $\dim \operatorname{Ker}(\varphi^m) + \dim \operatorname{Im}(\varphi^m) = \dim V$. Hence it equals V. □

Note:-

This decomposition is the key structural result for understanding non-diagonalizable operators. The generalized kernel $\operatorname{gker}(\varphi)$ captures the "nilpotent part" of φ, while $\operatorname{Im}(\varphi^m)$ is where φ acts invertibly. On $\operatorname{gker}(\varphi)$, repeated application of φ eventually kills everything; on $\operatorname{Im}(\varphi^m)$, the restriction of φ is an isomorphism.

Corollary 7.5.1 Restriction to image is an isomorphism

With notation as above, the restriction $\varphi|_{\operatorname{Im}(\varphi^m)} : \operatorname{Im}(\varphi^m) \to \operatorname{Im}(\varphi^m)$ is an isomorphism.

Proof: First, $\operatorname{Im}(\varphi^m)$ is φ-invariant since $\varphi(\operatorname{Im}(\varphi^m)) = \operatorname{Im}(\varphi^{m+1}) \subseteq \operatorname{Im}(\varphi^m)$. For injectivity: if $v \in \operatorname{Im}(\varphi^m)$ and $\varphi(v) = 0$, then $v \in \operatorname{Ker}(\varphi) \subseteq \operatorname{Ker}(\varphi^m)$. But $\operatorname{Ker}(\varphi^m) \cap \operatorname{Im}(\varphi^m) = \{0\}$, so $v = 0$. Since $\varphi|_{\operatorname{Im}(\varphi^m)}$ is injective on a finite-dimensional space, it's also surjective onto its image, which equals $\operatorname{Im}(\varphi^m)$ by φ-invariance. □

7.5.2 Generalized Eigenspaces

Now we extend the idea of generalized kernels to arbitrary eigenvalues, not just $\lambda = 0$.

Definition 7.5.2: Generalized eigenspace

Let $\varphi \in \operatorname{End}(V)$ and let λ be an eigenvalue of φ. The **generalized eigenspace** of φ corresponding to λ is

$$V_\lambda = \operatorname{Ker}(\varphi - \lambda \mathrm{id})^{\dim V} = \{v \in V : (\varphi - \lambda \mathrm{id})^n(v) = 0 \text{ for some } n \geqslant 1\}.$$

Equivalently, $V_\lambda = \operatorname{gker}(\varphi - \lambda \mathrm{id})$.

Note:-

The ordinary eigenspace $E(\lambda, \varphi) = \operatorname{Ker}(\varphi - \lambda \mathrm{id})$ is always contained in V_λ, but V_λ may be strictly larger. The elements of $V_\lambda \setminus E(\lambda, \varphi)$ are called **generalized eigenvectors**—they are not true eigenvectors, but they are "eventually killed" by $\varphi - \lambda \mathrm{id}$.

Example 7.5.2 (Generalized eigenspace of the shear)

Recall the shear $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ with matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. The only eigenvalue is $\lambda = 1$, and the eigenspace is $E(1, \varphi) = \operatorname{Ker}(A - I) = \operatorname{Ker} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \operatorname{Span}\{e_1\}$, which is one-dimensional.

However, $(A - I)^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0$.

So the generalized eigenspace is $V_1 = \operatorname{Ker}(A - I)^2 = \mathbb{R}^2$. The vector e_2 is a generalized eigenvector: $(A - I)e_2 = e_1 \neq 0$, but $(A - I)^2 e_2 = 0$.

Proposition 7.5.3 Basic properties of generalized eigenspaces

Let $\varphi \in \operatorname{End}(V)$ with eigenvalue λ. Then:

1. $E(\lambda, \varphi) \subseteq V_\lambda$.
2. V_λ is φ-invariant.
3. On V_λ, the operator $\varphi - \lambda\mathrm{id}$ is nilpotent.
4. If $\mu \neq \lambda$ are distinct eigenvalues, then $V_\lambda \cap V_\mu = \{0\}$.

Proof: (1) is immediate: if $(\varphi - \lambda\mathrm{id})(v) = 0$, then certainly $(\varphi - \lambda\mathrm{id})^n(v) = 0$ for all $n \geqslant 1$.

(2) Let $v \in V_\lambda$, so $(\varphi - \lambda\mathrm{id})^n(v) = 0$ for some n. Since φ commutes with $\varphi - \lambda\mathrm{id}$, we have

$$(\varphi - \lambda\mathrm{id})^n(\varphi(v)) = \varphi((\varphi - \lambda\mathrm{id})^n(v)) = \varphi(0) = 0.$$

Thus $\varphi(v) \in V_\lambda$.

(3) By definition, $V_\lambda = \mathrm{Ker}(\varphi - \lambda\mathrm{id})^m$ for some $m \leqslant \dim V$. So $(\varphi - \lambda\mathrm{id})^m$ restricted to V_λ is the zero map.

(4) Suppose $v \in V_\lambda \cap V_\mu$ with $\lambda \neq \mu$. Then there exist $j, k \geqslant 1$ such that $(\varphi - \lambda\mathrm{id})^j(v) = 0$ and $(\varphi - \mu\mathrm{id})^k(v) = 0$.

The polynomials $p(x) = (x - \lambda)^j$ and $q(x) = (x - \mu)^k$ are coprime (they share no common roots). By Bézout's identity, there exist polynomials $a(x), b(x) \in k[x]$ such that $a(x)p(x) + b(x)q(x) = 1$.

Evaluating at φ: $a(\varphi)p(\varphi) + b(\varphi)q(\varphi) = \mathrm{id}$. Applying to v:

$$v = a(\varphi)p(\varphi)(v) + b(\varphi)q(\varphi)(v) = a(\varphi)(0) + b(\varphi)(0) = 0.$$

$\square$

Note:-

Part (4) shows that generalized eigenspaces for distinct eigenvalues intersect trivially, just like ordinary eigenspaces. The proof uses the coprimality of $(x - \lambda)^j$ and $(x - \mu)^k$—this is a technique that will reappear in the generalized eigenspace decomposition.

7.5.3 The Generalized Eigenspace Decomposition

The following theorem is the main structural result for operators on finite-dimensional vector spaces over algebraically closed fields.

Theorem 7.5.1 Generalized eigenspace decomposition

Let V be a finite-dimensional vector space over an algebraically closed field k, and let $\varphi \in \mathrm{End}(V)$. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of φ. Then

$$V = V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_m}.$$

Moreover, $\sum_{i=1}^m \dim V_{\lambda_i} = \dim V$.

Proof: We proceed in two steps.

Step 1: The sum is direct. We already showed that $V_{\lambda_i} \cap V_{\lambda_j} = \{0\}$ for $i \neq j$. More generally, suppose $v_1 + \cdots + v_m = 0$ where $v_i \in V_{\lambda_i}$. We show each $v_i = 0$.

For each i, let n_i be such that $(\varphi - \lambda_i\mathrm{id})^{n_i}(v_i) = 0$. Define $p_i(x) = (x - \lambda_i)^{n_i}$ and

$$q_i(x) = \prod_{j \neq i} (x - \lambda_j)^{n_j}.$$

Since λ_i is not a root of $q_i(x)$, the polynomials $p_i(x)$ and $q_i(x)$ are coprime. By Bézout, there exist $a_i(x), b_i(x)$ with $a_i(x)p_i(x) + b_i(x)q_i(x) = 1$.

Now, $q_i(\varphi)(v_j) = 0$ for all $j \neq i$ (since p_j divides q_i). Applying $q_i(\varphi)$ to $v_1 + \cdots + v_m = 0$:

$$0 = q_i(\varphi)(v_1 + \cdots + v_m) = q_i(\varphi)(v_i).$$

Thus $v_i \in \operatorname{Ker}(q_i(\varphi)) \cap \operatorname{Ker}(p_i(\varphi)) = V_{\lambda_i} \cap \operatorname{Ker}(q_i(\varphi))$.

But from $a_i(\varphi)p_i(\varphi)(v_i) + b_i(\varphi)q_i(\varphi)(v_i) = v_i$ and both terms being zero, we get $v_i = 0$.

Step 2: The sum equals V. Define $p(x) = \prod_{i=1}^{m}(x - \lambda_i)^{n_i}$ where $n_i = \dim V_{\lambda_i}$ (we'll justify this choice shortly). We claim $p(\varphi) = 0$, i.e., p annihilates φ.

Consider the polynomial $q_i(x) = p(x)/(x-\lambda_i)^{n_i}$. These polynomials are pairwise coprime (each q_i excludes exactly the factor $(x - \lambda_i)^{n_i}$). Their gcd is 1, so there exist $r_i(x)$ with $\sum_i r_i(x)q_i(x) = 1$.

For any $v \in V$, write $v = \sum_i r_i(\varphi)q_i(\varphi)(v)$. We claim $r_i(\varphi)q_i(\varphi)(v) \in V_{\lambda_i}$. Indeed:

$$p_i(\varphi)(r_i(\varphi)q_i(\varphi)(v)) = r_i(\varphi)(p_i(\varphi)q_i(\varphi))(v) = r_i(\varphi)p(\varphi)(v).$$

Once we establish that $p(\varphi) = 0$ (which follows from Cayley-Hamilton, proved later, or can be verified by dimension counting), this shows v is a sum of elements from the V_{λ_i}.

Alternative dimension argument: Since k is algebraically closed, φ has at least one eigenvalue. Apply the decomposition $V = \operatorname{gker}(\varphi - \lambda_1 \mathrm{id}) \oplus \operatorname{Im}((\varphi - \lambda_1 \mathrm{id})^{n_1})$ where n_1 is the stabilization index. The first summand is V_{λ_1}. On the second summand, $\varphi - \lambda_1\mathrm{id}$ is invertible, so λ_1 is not an eigenvalue of φ restricted there. Proceed by induction on $\dim V$. □

Corollary 7.5.2 Dimension of generalized eigenspaces

With notation as above, if $\chi_\varphi(x) = \prod_{i=1}^{m}(x-\lambda_i)^{n_i}$ is the characteristic polynomial, then $\dim V_{\lambda_i} = n_i$.

Note:-

The exponent n_i in the characteristic polynomial is called the **algebraic multiplicity** of λ_i. The dimension of the ordinary eigenspace $\dim E(\lambda_i, \varphi)$ is called the **geometric multiplicity**. We always have:

$$1 \leqslant \dim E(\lambda_i, \varphi) \leqslant \dim V_{\lambda_i} = n_i.$$

The operator φ is diagonalizable if and only if geometric multiplicity equals algebraic multiplicity for every eigenvalue.

Example 7.5.3 (Decomposition for a 3×3 matrix)

Consider $\varphi : \mathbb{C}^3 \to \mathbb{C}^3$ with matrix

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

The characteristic polynomial is $\chi_A(x) = (x-2)^2(x-3)$, so $\lambda_1 = 2$ has algebraic multiplicity 2 and $\lambda_2 = 3$ has algebraic multiplicity 1.
For $\lambda_1 = 2$:

$$A - 2I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (A-2I)^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

So $V_2 = \operatorname{Ker}(A-2I)^2 = \operatorname{Span}\{e_1, e_2\}$, which has dimension 2.
For $\lambda_2 = 3$: $V_3 = \operatorname{Ker}(A - 3I) = \operatorname{Span}\{e_3\}$, dimension 1.
Indeed, $\mathbb{C}^3 = V_2 \oplus V_3$.

7.6 Nilpotent Operators

Nilpotent operators are the building blocks for understanding non-diagonalizable maps. On each generalized eigenspace V_λ, the operator $\varphi - \lambda\mathrm{id}$ is nilpotent, so understanding nilpotent operators is the key to understanding all operators.

7.6.1 Definition and Basic Properties

Definition 7.6.1: Nilpotent operator

An operator $\varphi \in \mathrm{End}(V)$ is **nilpotent** if $\varphi^m = 0$ for some positive integer m. The smallest such m is called the **index of nilpotency** (or **nilpotence index**) of φ.

Example 7.6.1 (Basic examples of nilpotent operators)

1. The zero operator is nilpotent with index 1.
2. The shift operator on k^n with matrix $\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$ is nilpotent with index n.
3. Any strictly upper-triangular matrix (zeros on and below the diagonal) is nilpotent.

Proposition 7.6.1 Characterizations of nilpotent operators

For $\varphi \in \mathrm{End}(V)$ with $\dim V = n$, the following are equivalent:

1. φ is nilpotent.
2. $\varphi^n = 0$.
3. The only eigenvalue of φ is 0 (if k is algebraically closed).
4. $V = \mathrm{gker}(\varphi)$.

Proof: (1) $\Rightarrow$ (2): If $\varphi^m = 0$, then the chain $\{0\} \subseteq \mathrm{Ker}(\varphi) \subseteq \mathrm{Ker}(\varphi^2) \subseteq \cdots$ stabilizes by step m, hence by step n (since dimensions are bounded by n). So $\mathrm{Ker}(\varphi^n) = \mathrm{gker}(\varphi)$. But $\varphi^m = 0$ means $\mathrm{Ker}(\varphi^m) = V$, so $\mathrm{gker}(\varphi) = V$, which means $\varphi^n = 0$.

(2) $\Rightarrow$ (3): Suppose λ is an eigenvalue with eigenvector $v \neq 0$. Then $\varphi(v) = \lambda v$, so $\varphi^n(v) = \lambda^n v$. If $\varphi^n = 0$, then $\lambda^n v = 0$, so $\lambda^n = 0$, hence $\lambda = 0$.

(3) $\Rightarrow$ (4): Over an algebraically closed field, by the generalized eigenspace decomposition, $V = \bigoplus_\lambda V_\lambda$. If the only eigenvalue is 0, then $V = V_0 = \mathrm{gker}(\varphi)$.

(4) $\Rightarrow$ (1): If $V = \mathrm{gker}(\varphi) = \mathrm{Ker}(\varphi^n)$, then $\varphi^n = 0$. $\square$

Note:-

Over a non-algebraically closed field, (3) needs modification: φ is nilpotent iff φ has no eigenvalues in any field extension, or equivalently, iff the characteristic polynomial is x^n.

Corollary 7.6.1 Nilpotent matrices are similar to strictly upper-triangular matrices

If φ is nilpotent, then there exists a basis in which the matrix of φ is strictly upper-triangular.

Proof: We prove this by induction on $n = \dim V$. If $n = 1$, then $\varphi = 0$ and we're done.

For $n > 1$: since φ is nilpotent and nonzero (otherwise trivial), $\mathrm{Ker}(\varphi) \neq \{0\}$. Pick $0 \neq v_1 \in \mathrm{Ker}(\varphi)$ and extend to a basis $\{v_1, v_2, \ldots, v_n\}$. The matrix of φ has first column all zeros.

Alternatively, consider $W = \mathrm{Im}(\varphi)$. Since φ is nilpotent and nonzero, $W \subsetneq V$ (otherwise φ would be surjective, hence bijective, contradicting nilpotency). The restriction $\varphi|_W : W \to W$ is well-defined (W is

φ-invariant) and nilpotent. By induction, choose a basis of W making $\varphi|_W$ strictly upper-triangular, extend to a basis of V, and the result follows. $\square$

7.6.2 The Nilpotent Basis Construction

The key to understanding nilpotent operators is building a basis adapted to the "chains" of vectors connected by φ.

Definition 7.6.2: Chains and Jordan chains

Let $\varphi \in \mathrm{End}(V)$ be nilpotent. A **chain** (or **Jordan chain**) of length m is a sequence of vectors $v_1, v_2, \ldots, v_m$ such that:

$$\varphi(v_m) = v_{m-1}, \quad \varphi(v_{m-1}) = v_{m-2}, \quad \ldots, \quad \varphi(v_2) = v_1, \quad \varphi(v_1) = 0.$$

The vector v_m is called the **head** of the chain, and v_1 is the **tail**.

Note:-

In a chain, $v_1 \in \mathrm{Ker}(\varphi)$, $v_2 \in \mathrm{Ker}(\varphi^2) \setminus \mathrm{Ker}(\varphi)$, and generally $v_j \in \mathrm{Ker}(\varphi^j) \setminus \mathrm{Ker}(\varphi^{j-1})$. The chain encodes how φ "moves" vectors toward the kernel.

Lemma 7.6.1 Key lemma for chain construction

Let φ be nilpotent. Suppose $U \subseteq \mathrm{Ker}(\varphi^{k+1})$ is a subspace satisfying $U \cap \mathrm{Ker}(\varphi^k) = \{0\}$. Then:

1. $\varphi|_U$ is injective.
2. $\varphi(U) \subseteq \mathrm{Ker}(\varphi^k)$.
3. $\varphi(U) \cap \mathrm{Ker}(\varphi^{k-1}) = \{0\}$ (if $k \geqslant 1$).

Proof: (1) If $v \in U$ and $\varphi(v) = 0$, then $v \in U \cap \mathrm{Ker}(\varphi) \subseteq U \cap \mathrm{Ker}(\varphi^k) = \{0\}$.

(2) For $v \in U \subseteq \mathrm{Ker}(\varphi^{k+1})$, we have $\varphi^{k+1}(v) = 0$, so $\varphi^k(\varphi(v)) = 0$, meaning $\varphi(v) \in \mathrm{Ker}(\varphi^k)$.

(3) If $\varphi(v) \in \mathrm{Ker}(\varphi^{k-1})$, then $\varphi^k(v) = \varphi^{k-1}(\varphi(v)) = 0$, so $v \in \mathrm{Ker}(\varphi^k)$. By assumption $U \cap \mathrm{Ker}(\varphi^k) = \{0\}$, so $v = 0$, hence $\varphi(v) = 0$. $\square$

Theorem 7.6.1 Structure theorem for nilpotent operators

Let $\varphi \in \mathrm{End}(V)$ be nilpotent with index m (so $\varphi^m = 0$ but $\varphi^{m-1} \neq 0$). Then there exists a basis of V of the form

$$\{v_{1,1}, \ldots, v_{1,m_1};\ v_{2,1}, \ldots, v_{2,m_2};\ \ldots;\ v_{\ell,1}, \ldots, v_{\ell,m_\ell}\}$$

where each sequence $(v_{i,1}, \ldots, v_{i,m_i})$ is a chain:

$$\varphi(v_{i,j}) = v_{i,j-1} \text{ for } j > 1, \quad \varphi(v_{i,1}) = 0.$$

The chain lengths satisfy $m = m_1 \geqslant m_2 \geqslant \cdots \geqslant m_\ell \geqslant 1$ and $\sum_{i=1}^{\ell} m_i = \dim V$.

Proof: We construct the basis by working down from $\mathrm{Ker}(\varphi^m) = V$.

Step 1: Let U_m be a complement of $\mathrm{Ker}(\varphi^{m-1})$ in $V = \mathrm{Ker}(\varphi^m)$. Pick a basis $\{v_{1,m}, \ldots, v_{k_m,m}\}$ of U_m. These will be the heads of chains of length m.

By the lemma, $\varphi(U_m) \subseteq \mathrm{Ker}(\varphi^{m-1})$ and $\varphi(U_m) \cap \mathrm{Ker}(\varphi^{m-2}) = \{0\}$. Define $v_{i,m-1} = \varphi(v_{i,m})$ for each i. These vectors are linearly independent (since $\varphi|_{U_m}$ is injective).

Step 2: Now work in $\mathrm{Ker}(\varphi^{m-1})$. We have $\varphi(U_m) \subseteq \mathrm{Ker}(\varphi^{m-1})$ with $\varphi(U_m) \cap \mathrm{Ker}(\varphi^{m-2}) = \{0\}$. Extend $\varphi(U_m)$ to a larger subspace U_{m-1} that complements $\mathrm{Ker}(\varphi^{m-2})$ in $\mathrm{Ker}(\varphi^{m-1})$. Pick additional basis vectors $\{v_{k_m+1,m-1}, \ldots, v_{k_{m-1},m-1}\}$ to span the complement of $\varphi(U_m)$ in U_{m-1}. These start new chains of length $m-1$.

Continue inductively: At each stage, we have chains of various lengths. Applying φ extends existing chains, and we add new chain heads as needed to complement $\operatorname{Ker}(\varphi^{j-1})$ in $\operatorname{Ker}(\varphi^j)$.

Termination: At the final step, we're in $\operatorname{Ker}(\varphi)$, and all chains terminate. The collection of all chain vectors forms a basis of V. □

Note:-

The chain lengths $(m_1, m_2, \ldots, m_\ell)$ form a partition of $n = \dim V$. This partition is an invariant of the nilpotent operator—two nilpotent operators are similar iff they have the same partition.

7.6.3 Jordan Blocks for Nilpotent Operators

The chain basis gives the matrix of a nilpotent operator a very specific form.

Definition 7.6.3: Nilpotent Jordan block

A **nilpotent Jordan block** of size m is the $m \times m$ matrix

$$J_m(0) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

This has 1's on the superdiagonal and 0's elsewhere.

Corollary 7.6.2 Matrix form of nilpotent operators

With respect to a chain basis ordered as $(v_{1,1}, \ldots, v_{1,m_1}, v_{2,1}, \ldots, v_{2,m_2}, \ldots)$, the matrix of a nilpotent operator φ is block diagonal:

$$\mathcal{M}(\varphi) = \begin{pmatrix} J_{m_1}(0) & & 0 \\ & J_{m_2}(0) & \\ 0 & & \ddots \end{pmatrix}.$$

Proof: Within each chain $(v_{i,1}, \ldots, v_{i,m_i})$, we have $\varphi(v_{i,j}) = v_{i,j-1}$ for $j > 1$ and $\varphi(v_{i,1}) = 0$. In matrix form with this basis ordering, this gives exactly $J_{m_i}(0)$ for the i-th block. The blocks are independent since chains span complementary subspaces. □

Example 7.6.2 (Nilpotent operator with partition $(3, 2, 1)$)

Consider $\varphi : k^6 \to k^6$ nilpotent with chain lengths $3, 2, 1$. A chain basis might be:

$$\{v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{3,1}\}$$

where $\varphi(v_{1,3}) = v_{1,2}$, $\varphi(v_{1,2}) = v_{1,1}$, $\varphi(v_{1,1}) = 0$, etc.
The matrix is:

$$\mathcal{M}(\varphi) = \begin{pmatrix} 0 & 1 & 0 & & & \\ 0 & 0 & 1 & & 0 & \\ 0 & 0 & 0 & & & \\ & & & 0 & 1 & \\ & 0 & & 0 & 0 & \\ & & & & & 0 \end{pmatrix} = J_3(0) \oplus J_2(0) \oplus J_1(0).$$

Proposition 7.6.2 Uniqueness of the Jordan form for nilpotent operators

The partition $(m_1, m_2, \ldots, m_\ell)$ associated to a nilpotent operator φ is uniquely determined by φ. Equivalently, two nilpotent operators are similar iff they have the same Jordan block sizes (counting multiplicity).

Proof: The partition can be recovered from dimensions of iterated kernels. Let $d_j = \dim \mathrm{Ker}(\varphi^j)$. Then:

$$d_1 - d_0 = d_1 = \text{number of chains} = \ell$$

$$d_2 - d_1 = \text{number of chains of length} \geqslant 2$$

$$d_j - d_{j-1} = \text{number of chains of length} \geqslant j.$$

From this sequence of differences, we can reconstruct the partition: the number of blocks of size exactly j is $(d_j - d_{j-1}) - (d_{j+1} - d_j)$. Since the d_j are determined by φ, so is the partition. □

Note:-

This dimension-counting argument is the key to proving uniqueness of Jordan form in general. The rank of $(\varphi - \lambda I)^j$ determines how many Jordan blocks for λ have size $\geqslant j$.

7.7 Jordan Normal Form

We now combine the generalized eigenspace decomposition with the structure theory of nilpotent operators to obtain the Jordan normal form—the "best possible" matrix representation for an operator over an algebraically closed field.

7.7.1 Jordan Blocks

Definition 7.7.1: Jordan block

A **Jordan block** of size m with eigenvalue λ is the $m \times m$ matrix

$$J_m(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix} = \lambda I_m + J_m(0).$$

This has λ on the diagonal, 1's on the superdiagonal, and 0's elsewhere.

Example 7.7.1 (Small Jordan blocks)

$$J_1(\lambda) = (\lambda), \quad J_2(\lambda) = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}, \quad J_3(\lambda) = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}.$$

A 1×1 Jordan block is just a scalar—this corresponds to a true eigenvector. Larger blocks correspond to generalized eigenvectors that are not true eigenvectors.

Proposition 7.7.1 Properties of Jordan blocks

Let $J = J_m(\lambda)$. Then:

1. The only eigenvalue of J is λ, with algebraic multiplicity m.

2. The eigenspace $E(\lambda, J) = \mathrm{Ker}(J - \lambda I) = \mathrm{Span}\{e_1\}$ has dimension 1.
3. The generalized eigenspace is all of k^m.
4. $J - \lambda I = J_m(0)$ is nilpotent with index m.

Proof: (1) The characteristic polynomial is $\det(J - xI) = (\lambda - x)^m$.
(2) $J - \lambda I = J_m(0)$ has kernel spanned by e_1 (the first standard basis vector).
(3) Since $(J - \lambda I)^m = J_m(0)^m = 0$, every vector is in $\mathrm{Ker}(J - \lambda I)^m$.
(4) $J_m(0)^{m-1} \neq 0$ (it maps $e_m \mapsto e_1$) but $J_m(0)^m = 0$. $\square$

Note:-

A Jordan block $J_m(\lambda)$ represents the action of an operator on a single chain of generalized eigenvectors. The chain is $(e_1, e_2, \ldots, e_m)$ where $Je_j = \lambda e_j + e_{j-1}$ for $j > 1$ and $Je_1 = \lambda e_1$. Equivalently, $(J - \lambda I)e_j = e_{j-1}$ for $j > 1$ and $(J - \lambda I)e_1 = 0$.

7.7.2 The Jordan Normal Form Theorem

Theorem 7.7.1 Jordan normal form

Let V be a finite-dimensional vector space over an algebraically closed field k, and let $\varphi \in \mathrm{End}(V)$. Then there exists a basis of V with respect to which the matrix of φ is block diagonal:

$$\mathcal{M}(\varphi) = \begin{pmatrix} J_{m_1}(\lambda_1) & & 0 \\ & J_{m_2}(\lambda_2) & \\ 0 & & \ddots \end{pmatrix}$$

where each block is a Jordan block. This matrix is called the **Jordan normal form** (or **Jordan canonical form**) of φ.

Proof: By the generalized eigenspace decomposition, $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_r}$ where $\lambda_1, \ldots, \lambda_r$ are the distinct eigenvalues. Each V_{λ_i} is φ-invariant, so we can work on each generalized eigenspace separately.

On V_{λ_i}, the operator $\psi_i = \varphi|_{V_{\lambda_i}} - \lambda_i \mathrm{id}$ is nilpotent (this was proved earlier). By the structure theorem for nilpotent operators, there exists a chain basis of V_{λ_i} such that ψ_i has matrix $J_{m_1}(0) \oplus J_{m_2}(0) \oplus \cdots$.

Since $\varphi|_{V_{\lambda_i}} = \psi_i + \lambda_i \mathrm{id}$, the matrix of $\varphi|_{V_{\lambda_i}}$ in this basis is

$$J_{m_1}(0) + \lambda_i I \oplus J_{m_2}(0) + \lambda_i I \oplus \cdots = J_{m_1}(\lambda_i) \oplus J_{m_2}(\lambda_i) \oplus \cdots$$

Combining the bases for all generalized eigenspaces gives a basis of V in which φ has the desired block diagonal form. $\square$

Note:-

The Jordan form groups blocks by eigenvalue: all $J_m(\lambda_i)$ blocks for a given λ_i appear together, corresponding to the decomposition of V_{λ_i} into chains. The total size of blocks with eigenvalue λ_i equals $\dim V_{\lambda_i}$ = algebraic multiplicity of λ_i.

7.7.3 Existence and Uniqueness

Theorem 7.7.2 Uniqueness of Jordan normal form

The Jordan normal form of φ is unique up to permutation of the blocks. That is, the multiset of Jordan blocks $\{J_{m_1}(\lambda_1), J_{m_2}(\lambda_2), \ldots\}$ is uniquely determined by φ.

Proof: For each eigenvalue λ, the Jordan blocks with eigenvalue λ correspond to the partition of $\dim V_\lambda$ given by the nilpotent operator $\varphi|_{V_\lambda} - \lambda\mathrm{id}$. We showed this partition is uniquely determined by dimensions of iterated kernels:

$$d_j(\lambda) = \dim \mathrm{Ker}((\varphi - \lambda\mathrm{id})^j).$$

The number of blocks $J_m(\lambda)$ of size exactly m is

$$(d_m(\lambda) - d_{m-1}(\lambda)) - (d_{m+1}(\lambda) - d_m(\lambda)) = 2d_m(\lambda) - d_{m-1}(\lambda) - d_{m+1}(\lambda).$$

Since these dimensions are intrinsic to φ, the block structure is uniquely determined. □

Corollary 7.7.1 Similarity criterion

Two operators $\varphi, \psi \in \mathrm{End}(V)$ are similar if and only if they have the same Jordan normal form (up to block ordering). Equivalently, $\varphi \sim \psi$ iff for each $\lambda \in k$ and each $j \geqslant 1$:

$$\dim \mathrm{Ker}((\varphi - \lambda\mathrm{id})^j) = \dim \mathrm{Ker}((\psi - \lambda\mathrm{id})^j).$$

Note:-

This gives a complete set of similarity invariants. Unlike trace and determinant alone (which don't distinguish all similarity classes), the full sequence of kernel dimensions completely determines the similarity class.

7.7.4 Computing Jordan Form: The Algorithm

Given a matrix A, here is the procedure to find its Jordan form:

Step 1: Find eigenvalues. Compute $\chi_A(x) = \det(xI - A)$ and factor it as $\prod_i (x - \lambda_i)^{n_i}$. The λ_i are eigenvalues with algebraic multiplicities n_i.

Step 2: For each eigenvalue λ, determine block sizes. Compute $d_j = \dim \mathrm{Ker}(A - \lambda I)^j$ for $j = 1, 2, \ldots$ until stabilization. The number of blocks of size $\geqslant j$ is $d_j - d_{j-1}$. Alternatively:

- Number of blocks $= d_1 =$ geometric multiplicity.
- Number of blocks of size $\geqslant 2$ is $d_2 - d_1$.
- Number of blocks of size exactly j is $(d_j - d_{j-1}) - (d_{j+1} - d_j)$.

Step 3: Assemble the Jordan form. List all Jordan blocks grouped by eigenvalue.

Example 7.7.2 (Computing Jordan form)

Let $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.

Step 1: $\chi_A(x) = (x-2)^3$, so $\lambda = 2$ with multiplicity 3.

Step 2:

$$A - 2I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad (A-2I)^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (A-2I)^3 = 0.$$

So $d_1 = 1$, $d_2 = 2$, $d_3 = 3$.
Number of blocks $= d_1 = 1$. Size of the single block $= 3$.
Step 3: Jordan form is $J_3(2)$. (The matrix is already in Jordan form!)

Example 7.7.3 (A more interesting example)

Let $A = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$.

Step 1: $\chi_A(x) = (x-3)^4$, so $\lambda = 3$ with multiplicity 4.
Step 2:

$$A - 3I = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

$\mathrm{Ker}(A - 3I) = \mathrm{Span}\{e_1, e_3\}$, so $d_1 = 2$. $(A-3I)^2 = 0$, so $d_2 = 4$.
Number of blocks $= 2$. Blocks of size $\geqslant 2$: $d_2 - d_1 = 2$. So both blocks have size 2.
Step 3: Jordan form is $J_2(3) \oplus J_2(3)$.

Example 7.7.4 (Multiple eigenvalues)

Let $A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$.

Step 1: $\chi_A(x) = (x-1)^2(x-2)^2$. Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 2$, each with multiplicity 2.
Step 2 for $\lambda = 1$:

$$A - I = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Restricted to generalized eigenspace (first two coordinates): $d_1 = 1$, $d_2 = 2$. One block of size 2: $J_2(1)$.
Step 2 for $\lambda = 2$:

$$A - 2I = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Restricted to generalized eigenspace (last two coordinates): $d_1 = 2$. Two blocks of size 1: $J_1(2) \oplus J_1(2)$.

Step 3: Jordan form is $J_2(1) \oplus J_1(2) \oplus J_1(2) = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$.

Note:-

The last example shows that $\lambda = 2$ contributes two 1×1 blocks because the eigenspace has dimension 2 (geometric = algebraic multiplicity). This is the diagonalizable case for that eigenvalue. In contrast, $\lambda = 1$ has a 2×2 block because geometric multiplicity (1) is less than algebraic multiplicity (2).

7.8 Characteristic and Minimal Polynomials

We now introduce two polynomials canonically associated to a linear operator. The characteristic polynomial encodes eigenvalues and their algebraic multiplicities; the minimal polynomial captures the "smallest" polynomial relation satisfied by the operator.

7.8.1 The Characteristic Polynomial

Definition 7.8.1: Characteristic polynomial

Let $\varphi \in \mathrm{End}(V)$ with $\dim V = n$, and let $A = \mathcal{M}(\varphi)$ be the matrix of φ with respect to some basis. The **characteristic polynomial** of φ is

$$\chi_\varphi(x) = \det(xI - \varphi) = \det(xI - A).$$

This is a monic polynomial of degree n.

Proposition 7.8.1 Basic properties of the characteristic polynomial

1. $\chi_\varphi(x)$ is independent of the choice of basis (hence well-defined for the operator φ).
2. $\chi_\varphi(x) = x^n - (\mathrm{Tr}\,\varphi)x^{n-1} + \cdots + (-1)^n \det(\varphi)$.
3. The roots of $\chi_\varphi(x)$ are exactly the eigenvalues of φ.
4. If k is algebraically closed, $\chi_\varphi(x) = \prod_{i=1}^m (x - \lambda_i)^{n_i}$ where λ_i are the distinct eigenvalues and $n_i = \dim V_{\lambda_i}$.

Proof: (1) If $B = P^{-1}AP$ is the matrix in another basis, then

$$\det(xI - B) = \det(xI - P^{-1}AP) = \det(P^{-1}(xI - A)P) = \det(xI - A).$$

(2) Expand $\det(xI - A)$ using the Leibniz formula. The leading term comes from the diagonal product $\prod_i (x - a_{ii}) = x^n - (\sum_i a_{ii})x^{n-1} + \cdots$. The constant term is $\det(-A) = (-1)^n \det(A)$.

(3) λ is a root iff $\det(\lambda I - A) = 0$ iff $\lambda I - A$ is singular iff $\mathrm{Ker}(\varphi - \lambda \mathrm{id}) \neq \{0\}$ iff λ is an eigenvalue.

(4) This follows from the generalized eigenspace decomposition: the algebraic multiplicity of λ_i equals $\dim V_{\lambda_i}$. $\square$

Example 7.8.1 (Characteristic polynomial of a 2×2 matrix)

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$\chi_A(x) = \det \begin{pmatrix} x - a & -b \\ -c & x - d \end{pmatrix} = (x-a)(x-d) - bc = x^2 - (a+d)x + (ad - bc) = x^2 - (\mathrm{Tr}\,A)x + \det A.$$

7.8.2 The Cayley-Hamilton Theorem

The Cayley-Hamilton theorem states that every operator satisfies its own characteristic polynomial.

Theorem 7.8.1 Cayley-Hamilton

Let $\varphi \in \mathrm{End}(V)$ with characteristic polynomial $\chi_\varphi(x)$. Then

$$\chi_\varphi(\varphi) = 0.$$

That is, substituting φ for x in $\chi_\varphi(x)$ gives the zero operator.

Proof: We give two proofs.

Proof 1 (via Jordan form, over algebraically closed k): It suffices to prove this for Jordan blocks, since the Jordan form is block diagonal and $\chi_\varphi = \prod \chi_{J_i}$.

For a single Jordan block $J = J_m(\lambda)$, we have $\chi_J(x) = (x-\lambda)^m$. Then

$$\chi_J(J) = (J - \lambda I)^m = J_m(0)^m = 0$$

since $J_m(0)$ is nilpotent with index m.

Proof 2 (general, using adjugate matrix): Let A be the matrix of φ. The adjugate (classical adjoint) matrix $\mathrm{adj}(xI - A)$ is a matrix whose entries are polynomials in x of degree at most $n-1$. Write

$$\mathrm{adj}(xI - A) = B_{n-1}x^{n-1} + B_{n-2}x^{n-2} + \cdots + B_1 x + B_0$$

where B_i are constant matrices.

By the adjugate identity: $(xI - A) \cdot \mathrm{adj}(xI - A) = \det(xI - A) \cdot I = \chi_A(x) \cdot I$.

Expanding both sides and comparing coefficients of powers of x gives a system of matrix equations. Multiplying the equation for x^k by A^k and summing yields $\chi_A(A) = 0$. (The details involve telescoping—each B_i appears twice with opposite signs.) □

Note:-

The Cayley-Hamilton theorem does <u>not</u> follow from "plugging in A for x in $\det(xI - A) = 0$." The determinant is a scalar-valued function, not a matrix-valued function, so this substitution is meaningless. The theorem requires proof.

Corollary 7.8.1 Bound on nilpotence index

If φ is nilpotent on an n-dimensional space, then $\varphi^n = 0$.

Proof: The only eigenvalue is 0, so $\chi_\varphi(x) = x^n$. By Cayley-Hamilton, $\varphi^n = 0$. □

Corollary 7.8.2 Expression for the inverse

If φ is invertible, then φ^{-1} can be expressed as a polynomial in φ.

Proof: Write $\chi_\varphi(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$. Since φ is invertible, 0 is not an eigenvalue, so $c_0 = (-1)^n \det(\varphi) \neq 0$.

By Cayley-Hamilton: $\varphi^n + c_{n-1}\varphi^{n-1} + \cdots + c_1\varphi + c_0 I = 0$.

Rearranging: $\varphi(\varphi^{n-1} + c_{n-1}\varphi^{n-2} + \cdots + c_1 I) = -c_0 I$.

Thus $\varphi^{-1} = -\frac{1}{c_0}(\varphi^{n-1} + c_{n-1}\varphi^{n-2} + \cdots + c_1 I)$. □

7.8.3 The Minimal Polynomial

Definition 7.8.2: Minimal polynomial

The **minimal polynomial** $\mu_\varphi(x)$ of $\varphi \in \mathrm{End}(V)$ is the unique monic polynomial of smallest degree such that $\mu_\varphi(\varphi) = 0$.

Proposition 7.8.2 Existence and uniqueness of the minimal polynomial

1. There exists a nonzero polynomial $p(x)$ with $p(\varphi) = 0$.
2. Among all such polynomials, there is a unique monic one of minimal degree.
3. $\mu_\varphi(x)$ divides every polynomial $p(x)$ satisfying $p(\varphi) = 0$.

Proof: (1) Cayley-Hamilton gives $\chi_\varphi(\varphi) = 0$.

(2) Let d be the minimal degree of a nonzero polynomial annihilating φ. If $p(x)$ and $q(x)$ are both monic of degree d with $p(\varphi) = q(\varphi) = 0$, then $(p - q)(\varphi) = 0$ and $\deg(p - q) < d$. By minimality, $p - q = 0$.

(3) Given $p(x)$ with $p(\varphi) = 0$, perform polynomial division: $p(x) = q(x)\mu_\varphi(x) + r(x)$ where $\deg r < \deg \mu_\varphi$. Then $r(\varphi) = p(\varphi) - q(\varphi)\mu_\varphi(\varphi) = 0 - 0 = 0$. By minimality of μ_φ, we must have $r = 0$, so $\mu_\varphi \mid p$. □

Theorem 7.8.2 Relationship between characteristic and minimal polynomials

Let $\varphi \in \mathrm{End}(V)$.

1. $\mu_\varphi(x)$ divides $\chi_\varphi(x)$.
2. $\mu_\varphi(x)$ and $\chi_\varphi(x)$ have the same roots (i.e., the same irreducible factors over k).
3. Over an algebraically closed field, if $\chi_\varphi(x) = \prod_i (x - \lambda_i)^{n_i}$, then $\mu_\varphi(x) = \prod_i (x - \lambda_i)^{m_i}$ where $1 \leqslant m_i \leqslant n_i$.

Proof: (1) By Cayley-Hamilton, $\chi_\varphi(\varphi) = 0$. By the divisibility property of μ_φ, we have $\mu_\varphi \mid \chi_\varphi$.

(2) If λ is a root of χ_φ, then λ is an eigenvalue: there exists $v \neq 0$ with $\varphi(v) = \lambda v$. For any polynomial p, $p(\varphi)(v) = p(\lambda)v$. If $p(\varphi) = 0$, then $p(\lambda) = 0$. In particular, $\mu_\varphi(\lambda) = 0$.

Conversely, if $\mu_\varphi(\lambda) = 0$, then $(x - \lambda) \mid \mu_\varphi(x)$, so $(x - \lambda) \mid \chi_\varphi(x)$ by (1), meaning λ is a root of χ_φ.

(3) From (1) and (2), the factorizations have the same roots with $m_i \leqslant n_i$. The exponent m_i is the size of the largest Jordan block with eigenvalue λ_i (proof below). □

Proposition 7.8.3 Minimal polynomial and Jordan block sizes

The exponent of $(x - \lambda)$ in $\mu_\varphi(x)$ equals the size of the largest Jordan block with eigenvalue λ.

Proof: On the generalized eigenspace V_λ, the operator $\varphi - \lambda\mathrm{id}$ is nilpotent. Its nilpotence index is the size of the largest Jordan block (the longest chain). Call this m_λ.

For $(\varphi - \lambda\mathrm{id})^k$ to annihilate V_λ, we need $k \geqslant m_\lambda$. The smallest such k is m_λ.

Since $V = \bigoplus_\lambda V_\lambda$ and these are φ-invariant, $p(\varphi) = 0$ iff $p(\varphi)|_{V_\lambda} = 0$ for all λ. The latter requires $(x - \lambda)^{m_\lambda} \mid p(x)$. Thus $\mu_\varphi(x) = \prod_\lambda (x - \lambda)^{m_\lambda}$. □

7.8.4 Diagonalizability Criterion

Theorem 7.8.3 Diagonalizability via minimal polynomial

An operator $\varphi \in \mathrm{End}(V)$ is diagonalizable if and only if $\mu_\varphi(x)$ splits into distinct linear factors:

$$\mu_\varphi(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_m)$$

where $\lambda_1, \ldots, \lambda_m$ are distinct.

Proof: ($\Rightarrow$) If φ is diagonalizable, all Jordan blocks are 1×1. The largest block for each eigenvalue has size 1, so each $(x - \lambda_i)$ appears with exponent 1 in μ_φ.

($\Leftarrow$) If $\mu_\varphi(x) = \prod_i (x-\lambda_i)$ with distinct λ_i, then each Jordan block has size 1 (since the exponent of $(x-\lambda_i)$ in μ_φ equals the largest block size for λ_i). So φ has a basis of eigenvectors. $\square$

Corollary 7.8.3 Diagonalizability of operators with distinct eigenvalues

If φ has $n = \dim V$ distinct eigenvalues, then φ is diagonalizable.

Proof: The characteristic polynomial has n distinct roots, so $\chi_\varphi(x) = \prod_{i=1}^n (x-\lambda_i)$. Since $\mu_\varphi \mid \chi_\varphi$ and they have the same roots, μ_φ also has distinct linear factors. $\square$

Example 7.8.2 (Minimal polynomial examples)

1. For $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I$: $\chi_A(x) = (x-2)^2$, but $\mu_A(x) = x - 2$ since $(A - 2I) = 0$.
2. For $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$: $\chi_A(x) = (x-2)^2$. Since $A - 2I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0$, we need $\mu_A(x) = (x-2)^2 = \chi_A(x)$.
3. For $A = \operatorname{diag}(1,1,2,2,2)$: $\chi_A(x) = (x-1)^2(x-2)^3$, but $\mu_A(x) = (x-1)(x-2)$ since $(A-I)(A-2I) = 0$.

7.8.5 Exercises

Question 2: Exercises on advanced spectral theory

1. Let $\varphi : \mathbb{C}^4 \to \mathbb{C}^4$ have characteristic polynomial $\chi_\varphi(x) = (x-1)^2(x-2)^2$. List all possible Jordan normal forms for φ.
2. Prove that if A and B are similar matrices, then $\mu_A = \mu_B$.
3. Find the Jordan normal form and minimal polynomial of $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.
4. Let $\varphi \in \operatorname{End}(V)$ satisfy $\varphi^2 = \varphi$. Prove that φ is diagonalizable and find its possible eigenvalues.
5. Suppose $\varphi \in \operatorname{End}(V)$ has minimal polynomial $\mu_\varphi(x) = (x-1)(x-2)(x-3)$. What can you conclude about φ?

Solution

(1) The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$, each with algebraic multiplicity 2. For each eigenvalue, the Jordan blocks must have sizes summing to 2. Possibilities for each:

- One block of size 2: $J_2(\lambda)$
- Two blocks of size 1: $J_1(\lambda) \oplus J_1(\lambda)$

Combining independently for each eigenvalue, the possible Jordan forms are:

(i) $J_2(1) \oplus J_2(2)$

(ii) $J_2(1) \oplus J_1(2) \oplus J_1(2)$

(iii) $J_1(1) \oplus J_1(1) \oplus J_2(2)$

(iv) $J_1(1) \oplus J_1(1) \oplus J_1(2) \oplus J_1(2)$ (diagonal matrix)

(2) If $B = P^{-1}AP$, then for any polynomial p:

$$p(B) = p(P^{-1}AP) = P^{-1}p(A)P.$$

So $p(B) = 0$ iff $p(A) = 0$. The set of annihilating polynomials is the same, hence the minimal polynomial (the monic generator of this ideal) is the same.
(3) The matrix is already in Jordan form: $A = J_3(0)$.
Characteristic polynomial: $\chi_A(x) = x^3$.

Minimal polynomial: We need the smallest k with $A^k = 0$. Compute $A^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq 0$ and $A^3 = 0$. So $\mu_A(x) = x^3 = \chi_A(x)$.
(4) The condition $\varphi^2 = \varphi$ means $\varphi^2 - \varphi = 0$, so $\varphi(\varphi - \mathrm{id}) = 0$. Thus $p(x) = x(x-1) = x^2 - x$ annihilates φ.
The minimal polynomial divides $x(x-1)$. Since $x(x-1)$ has distinct roots, so does μ_φ. By the diagonalizability criterion, φ is diagonalizable.
The possible eigenvalues are roots of μ_φ, which divides $x(x-1)$. So eigenvalues are among $\{0, 1\}$.
(Such operators are called **projections** or **idempotents**. They decompose $V = \mathrm{Ker}(\varphi) \oplus \mathrm{Im}(\varphi)$ and project onto $\mathrm{Im}(\varphi)$ along $\mathrm{Ker}(\varphi)$.)
(5) Since $\mu_\varphi(x) = (x-1)(x-2)(x-3)$ has three distinct linear factors:

- φ is diagonalizable (by the criterion).
- The eigenvalues are exactly $1, 2, 3$ (roots of μ_φ).
- $\dim V \geqslant 3$ (at least one eigenvector per eigenvalue).
- $V = E(1, \varphi) \oplus E(2, \varphi) \oplus E(3, \varphi)$.

The eigenspace dimensions can be any positive integers summing to $\dim V$, but we need at least one of each.

Lecture 8.

Categories: An Interlude

Many otherwise disparate fields of mathematics share fundamental structures. The language of categories systematizes these common patterns. While category theory won't answer questions specific to a particular area, it suggests useful ways to think about problems by analogy with other settings. We've already encountered categorical ideas implicitly: the universal property of quotient spaces, the behavior of dual maps, and the isomorphism theorems all fit naturally into this framework.

8.1 The Language of Categories

8.1.1 Objects and Morphisms

The basic data of a category consists of objects and arrows between them, subject to axioms that capture the essential features of function composition.

Definition 8.1.1: Category

A **category** $\mathcal{C}$ consists of:

(1) A collection of **objects**, denoted $\mathrm{Ob}(\mathcal{C})$.

(2) For every pair of objects $A, B \in \mathrm{Ob}(\mathcal{C})$, a collection of **morphisms** from A to B, denoted $\mathrm{Mor}(A, B)$ or $\mathrm{Hom}_{\mathcal{C}}(A, B)$.

(3) A **composition rule**: for every triple $A, B, C \in \mathrm{Ob}(\mathcal{C})$, a map

$$\mathrm{Mor}(A, B) \times \mathrm{Mor}(B, C) \longrightarrow \mathrm{Mor}(A, C)$$

sending (f, g) to $g \circ f$.

These must satisfy two axioms:

(i) **Associativity:** For any $f \in \mathrm{Mor}(A, B)$, $g \in \mathrm{Mor}(B, C)$, $h \in \mathrm{Mor}(C, D)$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

(ii) **Identity:** For every object A, there exists $\mathrm{id}_A \in \mathrm{Mor}(A, A)$ such that for any $f \in \mathrm{Mor}(A, B)$,

$$f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f.$$

Note:-

The identity morphism id_A is unique: if id'_A also satisfies the identity axiom, then $\mathrm{id}'_A = \mathrm{id}'_A \circ \mathrm{id}_A = \mathrm{id}_A$.

We write $f : A \to B$ to indicate $f \in \mathrm{Mor}(A, B)$, and call A the **domain** (or source) and B the **codomain** (or target) of f.

Example 8.1.1 (The category of sets)

The category **Set** has:

- Objects: all sets.

- Morphisms: $\mathrm{Mor}(A, B) = \{f : A \to B \mid f \text{ is a function}\}$.
- Composition: ordinary function composition.

Variants include:

- **FinSet**: objects are finite sets, morphisms are functions.
- $\mathbf{Set}_*$: **pointed sets**. Objects are pairs (A, a) with $a \in A$ a distinguished element. Morphisms $(A, a) \to (B, b)$ are functions $f : A \to B$ with $f(a) = b$.

Example 8.1.2 (The category of groups)
The category **Grp** has:

- Objects: all groups.
- Morphisms: group homomorphisms.
- Composition: composition of homomorphisms.

The identity axiom holds because $\mathrm{id}_G : G \to G$ is a homomorphism. Variants:

- **Ab**: abelian groups and group homomorphisms.
- **FinGrp**: finite groups and homomorphisms.

Similarly, we have categories **Ring** (rings and ring homomorphisms) and **CRing** (commutative rings).

Example 8.1.3 (The category of vector spaces)
Fix a field k. The category $\mathbf{Vect}_k$ has:

- Objects: vector spaces over k.
- Morphisms: k-linear maps.
- Composition: composition of linear maps.

The subcategory $\mathbf{FinVect}_k$ consists of finite-dimensional vector spaces. We've been studying $\mathbf{Vect}_k$ all along: $\mathrm{Hom}(V, W)$ is exactly $\mathrm{Mor}(V, W)$ in this category.

Example 8.1.4 (The category of topological spaces)
The category **Top** has:

- Objects: topological spaces.
- Morphisms: continuous maps.
- Composition: composition of continuous maps.

The pointed variant $\mathbf{Top}_*$ has objects (X, x_0) (a space with basepoint) and morphisms being continuous maps preserving basepoints. This category is fundamental in algebraic topology.

Note:-

In all these examples, objects are "decorated sets"—sets equipped with additional structure (a group operation, a topology, a vector space structure)—and morphisms are functions that respect this structure. This is typical of categories arising in practice, though the abstract definition permits more exotic examples. For instance, any poset $(P, \leqslant)$ defines a category: objects are elements of P, and $\mathrm{Mor}(a, b)$ contains a single element if $a \leqslant b$ and is empty otherwise. Composition is forced by transitivity.

Example 8.1.5 (Posets as categories)
Let $(P, \leqslant)$ be a partially ordered set. Define a category $\mathcal{C}_P$ by:

- $\mathrm{Ob}(\mathcal{C}_P) = P$.
- $\mathrm{Mor}(a,b) = \begin{cases} \{*\} & \text{if } a \leqslant b \\ \varnothing & \text{otherwise} \end{cases}$
- Composition: the unique element of $\mathrm{Mor}(a,b) \times \mathrm{Mor}(b,c)$ maps to the unique element of $\mathrm{Mor}(a,c)$ (by transitivity).

The identity axiom uses reflexivity ($a \leqslant a$), and associativity is trivial since all compositions are unique when defined.

8.1.2 Isomorphisms

The notion of "sameness" in a category is captured by isomorphisms.

Definition 8.1.2: Isomorphism

Let $\mathcal{C}$ be a category and $A, B \in \mathrm{Ob}(\mathcal{C})$. A morphism $f \in \mathrm{Mor}(A,B)$ is an **isomorphism** if there exists $g \in \mathrm{Mor}(B,A)$ such that
$$g \circ f = \mathrm{id}_A \quad \text{and} \quad f \circ g = \mathrm{id}_B.$$
We write $A \cong B$ if such an isomorphism exists, and say A and B are **isomorphic**.

Proposition 8.1.1 Uniqueness of inverses
If $f : A \to B$ is an isomorphism, then the morphism g in the definition is unique. We denote it f^{-1} and call it the **inverse** of f.

Proof: Suppose $g, g' \in \mathrm{Mor}(B,A)$ both satisfy the conditions. Then
$$g = g \circ \mathrm{id}_B = g \circ (f \circ g') = (g \circ f) \circ g' = \mathrm{id}_A \circ g' = g'.$$
□

Proposition 8.1.2 Properties of isomorphisms
Let $\mathcal{C}$ be a category.

(i) For any object A, id_A is an isomorphism with $\mathrm{id}_A^{-1} = \mathrm{id}_A$.

(ii) If $f : A \to B$ is an isomorphism, then $f^{-1} : B \to A$ is an isomorphism with $(f^{-1})^{-1} = f$.

(iii) If $f : A \to B$ and $g : B \to C$ are isomorphisms, then $g \circ f : A \to C$ is an isomorphism with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof: (i) $\mathrm{id}_A \circ \mathrm{id}_A = \mathrm{id}_A$.
(ii) By definition, $f^{-1} \circ f = \mathrm{id}_A$ and $f \circ f^{-1} = \mathrm{id}_B$. This says exactly that f is the inverse of f^{-1}.
(iii) Compute:
$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \mathrm{id}_B \circ f = f^{-1} \circ f = \mathrm{id}_A.$$
Similarly, $(g \circ f) \circ (f^{-1} \circ g^{-1}) = \mathrm{id}_C$. □

Example 8.1.6 (Isomorphisms in familiar categories)

- In **Set**: isomorphisms are bijections.
- In **Grp**: isomorphisms are group isomorphisms (bijective homomorphisms).
- In $\mathbf{Vect}_k$: isomorphisms are invertible linear maps.
- In **Top**: isomorphisms are homeomorphisms (continuous bijections with continuous inverse).

In $\mathbf{FinVect}_k$, two objects are isomorphic iff they have the same dimension. In **FinSet**, two objects are isomorphic iff they have the same cardinality.

Note:-

In **Top**, a bijective continuous map need not be an isomorphism—the inverse may fail to be continuous. For example, $\mathrm{id} : (\mathbb{R}, \text{discrete}) \to (\mathbb{R}, \text{standard})$ is a continuous bijection but not a homeomorphism. This illustrates that "isomorphism" in a category means more than just bijection on underlying sets.

8.1.3 Automorphism Groups

The self-isomorphisms of an object form a group, connecting category theory back to group theory.

Definition 8.1.3: Automorphism group

For an object A in a category $\mathcal{C}$, the **automorphism group** of A is

$$\mathrm{Aut}(A) = \{f \in \mathrm{Mor}(A, A) : f \text{ is an isomorphism}\}.$$

Theorem 8.1.1 Automorphisms form a group

$\mathrm{Aut}(A)$, equipped with composition, is a group.

Proof:

- **Closure:** If $f, g \in \mathrm{Aut}(A)$, then $g \circ f$ is an isomorphism by the previous proposition.
- **Associativity:** Inherited from the category axiom.
- **Identity:** $\mathrm{id}_A \in \mathrm{Aut}(A)$ is the identity element.
- **Inverses:** If $f \in \mathrm{Aut}(A)$, then $f^{-1} \in \mathrm{Aut}(A)$. $\square$

Example 8.1.7 (Automorphism groups in practice)

- In **FinSet**: if $|A| = n$, then $\mathrm{Aut}(A) \cong S_n$, the symmetric group.
- In $\mathbf{FinVect}_k$: if $\dim V = n$, then $\mathrm{Aut}(V) \cong GL_n(k)$, the general linear group.
- In **Grp**: $\mathrm{Aut}(G)$ is the group of group automorphisms. For $G = \mathbb{Z}/n$, we have $\mathrm{Aut}(\mathbb{Z}/n) \cong (\mathbb{Z}/n)^\times$.

Exercise 8.1.1 Isomorphic objects have isomorphic automorphism groups

Let $f : A \to B$ be an isomorphism in a category $\mathcal{C}$. Show that the map

$$c_f : \mathrm{Aut}(A) \to \mathrm{Aut}(B), \quad g \mapsto f \circ g \circ f^{-1}$$

is a group isomorphism.

Solution

Well-defined: If $g \in \mathrm{Aut}(A)$, then $f \circ g \circ f^{-1}$ is a composition of isomorphisms, hence an isomorphism $B \to B$.
Homomorphism: For $g, h \in \mathrm{Aut}(A)$:

$$c_f(g \circ h) = f \circ (g \circ h) \circ f^{-1} = (f \circ g \circ f^{-1}) \circ (f \circ h \circ f^{-1}) = c_f(g) \circ c_f(h).$$

Bijective: The map $c_{f^{-1}} : \mathrm{Aut}(B) \to \mathrm{Aut}(A)$ is an inverse:

$$c_{f^{-1}}(c_f(g)) = f^{-1} \circ (f \circ g \circ f^{-1}) \circ f = g.$$

Similarly, $c_f(c_{f^{-1}}(h)) = h$.

Note:-

This exercise shows that the automorphism group is an "isomorphism invariant": isomorphic objects have isomorphic automorphism groups. The converse is false—non-isomorphic objects can have isomorphic automorphism groups. For instance, in **Set**, any two 3-element sets have automorphism group S_3, but they are only isomorphic if we consider them as abstract sets (which they always are).

A category where every morphism is an isomorphism has a special name.

Definition 8.1.4: Groupoid

A **groupoid** is a category in which every morphism is an isomorphism.

Note:-

A groupoid generalizes the notion of a group: a group is precisely a groupoid with exactly one object. In a groupoid, morphisms can be composed only when the target of one matches the source of the other, but when composition is defined, it has inverses. The fundamental groupoid of a topological space, with objects being points and morphisms being homotopy classes of paths, is a key example in algebraic topology.

Example 8.1.8 (Groupoids)

- The category whose objects are finite sets and whose morphisms are only bijections (not all functions) is a groupoid.
- Any group G defines a groupoid $\mathbf{B}G$ with one object $*$ and $\mathrm{Mor}(*, *) = G$.
- The category whose objects are groups and whose morphisms are only group isomorphisms is a groupoid.

8.2 Universal Properties

Many constructions in mathematics are characterized not by explicit formulas but by universal properties—conditions that specify the construction uniquely up to isomorphism. We've already seen this with quotient spaces: the quotient V/U is characterized by factoring any linear map that kills U. Products and coproducts provide two more fundamental examples.

8.2.1 Products and Coproducts

Definition 8.2.1: Product

Let $\mathcal{C}$ be a category and $A, B \in \mathrm{Ob}(\mathcal{C})$. A **product** of A and B is an object P together with morphisms $\pi_1 : P \to A$ and $\pi_2 : P \to B$ (called **projections**) satisfying the following **universal property**:
For any object T and morphisms $\alpha : T \to A$, $\beta : T \to B$, there exists a <u>unique</u> morphism $\varphi : T \to P$ such that $\pi_1 \circ \varphi = \alpha$ and $\pi_2 \circ \varphi = \beta$.

$$\begin{tikzcd}
T \arrow[d, dashed, "\exists!\,\varphi"'] \arrow[dr, "\alpha"] \arrow[drr, bend left, "\beta"] & & \\
P \arrow[r, "\pi_1"'] & A & B \arrow[l, "\pi_2"]
\end{tikzcd}$$

We write $P = A \times B$ and $\varphi = (\alpha, \beta)$ for the unique induced map.

Note:-

The universal property says: to give a map into the product is the same as giving a map into each factor. This is the categorical abstraction of the familiar fact that a function $f : T \to A \times B$ is determined by its components $f_1 = \pi_1 \circ f$ and $f_2 = \pi_2 \circ f$.

The dual notion reverses all arrows.

Definition 8.2.2: Coproduct

Let $\mathcal{C}$ be a category and $A, B \in \mathrm{Ob}(\mathcal{C})$. A **coproduct** (or **sum**) of A and B is an object S together with morphisms $\iota_1 : A \to S$ and $\iota_2 : B \to S$ (called **inclusions** or **coprojections**) satisfying:
For any object T and morphisms $\alpha : A \to T$, $\beta : B \to T$, there exists a <u>unique</u> morphism $\varphi : S \to T$ such that $\varphi \circ \iota_1 = \alpha$ and $\varphi \circ \iota_2 = \beta$.

$$\begin{tikzcd}
A \arrow[r, "\iota_1"] \arrow[dr, "\alpha"'] & S \arrow[d, dashed, "\exists!\,\varphi"] & B \arrow[l, "\iota_2"'] \arrow[dl, "\beta"] \\
& T &
\end{tikzcd}$$

We write $S = A \amalg B$ or $A + B$ and $\varphi = [\alpha, \beta]$ for the unique induced map.

Note:-

The universal property of the coproduct says: to give a map out of the coproduct is the same as giving a map out of each summand. This duality between products and coproducts—reversing all arrows—is a general phenomenon in category theory.

Theorem 8.2.1 Uniqueness of products and coproducts

If a product (resp. coproduct) of A and B exists, it is unique up to unique isomorphism. That is, if (P, π_1, π_2) and (P', π_1', π_2') are both products of A and B, then there is a unique isomorphism $\psi : P \to P'$ with $\pi_1' \circ \psi = \pi_1$ and $\pi_2' \circ \psi = \pi_2$.

Proof: By the universal property of P', applied to the maps $\pi_1 : P \to A$ and $\pi_2 : P \to B$, there exists a unique $\psi : P \to P'$ with $\pi_1' \circ \psi = \pi_1$ and $\pi_2' \circ \psi = \pi_2$.

By the universal property of P, applied to $\pi_1' : P' \to A$ and $\pi_2' : P' \to B$, there exists a unique $\psi' : P' \to P$ with $\pi_1 \circ \psi' = \pi_1'$ and $\pi_2 \circ \psi' = \pi_2'$.

Now $\psi' \circ \psi : P \to P$ satisfies $\pi_1 \circ (\psi' \circ \psi) = \pi_1$ and $\pi_2 \circ (\psi' \circ \psi) = \pi_2$. But id_P also satisfies these equations. By uniqueness in the universal property of P, we have $\psi' \circ \psi = \mathrm{id}_P$. Similarly, $\psi \circ \psi' = \mathrm{id}_{P'}$.

The proof for coproducts is dual (reverse all arrows). $\square$

Note:-

This proof pattern—using universal properties to construct inverse maps—is standard in category theory. The key is that universal properties give uniqueness, which forces the composites to be identities.

Products and coproducts need not exist in every category. When working in a new category, one of the first questions is whether these constructions exist and what form they take.

8.2.2 Examples in $\mathbf{Vect}_k$

Example 8.2.1 (Products and coproducts in **Set**)

In the category of sets:

- The **product** $A \times B$ is the Cartesian product $\{(a, b) : a \in A, b \in B\}$ with projections $\pi_1(a, b) = a$ and $\pi_2(a, b) = b$.
- The **coproduct** $A \amalg B$ is the disjoint union $A \sqcup B = \{(a, 1) : a \in A\} \cup \{(b, 2) : b \in B\}$ with inclusions $\iota_1(a) = (a, 1)$ and $\iota_2(b) = (b, 2)$.

The universal properties are immediate from the definitions of functions on Cartesian products and disjoint unions.

Example 8.2.2 (Products and coproducts in **Grp**)

In the category of groups:

- The **product** $G \times H$ is the direct product with componentwise operations and projections.
- The **coproduct** $G * H$ is the **free product**—a more complicated construction involving alternating words from G and H.

In the subcategory **Ab** of abelian groups, both product and coproduct are the direct sum $G \oplus H$ (with the same underlying group but different universal properties).

Example 8.2.3 (Products and coproducts in $\mathbf{Vect}_k$)

In the category of finite-dimensional vector spaces over k, both the product and coproduct of V and W are realized by the **direct sum** $V \oplus W$.
As a product: Define $\pi_1 : V \oplus W \to V$ by $\pi_1(v, w) = v$ and $\pi_2 : V \oplus W \to W$ by $\pi_2(v, w) = w$. Given linear maps $\alpha : T \to V$ and $\beta : T \to W$, define $\varphi : T \to V \oplus W$ by $\varphi(t) = (\alpha(t), \beta(t))$. This is the unique map with $\pi_1 \circ \varphi = \alpha$ and $\pi_2 \circ \varphi = \beta$.
As a coproduct: Define $\iota_1 : V \to V \oplus W$ by $\iota_1(v) = (v, 0)$ and $\iota_2 : W \to V \oplus W$ by $\iota_2(w) = (0, w)$. Given linear maps $\alpha : V \to T$ and $\beta : W \to T$, define $\varphi : V \oplus W \to T$ by $\varphi(v, w) = \alpha(v) + \beta(w)$. This is the unique map with $\varphi \circ \iota_1 = \alpha$ and $\varphi \circ \iota_2 = \beta$.

Note:-

The coincidence of products and coproducts in $\mathbf{Vect}_k$ (and **Ab**) reflects the additive structure of these categories. In categories without such structure (like **Set** or **Grp**), products and coproducts are genuinely different constructions.

Exercise 8.2.1 Products and coproducts exercises

(1) Verify explicitly that the Cartesian product satisfies the universal property of products in **Set**.

(2) Show that in **Ab**, the coproduct of $\mathbb{Z}$ and $\mathbb{Z}$ is $\mathbb{Z}^2$, while in **Grp**, the coproduct is the free group F_2 on two generators.

(3) Let V and W be finite-dimensional vector spaces over k. Show that $\dim(V \oplus W) = \dim V + \dim W$ using the universal property (not by direct computation).

Solution

(1) Let A, B be sets and let T be any set with functions $\alpha : T \to A$ and $\beta : T \to B$. We must show there exists a unique function $\varphi : T \to A \times B$ with $\pi_1 \circ \varphi = \alpha$ and $\pi_2 \circ \varphi = \beta$.
Existence: Define $\varphi(t) = (\alpha(t), \beta(t))$. Then $\pi_1(\varphi(t)) = \alpha(t)$ and $\pi_2(\varphi(t)) = \beta(t)$.
Uniqueness: If $\psi : T \to A \times B$ satisfies $\pi_1 \circ \psi = \alpha$ and $\pi_2 \circ \psi = \beta$, then for any $t \in T$, $\psi(t) = (\pi_1(\psi(t)), \pi_2(\psi(t))) = (\alpha(t), \beta(t)) = \varphi(t)$.
(2) In **Ab**: The coproduct of $\mathbb{Z}$ and $\mathbb{Z}$ has inclusions $\iota_1, \iota_2 : \mathbb{Z} \to S$ such that any pair of homomorphisms $\alpha, \beta : \mathbb{Z} \to A$ (for A abelian) factors uniquely through S. Taking $S = \mathbb{Z}^2$ with $\iota_1(n) = (n, 0)$ and $\iota_2(n) = (0, n)$, the map $\varphi : \mathbb{Z}^2 \to A$ defined by $\varphi(m, n) = m \cdot \alpha(1) + n \cdot \beta(1)$ is the unique such factorization.
In **Grp**: The coproduct must handle non-commuting images. Given any group G and homomorphisms $\alpha, \beta : \mathbb{Z} \to G$, the elements $\alpha(1)$ and $\beta(1)$ generate a subgroup that need not be abelian. The free group $F_2 = \langle a, b \rangle$ with $\iota_1(1) = a$ and $\iota_2(1) = b$ satisfies the universal property: any choice of $g = \alpha(1)$ and $h = \beta(1)$ in G extends uniquely to a homomorphism $F_2 \to G$.
(3) By the universal property of coproducts, $\mathrm{Hom}(V \oplus W, T) \cong \mathrm{Hom}(V, T) \times \mathrm{Hom}(W, T)$ for any T. Taking $T = k$:

$$(V \oplus W)^* \cong V^* \times W^*.$$

Thus $\dim(V \oplus W)^* = \dim V^* + \dim W^*$. Since $\dim U^* = \dim U$ for finite-dimensional spaces, we get $\dim(V \oplus W) = \dim V + \dim W$.

Note:-

The argument in (3) illustrates a general principle: universal properties translate into isomorphisms of Hom-sets. The functor $\mathrm{Hom}(-, T)$ sends coproducts to products, which is why $\mathrm{Hom}(V \oplus W, T) \cong \mathrm{Hom}(V, T) \times \mathrm{Hom}(W, T)$. This is part of a larger story involving adjoint functors.

Lecture 9.

Inner Product Spaces

So far we have studied vector spaces equipped only with their linear structure. Adding geometric notions—length, angle, orthogonality—requires additional data: an inner product. This structure transforms abstract linear algebra into something closer to Euclidean geometry, while revealing that many geometric intuitions extend far beyond $\mathbb{R}^n$.

We begin with the more general notion of bilinear forms, then specialize to the symmetric positive-definite case (inner products over $\mathbb{R}$) and its complex analogue (Hermitian inner products over $\mathbb{C}$).

9.1 Inner Products

9.1.1 Bilinear Forms

Before defining inner products, we study the broader class of bilinear forms. This generality clarifies which properties of inner products come from bilinearity alone and which require symmetry or positivity.

Definition 9.1.1: Bilinear form

Let V be a vector space over a field k. A **bilinear form** on V is a map $b : V \times V \to k$ that is linear in each variable:

(i) $b(\lambda v, w) = \lambda b(v, w) = b(v, \lambda w)$ for all $\lambda \in k$, $v, w \in V$.

(ii) $b(u + v, w) = b(u, w) + b(v, w)$ for all $u, v, w \in V$.

(iii) $b(u, v + w) = b(u, v) + b(u, w)$ for all $u, v, w \in V$.

Note:-

A bilinear form is not a linear map $V \times V \to k$. If we view $V \times V$ as a vector space (the external direct sum $V \oplus V$), a linear map would satisfy $b(\lambda(v, w)) = \lambda b(v, w)$, i.e., $b(\lambda v, \lambda w) = \lambda b(v, w)$. But bilinearity gives $b(\lambda v, \lambda w) = \lambda^2 b(v, w)$. The tensor product $V \otimes V$ is the correct domain for linearizing bilinear maps.

Definition 9.1.2: Symmetric and skew-symmetric forms

A bilinear form b is:

- **Symmetric** if $b(v, w) = b(w, v)$ for all $v, w \in V$.
- **Skew-symmetric** (or **alternating**) if $b(v, w) = -b(w, v)$ for all $v, w \in V$.

Example 9.1.1 (The standard dot product)

On k^n, the map $b(v, w) = \sum_{i=1}^{n} v_i w_i$ is a symmetric bilinear form. This is the "dot product" familiar from calculus.

Example 9.1.2 (A skew-symmetric form)

On k^2, define $b((x_1, x_2), (y_1, y_2)) = x_1y_2 - x_2y_1$. This equals $\det \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix}$ and is skew-symmetric. Geometrically, it computes signed area.

Matrix Representation

Given a basis $\{e_1, \ldots, e_n\}$ of V, a bilinear form b is determined by the values $a_{ij} = b(e_i, e_j)$. By bilinearity:

$$b\left(\sum_i x_i e_i, \sum_j y_j e_j\right) = \sum_{i,j} x_i y_j \, b(e_i, e_j) = \sum_{i,j} a_{ij} x_i y_j.$$

Writing $x = (x_1, \ldots, x_n)^T$ and $y = (y_1, \ldots, y_n)^T$ as column vectors, this is:

$$b(x, y) = x^T A y$$

where $A = (a_{ij})$ is the **matrix of** b with respect to the chosen basis.

Proposition 9.1.1 Symmetry and the matrix

(i) b is symmetric if and only if $A = A^T$.

(ii) b is skew-symmetric if and only if $A = -A^T$.

Proof: We have $b(v, w) = v^T A w$ and $b(w, v) = w^T A v = (w^T A v)^T = v^T A^T w$. So $b(v, w) = b(w, v)$ for all v, w iff $v^T A w = v^T A^T w$ for all v, w, iff $A = A^T$. The skew-symmetric case is similar. □

Bilinear Forms and the Dual Space

There is a natural connection between bilinear forms and linear maps to the dual space.

Proposition 9.1.2 The map $\varphi_b : V \to V^*$

Given a bilinear form $b : V \times V \to k$, define $\varphi_b : V \to V^*$ by

$$\varphi_b(v) = b(v, \cdot) \in V^*,$$

i.e., $\varphi_b(v)$ is the linear functional $w \mapsto b(v, w)$. Then φ_b is a linear map.

Proof: For $v_1, v_2 \in V$ and $\lambda \in k$:

$$\varphi_b(v_1 + \lambda v_2)(w) = b(v_1 + \lambda v_2, w) = b(v_1, w) + \lambda b(v_2, w) = (\varphi_b(v_1) + \lambda \varphi_b(v_2))(w).$$

So $\varphi_b(v_1 + \lambda v_2) = \varphi_b(v_1) + \lambda \varphi_b(v_2)$. □

Conversely, any linear map $\varphi : V \to V^*$ determines a bilinear form $b(v, w) = \varphi(v)(w)$.

Theorem 9.1.1 Bilinear forms and $\mathrm{Hom}(V, V^*)$

The map $b \mapsto \varphi_b$ defines an isomorphism of vector spaces

$$B(V) \xrightarrow{\sim} \mathrm{Hom}(V, V^*)$$

where $B(V)$ denotes the space of bilinear forms on V.

Definition 9.1.3: Rank and nondegeneracy

The **rank** of a bilinear form b is $\operatorname{rank}(\varphi_b) = \dim(\operatorname{Im}(\varphi_b))$. Equivalently, it is the rank of the matrix A in any basis.
The form b is **nondegenerate** if φ_b is an isomorphism, i.e., if $\ker(\varphi_b) = \{0\}$. Equivalently, b is nondegenerate iff A is invertible.

Note:-

For V finite-dimensional, $\dim V = \dim V^*$, so φ_b is an isomorphism iff it is injective iff it is surjective. Nondegeneracy means: if $b(v, w) = 0$ for all $w \in V$, then $v = 0$.

Proposition 9.1.3 Dimension of the space of bilinear forms

If $\dim V = n$, then $\dim B(V) = n^2$.

Proof: We have $B(V) \cong \operatorname{Hom}(V, V^*)$. Since $\dim V^* = n$, we get $\dim \operatorname{Hom}(V, V^*) = n \cdot n = n^2$. Alternatively, a bilinear form is determined by an $n \times n$ matrix. □

9.1.2 Inner Products over the reals

To do geometry, we need more than bilinearity. The key additional properties are symmetry (so that $b(v, w) = b(w, v)$) and positive-definiteness (so that "length" is always nonnegative).

Definition 9.1.4: Inner product (real case)

Let V be a vector space over $\mathbb{R}$. An **inner product** on V is a bilinear form $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying:

(i) **Symmetry:** $\langle v, w \rangle = \langle w, v \rangle$ for all $v, w \in V$.

(ii) **Positive-definiteness:** $\langle v, v \rangle \geqslant 0$ for all $v \in V$, with equality iff $v = 0$.

A vector space equipped with an inner product is called an **inner product space**.

Note:-

Positive-definiteness only makes sense over an ordered field where "$\geqslant 0$" is meaningful. In practice, this means $\mathbb{R}$. We cannot directly define inner products over $\mathbb{C}$ or finite fields using this definition—we will see the complex workaround shortly.

Example 9.1.3 (The standard inner product on $\mathbb{R}^n$)

The dot product

$$\langle v, w \rangle = \sum_{i=1}^{n} v_i w_i = v^T w$$

is an inner product. Symmetry is clear, and $\langle v, v \rangle = \sum v_i^2 \geqslant 0$ with equality iff all $v_i = 0$.

Example 9.1.4 (A weighted inner product)

On $\mathbb{R}^n$, for positive constants $c_1, \ldots, c_n > 0$, define

$$\langle v, w \rangle = \sum_{i=1}^{n} c_i v_i w_i.$$

This is an inner product (check!). Different weights emphasize different coordinates.

Proposition 9.1.4 Inner products are nondegenerate

Every inner product is nondegenerate.

Proof: Suppose $\langle v, w\rangle = 0$ for all $w \in V$. Taking $w = v$, we get $\langle v, v\rangle = 0$. By positive-definiteness, $v = 0$. $\square$

Note:-

The converse is false: a nondegenerate symmetric bilinear form need not be positive-definite. For example, the Minkowski metric $b(v, w) = v_1w_1 - v_2w_2 - v_3w_3 - v_4w_4$ on $\mathbb{R}^4$ is nondegenerate and symmetric, but $b(v, v)$ can be negative.

9.1.3 Hermitian Inner Products over the complexes

Why can't we directly use the same definition over $\mathbb{C}$? If b is a bilinear form on a complex vector space with $b(v, v) \geqslant 0$, consider $b(iv, iv) = i^2 b(v, v) = -b(v, v)$. So if $b(v, v) > 0$, then $b(iv, iv) < 0$. Bilinearity and positive-definiteness are incompatible over $\mathbb{C}$.

The solution is to weaken bilinearity: we require linearity in one variable and conjugate-linearity in the other.

Definition 9.1.5: Sesquilinear form

A map $H : V \times V \to \mathbb{C}$ on a complex vector space V is **sesquilinear** if:

(i) $H(u + v, w) = H(u, w) + H(v, w)$ and $H(u, v + w) = H(u, v) + H(u, w)$.

(ii) $H(\lambda u, v) = \lambda H(u, v)$ for $\lambda \in \mathbb{C}$.

(iii) $H(u, \lambda v) = \bar{\lambda} H(u, v)$ for $\lambda \in \mathbb{C}$.

That is, H is linear in the first argument and conjugate-linear in the second.

Note:-

Some authors (including Axler) use the opposite convention: conjugate-linear in the first argument, linear in the second. The choice is a matter of convention; we follow Artin here.

Definition 9.1.6: Hermitian form

A sesquilinear form H is **Hermitian** (or **conjugate-symmetric**) if

$$H(u, v) = \overline{H(v, u)}$$

for all $u, v \in V$.

Note:-

Conjugate-symmetry implies $H(v, v) = \overline{H(v, v)}$, so $H(v, v) \in \mathbb{R}$ for all v. This makes positive-definiteness meaningful.

Definition 9.1.7: Hermitian inner product

A **Hermitian inner product** on a complex vector space V is a sesquilinear form $H : V \times V \to \mathbb{C}$ that is:

(i) **Conjugate-symmetric:** $H(u, v) = \overline{H(v, u)}$.

(ii) **Positive-definite:** $H(v, v) \geqslant 0$ for all v, with equality iff $v = 0$.

Example 9.1.5 (The standard Hermitian inner product on $\mathbb{C}^n$)

Define

$$H(z, w) = \sum_{i=1}^{n} z_i \overline{w_i} = z^* w$$

where $z^* = \overline{z}^T$ is the conjugate transpose. Then:

- $H(w, z) = \sum w_i \overline{z_i} = \overline{\sum z_i \overline{w_i}} = \overline{H(z, w)}$.
- $H(z, z) = \sum |z_i|^2 \geqslant 0$, with equality iff $z = 0$.

Proposition 9.1.5 The map $\varphi_H : V \to V^*$

For a Hermitian form H, the map $\varphi_H(v) = H(v, \cdot)$ is a **conjugate-linear** map $V \to V^*$:

$$\varphi_H(\lambda v) = \overline{\lambda} \varphi_H(v).$$

If H is positive-definite, then φ_H is injective (and hence bijective when V is finite-dimensional).

Note:-

Various properties carry over from the real case. A positive-definite Hermitian form is nondegenerate. Given a subspace $W \subseteq V$, its orthogonal complement $W^\perp = \{v \in V : H(v, w) = 0 \ \forall w \in W\}$ is a subspace, and $V = W \oplus W^\perp$ when H is positive-definite.

9.1.4 Norms and the Cauchy-Schwarz Inequality

An inner product induces a notion of length via the norm.

Definition 9.1.8: Norm

Let V be an inner product space (real or Hermitian). The **norm** of a vector $v \in V$ is

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

This is well-defined since $\langle v, v \rangle \geqslant 0$.

Definition 9.1.9: Orthogonality

Vectors $v, w \in V$ are **orthogonal**, written $v \perp w$, if $\langle v, w \rangle = 0$.

Proposition 9.1.6 Basic properties of the norm

(i) $\|v\| \geqslant 0$, with equality iff $v = 0$.

(ii) $\|\lambda v\| = |\lambda| \|v\|$ for $\lambda \in k$.

(iii) (Pythagorean theorem) If $v \perp w$, then $\|v + w\|^2 = \|v\|^2 + \|w\|^2$.

Proof: (i) and (ii) follow from the definitions. For (iii):

$$\|v + w\|^2 = \langle v + w, v + w \rangle = \|v\|^2 + \langle v, w \rangle + \langle w, v \rangle + \|w\|^2 = \|v\|^2 + \|w\|^2$$

since $\langle v, w \rangle = 0 = \langle w, v \rangle$ (using symmetry/conjugate-symmetry). $\square$

The Pythagorean theorem suggests that orthogonal vectors behave like legs of a right triangle. More generally:

$$\|v-w\|^2 = \|v\|^2 + \|w\|^2 - 2\,\mathrm{Re}\langle v, w\rangle.$$

This motivates defining the angle between nonzero vectors. But first we need to know that $|\langle v, w\rangle| \leqslant \|v\|\|w\|$ so that the ratio lies in $[-1, 1]$.

Theorem 9.1.2 Cauchy–Schwarz inequality

For all u, v in an inner product space,

$$|\langle u, v\rangle| \leqslant \|u\|\|v\|,$$

with equality iff u and v are linearly dependent.

Proof: The inequality is trivial if $u = 0$. Assume $u \neq 0$; by scaling, we may assume $\|u\| = 1$.

Decompose v along u: let $S = \mathrm{span}(u)$. Since $\langle \cdot, \cdot \rangle$ is nondegenerate and $\dim S = 1$, we have $V = S \oplus S^{\perp}$. Write

$$v = v_1 + v_2, \quad v_1 = \langle v, u\rangle u \in S, \quad v_2 = v - \langle v, u\rangle u \in S^{\perp}.$$

Then $v_1 \perp v_2$, so by the Pythagorean theorem:

$$\|v\|^2 = \|v_1\|^2 + \|v_2\|^2 \geqslant \|v_1\|^2 = |\langle v, u\rangle|^2\|u\|^2 = |\langle v, u\rangle|^2.$$

Taking square roots gives $\|v\| \geqslant |\langle v, u\rangle| = |\langle u, v\rangle|$, which is the inequality for $\|u\| = 1$.

Equality holds iff $v_2 = 0$, i.e., $v \in S = \mathrm{span}(u)$. □

Definition 9.1.10: Angle between vectors

For nonzero vectors v, w in a real inner product space, the **angle** θ between them is defined by

$$\cos\theta = \frac{\langle v, w\rangle}{\|v\|\|w\|}, \quad \theta \in [0, \pi].$$

The Cauchy–Schwarz inequality ensures this ratio lies in $[-1, 1]$.

Corollary 9.1.1 Triangle inequality

For all u, v in an inner product space,

$$\|u + v\| \leqslant \|u\| + \|v\|.$$

Proof:

$$\begin{aligned} \|u+v\|^2 &= \|u\|^2 + 2\,\mathrm{Re}\langle u, v\rangle + \|v\|^2 \\ &\leqslant \|u\|^2 + 2|\langle u, v\rangle| + \|v\|^2 \\ &\leqslant \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2. \end{aligned}$$

□

9.1.5 Orthogonality and Orthogonal Complements

Definition 9.1.11: Orthogonal complement

Let V be an inner product space and $S \subseteq V$ a subset. The **orthogonal complement** of S is

$$S^{\perp} = \{v \in V : \langle v, w\rangle = 0 \ \forall w \in S\}.$$

Proposition 9.1.7 Basic properties of orthogonal complements

(i) $S^\perp$ is a subspace of V (even if S is not).

(ii) $S \subseteq T \implies T^\perp \subseteq S^\perp$.

(iii) $S \subseteq (S^\perp)^\perp$.

(iv) $\{0\}^\perp = V$ and $V^\perp = \{0\}$.

Proof: (i) If $v_1, v_2 \in S^\perp$ and $\lambda \in k$, then for any $w \in S$:

$$\langle v_1 + \lambda v_2, w\rangle = \langle v_1, w\rangle + \lambda\langle v_2, w\rangle = 0 + \lambda \cdot 0 = 0.$$

(ii)–(iv) are immediate from the definition. □

Note:-

For a general bilinear form (not necessarily symmetric), we must distinguish "left-orthogonal" ($b(v, w) = 0$ for all $w \in S$) from "right-orthogonal" ($b(w, v) = 0$ for all $w \in S$). For symmetric forms and inner products, these coincide.

The key result is that an inner product space decomposes as $V = S \oplus S^\perp$ for any subspace S.

Lemma 9.1.1 Dimension of orthogonal complement

Let b be a nondegenerate bilinear form on a finite-dimensional space V, and let $S \subseteq V$ be a subspace. Then

$$\dim S^\perp = \dim V - \dim S.$$

If b is an inner product (positive-definite), then $S \cap S^\perp = \{0\}$.

Proof: Consider the composition

$$V \xrightarrow{\varphi_b} V^* \xrightarrow{r} S^*$$

where $\varphi_b(v) = b(v, \cdot)$ and $r : V^* \to S^*$ is restriction. The kernel of $r \circ \varphi_b$ is exactly $S^\perp$.

By rank-nullity: $\dim S^\perp = \dim V - \operatorname{rank}(r \circ \varphi_b)$.

Since φ_b is an isomorphism (nondegeneracy) and r is surjective, $r \circ \varphi_b$ is surjective. Thus $\operatorname{rank}(r \circ \varphi_b) = \dim S^* = \dim S$, giving $\dim S^\perp = \dim V - \dim S$.

For the second claim: if $v \in S \cap S^\perp$, then $\langle v, v\rangle = 0$. By positive-definiteness, $v = 0$. □

Theorem 9.1.3 Orthogonal decomposition

Let V be a finite-dimensional inner product space and $S \subseteq V$ a subspace. Then

$$V = S \oplus S^\perp.$$

Proof: By the lemma, $\dim S + \dim S^\perp = \dim V$ and $S \cap S^\perp = \{0\}$. The dimension formula for sums gives

$$\dim(S + S^\perp) = \dim S + \dim S^\perp - \dim(S \cap S^\perp) = \dim V.$$

So $S + S^\perp = V$, and the sum is direct since $S \cap S^\perp = \{0\}$. □

Example 9.1.6 (Standard orthogonal complement in $\mathbb{R}^n$)

Let $V = \mathbb{R}^n$ with the standard inner product $\langle v, w\rangle = \sum v_i w_i$. If S is a subspace, then $S^\perp$ is the "usual" orthogonal complement: vectors perpendicular to every vector in S.

For instance, if $S = \operatorname{span}((1, 0, 0))$ in $\mathbb{R}^3$, then $S^\perp = \{(0, y, z) : y, z \in \mathbb{R}\}$.

Example 9.1.7 (Skew-symmetric forms can have $S^\perp = S$)

Let $V = k^2$ with the skew-symmetric form $b((x_1, x_2), (y_1, y_2)) = x_1 y_2 - x_2 y_1$. Let S be any 1-dimensional subspace spanned by a nonzero vector v.
For any $w \in k^2$, we have $b(v, w) = 0 \iff \det(v \mid w) = 0 \iff w \in \operatorname{span}(v) = S$.
So $S^\perp = S$. In particular, $S \cap S^\perp = S \neq \{0\}$, and $V \neq S \oplus S^\perp$. This cannot happen for inner products.

9.2 Orthonormal Bases

9.2.1 The Gram-Schmidt Process

In an inner product space, we can ask for bases that respect the geometry: bases whose vectors are mutually orthogonal and have unit length.

Definition 9.2.1: Orthonormal basis

Let V be a finite-dimensional inner product space. A basis $\{v_1, \dots, v_n\}$ of V is **orthonormal** if

$$\langle v_i, v_j \rangle = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Equivalently, $\|v_i\| = 1$ for all i, and $v_i \perp v_j$ for $i \neq j$.

Note:-

In an orthonormal basis, the inner product takes a simple form: if $v = \sum x_i e_i$ and $w = \sum y_j e_j$, then

$$\langle v, w \rangle = \sum_{i,j} x_i \overline{y_j} \langle e_i, e_j \rangle = \sum_i x_i \overline{y_i}.$$

For real spaces, this is just $x^T y$. For complex spaces with Hermitian inner product, it is $x^* y$.

Theorem 9.2.1 Existence of orthonormal bases

Every finite-dimensional inner product space has an orthonormal basis.

We give two proofs: one by induction, one constructive (Gram–Schmidt).

Proof 1 (Induction on dimension): Let $\dim V = n$. If $n = 0$, the empty set is an orthonormal basis. For $n \geqslant 1$:

Pick any nonzero $v \in V$ and set $v_1 = v/\|v\|$, so $\|v_1\| = 1$. Let $S = \operatorname{span}(v_1)$. By the orthogonal decomposition theorem, $V = S \oplus S^\perp$, and $\dim S^\perp = n - 1$.

The restriction of $\langle \cdot, \cdot \rangle$ to $S^\perp$ is still an inner product (check positive-definiteness). By induction, $S^\perp$ has an orthonormal basis $\{v_2, \dots, v_n\}$.

Then $\{v_1, v_2, \dots, v_n\}$ is an orthonormal basis of V: the v_i are orthonormal in $S^\perp$, and $v_1 \perp v_i$ for $i \geqslant 2$ since $v_i \in S^\perp$. □

Proof 2 (Gram–Schmidt): Start with any basis $\{w_1, \dots, w_n\}$ of V. We inductively construct orthonormal vectors $v_1, \dots, v_n$ such that $\operatorname{span}(v_1, \dots, v_k) = \operatorname{span}(w_1, \dots, w_k)$ for each k.

Step 1: Set $v_1 = w_1/\|w_1\|$.

Step j: Given orthonormal $v_1, \dots, v_{j-1}$, define

$$u_j = w_j - \sum_{i=1}^{j-1} \langle w_j, v_i \rangle v_i.$$

This subtracts from w_j its projections onto each v_i, leaving a vector orthogonal to $\operatorname{span}(v_1, \ldots, v_{j-1})$. Since $w_j \notin \operatorname{span}(w_1, \ldots, w_{j-1}) = \operatorname{span}(v_1, \ldots, v_{j-1})$ (linear independence), we have $u_j \neq 0$. Set

$$v_j = \frac{u_j}{\|u_j\|}.$$

By construction, $v_j \perp v_i$ for $i < j$, and $\|v_j\| = 1$. The span condition holds since $v_j \in \operatorname{span}(w_1, \ldots, w_j)$ and $w_j \in \operatorname{span}(v_1, \ldots, v_j)$. □

Corollary 9.2.1 Isomorphism with standard space

Every n-dimensional real inner product space is isomorphic to $(\mathbb{R}^n, \text{standard inner product})$ as an inner product space. Similarly, every n-dimensional Hermitian inner product space is isomorphic to $(\mathbb{C}^n, \text{standard Hermitian product})$.

Proof: Let $\{v_1, \ldots, v_n\}$ be an orthonormal basis. The coordinate map $\varphi : V \to k^n$ sending $v = \sum a_i v_i$ to $(a_1, \ldots, a_n)$ is a linear isomorphism. Moreover,

$$\langle v, w \rangle = \langle \sum a_i v_i, \sum b_j v_j \rangle = \sum_{i,j} a_i \overline{b_j} \delta_{ij} = \sum_i a_i \overline{b_i}$$

which is the standard inner product of the coordinate vectors. □

Note:-

This says that all n-dimensional inner product spaces (over the same field) are "the same." The choice of orthonormal basis corresponds to the choice of isomorphism with standard space. Different orthonormal bases give different coordinate systems, all equally valid.

Example 9.2.1 (Gram–Schmidt in $\mathbb{R}^3$)

Apply Gram–Schmidt to the basis $w_1 = (1, 1, 0)$, $w_2 = (1, 0, 1)$, $w_3 = (0, 1, 1)$ of $\mathbb{R}^3$.
Step 1: $\|w_1\| = \sqrt{2}$, so $v_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$.
Step 2: $\langle w_2, v_1 \rangle = \frac{1}{\sqrt{2}}(1 + 0 + 0) = \frac{1}{\sqrt{2}}$.

$$u_2 = w_2 - \frac{1}{\sqrt{2}} v_1 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right).$$

$\|u_2\|^2 = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$, so $v_2 = \sqrt{\frac{2}{3}}\left(\frac{1}{2}, -\frac{1}{2}, 1\right) = \frac{1}{\sqrt{6}}(1, -1, 2)$.
Step 3: Compute projections and proceed similarly to get v_3.

9.2.2 Orthogonal Projections

The orthogonal decomposition $V = S \oplus S^\perp$ gives a canonical way to project onto S.

Definition 9.2.2: Orthogonal projection

Let V be an inner product space and $S \subseteq V$ a subspace. The **orthogonal projection** onto S is the linear map $\pi_S : V \to V$ defined by: for $v = v_S + v_{S^\perp}$ with $v_S \in S$ and $v_{S^\perp} \in S^\perp$,

$$\pi_S(v) = v_S.$$

Proposition 9.2.1 Properties of orthogonal projections

Let $\pi_S : V \to V$ be the orthogonal projection onto S.

(i) $\operatorname{Im}(\pi_S) = S$ and $\operatorname{Ker}(\pi_S) = S^\perp$.

(ii) $\pi_S^2 = \pi_S$ (i.e., π_S is idempotent).

(iii) $\pi_S^* = \pi_S$ (i.e., π_S is self-adjoint; see §9.3).

(iv) $\pi_{S^\perp} = \mathrm{id} - \pi_S$.

Proof: (i) follows from the definition. For (ii): if $v = v_S + v_{S^\perp}$, then $\pi_S(v) = v_S \in S$, so $\pi_S(\pi_S(v)) = \pi_S(v_S) = v_S$. (iv) is immediate from $v = \pi_S(v) + \pi_{S^\perp}(v)$. □

Proposition 9.2.2 Formula in orthonormal basis

If $\{e_1, \dots, e_k\}$ is an orthonormal basis of S, then

$$\pi_S(v) = \sum_{i=1}^{k} \langle v, e_i\rangle e_i.$$

Proof: Let $w = \sum_{i=1}^{k} \langle v, e_i\rangle e_i \in S$. We show $v - w \in S^\perp$: for any $j \leqslant k$,

$$\langle v - w, e_j\rangle = \langle v, e_j\rangle - \sum_i \langle v, e_i\rangle\langle e_i, e_j\rangle = \langle v, e_j\rangle - \langle v, e_j\rangle = 0.$$

Since $\{e_j\}$ spans S, we have $v - w \perp S$, so $v - w \in S^\perp$. Thus $w = \pi_S(v)$. □

Proposition 9.2.3 Best approximation

For $v \in V$ and $S \subseteq V$ a subspace, the projection $\pi_S(v)$ is the unique vector in S minimizing $\|v - s\|$ over $s \in S$.

Proof: For any $s \in S$, write $s = \pi_S(v) + (s - \pi_S(v))$ where $s - \pi_S(v) \in S$. Then:

$$\|v - s\|^2 = \|(v - \pi_S(v)) - (s - \pi_S(v))\|^2.$$

Since $v - \pi_S(v) \in S^\perp$ and $s - \pi_S(v) \in S$, these are orthogonal:

$$\|v - s\|^2 = \|v - \pi_S(v)\|^2 + \|s - \pi_S(v)\|^2 \geqslant \|v - \pi_S(v)\|^2,$$

with equality iff $s = \pi_S(v)$. □

9.3 The Adjoint

9.3.1 Definition and Existence

Given a linear operator on an inner product space, we can define its "adjoint" by transposing the role of the two arguments in the inner product.

Definition 9.3.1: Adjoint operator

Let $(V, \langle\cdot,\cdot\rangle)$ be a finite-dimensional inner product space and $T : V \to V$ a linear operator. The **adjoint** of T is the unique linear operator $T^* : V \to V$ satisfying

$$\langle v, T(w)\rangle = \langle T^*(v), w\rangle$$

for all $v, w \in V$.

Theorem 9.3.1 Existence and uniqueness of the adjoint

For any linear operator T on a finite-dimensional inner product space, the adjoint T^* exists and is unique.

Proof (Existence): Fix $v \in V$. The map $w \mapsto \langle v, T(w) \rangle$ is a linear functional on V.

Since the inner product is nondegenerate, the map $\varphi : V \to V^*$ defined by $\varphi(u) = \langle u, \cdot \rangle$ is an isomorphism. Thus there exists a unique $u \in V$ with $\langle u, w \rangle = \langle v, T(w) \rangle$ for all w. Define $T^*(v) = u$.

We must check T^* is linear. For $v_1, v_2 \in V$ and $\lambda \in k$:

$$\langle T^*(v_1 + \lambda v_2), w \rangle = \langle v_1 + \lambda v_2, T(w) \rangle = \langle v_1, T(w) \rangle + \lambda \langle v_2, T(w) \rangle$$

$$= \langle T^*(v_1), w \rangle + \lambda \langle T^*(v_2), w \rangle = \langle T^*(v_1) + \lambda T^*(v_2), w \rangle.$$

Since this holds for all w, we get $T^*(v_1 + \lambda v_2) = T^*(v_1) + \lambda T^*(v_2)$.

Uniqueness: If S also satisfies $\langle v, T(w) \rangle = \langle S(v), w \rangle$, then $\langle T^*(v) - S(v), w \rangle = 0$ for all w, so $T^*(v) = S(v)$ by nondegeneracy. $\square$

Note:-

Alternatively, the adjoint can be described as follows. The inner product gives an isomorphism $\varphi : V \xrightarrow{\sim} V^*$, $\varphi(v) = \langle v, \cdot \rangle$. The transpose $T^t : V^* \to V^*$ is defined by $T^t(\ell) = \ell \circ T$. Then

$$T^* = \varphi^{-1} \circ T^t \circ \varphi.$$

This shows the adjoint is the “conjugate” of the transpose by the inner-product isomorphism.

9.3.2 Properties of the Adjoint

Proposition 9.3.1 Algebraic properties of the adjoint

Let $S, T : V \to V$ be linear operators on an inner product space, and $\lambda \in k$. Then:

(i) $(T^*)^* = T$.

(ii) $(S + T)^* = S^* + T^*$.

(iii) $(\lambda T)^* = \overline{\lambda} T^*$.

(iv) $(ST)^* = T^* S^*$.

(v) $(\mathrm{id})^* = \mathrm{id}$.

Proof: Each follows from manipulating the defining equation $\langle v, T(w) \rangle = \langle T^*(v), w \rangle$.

(i) $\langle v, T^*(w) \rangle = \overline{\langle T^*(w), v \rangle} = \overline{\langle w, T(v) \rangle} = \langle T(v), w \rangle$. So $(T^*)^* = T$.

(iv) $\langle v, (ST)(w) \rangle = \langle v, S(T(w)) \rangle = \langle S^*(v), T(w) \rangle = \langle T^*(S^*(v)), w \rangle = \langle (T^* S^*)(v), w \rangle$. $\square$

Note:-

Property (iv) shows that taking adjoints reverses order, just like taking transposes or inverses. The map $T \mapsto T^*$ is a conjugate-linear anti-automorphism of $\mathrm{End}(V)$.

Matrix of the Adjoint

Theorem 9.3.2 Matrix of the adjoint

Let $\{e_1, \dots, e_n\}$ be an orthonormal basis of V, and let $M = M(T)$ be the matrix of T in this basis. Then the matrix of T^* is:

$$M(T^*) = \begin{cases} M^T & \text{(real inner product)} \\ M^* = \overline{M}^T & \text{(Hermitian inner product)} \end{cases}$$

where M^* denotes the conjugate transpose.

Proof: The (i,j)-entry of $M(T)$ is the coefficient of e_i in $T(e_j)$, which equals $\langle T(e_j), e_i\rangle = \overline{\langle e_i, T(e_j)\rangle}$.
The (i,j)-entry of $M(T^*)$ is $\langle T^*(e_j), e_i\rangle$.
Using the defining property:

$$\langle T^*(e_j), e_i\rangle = \langle e_j, T(e_i)\rangle = \overline{\langle T(e_i), e_j\rangle}.$$

The (j,i)-entry of $M(T)$ is $\langle T(e_i), e_j\rangle$, so

$$M(T^*)_{ij} = \overline{M(T)_{ji}} = (M^*)_{ij}.$$

□

Note:-

In an orthonormal basis, self-adjoint operators ($T^* = T$) correspond to symmetric matrices (real case) or Hermitian matrices (complex case, $M = M^*$).

Proposition 9.3.2 Kernel and image of the adjoint

For $T : V \to V$:

(i) $\mathrm{Ker}(T^*) = \mathrm{Im}(T)^\perp$.

(ii) $\mathrm{Im}(T^*) = \mathrm{Ker}(T)^\perp$.

Proof: (i) $v \in \mathrm{Ker}(T^*) \iff T^*(v) = 0 \iff \langle T^*(v), w\rangle = 0\ \forall w \iff \langle v, T(w)\rangle = 0\ \forall w \iff v \perp \mathrm{Im}(T)$.
(ii) Apply (i) to T^*: $\mathrm{Ker}((T^*)^*) = \mathrm{Im}(T^*)^\perp$, i.e., $\mathrm{Ker}(T) = \mathrm{Im}(T^*)^\perp$. Taking orthogonal complements: $\mathrm{Ker}(T)^\perp = (\mathrm{Im}(T^*)^\perp)^\perp = \mathrm{Im}(T^*)$ (using $(S^\perp)^\perp = S$ in finite dimensions). □

Corollary 9.3.1 Rank equality

$\mathrm{rank}(T) = \mathrm{rank}(T^*)$.

Proof: $\mathrm{rank}(T^*) = \dim \mathrm{Im}(T^*) = \dim \mathrm{Ker}(T)^\perp = \dim V - \dim \mathrm{Ker}(T) = \mathrm{rank}(T)$. □

9.4 Self-Adjoint and Normal Operators

9.4.1 The Spectral Theorem (Real Symmetric Case)

We now study operators with special relationships to their adjoints. The most important case is when $T = T^*$.

Definition 9.4.1: Self-adjoint operator

An operator $T : V \to V$ on an inner product space is **self-adjoint** (or **Hermitian**) if $T^* = T$, i.e.,

$$\langle v, T(w)\rangle = \langle T(v), w\rangle$$

for all $v, w \in V$.

Note:-

In an orthonormal basis, T is self-adjoint iff its matrix is symmetric (real case) or Hermitian (complex case, $M = \overline{M}^T$). Self-adjoint operators need not be invertible: for example, the zero operator is self-adjoint.

Proposition 9.4.1 Invariance of orthogonal complements

Let $T : V \to V$ be self-adjoint and $S \subseteq V$ a T-invariant subspace (i.e., $T(S) \subseteq S$). Then $S^\perp$ is also T-invariant.

Proof: Let $v \in S^\perp$. For any $w \in S$, we have $T(w) \in S$ (since S is T-invariant), so:

$$\langle T(v), w\rangle = \langle v, T(w)\rangle = 0$$

since $v \perp T(w)$. Thus $T(v) \perp S$, i.e., $T(v) \in S^\perp$. □

Note:-

This is the key property that makes self-adjoint operators diagonalizable: once we find an eigenvector, its orthogonal complement is invariant, and we can proceed by induction.

To prove the spectral theorem, we need to establish that self-adjoint operators have real eigenvalues and that eigenvectors exist over $\mathbb{R}$.

Lemma 9.4.1 Positivity lemma

Let T be self-adjoint on a real inner product space. For any $a > 0$, the operator $T^2 + a \cdot \mathrm{id}$ is invertible.

Proof: We show $\mathrm{Ker}(T^2 + a \cdot \mathrm{id}) = \{0\}$. For $v \neq 0$:

$$\langle (T^2 + a \cdot \mathrm{id})(v), v\rangle = \langle T^2(v), v\rangle + a\langle v, v\rangle = \langle T(v), T(v)\rangle + a\|v\|^2 = \|T(v)\|^2 + a\|v\|^2 > 0.$$

So $(T^2 + a \cdot \mathrm{id})(v) \neq 0$ for $v \neq 0$, i.e., $T^2 + a \cdot \mathrm{id}$ is injective, hence invertible. □

Corollary 9.4.1 Quadratics without real roots

If $p(x) = x^2 + bx + c \in \mathbb{R}^x$ has no real roots (i.e., $b^2 - 4c < 0$), and T is self-adjoint, then $p(T) = T^2 + bT + c \cdot \mathrm{id}$ is invertible.

Proof: Complete the square: $T^2 + bT + c \cdot \mathrm{id} = (T + \frac{b}{2}\mathrm{id})^2 + (c - \frac{b^2}{4})\mathrm{id}$.

Since $b^2 - 4c < 0$, we have $a = c - \frac{b^2}{4} > 0$. The operator $S = T + \frac{b}{2}\mathrm{id}$ is self-adjoint (check: $S^* = T^* + \frac{b}{2}\mathrm{id} = T + \frac{b}{2}\mathrm{id} = S$). By the lemma, $S^2 + a \cdot \mathrm{id}$ is invertible. □

Theorem 9.4.1 Spectral theorem (real self-adjoint)

Let $T : V \to V$ be a self-adjoint operator on a finite-dimensional real inner product space. Then:

(i) All eigenvalues of T are real.

(ii) T is diagonalizable.

(iii) V has an orthonormal basis consisting of eigenvectors of T.

Proof: We first show that T has an eigenvector. Since V is finite-dimensional, choose any nonzero $v \in V$. The vectors $v, T(v), T^2(v), \ldots, T^n(v)$ (where $n = \dim V$) are linearly dependent, so there exists a nonzero polynomial $p \in \mathbb{R}^x$ with $p(T)(v) = 0$.

Factor p over $\mathbb{R}$: since $\mathbb{R}^x$ has unique factorization, we can write

$$p(x) = c \prod_i (x - \lambda_i) \prod_j (x^2 + b_j x + c_j)$$

where the linear factors have real roots and the quadratic factors are irreducible (no real roots, i.e., $b_j^2 - 4c_j < 0$).

By the corollary, each $(T^2 + b_j T + c_j \cdot \mathrm{id})$ is invertible. So

$$0 = p(T)(v) = c \prod_i (T - \lambda_i \cdot \mathrm{id}) \prod_j (T^2 + b_j T + c_j \cdot \mathrm{id})(v).$$

Since the quadratic factors are invertible, we can remove them:

$$\prod_i (T - \lambda_i \cdot \mathrm{id})(v') = 0$$

for some nonzero v'.

At least one factor $(T - \lambda_i \cdot \mathrm{id})$ must have nontrivial kernel (otherwise their product would be invertible). Thus T has an eigenvalue $\lambda_i \in \mathbb{R}$.

(i) Eigenvalues are real: This is automatic since we found the eigenvalue in $\mathbb{R}$.

(iii) Orthonormal eigenbasis: We proceed by induction on $\dim V$.

Base case: $\dim V = 1$. Any nonzero vector is an eigenvector; normalize it.

Inductive step: Let v_1 be an eigenvector with $T(v_1) = \lambda_1 v_1$, normalized so $\|v_1\| = 1$. Let $S = \mathrm{span}(v_1)$. Since $T(S) \subseteq S$, the orthogonal complement $S^\perp$ is T-invariant.

The restriction $T|_{S^\perp} : S^\perp \to S^\perp$ is self-adjoint (the inner product on $S^\perp$ is the restriction of the inner product on V). By induction, $S^\perp$ has an orthonormal eigenbasis $\{v_2, \ldots, v_n\}$.

Then $\{v_1, v_2, \ldots, v_n\}$ is an orthonormal eigenbasis of V: the v_i are orthonormal, and each is an eigenvector of T.

(ii) Diagonalizable: In an eigenbasis, T has a diagonal matrix. □

Note:-

The spectral theorem says that self-adjoint operators on real inner product spaces are "as nice as possible": they are diagonalizable, with real eigenvalues, and the eigenspaces are mutually orthogonal. The matrix in an orthonormal eigenbasis is $\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

Corollary 9.4.2 Eigenvectors for distinct eigenvalues are orthogonal

If T is self-adjoint and $T(v) = \lambda v$, $T(w) = \mu w$ with $\lambda \neq \mu$, then $v \perp w$.

Proof: $\lambda\langle v, w\rangle = \langle \lambda v, w\rangle = \langle T(v), w\rangle = \langle v, T(w)\rangle = \langle v, \mu w\rangle = \mu\langle v, w\rangle$. So $(\lambda - \mu)\langle v, w\rangle = 0$, and $\lambda \neq \mu$ implies $\langle v, w\rangle = 0$. □

Skew-Adjoint Operators

The "opposite" of self-adjoint is skew-adjoint.

Definition 9.4.2: Skew-adjoint operator

An operator $T : V \to V$ on an inner product space is **skew-adjoint** (or **anti-self-adjoint**) if $T^* = -T$, i.e.,

$$\langle v, T(w) \rangle = -\langle T(v), w \rangle$$

for all $v, w \in V$.

Note:-

In an orthonormal basis, T is skew-adjoint iff its matrix M satisfies $M^T = -M$ (real case) or $M^* = -M$ (complex case). Such matrices are called **skew-symmetric** or **skew-Hermitian**.

Proposition 9.4.2 Properties of skew-adjoint operators

Let $T : V \to V$ be skew-adjoint on a real inner product space.

(i) $\langle T(v), v \rangle = 0$ for all $v \in V$.

(ii) If λ is an eigenvalue of T (in $\mathbb{C}$), then λ is purely imaginary.

(iii) $I + T$ is invertible.

Proof: (i) $\langle T(v), v \rangle = -\langle v, T(v) \rangle = -\overline{\langle T(v), v \rangle}$. Over $\mathbb{R}$, this gives $2\langle T(v), v \rangle = 0$.

(ii) If $T(v) = \lambda v$ with $v \neq 0$, then $\lambda \langle v, v \rangle = \langle T(v), v \rangle = 0$ by (i), so $\lambda = 0$. More generally, working in the complexification, if $\lambda = a + bi$ is an eigenvalue, the skew-adjoint condition forces $a = 0$.

(iii) Suppose $(I + T)(v) = 0$, i.e., $T(v) = -v$. Then $-\|v\|^2 = \langle T(v), v \rangle = 0$ by (i), so $v = 0$. Thus $I + T$ is injective, hence invertible. $\square$

Proposition 9.4.3 Cayley transform

Let T be skew-adjoint on a real inner product space. Then $S = (I - T)(I + T)^{-1}$ is orthogonal (i.e., $S^* = S^{-1}$).

Proof: Since $T^* = -T$, we have $(I + T)^* = I - T$ and $(I - T)^* = I + T$. Also, $I - T$ and $I + T$ commute (both are polynomials in T).

Then:

$$S^* = ((I + T)^{-1})^*(I - T)^* = ((I + T)^*)^{-1}(I - T)^* = (I - T)^{-1}(I + T).$$

And:

$$S^{-1} = (I + T)(I - T)^{-1}.$$

Since $I - T$ and $I + T$ commute, $(I - T)^{-1}(I + T) = (I + T)(I - T)^{-1}$. Thus $S^* = S^{-1}$. $\square$

9.4.2 Normal Operators on Complex Spaces

Over $\mathbb{C}$, the spectral theorem extends to a larger class of operators: those that commute with their adjoint.

Definition 9.4.3: Normal operator

An operator $T : V \to V$ on an inner product space is **normal** if

$$TT^* = T^*T.$$

Example 9.4.1 (Classes of normal operators)

- Self-adjoint operators ($T^* = T$) are normal: $TT^* = T^2 = T^*T$.

- Skew-adjoint operators ($T^* = -T$) are normal: $TT^* = -T^2 = T^*T$.
- Unitary operators ($T^* = T^{-1}$) are normal: $TT^* = I = T^*T$.

Lemma 9.4.2 Characterization of normality
An operator T is normal if and only if $\|T(v)\| = \|T^*(v)\|$ for all $v \in V$.

Proof: $\|T(v)\|^2 = \langle T(v), T(v)\rangle = \langle T^*T(v), v\rangle$ and $\|T^*(v)\|^2 = \langle T^*(v), T^*(v)\rangle = \langle TT^*(v), v\rangle$.
So $\|T(v)\| = \|T^*(v)\|$ for all v iff $\langle (T^*T - TT^*)(v), v\rangle = 0$ for all v.
For self-adjoint operators (and $T^*T - TT^*$ is self-adjoint), this implies $T^*T - TT^* = 0$. Indeed, if S is self-adjoint with $\langle S(v), v\rangle = 0$ for all v, then for any v, w:

$$0 = \langle S(v+w), v+w\rangle = \langle S(v), w\rangle + \langle S(w), v\rangle = 2\,\mathrm{Re}\langle S(v), w\rangle,$$

and similarly with $v + iw$ gives the imaginary part, so $\langle S(v), w\rangle = 0$ for all v, w, hence $S = 0$. □

Proposition 9.4.4 Normal operators and eigenspaces
Let T be normal.

(i) If $T(v) = \lambda v$, then $T^*(v) = \bar{\lambda} v$.

(ii) Eigenvectors for distinct eigenvalues are orthogonal.

(iii) $\mathrm{Ker}(T - \lambda I) = \mathrm{Ker}(T^* - \bar{\lambda} I)$.

Proof: (i) Note that $T - \lambda I$ is normal (check: $(T - \lambda I)(T - \lambda I)^* = (T - \lambda I)(T^* - \bar{\lambda} I)$, and these commute since T and T^* commute).
By the lemma, $\|T(v) - \lambda v\| = \|T^*(v) - \bar{\lambda} v\|$. If $T(v) = \lambda v$, the left side is zero, so $T^*(v) = \bar{\lambda} v$.
(ii) If $T(v) = \lambda v$ and $T(w) = \mu w$ with $\lambda \neq \mu$:

$$\lambda\langle v, w\rangle = \langle T(v), w\rangle = \langle v, T^*(w)\rangle = \langle v, \bar{\mu} w\rangle = \mu\langle v, w\rangle.$$

So $(\lambda - \mu)\langle v, w\rangle = 0$, giving $v \perp w$.
(iii) Follows from (i): $v \in \mathrm{Ker}(T - \lambda I) \iff T(v) = \lambda v \iff T^*(v) = \bar{\lambda} v \iff v \in \mathrm{Ker}(T^* - \bar{\lambda} I)$. □

Theorem 9.4.2 Complex spectral theorem
Let $T : V \to V$ be a normal operator on a finite-dimensional complex inner product space (with Hermitian inner product). Then V has an orthonormal basis consisting of eigenvectors of T.

Proof: Since $\mathbb{C}$ is algebraically closed, T has an eigenvalue λ. Let v_1 be a unit eigenvector with $T(v_1) = \lambda v_1$.
Let $S = \mathrm{span}(v_1)$. We show $S^\perp$ is T-invariant. By part (i) of the proposition, $T^*(v_1) = \bar{\lambda} v_1$, so S is also T^*-invariant.
For $w \in S^\perp$ and any $s \in S$:

$$\langle T(w), s\rangle = \langle w, T^*(s)\rangle.$$

Since $T^*(s) \in S$ and $w \perp S$, we get $\langle T(w), s\rangle = 0$. Thus $T(w) \perp S$, i.e., $T(w) \in S^\perp$.
The restriction $T|_{S^\perp}$ is normal (the inner product restricts, and $(T|_{S^\perp})^* = T^*|_{S^\perp}$). By induction on dimension, $S^\perp$ has an orthonormal eigenbasis $\{v_2, \ldots, v_n\}$. Then $\{v_1, \ldots, v_n\}$ is an orthonormal eigenbasis of V. □

Note:-

The complex spectral theorem is stronger than the real one: any normal operator (not just self-adjoint) is diagonalizable. The price is working over $\mathbb{C}$ with Hermitian inner products.
For self-adjoint T, the eigenvalues are real. For skew-adjoint T, they are purely imaginary. For unitary T, they lie on the unit circle $|\lambda| = 1$.

9.5 Orthogonal and Unitary Operators

9.5.1 Definitions and Characterizations

Orthogonal (real) and unitary (complex) operators are the isometries of inner product spaces: they preserve lengths and angles.

Definition 9.5.1: Orthogonal operator

Let $(V, \langle\cdot,\cdot\rangle)$ be a real inner product space. An operator $T : V \to V$ is **orthogonal** if

$$\langle T(u), T(v)\rangle = \langle u, v\rangle$$

for all $u, v \in V$.

Definition 9.5.2: Unitary operator

Let (V, H) be a complex inner product space (with Hermitian inner product H). An operator $T : V \to V$ is **unitary** if

$$H(T(u), T(v)) = H(u, v)$$

for all $u, v \in V$.

Note:-

We use "orthogonal" over $\mathbb{R}$ and "unitary" over $\mathbb{C}$, though the definitions are parallel. Some authors use "orthogonal" for both.

Theorem 9.5.1 Characterizations of orthogonal/unitary operators

For an operator $T : V \to V$ on a finite-dimensional inner product space, the following are equivalent:

(i) T is orthogonal (resp. unitary).

(ii) $T^*T = I$ (equivalently, $T^* = T^{-1}$).

(iii) $TT^* = I$.

(iv) $\|T(v)\| = \|v\|$ for all $v \in V$.

(v) T maps some orthonormal basis to an orthonormal basis.

(vi) T maps every orthonormal basis to an orthonormal basis.

Proof: (i) $\Rightarrow$ (ii): $\langle T^*T(u), v\rangle = \langle T(u), T(v)\rangle = \langle u, v\rangle = \langle I(u), v\rangle$ for all u, v. By nondegeneracy, $T^*T = I$.
(ii) $\Rightarrow$ (iii): $T^*T = I$ implies T is invertible with $T^{-1} = T^*$. Then $TT^* = T(T^{-1}) = I$.
(ii) $\Rightarrow$ (iv): $\|T(v)\|^2 = \langle T(v), T(v)\rangle = \langle T^*T(v), v\rangle = \langle v, v\rangle = \|v\|^2$.
(iv) $\Rightarrow$ (i): Use the polarization identity. Over $\mathbb{R}$:

$$\langle u, v\rangle = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2).$$

If T preserves norms: $\langle T(u), T(v)\rangle = \frac{1}{4}(\|T(u)+T(v)\|^2 - \|T(u)-T(v)\|^2) = \frac{1}{4}(\|T(u+v)\|^2 - \|T(u-v)\|^2) = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2) = \langle u, v\rangle$.

(i) $\Rightarrow$ (vi): If $\{e_1, \dots, e_n\}$ is orthonormal, then $\langle T(e_i), T(e_j)\rangle = \langle e_i, e_j\rangle = \delta_{ij}$.

(vi) $\Rightarrow$ (v): Trivial.

(v) $\Rightarrow$ (ii): Let $\{e_i\}$ be orthonormal with $\{T(e_i)\}$ orthonormal. For any $v = \sum a_i e_i$:

$$T^*T(v) = T^*(\sum a_i T(e_i)).$$

Since $\{T(e_i)\}$ is orthonormal, $\langle T(e_i), T(e_j)\rangle = \delta_{ij}$, so $T^*(T(e_i)) = e_i$ (the dual basis property in an orthonormal basis). Thus $T^*T(v) = \sum a_i e_i = v$. ☐

Corollary 9.5.1 Orthogonal/unitary operators are invertible

If T is orthogonal or unitary, then $T^{-1} = T^*$, which is also orthogonal/unitary.

Corollary 9.5.2 Orthogonal/unitary operators form a group

The set of orthogonal (resp. unitary) operators on V forms a subgroup of GL(V).

Proof: Identity: $I^* = I$, so $I^*I = I$. Closure: if $S^*S = I$ and $T^*T = I$, then $(ST)^*(ST) = T^*S^*ST = T^*T = I$. Inverses: $(T^{-1})^* = (T^*)^* = T$, so $(T^{-1})^*(T^{-1}) = TT^{-1} = I$. ☐

Matrix Characterization

Proposition 9.5.1 Orthogonal and unitary matrices

Let $\{e_1, \dots, e_n\}$ be an orthonormal basis of V, and let M be the matrix of T in this basis.

(i) (Real case) T is orthogonal iff $M^TM = I$, i.e., M is an **orthogonal matrix**.

(ii) (Complex case) T is unitary iff $M^*M = I$, i.e., M is a **unitary matrix**.

Equivalently: the columns of M form an orthonormal set (with respect to the standard inner product on k^n).

Proof: In an orthonormal basis, $M(T^*) = M(T)^*$. So $T^*T = I$ iff $M^*M = I$.

The columns of M are $Me_1, \dots, Me_n$ (standard basis vectors). The (i, j)-entry of M^*M is $(M^*M)_{ij} = \sum_k \overline{M_{ki}} M_{kj} = \langle \mathrm{col}_i(M), \mathrm{col}_j(M)\rangle$. So $M^*M = I$ iff columns are orthonormal. ☐

Corollary 9.5.3 Rows of orthogonal/unitary matrices

If M is orthogonal (resp. unitary), then the rows of M also form an orthonormal set.

Proof: $M^*M = I$ implies M is invertible with $M^{-1} = M^*$. Thus $MM^* = I$, which says the columns of M^* (= rows of M) are orthonormal. ☐

Proposition 9.5.2 Determinant of orthogonal/unitary operators

If T is orthogonal or unitary, then $|\det(T)| = 1$.
For orthogonal T (real case), $\det(T) = \pm 1$.

Proof: $\det(T^*T) = \det(I) = 1$. But $\det(T^*T) = \det(T^*)\det(T) = \overline{\det(T)}\det(T) = |\det(T)|^2$.

Over $\mathbb{R}$, $\det(T^T) = \det(T)$, so $\det(T)^2 = 1$, giving $\det(T) = \pm 1$. ☐

9.5.2 Reflections

A particularly simple class of orthogonal operators are reflections across hyperplanes.

Definition 9.5.3: Reflection

Let V be a real inner product space. A **reflection** is an orthogonal operator $P : V \to V$ such that $P^2 = I$ and $\dim \operatorname{Ker}(P + I) = 1$.

Note:-

The condition $P^2 = I$ says P is an involution. Since P is orthogonal, it is also self-adjoint ($P^* = P^{-1} = P$). The eigenvalues of P are ± 1 (since $P^2 = I$), and $\dim \operatorname{Ker}(P + I) = 1$ means the -1-eigenspace is 1-dimensional.

Proposition 9.5.3 Formula for reflections

Every reflection can be written as

$$P(x) = x - 2\langle x, v\rangle v$$

where $v \in V$ with $\|v\| = 1$. Geometrically, P reflects across the hyperplane $v^\perp$.

Proof: Define $P(x) = x - 2\langle x, v\rangle v$. We verify P is a reflection.

Linear: Clear from the formula.

Orthogonal:

$$\begin{aligned}\langle P(x), P(y)\rangle &= \langle x - 2\langle x, v\rangle v, y - 2\langle y, v\rangle v\rangle \\ &= \langle x, y\rangle - 2\langle x, v\rangle\langle v, y\rangle - 2\langle y, v\rangle\langle x, v\rangle + 4\langle x, v\rangle\langle y, v\rangle\|v\|^2 \\ &= \langle x, y\rangle - 4\langle x, v\rangle\langle y, v\rangle + 4\langle x, v\rangle\langle y, v\rangle = \langle x, y\rangle.\end{aligned}$$

Involution: $P(P(x)) = P(x) - 2\langle P(x), v\rangle v = x - 2\langle x, v\rangle v - 2(\langle x, v\rangle - 2\langle x, v\rangle)v = x$.

Eigenspaces: $P(v) = v - 2\|v\|^2 v = -v$, so v spans the -1-eigenspace. If $w \perp v$, then $P(w) = w$, so $v^\perp$ is the $+1$-eigenspace.

Conversely, any reflection P has $\operatorname{Ker}(P + I)$ one-dimensional. Let v span this kernel with $\|v\| = 1$. Then $P(v) = -v$, and on $v^\perp$, P acts as $+I$ (since P is diagonalizable with eigenvalues ± 1, and the -1-eigenspace is spanned by v). For $x = \langle x, v\rangle v + (x - \langle x, v\rangle v)$ with the second term in $v^\perp$:

$$P(x) = -\langle x, v\rangle v + (x - \langle x, v\rangle v) = x - 2\langle x, v\rangle v.$$

□

Proposition 9.5.4 Reflections map equal-length vectors

Given any two vectors $x, y \in V$ with $\|x\| = \|y\|$, there exists a reflection P with $P(x) = y$.

Proof: **Case 1: $x = y$.** If $x = 0$, any reflection fixes the origin, so any reflection works. If $x \neq 0$ (and $\dim V \geqslant 2$), choose any unit vector v orthogonal to x and set $P(z) = z - 2\langle z, v\rangle v$. Then $P(x) = x - 2\langle x, v\rangle v = x = y$.

Case 2: $x \neq y$. Let $v = \dfrac{x - y}{\|x - y\|}$ and define $P(z) = z - 2\langle z, v\rangle v$.

Compute:

$$P(x) = x - 2\langle x, v\rangle v = x - \frac{2\langle x,\, x - y\rangle}{\|x - y\|^2}(x - y).$$

Since $\|x\| = \|y\|$, we have $\|x\|^2 = \|y\|^2$, so

$$\langle x,\, x - y\rangle = \|x\|^2 - \langle x, y\rangle = \tfrac{1}{2}\big(\|x\|^2 + \|y\|^2 - 2\langle x, y\rangle\big) = \tfrac{1}{2}\|x - y\|^2.$$

Substituting:

$$P(x) = x - \frac{2 \cdot \frac{1}{2}\|x-y\|^2}{\|x-y\|^2}(x-y) = x-(x-y) = y.$$

□

Theorem 9.5.2 Orthogonal operators as products of reflections

Every orthogonal operator on $\mathbb{R}^n$ is a product of at most n reflections.

Proof: Induction on n. For $n = 0$, the identity is the empty product.

For $n \geqslant 1$: Let T be orthogonal. Pick a unit vector e_1. Then $\|T(e_1)\| = \|e_1\| = 1$.

By the previous proposition, there exists a reflection P_1 with $P_1(T(e_1)) = e_1$. Let $T_1 = P_1 T$. Then T_1 is orthogonal and $T_1(e_1) = e_1$.

Since $T_1(e_1) = e_1$, the subspace $e_1^\perp$ is T_1-invariant (if $w \perp e_1$, then $\langle T_1(w), e_1\rangle = \langle w, T_1^*(e_1)\rangle = \langle w, T_1^{-1}(e_1)\rangle = \langle w, e_1\rangle = 0$).

The restriction $T_1|_{e_1^\perp}$ is orthogonal on the $(n-1)$-dimensional space $e_1^\perp$. By induction, $T_1|_{e_1^\perp} = Q_{n-1}\cdots Q_2$ for reflections Q_i on $e_1^\perp$.

Extend each Q_i to V by setting $\tilde{Q}_i(e_1) = e_1$ and $\tilde{Q}_i|_{e_1^\perp} = Q_i$. Then $\tilde{Q}_i$ is a reflection on V (the -1-eigenspace of Q_i in $e_1^\perp$ becomes the -1-eigenspace of $\tilde{Q}_i$ in V).

We have $T_1 = \tilde{Q}_{n-1}\cdots\tilde{Q}_2$, so $P_1 T = \tilde{Q}_{n-1}\cdots\tilde{Q}_2$, giving

$$T = P_1^{-1}\tilde{Q}_{n-1}\cdots\tilde{Q}_2 = P_1\tilde{Q}_{n-1}\cdots\tilde{Q}_2$$

(since $P_1^2 = I$). This is a product of at most $1 + (n-1) = n$ reflections. □

9.5.3 The Classical Matrix Groups

The orthogonal and unitary operators form important matrix groups.

Definition 9.5.4: Orthogonal and special orthogonal groups

- $O(n) = \{A \in GL_n(\mathbb{R}) : A^T A = I\}$, the **orthogonal group**.
- $SO(n) = \{A \in O(n) : \det(A) = 1\}$, the **special orthogonal group**.

Definition 9.5.5: Unitary and special unitary groups

- $U(n) = \{A \in GL_n(\mathbb{C}) : A^* A = I\}$, the **unitary group**.
- $SU(n) = \{A \in U(n) : \det(A) = 1\}$, the **special unitary group**.

Proposition 9.5.5 Index of $SO(n)$ in $O(n)$

$SO(n)$ is a normal subgroup of $O(n)$ of index 2. We have an exact sequence

$$1 \to SO(n) \to O(n) \xrightarrow{\det} \{\pm 1\} \to 1.$$

Proof: The determinant map $\det : O(n) \to \{\pm1\}$ is a surjective homomorphism (surjective since $\det(I) = 1$ and $\det(-I) = (-1)^n$, or use a reflection with $\det = -1$). Its kernel is $SO(n)$. □

Note:-

Elements of $SO(n)$ are called **rotations** or **proper orthogonal transformations**. They preserve orientation. Elements of $O(n) \setminus SO(n)$ reverse orientation.

Example 9.5.1 (Low-dimensional cases)

- $O(1) = \{\pm 1\} \cong \mathbb{Z}/2\mathbb{Z}$, $SO(1) = \{1\}$.
- $SO(2)$ consists of rotation matrices $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$. Thus $SO(2) \cong S^1$ (the circle group) via $\theta \mapsto e^{i\theta}$.
- $U(1) = \{z \in \mathbb{C} : |z| = 1\} \cong S^1$.

Theorem 9.5.3 Structure of orthogonal operators

Let $T \in O(V)$ where V is a finite-dimensional real inner product space. Then V decomposes as an orthogonal direct sum

$$V = V_1 \oplus V_{-1} \oplus W_1 \oplus \cdots \oplus W_k$$

where $V_1 = \mathrm{Ker}(T - I)$ is the $+1$-eigenspace, $V_{-1} = \mathrm{Ker}(T + I)$ is the -1-eigenspace, and each W_j is a 2-dimensional T-invariant subspace on which T acts as rotation by angle $\theta_j \neq 0, \pi$.

Proof: The characteristic polynomial of T has real coefficients. Its roots (possibly complex) come in conjugate pairs. If $\lambda = e^{i\theta}$ is an eigenvalue with $|\lambda| = 1$ and $\lambda \notin \mathbb{R}$, then $\overline{\lambda} = e^{-i\theta}$ is also an eigenvalue.

Let $p(x) = (x - e^{i\theta})(x - e^{-i\theta}) = x^2 - 2\cos\theta \cdot x + 1$, an irreducible quadratic over $\mathbb{R}$. By the real Jordan form theory (or direct argument), $\mathrm{Ker}(p(T))$ is an even-dimensional T-invariant subspace.

Decompose V into generalized eigenspaces for real eigenvalues (± 1) and for conjugate pairs of complex eigenvalues. Since T is orthogonal and hence normal, it is diagonalizable over $\mathbb{C}$, so generalized eigenspaces are actual eigenspaces.

For the conjugate pair $e^{\pm i\theta}$, the real subspace $W = \mathrm{Ker}((T - e^{i\theta}I)(T - e^{-i\theta}I)) \cap V$ is T-invariant. Choose a unit eigenvector $v \in W_{\mathbb{C}}$ for $e^{i\theta}$. Then $v + \overline{v}$ and $i(v - \overline{v})$ are real vectors spanning a 2-dimensional T-invariant subspace on which T acts as rotation by θ.

Repeating this decomposition and using orthogonality of distinct eigenspaces gives the result. ☐

Note:-

In matrix form, choosing an orthonormal basis adapted to this decomposition, T is block-diagonal:

$$T \sim \begin{pmatrix} I_a & & & & \\ & -I_b & & & \\ & & R_{\theta_1} & & \\ & & & \ddots & \\ & & & & R_{\theta_k} \end{pmatrix}$$

where $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.
This is the “real canonical form” for orthogonal matrices.

Hermitian Forms and Real Forms

When a complex vector space is viewed as a real vector space, a Hermitian form decomposes into real and imaginary parts with specific properties.

Proposition 9.5.6 Decomposition of Hermitian forms

Let V be a complex vector space with Hermitian form H, and let W denote V viewed as a real vector space (of twice the dimension). Define $G, B : W \times W \to \mathbb{R}$ by

$$G(u, v) = \mathrm{Re}\, H(u, v), \qquad B(u, v) = \mathrm{Im}\, H(u, v).$$

Let $J : W \to W$ be scalar multiplication by i. Then:

(i) G is a symmetric bilinear form on W.

(ii) B is a skew-symmetric bilinear form on W.

(iii) $G(u, v) = B(Ju, v)$ for all $u, v \in W$.

(iv) H is nondegenerate iff G is nondegenerate iff B is nondegenerate.

Proof: (i) G is $\mathbb{R}$-bilinear since H is sesquilinear and Re is $\mathbb{R}$-linear. For symmetry:

$$G(v, u) = \operatorname{Re} H(v, u) = \operatorname{Re} \overline{H(u, v)} = \operatorname{Re} H(u, v) = G(u, v).$$

(ii) Similarly, $B(v, u) = \operatorname{Im} H(v, u) = \operatorname{Im} \overline{H(u, v)} = -\operatorname{Im} H(u, v) = -B(u, v)$.

(iii) $B(Ju, v) = \operatorname{Im} H(iu, v) = \operatorname{Im}(i \cdot H(u, v)) = \operatorname{Re} H(u, v) = G(u, v)$.

(iv) Suppose H is nondegenerate and $G(u, v) = 0$ for all v. Then $B(Ju, v) = G(u, v) = 0$ for all v, so $B(w, v) = 0$ for all w, v (since J is surjective). Thus $H(u, v) = G(u, v) + iB(u, v) = 0$ for all v, giving $u = 0$. The other implications are similar. □

Note:-

A linear operator $T : W \to W$ is $\mathbb{C}$-linear (i.e., $T \circ J = J \circ T$) iff it comes from a $\mathbb{C}$-linear operator on V. The proposition implies: for nondegenerate H, any two of the following imply the third:

(i) T preserves G.

(ii) T preserves B.

(iii) T is $\mathbb{C}$-linear.

This gives the relation $U(n) = O(2n) \cap GL_n(\mathbb{C}) = Sp(2n, \mathbb{R}) \cap GL_n(\mathbb{C}) = O(2n) \cap Sp(2n, \mathbb{R})$.

Exercise 9.5.1 Inner product spaces exercises

(1) Let $V \subset \mathbb{R}^x$ be the space of polynomials of degree at most 2, with inner product $\langle f, g \rangle = \int_0^1 f(x)g(x)\,dx$. Apply Gram–Schmidt to the basis $\{1, x, x^2\}$ to find an orthonormal basis.

(2) Let $T : V \to V$ be self-adjoint on a real inner product space, and suppose $v \in V$ with $\|v\| = 1$ maximizes the Rayleigh quotient $R(w) = \langle T(w), w \rangle / \|w\|^2$ among unit vectors. Prove that v is an eigenvector of T. (Do not use the spectral theorem.)

(3) Let $T : V \to V$ be an invertible linear operator on a finite-dimensional real inner product space. Show that T maps the unit sphere $S = \{v : \|v\| = 1\}$ to an ellipsoid. (Hint: consider T^*T.)

(4) For an invertible operator T, define the **dilatation** $K(T) = \frac{\sup_{\|v\|=1} \|Tv\|}{\inf_{\|v\|=1} \|Tv\|}$. Show that $K(T)^2$ equals the ratio of the largest to smallest eigenvalues of T^*T.

(5) Let A be the matrix of a (symmetric positive-definite) inner product on $\mathbb{R}^n$ with respect to the standard basis, so the inner product is $\langle x, y \rangle = x^T A y$. For $n = 2$, prove that the largest entry of A occurs on the diagonal.

Solution

(1) Start with $w_1 = 1$, $w_2 = x$, $w_3 = x^2$.
$\langle w_1, w_1 \rangle = \int_0^1 1\,dx = 1$, so $v_1 = 1$.
$\langle w_2, v_1 \rangle = \int_0^1 x\,dx = \frac{1}{2}$. $u_2 = x - \frac{1}{2} \cdot 1 = x - \frac{1}{2}$. $\|u_2\|^2 = \int_0^1 (x - \frac{1}{2})^2 dx = \int_0^1 (x^2 - x + \frac{1}{4}) dx = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12}$.
So $v_2 = \sqrt{12}(x - \frac{1}{2}) = 2\sqrt{3}(x - \frac{1}{2})$.

$\langle w_3, v_1\rangle = \int_0^1 x^2 dx = \frac{1}{3}$. $\langle w_3, v_2\rangle = 2\sqrt{3}\int_0^1 x^2(x-\frac{1}{2})dx = 2\sqrt{3}(\frac{1}{4}-\frac{1}{6}) = 2\sqrt{3}\cdot\frac{1}{12} = \frac{\sqrt{3}}{6}$. $u_3 = x^2 - \frac{1}{3} - \frac{\sqrt{3}}{6}\cdot 2\sqrt{3}(x-\frac{1}{2}) = x^2 - \frac{1}{3} - (x - \frac{1}{2}) = x^2 - x + \frac{1}{6}$. $\|u_3\|^2 = \int_0^1 (x^2 - x + \frac{1}{6})^2 dx = \frac{1}{180}$. So $v_3 = \sqrt{180}(x^2 - x + \frac{1}{6}) = 6\sqrt{5}(x^2 - x + \frac{1}{6})$.
The orthonormal basis is $\{1,\ 2\sqrt{3}(x-\frac{1}{2}),\ 6\sqrt{5}(x^2 - x + \frac{1}{6})\}$.
(2) Let $f(w) = \langle T(w), w\rangle/\|w\|^2$. For $u \perp v$ and $t \in \mathbb{R}$ small, consider $w_t = v + tu$. Since $\|v\| = 1$ and $u \perp v$:

$$\|w_t\|^2 = 1 + t^2\|u\|^2.$$

$$\langle T(w_t), w_t\rangle = \langle T(v), v\rangle + t(\langle T(v), u\rangle + \langle T(u), v\rangle) + t^2\langle T(u), u\rangle.$$

Since T is self-adjoint, $\langle T(v), u\rangle = \langle v, T(u)\rangle = \overline{\langle T(u), v\rangle} = \langle T(u), v\rangle$ (real), so:

$$\langle T(w_t), w_t\rangle = \langle T(v), v\rangle + 2t\langle T(v), u\rangle + t^2\langle T(u), u\rangle.$$

Thus:

$$f(w_t) = \frac{\langle T(v), v\rangle + 2t\langle T(v), u\rangle + O(t^2)}{1 + t^2\|u\|^2}.$$

At $t = 0$, $f(v) = \langle T(v), v\rangle$ is a maximum, so $\frac{d}{dt}|_{t=0} f(w_t) = 0$.
Differentiating: $\frac{d}{dt}|_{t=0} f(w_t) = 2\langle T(v), u\rangle$. So $\langle T(v), u\rangle = 0$ for all $u \perp v$.
Thus $T(v) \perp v^\perp$, meaning $T(v) \in (v^\perp)^\perp = \mathrm{span}(v)$. So $T(v) = \lambda v$ for some λ.
(3) We have $T(S) = \{w : \|T^{-1}(w)\| = 1\}$. Now

$$\|T^{-1}(w)\|^2 = \langle T^{-1}(w),\ T^{-1}(w)\rangle = \langle (T^{-1})^* T^{-1}(w),\ w\rangle = \langle Q(w),\ w\rangle,$$

where $Q = (T^{-1})^* T^{-1}$ is self-adjoint and positive-definite.
By the spectral theorem, there exists an orthonormal eigenbasis $\{u_1, \ldots, u_n\}$ with $Q(u_i) = \mu_i u_i$, where $\mu_i > 0$.
For $w = \sum x_i u_i$: $\langle Q(w), w\rangle = \sum \mu_i x_i^2$.
So $T(S) = \{w : \sum \mu_i x_i^2 = 1\}$. Setting $a_i = 1/\sqrt{\mu_i}$, this is $\sum x_i^2/a_i^2 = 1$, an ellipsoid with semi-axes a_i along u_i.
(4) We have $\|T(v)\|^2 = \langle T(v), T(v)\rangle = \langle T^*T(v), v\rangle$. The operator T^*T is self-adjoint and positive-definite (since T is invertible).
By the spectral theorem, let $\{e_i\}$ be an orthonormal eigenbasis with $T^*T(e_i) = \lambda_i e_i$, $\lambda_i > 0$. For $v = \sum a_i e_i$ with $\|v\| = 1$ (i.e., $\sum a_i^2 = 1$):

$$\|T(v)\|^2 = \langle T^*T(v), v\rangle = \sum \lambda_i a_i^2.$$

This is maximized when all weight is on the largest eigenvalue: $\sup = \lambda_{\max}$. It is minimized when all weight is on the smallest: $\inf = \lambda_{\min}$.
Thus $K(T)^2 = \frac{\lambda_{\max}}{\lambda_{\min}}$.
(5) For $n = 2$, let $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ with $a, c > 0$ (positive-definiteness requires positive diagonal) and $ac > b^2$.
We show $\max\{a, c\} \geqslant |b|$. Suppose not: $|b| > a$ and $|b| > c$. Then $b^2 > ac$, contradicting $ac > b^2$.
So the largest entry in magnitude is on the diagonal. Since $a, c > 0$ and $ac > b^2 \geqslant 0$, the largest entry is $\max\{a, c\}$, on the diagonal.

Lecture 10.

Modules and Finitely Generated Abelian Groups

We now take a detour from the purely linear-algebraic material to study modules—a generalization of vector spaces where the scalars come from a ring rather than a field. The payoff is immediate: the classification of finitely generated abelian groups reduces to "linear algebra over $\mathbb{Z}$."

Recall from **Lecture 2** that every abelian group $(G, +)$ is a $\mathbb{Z}$-module: the action $\mathbb{Z} \times G \to G$ sends $(n, g) \mapsto ng$ (where $ng = g + g + \cdots + g$ for $n > 0$, $0 \cdot g = 0$, and $ng = -(|n|g)$ for $n < 0$). This observation is the bridge between group theory and module theory.

10.1 Modules over a Ring

10.1.1 Definitions and Examples

Definition 10.1.1: Module

Let R be a ring (with unity 1). A (left) **R-module** is an abelian group $(M, +)$ equipped with a scalar multiplication $R \times M \to M$, $(r, m) \mapsto r \cdot m$, satisfying:

(i) $r(m_1 + m_2) = rm_1 + rm_2$ for all $r \in R$, $m_1, m_2 \in M$.

(ii) $(r_1 + r_2)m = r_1m + r_2m$ for all $r_1, r_2 \in R$, $m \in M$.

(iii) $(r_1r_2)m = r_1(r_2m)$ for all $r_1, r_2 \in R$, $m \in M$.

(iv) $1 \cdot m = m$ for all $m \in M$.

Example 10.1.1 (Vector spaces)

If $R = k$ is a field, then a k-module is the same thing as a vector space over k. So modules generalize vector spaces by allowing the scalars to come from a ring.

Example 10.1.2 (Abelian groups as $\mathbb{Z}$-modules)

Every abelian group $(G, +)$ is a $\mathbb{Z}$-module via $n \cdot g = ng$. Conversely, every $\mathbb{Z}$-module is an abelian group (just forget the $\mathbb{Z}$-action, which is determined by the group structure anyway). The categories of abelian groups and $\mathbb{Z}$-modules are the same.

Example 10.1.3 (The ring as a module over itself)

R is an R-module via left multiplication: $r \cdot s = rs$. This is a free module of rank 1.

Example 10.1.4 (R-modules from ring homomorphisms)

If $\varphi : R \to S$ is a ring homomorphism, then any S-module M becomes an R-module via $r \cdot m = \varphi(r)m$. This is called restriction of scalars.

Example 10.1.5 (Polynomial modules)

Let $R = k[x]$ and V a vector space over k. Any linear operator $T : V \to V$ makes V into a $k[x]$-module via $p(x) \cdot v = p(T)(v)$. Different operators give different module structures on the same underlying vector space.

10.1.2 Submodules and Ideals

Definition 10.1.2: Submodule

A **submodule** of an R-module M is a subset $N \subseteq M$ that is itself an R-module under the inherited operations. Equivalently, N is a subgroup of $(M, +)$ that is closed under scalar multiplication: $rn \in N$ for all $r \in R$, $n \in N$.

Definition 10.1.3: Ideal

A submodule of R (viewed as a module over itself) is called a **left ideal**. When R is commutative (as will be the case for us), left ideals are automatically **two-sided ideals**: subsets $I \subseteq R$ satisfying:

(i) I is an abelian subgroup of $(R, +)$.

(ii) $rI \subseteq I$ and $Ir \subseteq I$ for all $r \in R$.

Example 10.1.6 (Ideals in $\mathbb{Z}$ and $k[x]$)

The ideals of $\mathbb{Z}$ are exactly the subgroups $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ for $n \geqslant 0$. Similarly, the ideals of $k[x]$ are $p(x) \cdot k[x] = \{p(x)q(x) : q(x) \in k[x]\}$. In both cases, every ideal is generated by a single element. Rings with this property are called **principal ideal domains** (PIDs). This is closely related to the Euclidean division algorithm: $\mathrm{span}(p, q) = \mathrm{span}(\gcd(p, q))$.

Note:-

The quotient of an R-module M by a submodule N is again an R-module, with $r \cdot (m + N) = rm + N$. For example, $\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/n$ as a $\mathbb{Z}$-module, and $k[x]/(p(x)) \cdot k[x]$ is a $k[x]$-module. The quotient of R by an ideal is not just an R-module but a ring in its own right.

10.1.3 General Facts about Modules: Nothing Is True

Over a field, every vector space has a basis, every linearly independent set extends to one, and every spanning set contains one. Over a general ring, all of these fail.

Example 10.1.7 (Bases need not exist)

The $\mathbb{Z}$-module $M = \mathbb{Z}/n$ (for $n \geqslant 2$) has no basis. Any map $\varphi : \mathbb{Z} \to M$ sending $1 \mapsto m$ cannot be injective (since $nm = 0$ in M but $n \neq 0$ in $\mathbb{Z}$). So no single-element set is linearly independent, and no basis can exist.

Example 10.1.8 (Linearly independent sets may not be part of a basis)

Consider $M = \mathbb{Z}$ as a $\mathbb{Z}$-module. The element $2 \in \mathbb{Z}$ is linearly independent (if $a \cdot 2 = 0$ then $a = 0$). But $\{2\}$ cannot be extended to a basis of $\mathbb{Z}$: the only bases of $\mathbb{Z}$ are $\{1\}$ and $\{-1\}$, and no set containing 2 is a basis.

Example 10.1.9 (Spanning sets need not contain a basis)

The $\mathbb{Z}$-module $M = \mathbb{Z}$ is spanned by $\{4, 5\}$ (since $\gcd(4, 5) = 1$, so $1 = 5 \cdot 5 - 6 \cdot 4 \in \operatorname{span}(4, 5)$). But $\{4, 5\}$ is not linearly independent ($5 \cdot 4 - 4 \cdot 5 = 0$), and neither subset $\{4\}$ or $\{5\}$ spans $\mathbb{Z}$. So the spanning set contains no basis.

Example 10.1.10 (Submodules of finitely generated modules need not be finitely generated)

Let $R = k[x_1, x_2, \ldots]$ (polynomials in infinitely many variables) and $M = R$ as an R-module. Then M is generated by $\{1\}$. But the ideal $M' = \{f \in R : f(0) = 0\}$ (polynomials with zero constant term) is a submodule not finitely generated: any finite subset only involves finitely many variables x_i, so cannot span x_j for large j.

By contrast, this pathology does not occur over Noetherian rings (including $\mathbb{Z}$ and $k[x_1, \ldots, x_n]$).

10.1.4 Module Homomorphisms

Definition 10.1.4: Module homomorphism

Let M, N be R-modules. A map $\varphi \in \operatorname{Hom}_R(M, N)$ is a module homomorphism if

$$\varphi(u + w) = \varphi(u) + \varphi(w) \quad \text{and} \quad \varphi(av) = a\varphi(v)$$

for all $u, w \in M$, $a \in R$.

Note:-

When R is commutative, $\operatorname{Hom}_R(M, N)$ is itself an R-module: $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$ and $(a\varphi)(v) = a\varphi(v)$. (For noncommutative R, this scalar multiplication need not yield an R-module homomorphism, so $\operatorname{Hom}_R(M, N)$ is only an abelian group.) For free modules, things work as expected: $\operatorname{Hom}_R(R^m, R^n) \cong R^{m \times n}$ (determined by images of basis vectors, just like linear maps between vector spaces). But for non-free modules, $\operatorname{Hom}_R(M, N)$ can vanish even when both M and N are nonzero. For instance, $\operatorname{Hom}_R(k, k[x]) = 0$ when $R = k[x]$ and $M = k = k[x]/(x)$: any φ must satisfy $x \cdot \varphi(1) = \varphi(x \cdot 1) = \varphi(0) = 0$, so $\varphi(1)$ is killed by x in $k[x]$, forcing $\varphi(1) = 0$.

10.1.5 Free Modules and Direct Sums

Definition 10.1.5: Free module

An R-module M is free if it has a basis: a set $\{e_i\}_{i \in I}$ such that every element of M can be written uniquely as a finite R-linear combination $\sum r_i e_i$. Equivalently, $M \cong R^{(I)} = \bigoplus_{i \in I} R$. The rank of a free module is $|I|$ (well-defined when R is commutative).

Note:-

Over a field, every module (= vector space) is free. Over $\mathbb{Z}$, free modules are the groups $\mathbb{Z}^n$. The module $\mathbb{Z}/n$ is <u>not</u> free for $n \geqslant 2$.

Definition 10.1.6: Direct sum of modules

The direct sum $M_1 \oplus M_2$ is the set of pairs (m_1, m_2) with componentwise operations. More generally, $\bigoplus_{i \in I} M_i$ consists of tuples $(m_i)_{i \in I}$ with only finitely many nonzero entries. This has the universal property: maps out of $\bigoplus M_i$ correspond to families of maps out of each M_i.

Definition 10.1.7: Exact sequences of modules

A sequence of R-module homomorphisms

$$\cdots \to M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1} \to \cdots$$

is exact at M_i if $\mathrm{Im}(f_{i-1}) = \mathrm{Ker}(f_i)$. A short exact sequence $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ means f is injective, g is surjective, and $\mathrm{Im}(f) = \mathrm{Ker}(g)$, i.e., $C \cong B/f(A)$. A short exact sequence splits if $B \cong A \oplus C$. Over a PID, every short exact sequence of finitely generated modules with C free splits (analogous to Maschke's theorem).

10.2 Classification of Finitely Generated Abelian Groups

The main theorem of this lecture is:

Theorem 10.2.1 Classification of finitely generated abelian groups

Every finitely generated abelian group G is isomorphic to a product of cyclic groups:

$$G \cong \mathbb{Z}/n_1 \times \mathbb{Z}/n_2 \times \cdots \times \mathbb{Z}/n_k \times \mathbb{Z}^\ell$$

where $n_i \geqslant 2$ and $\ell \geqslant 0$. Moreover, using the Chinese Remainder Theorem ($\mathbb{Z}/mn \cong \mathbb{Z}/m \times \mathbb{Z}/n$ when $\gcd(m, n) = 1$), we can arrange each n_i to be a prime power. In this form, the decomposition is unique up to reordering.

The proof reduces to "linear algebra over $\mathbb{Z}$," following Artin §14.4–14.7. The strategy has two ingredients: first, express any finitely generated $\mathbb{Z}$-module as $\mathbb{Z}^n/\mathrm{Im}(T)$ for some $T \in \mathrm{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$; second, diagonalize T by changes of basis over $\mathbb{Z}$.

10.2.1 Step 1: Presenting a Module as a Quotient

Proposition 10.2.1 Presentation of finitely generated $\mathbb{Z}$-modules

If M is a finitely generated $\mathbb{Z}$-module, then there exist $m, n \geqslant 0$ and $T \in \mathrm{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$ such that

$$M \cong \mathbb{Z}^n/\mathrm{Im}(T).$$

Equivalently, there is an exact sequence $\mathbb{Z}^m \xrightarrow{T} \mathbb{Z}^n \to M \to 0$.

This says: quotient by a $\mathbb{Z}$-submodule of $\mathbb{Z}^n$ is the same as quotient of an abelian group by a subgroup. We need a lemma:

Lemma 10.2.1 Submodules of $\mathbb{Z}^n$ are free

Any submodule of $\mathbb{Z}^n$ is finitely generated and free of rank $\leqslant n$.

Proof: By induction on n. For $n = 1$: subgroups of $(\mathbb{Z}, +)$ are $\{0\}$ or $\mathbb{Z}a$ for some $a \in \mathbb{Z} \setminus \{0\}$, both free of rank $\leqslant 1$.

Assume the result for $\mathbb{Z}^{n-1}$, and let $N \subseteq \mathbb{Z}^n$ be a submodule. The projection $\pi : \mathbb{Z}^n \to \mathbb{Z}^{n-1}$ sending $(a_1, \ldots, a_n) \mapsto (a_2, \ldots, a_n)$ restricts to a homomorphism $\pi|_N : N \to \mathbb{Z}^{n-1}$.

The image $\mathrm{Im}(\pi|_N)$ is a submodule of $\mathbb{Z}^{n-1}$, hence free and finitely generated by induction. The kernel $\mathrm{Ker}(\pi|_N) = N \cap (\mathbb{Z} \times 0 \times \cdots \times 0)$ is a subgroup of $\mathbb{Z}$, hence free of rank 0 or 1.

If $\mathrm{Ker}(\pi|_N)$ and $\mathrm{Im}(\pi|_N)$ are both finitely generated (and free), then so is N. Let $e_1, \ldots, e_k$ be generators of $\mathrm{Ker}(\pi|_N)$ (a basis) and $g_1 = \pi(f_1), \ldots, g_m = \pi(f_m)$ generators of $\mathrm{Im}(\pi|_N)$. For any $x \in N$, write

$\pi(x) = \sum a_i g_i$, so $\pi(x - \sum a_i f_i) = 0$, meaning $x - \sum a_i f_i \in \mathrm{Ker}(\pi|_N) = \mathrm{span}(e_1, \ldots, e_k)$. Thus $x \in \mathrm{span}(e_1, \ldots, e_k, f_1, \ldots, f_m)$, so $\{e_i, f_j\}$ generates N. That this set is a basis (i.e., free) follows from the directness of $\mathrm{Ker} \cap \mathrm{Im} = 0$. ☐

Proof of the proposition: Let M be finitely generated with generators $e_1, \ldots, e_n$. Define $\varphi : \mathbb{Z}^n \to M$ by $(a_1, \ldots, a_n) \mapsto \sum a_i e_i$. This is surjective. The kernel $N = \mathrm{Ker}(\varphi) \subseteq \mathbb{Z}^n$ is a submodule, hence finitely generated by the lemma. Let $f_1, \ldots, f_m$ be generators of N. Define $T : \mathbb{Z}^m \to \mathbb{Z}^n$ by $T(b_i) = f_i$. Then $\mathrm{Ker}(\varphi) = \mathrm{Im}(T)$, giving the exact sequence

$$\mathbb{Z}^m \xrightarrow{T} \mathbb{Z}^n \xrightarrow{\varphi} M \to 0$$

and thus $M \cong \mathbb{Z}^n / \mathrm{Im}(T)$. ☐

10.2.2 Step 2: Diagonalization over $\mathbb{Z}$ (Smith Normal Form)

The key difference from linear algebra over a field: we cannot divide freely in $\mathbb{Z}$. The substitute is the Euclidean algorithm, which governs what "row and column operations" are available.

Definition 10.2.1: Divisibility

The **divisibility** of a nonzero element $x = (a_1, \ldots, a_n) \in \mathbb{Z}^n$ is the largest $d \in \mathbb{Z}_+$ such that $x = dy$ for some $y \in \mathbb{Z}^n$. Equivalently, $d = \gcd(a_1, \ldots, a_n)$. An element is **primitive** if its divisibility is 1.

Lemma 10.2.2 Primitive elements and bases

An element of $\mathbb{Z}^n$ can be part of a basis if and only if it is primitive. More generally, $v \in \mathbb{Z}^n$ with divisibility d satisfies $v = d \cdot w$ where w is a basis element.

Proof: Elements of a basis $(e_1, \ldots, e_n)$ are primitive: linear independence prevents $e_i = d(\sum a_j e_j)$ for $d > 1$.

Conversely, let $v = a_1 e_1 + \cdots + a_n e_n$ be primitive. Without loss of generality, $a_1 \neq 0$ and $|a_1| = \min\{|a_i| : a_i \neq 0\}$. For each $k \geqslant 2$, write $a_k = q_k a_1 + r_k$ by Euclidean division. Change basis to $(e_1' = e_1 + \sum_{k \geqslant 2} q_k e_k,\ e_2, \ldots, e_n)$. In this basis, $v = a_1 e_1' + r_2 e_2 + \cdots + r_n e_n$ with $0 \leqslant r_k < |a_1|$.

Repeating this process (which strictly decreases the minimum nonzero coefficient at each step) terminates in finitely many steps with $v = d \cdot e_1''$ for some basis vector e_1''. Since v is primitive, $d = 1$. ☐

Theorem 10.2.2 Smith Normal Form

For any $T \in \mathrm{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$, there exist bases $(e_1, \ldots, e_m)$ of $\mathbb{Z}^m$ and $(f_1, \ldots, f_n)$ of $\mathbb{Z}^n$, and positive integers $d_1, \ldots, d_r$ (where $r \leqslant \min(m, n)$ is the rank of T), such that

$$T(e_i) = \begin{cases} d_i f_i & \text{if } 1 \leqslant i \leqslant r \\ 0 & \text{if } i > r. \end{cases}$$

In matrix form:

$$\mathcal{M}(T) = \begin{pmatrix} d_1 & & & 0 & \\ & d_2 & & & \\ & & \ddots & & \\ & & & d_r & \\ 0 & & & & 0 \end{pmatrix}.$$

Moreover, the construction gives $d_1 = \min\{\mathrm{div}(T(x)) : x \notin \mathrm{Ker}(T)\} = \gcd$ of all entries of $\mathcal{M}(T)$.

Proof: By induction on m.

If $T = 0$, the statement is trivial for any m, n.

Case $m = 1$: T is determined by $T(1) \in \mathbb{Z}^n$. Let $d = \operatorname{div}(T(1))$. By the lemma, $T(1) = df_1$ where f_1 can be extended to a basis of $\mathbb{Z}^n$. Take $e_1 = 1$.

Inductive step: Assume $T \neq 0$. Let $d_1 = \min\{\operatorname{div}(T(x)) : x \notin \operatorname{Ker}(T)\}$, and choose e_1 with $\operatorname{div}(T(e_1)) = d_1$. Since $T(e_1)$ has divisibility d_1, write $T(e_1) = d_1 f_1$ with f_1 primitive, and extend f_1 to a basis $(f_1, \ldots, f_n)$ of $\mathbb{Z}^n$. Extend e_1 to a basis $(e_1, \ldots, e_m)$ of $\mathbb{Z}^m$ (using the lemma, since e_1 is primitive: if $e_1 = d \cdot e_1'$ then $\operatorname{div}(T(e_1')) = d_1/d$, contradicting minimality unless $d = 1$).

In the bases (e_i) and (f_j), the matrix of T has the form

$$\mathcal{M}(T) = \begin{pmatrix} d_1 & a_2 & \cdots & a_m \\ 0 & & & \\ \vdots & & \mathcal{M}(T') & \\ 0 & & & \end{pmatrix}$$

where T' is the restriction to $\operatorname{span}(e_2, \ldots, e_m)$ composed with projection to $\operatorname{span}(f_2, \ldots, f_n)$.

For $j \geqslant 2$, write $a_j = q_j d_1 + r_j$ with $0 \leqslant r_j < d_1$. Change basis of $\mathbb{Z}^m$ to $(e_1, e_2' = e_2 - q_2 e_1, \ldots, e_m' = e_m - q_m e_1)$. The matrix becomes

$$\begin{pmatrix} d_1 & r_2 & \cdots & r_m \\ 0 & & & \\ \vdots & & \mathcal{M}(T') & \\ 0 & & & \end{pmatrix}.$$

If any $r_j \neq 0$, then $\operatorname{div}(T(e_j')) \leqslant r_j < d_1$, contradicting the minimality of d_1. So $r_j = 0$ for all $j \geqslant 2$, and the matrix is

$$\begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & \mathcal{M}(T') & \\ 0 & & & \end{pmatrix}.$$

By induction on m, we can diagonalize T' by further basis changes in $\operatorname{span}(e_2, \ldots, e_m)$ and $\operatorname{span}(f_2, \ldots, f_n)$, giving the desired diagonal form. $\square$

Note:-

The Smith Normal Form is the integer analogue of row reduction. Over a field, Gaussian elimination diagonalizes any matrix (in fact reduces it to have 1's and 0's on the diagonal). Over $\mathbb{Z}$, the Euclidean algorithm plays the role of division, and the diagonal entries d_i need not be 1. The construction actually forces $d_1 \mid d_2 \mid \cdots \mid d_r$ (the invariant factor divisibility chain), though we do not prove this here.

10.2.3 Step 3: Completing the Classification

Proof of the classification theorem: By Proposition 1 (presentation as quotient), any finitely generated abelian group M satisfies $M \cong \mathbb{Z}^n / \operatorname{Im}(T)$ for some $T \in \operatorname{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$. By the Smith Normal Form, after choosing appropriate bases, $\operatorname{Im}(T)$ is spanned by $d_1 f_1, \ldots, d_r f_r$ for some positive integers $d_i > 0$ and $r \leqslant n$. Therefore

$$M \cong \mathbb{Z}^n / \operatorname{span}(d_1 f_1, \ldots, d_r f_r) \cong \mathbb{Z}/d_1 \times \cdots \times \mathbb{Z}/d_r \times \mathbb{Z}^{n-r}.$$

The factors $\mathbb{Z}/d_i$ with $d_i = 1$ are trivial and can be dropped. Relabeling the remaining $d_i \geqslant 2$ as $n_1, \ldots, n_k$ and setting $\ell = n - r$ gives the decomposition

$$M \cong \mathbb{Z}/n_1 \times \cdots \times \mathbb{Z}/n_k \times \mathbb{Z}^\ell.$$

By the Chinese Remainder Theorem, each $\mathbb{Z}/n_i$ with $n_i = p_1^{a_1} \cdots p_s^{a_s}$ decomposes as $\mathbb{Z}/p_1^{a_1} \times \cdots \times \mathbb{Z}/p_s^{a_s}$. Rearranging gives the prime power form. $\square$

Example 10.2.1 (Finitely generated abelian groups of small order)

- Order 4: $\mathbb{Z}/4$ or $\mathbb{Z}/2 \times \mathbb{Z}/2$. These are not isomorphic (the first has an element of order 4, the second does not).
- Order 6: $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$ (only one, since $\gcd(2, 3) = 1$).
- Order 8: $\mathbb{Z}/8$, $\mathbb{Z}/4 \times \mathbb{Z}/2$, or $\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/2$.
- Order 12: $\mathbb{Z}/12 \cong \mathbb{Z}/4 \times \mathbb{Z}/3$, or $\mathbb{Z}/2 \times \mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/3$.

Note:-

The number ℓ is the **free rank** (or **Betti number**) of M, and the integers $n_1, \dots, n_k$ (in prime power form) are the **elementary divisors**. These invariants completely determine M up to isomorphism. The free rank counts the number of "copies of $\mathbb{Z}$," while the elementary divisors encode the torsion (elements of finite order).

Exercise 10.2.1 Modules and finitely generated abelian groups exercises

(1) Classify all abelian groups of order 36 up to isomorphism.

(2) Let $T : \mathbb{Z}^2 \to \mathbb{Z}^2$ be the linear map with matrix $\begin{pmatrix} 6 & 4 \\ 3 & 5 \end{pmatrix}$. Compute the Smith Normal Form of T and determine the group $\mathbb{Z}^2/\operatorname{Im}(T)$.

(3) Prove that a finitely generated abelian group is free (i.e., isomorphic to $\mathbb{Z}^n$ for some n) if and only if it is torsion-free (i.e., the only element of finite order is 0).

(4) Let G be a finitely generated abelian group with $|G| = p^n$ for a prime p. Show that $G \cong \mathbb{Z}/p^{a_1} \times \cdots \times \mathbb{Z}/p^{a_k}$ where $a_1 + \cdots + a_k = n$ and $a_i \geqslant 1$. How many such groups are there (as a function of the partition structure of n)?

(5) Show that $\mathbb{Z}/m \otimes_{\mathbb{Z}} \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n)$. (Hint: use the presentation $\mathbb{Z}/m = \mathbb{Z}/m\mathbb{Z}$ and the right-exactness of the tensor product.)

Solution

(1) We have $36 = 2^2 \cdot 3^2$. For the 2-part: $\mathbb{Z}/4$ or $\mathbb{Z}/2 \times \mathbb{Z}/2$. For the 3-part: $\mathbb{Z}/9$ or $\mathbb{Z}/3 \times \mathbb{Z}/3$. Taking all combinations:

$$\mathbb{Z}/4 \times \mathbb{Z}/9 \cong \mathbb{Z}/36, \quad \mathbb{Z}/4 \times \mathbb{Z}/3 \times \mathbb{Z}/3, \quad \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/9, \quad \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/3.$$

So there are exactly 4 abelian groups of order 36.

(2) Start with $A = \begin{pmatrix} 6 & 4 \\ 3 & 5 \end{pmatrix}$. The gcd of all entries is 1, but let us apply row/column operations over $\mathbb{Z}$.

$R_1 \to R_1 - 2R_2$: $\begin{pmatrix} 0 & -6 \\ 3 & 5 \end{pmatrix}$. Swap rows: $\begin{pmatrix} 3 & 5 \\ 0 & -6 \end{pmatrix}$. $C_2 \to C_2 - C_1$: $\begin{pmatrix} 3 & 2 \\ 0 & -6 \end{pmatrix}$. $C_1 \to C_1 - C_2$: $\begin{pmatrix} 1 & 2 \\ 6 & -6 \end{pmatrix}$. $C_2 \to C_2 - 2C_1$: $\begin{pmatrix} 1 & 0 \\ 6 & -18 \end{pmatrix}$. $R_2 \to R_2 - 6R_1$: $\begin{pmatrix} 1 & 0 \\ 0 & -18 \end{pmatrix}$.

Smith Normal Form: $\operatorname{diag}(1, 18)$. So $\mathbb{Z}^2/\operatorname{Im}(T) \cong \mathbb{Z}/1 \times \mathbb{Z}/18 \cong \mathbb{Z}/18 \cong \mathbb{Z}/2 \times \mathbb{Z}/9$.

Check: $\det(A) = 30 - 12 = 18$, and $|\mathbb{Z}^2/\operatorname{Im}(T)| = |\det(T)| = 18$. Consistent.

(3) One direction: if $G \cong \mathbb{Z}^n$, then G is torsion-free (only $0 \in \mathbb{Z}^n$ has finite order).

Conversely, if G is finitely generated and torsion-free, write $G \cong \mathbb{Z}/n_1 \times \cdots \times \mathbb{Z}/n_k \times \mathbb{Z}^\ell$ by the classification theorem. Each $\mathbb{Z}/n_i$ contributes a nonzero element of finite order (namely $\bar{1}$, which has order n_i). Since G is torsion-free, $k = 0$, so $G \cong \mathbb{Z}^\ell$.

(4) Since $|G| = p^n$, the classification gives $G \cong \mathbb{Z}/p^{a_1} \times \cdots \times \mathbb{Z}/p^{a_k}$ with $a_i \geqslant 1$ and $p^{a_1} \cdots p^{a_k} = p^n$, i.e., $a_1 + \cdots + a_k = n$. The number of such groups equals the number of partitions of n. For example, p^4 gives partitions $(4), (3,1), (2,2), (2,1,1), (1,1,1,1)$, so 5 groups.

(5) From the short exact sequence $0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \to \mathbb{Z}/m \to 0$, tensor with $\mathbb{Z}/n$. Right-exactness gives:

$$\mathbb{Z}/n \xrightarrow{\times m} \mathbb{Z}/n \to \mathbb{Z}/m \otimes \mathbb{Z}/n \to 0.$$

The map $\times m : \mathbb{Z}/n \to \mathbb{Z}/n$ has image $m\mathbb{Z}/n$, which is the cyclic subgroup of order $n/\gcd(m,n)$, i.e., $m\mathbb{Z}/n \cong \mathbb{Z}/(n/\gcd(m,n))$. The quotient therefore has order $\gcd(m,n)$:

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong (\mathbb{Z}/n)/(m\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m,n).$$

Lecture 11.

Group Actions, Conjugacy, and the Sylow Theorems

We return to group theory, now armed with the machinery of group actions (introduced in **Lecture 3**). Understanding what sets a group G acts on—and how—reveals the internal structure of G itself. The main applications are the class equation, the Sylow theorems, and the classification of finite subgroups of SO(3).

11.1 Group Actions: Orbits, Stabilizers, and Counting

11.1.1 Recap and Definitions

Recall from **Lecture 3**: an action of a group G on a set S is a homomorphism $\rho : G \to \mathrm{Perm}(S)$. Equivalently, a map $G \times S \to S$, $(g, s) \mapsto g \cdot s$, satisfying $e \cdot s = s$ and $(gh) \cdot s = g \cdot (h \cdot s)$ for all $g, h \in G$, $s \in S$.

Definition 11.1.1: Faithful action

An action $\rho : G \to \mathrm{Perm}(S)$ is **faithful** if ρ is injective. Otherwise, the group that "really" acts on S is $G/\operatorname{Ker}(\rho)$.

Definition 11.1.2: Orbit

The orbit of $s \in S$ under G is $\mathcal{O}_s = G \cdot s = \{g \cdot s : g \in G\} \subseteq S$.

The relation $s \sim t \iff \exists g \in G$ with $g \cdot s = t$ is an equivalence relation on S: reflexivity from $e \cdot s = s$, symmetry from $g \cdot s = t \implies s = g^{-1} \cdot t$, and transitivity from $g \cdot s = t$, $h \cdot t = u \implies (hg) \cdot s = u$. The orbits are the equivalence classes, giving a partition $S = \bigsqcup \mathcal{O}_s$.

Definition 11.1.3: Transitive action

An action is **transitive** if there is only one orbit, i.e., for all $s, t \in S$ there exists $g \in G$ with $g \cdot s = t$.

Note:-

Any G-action on S restricts to a transitive G-action on each orbit. So every group action decomposes into a disjoint union of transitive actions.

Definition 11.1.4: Stabilizer and fixed points

- The **stabilizer** of $s \in S$ is $\mathrm{Stab}(s) = \{g \in G : g \cdot s = s\}$. This is a subgroup of G.
- The **fixed points** of $g \in G$ are the subset $S^g = \{s \in S : g \cdot s = s\}$.

Proposition 11.1.1 Conjugate stabilizers

If $s' = g \cdot s$, then $\mathrm{Stab}(s') = g\,\mathrm{Stab}(s)\,g^{-1}$. In particular, elements in the same orbit have conjugate stabilizers.

Proof: If $h \in \mathrm{Stab}(s)$, then $(ghg^{-1}) \cdot s' = (ghg^{-1})(g \cdot s) = g(h \cdot s) = g \cdot s = s'$, so $g\,\mathrm{Stab}(s)g^{-1} \subseteq \mathrm{Stab}(s')$. The reverse inclusion follows by applying the same argument with g^{-1}. $\square$

11.1.2 The Orbit-Stabilizer Theorem

Example 11.1.1 (The coset action)
Given a subgroup $H \leqslant G$, the set of left cosets $G/H = \{aH : a \in G\}$ carries a G-action by left multiplication: $g \cdot [aH] = [gaH]$. This action is transitive (a^{-1} maps $[aH]$ to $[H]$). The stabilizer of $[H]$ is H itself, and $\mathrm{Stab}([aH]) = aHa^{-1}$.

The coset action is the universal example of a transitive action:

Theorem 11.1.1 Orbit-Stabilizer Theorem
If G acts on S, let $s \in S$ and $H = \mathrm{Stab}(s)$. Then the map

$$\varepsilon : G/H \to \mathcal{O}_s, \qquad [aH] \mapsto a \cdot s$$

is a bijection that intertwines the G-actions: $\varepsilon(g \cdot [aH]) = g \cdot \varepsilon([aH])$.

Proof: Well-defined: if $a' = ah \in aH$, then $a' \cdot s = a \cdot (h \cdot s) = a \cdot s$.
Surjective: by definition of orbit, every element of $\mathcal{O}_s$ is $a \cdot s$ for some a.
Injective: $a' \cdot s = a \cdot s \iff a^{-1}a' \cdot s = s \iff a^{-1}a' \in \mathrm{Stab}(s) = H \iff a' \in aH$.
Equivariance: $\varepsilon(g \cdot [aH]) = \varepsilon([gaH]) = (ga) \cdot s = g \cdot (a \cdot s) = g \cdot \varepsilon([aH])$. □

Corollary 11.1.1 Counting formula
If G and S are finite, then $|\mathcal{O}_s| = |G|/|\mathrm{Stab}(s)|$ and $|S| = \sum_{\text{orbits}} |\mathcal{O}_s|$.

Example 11.1.2 (Rotational symmetries of the tetrahedron)
Let G be the group of rotational symmetries of a regular tetrahedron, acting on $S = \{4 \text{ faces}\}$. The action is transitive (any face can be mapped to any other), so $|\mathcal{O}| = |S| = 4$. The stabilizer of a face consists of rotations fixing that face: rotations about the axis through the face and the opposite vertex, giving $|\mathrm{Stab}| = 3$. Thus $|G| = |\mathcal{O}| \cdot |\mathrm{Stab}| = 4 \cdot 3 = 12$. In fact $G \cong A_4$: the identity, 8 elements of order 3 (rotations by 120° and 240° about each vertex-to-face axis), and 3 elements of order 2 (rotations by 180° about each edge-midpoint axis, corresponding to (12)(34) etc.).

11.1.3 Burnside's Lemma

Theorem 11.1.2 Burnside's Lemma (Cauchy–Frobenius)
Let G be a finite group acting on a finite set S. The number of orbits is

$$\#\text{orbits} = \frac{1}{|G|} \sum_{g \in G} |S^g|$$

where $S^g = \{s \in S : g \cdot s = s\}$ is the set of fixed points of g. In words: the number of orbits equals the average number of fixed points of elements of G.

Proof: Count the set $\Sigma = \{(g, s) \in G \times S : g \cdot s = s\}$ in two ways.
Summing over G: $|\Sigma| = \sum_{g \in G} |S^g|$.
Summing over S: $|\Sigma| = \sum_{s \in S} |\mathrm{Stab}(s)|$. Group this by orbits. All elements in an orbit $\mathcal{O}$ have conjugate

stabilizers of the same size $|\mathrm{Stab}(s)| = |G|/|\mathcal{O}|$. So

$$|\Sigma| = \sum_{\mathcal{O}} |\mathcal{O}| \cdot \frac{|G|}{|\mathcal{O}|} = \sum_{\mathcal{O}} |G| = |G| \cdot (\#\text{orbits}).$$

Equating: $\sum_{g \in G} |S^g| = |G| \cdot (\#\text{orbits})$. □

Example 11.1.3 (Coloring a tetrahedron)

How many ways to color the faces of a regular tetrahedron with 3 colors, up to rotational symmetry? Here $S = \{\text{3-colorings of 4 faces}\}$ with $|S| = 3^4 = 81$, and $G = A_4$ (12 rotations).

- e (identity): all 81 colorings are fixed. $|S^e| = 81$.
- 8 rotations of order 3 (120° about vertex-to-face axes): 3 faces must share a color, the 4th is free. $|S^g| = 3 \times 3 = 9$ each.
- 3 rotations of order 2 (180° about edge-midpoint axes): pairs of faces must match. $|S^g| = 3 \times 3 = 9$ each.

So $\#\text{orbits} = \frac{1}{12}(81 + 8 \cdot 9 + 3 \cdot 9) = \frac{180}{12} = 15$.

Note:-

Burnside's lemma extends to more complex situations. For coloring the edges of a tetrahedron with 3 colors: $\frac{1}{12}(3^6 + 8 \cdot 9 + 3 \cdot 3^4) = \frac{1}{12}(729 + 72 + 243) = 87$.

11.2 Actions of G on Itself

The two most important actions of a group on itself are left multiplication and conjugation.

11.2.1 Left Multiplication and Cayley's Theorem

The left multiplication action $g \cdot h = gh$ is transitive with $\mathrm{Stab}(h) = \{e\}$ for all h, and no fixed points except when $g = e$. Since the action is faithful, we recover:

Theorem 11.2.1 Cayley's Theorem

Every finite group G is isomorphic to a subgroup of S_n, where $n = |G|$.

This is not very useful for understanding the structure of G in practice. The conjugation action is far more revealing.

11.2.2 Conjugation, the Class Equation, and p-Groups

G acts on itself by conjugation: $g \cdot h = ghg^{-1}$. This is a group homomorphism $G \to \mathrm{Aut}(G) \subset \mathrm{Perm}(G)$.

Definition 11.2.1: Conjugacy class, centralizer, center

- The orbits of the conjugation action are the **conjugacy classes** $C_h = \{ghg^{-1} : g \in G\}$.
- The stabilizer of h is the **centralizer** $\mathcal{Z}(h) = \{g \in G : gh = hg\}$, the subgroup of elements commuting with h.
- The **center** of G is $Z(G) = \bigcap_{h \in G} \mathcal{Z}(h) = \{g \in G : gh = hg \ \forall h \in G\}$, the kernel of the conjugation action.

Note:-

The action is trivial (every element is a fixed point) iff G is abelian. The action is faithful iff $Z(G) = \{e\}$.

By the orbit-stabilizer theorem, $|C_h| = |G|/|\mathcal{Z}(h)|$, so the size of each conjugacy class divides $|G|$. Moreover, $|C_h| = 1$ iff $h \in Z(G)$. The conjugacy classes partition G, giving:

Theorem 11.2.2 The Class Equation

For a finite group G,

$$|G| = \sum_{C} |C| = |Z(G)| + \sum_{\substack{C \\ |C|>1}} |C|$$

where the sum runs over conjugacy classes C, and the second sum runs over non-central classes. Each $|C|$ divides $|G|$.

Theorem 11.2.3 Groups of order p^2 are abelian

If $|G| = p^2$ for p prime, then G is abelian.

Proof: Conjugacy classes have sizes dividing p^2, so $|C| \in \{1, p, p^2\}$. The class equation gives $p^2 = |Z(G)| + \sum_{|C|>1} |C|$. Each non-central class has size p or p^2 (the latter is impossible since it would mean $C = G$, but then $|\mathcal{Z}(h)| = 1$, contradicting $h \in \mathcal{Z}(h)$). So each non-central class has size p.

Since $Z(G)$ is a subgroup of G, $|Z(G)|$ divides p^2: it is 1, p, or p^2. If $|Z(G)| = p^2$ we are done (G is abelian).

If $|Z(G)| = p$: pick $g \notin Z(G)$. Then g commutes with itself and with $Z(G)$, so $\mathcal{Z}(g) \supseteq Z(G) \cup \{g\}$, giving $|\mathcal{Z}(g)| > p$. But $\mathcal{Z}(g)$ is a subgroup, so $|\mathcal{Z}(g)|$ divides p^2. The only option is $|\mathcal{Z}(g)| = p^2$, meaning $\mathcal{Z}(g) = G$, i.e., $g \in Z(G)$. Contradiction.

If $|Z(G)| = 1$: the class equation gives $p^2 = 1 + kp$ for some k, so $p \mid 1$. Impossible.

So $|Z(G)| = p^2$ and G is abelian. The only groups of order p^2 are $\mathbb{Z}/p^2$ and $\mathbb{Z}/p \times \mathbb{Z}/p$. $\square$

Proposition 11.2.1 There are exactly 5 groups of order 8

The three abelian groups $\mathbb{Z}/8$, $\mathbb{Z}/4 \times \mathbb{Z}/2$, $(\mathbb{Z}/2)^3$, plus two non-abelian groups: the dihedral group D_4 (symmetries of the square) and the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$.

Note:-

Two approaches to proving this. The "by hand" method: if $|G| = 8$ and G is not abelian, not every element has $g^2 = 1$ (else G would be abelian), so there exists a of order 4. The subgroup $\langle a \rangle$ has index 2 and is therefore normal. Work out the possibilities for multiplication by an element $b \notin \langle a \rangle$ with $ab \neq ba$.
The class equation method: $8 = \sum |C|$ with $|C| \in \{1, 2, 4, 8\}$ and $|C_e| = 1$. Since G is non-abelian, $|Z(G)| \neq 8$. By the p^2 argument, $|Z(G)| \neq 4$ (else $G/Z(G)$ is cyclic, forcing G abelian). So $|Z(G)| = 2$. For $g \notin Z(G)$, we have $\mathcal{Z}(g) \supsetneq Z(G)$ but $\mathcal{Z}(g) \neq G$, so $|\mathcal{Z}(g)| = 4$ and $|C_g| = 2$. The class equation is $8 = 1 + 1 + 2 + 2 + 2$ (2 central elements, 3 conjugacy classes of size 2). Then work out the possibilities.

11.3 The Sylow Theorems

The Sylow theorems are the most powerful general tools for understanding finite groups. They guarantee the existence of subgroups of prime-power order and constrain how many there can be.

Definition 11.3.1: Sylow p-subgroup

Let G be a finite group and p a prime dividing $|G|$. Write $|G| = p^e m$ where $p \nmid m$. A **Sylow p-subgroup** of G is a subgroup $H \leqslant G$ of order $|H| = p^e$.

Note:-

The converse to Lagrange's theorem fails in general: if $k \mid |G|$, there need not exist a subgroup of order k. For instance, A_4 has no subgroup of order 6, and A_5 has no subgroup of order 30. The Sylow theorems say this failure does not occur for prime-power divisors.

Theorem 11.3.1 Sylow Theorems (Sylow, 1872)

Let G be a finite group with $|G| = p^e m$, $p \nmid m$.

(1) For every prime p dividing $|G|$, a Sylow p-subgroup exists.

(2) All Sylow p-subgroups are conjugate: if $H, H' \leqslant G$ are both Sylow p-subgroups, then $H' = gHg^{-1}$ for some $g \in G$. Moreover, any subgroup $K \leqslant G$ with $|K|$ a power of p is contained in some Sylow p-subgroup.

(3) Let s_p be the number of Sylow p-subgroups. Then $s_p \equiv 1 \pmod p$ and $s_p \mid |G|$ (equivalently, $s_p \mid m = |G|/p^e$).

Example 11.3.1 (Groups of order 15)

Let $|G| = 15 = 3 \cdot 5$. By Sylow (3): $s_3 \mid 5$ and $s_3 \equiv 1 \pmod 3$, so $s_3 = 1$. Similarly, $s_5 \mid 3$ and $s_5 \equiv 1 \pmod 5$, so $s_5 = 1$. Both Sylow subgroups are normal (unique Sylow subgroups are always normal, since conjugation permutes Sylow subgroups). Let H be the Sylow 3-subgroup and K the Sylow 5-subgroup. Then $H \cap K = \{e\}$ (orders are coprime) and $|HK| = |H||K|/|H \cap K| = 15 = |G|$. So $G \cong H \times K \cong \mathbb{Z}/3 \times \mathbb{Z}/5 \cong \mathbb{Z}/15$. Every group of order 15 is cyclic.

11.4 Finite Subgroups of SO(3)

Using group actions and counting, we can classify all finite subgroups of the rotation group SO(3).

Recall from **Lecture 9**: every element $T \in SO(3)$, $T \neq \mathrm{id}$, is a rotation about some axis, hence fixes exactly two unit vectors $\pm v$ (the "poles" of T). For a finite subgroup $G \subset SO(3)$, let $P = \{v \in \mathbb{R}^3 : \|v\| = 1, \exists g \neq e \text{ in } G \text{ with } g \cdot v = v\}$ be the set of all poles. Since $h(gv) = (hgh^{-1})(hv)$, the set P is G-invariant, so G acts on P.

For each pole $p \in P$, the stabilizer $\mathrm{Stab}(p)$ is a cyclic subgroup of rotations about the axis $\pm p$, of some order $r_p \geqslant 2$. The proof proceeds by counting the set $\Sigma = \{(g, p) \in (G \setminus \{e\}) \times P : g \cdot p = p\}$:

Summing over G: each $g \neq e$ has exactly 2 poles, so $|\Sigma| = 2(|G| - 1)$.

Summing over P: for each pole p, there are $r_p - 1$ non-identity rotations fixing p, so $|\Sigma| = \sum_{p \in P}(r_p - 1)$. Grouping by orbits $\mathcal{O}_i$ (elements in the same orbit have the same $r_p = r_i$):

$$2(|G| - 1) = \sum_{\text{orbits}} |\mathcal{O}_i|(r_i - 1) = \sum_i \frac{|G|}{r_i}(r_i - 1).$$

Dividing by $|G|$:

$$2 - \frac{2}{|G|} = \sum_i \left(1 - \frac{1}{r_i}\right).$$

Since each $r_i \geqslant 2$, each term is $\geqslant 1/2$. The left side is < 2, so there are at most 3 orbits (since $3 \cdot 1/2 < 2$ but $4 \cdot 1/2 = 2$).

Analyzing the cases with $2 \leqslant r_1 \leqslant r_2 \leqslant r_3$ and the equation $2 - 2/|G| = \sum(1 - 1/r_i)$:

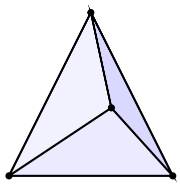
2 orbits: $2 - 2/|G| = (1 - 1/r_1) + (1 - 1/r_2)$. Since $1/r_1 + 1/r_2 > 2/|G| \geqslant 0$ and each $r_i \geqslant 2$, one finds $r_1 = r_2 = |G|$. The two poles form one orbit of size 2, with stabilizer of order $|G|$. So G is a cyclic group of rotations about a single axis: $G \cong \mathbb{Z}/n$.

3 orbits: $1/r_1 + 1/r_2 + 1/r_3 = 1 + 2/|G| > 1$. With $2 \leqslant r_1 \leqslant r_2 \leqslant r_3$, necessarily $r_1 = 2$ (else $\sum 1/r_i \leqslant 1$). Then $1/r_2 + 1/r_3 = 1/2 + 2/|G| > 1/2$, forcing $r_2 \leqslant 3$.

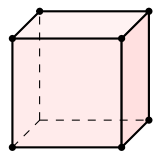
- $(r_1, r_2, r_3) = (2, 2, n)$: $|G| = 2n$. This gives the dihedral group D_n.
- $(2, 3, 3)$: $|G| = 12$. Rotational symmetries of the tetrahedron, $G \cong A_4$.
- $(2, 3, 4)$: $|G| = 24$. Rotational symmetries of the cube (equivalently, octahedron), $G \cong S_4$.
- $(2, 3, 5)$: $|G| = 60$. Rotational symmetries of the icosahedron (equivalently, dodecahedron), $G \cong A_5$.

Theorem 11.4.1 Classification of finite subgroups of $SO(3)$

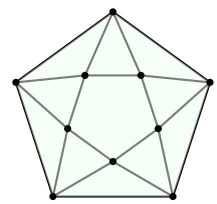
The finite subgroups of $SO(3)$ are exactly: the cyclic groups $\mathbb{Z}/n$, the dihedral groups D_n, the tetrahedral group A_4, the octahedral group S_4, and the icosahedral group A_5.



A_4, $|G| = 12$
$(2, 3, 3)$



S_4, $|G| = 24$
$(2, 3, 4)$



A_5, $|G| = 60$
$(2, 3, 5)$

Note:-

The cube and octahedron are dual polyhedra (vertices of one correspond to centers of faces of the other), so they share the same symmetry group. Similarly the dodecahedron and icosahedron are dual, sharing A_5. For regular polyhedra, the poles at midpoints of edges have $r = 2$, poles at vertices have $r = \#$faces meeting at a vertex, and poles at centers of faces have $r = \#$edges of the face.

11.5 Conjugacy in S_n and the Alternating Group

11.5.1 Conjugacy Classes in S_n

Proposition 11.5.1 Conjugation permutes cycle entries

Let $\sigma = (a_1\, a_2 \cdots a_k)$ be a k-cycle and $\tau \in S_n$ any permutation. Then $\tau\sigma\tau^{-1} = (\tau(a_1)\, \tau(a_2) \cdots \tau(a_k))$.

Proof: Compute: $\tau\sigma\tau^{-1}$ maps $\tau(a_i) \mapsto \tau(a_{i+1})$ (indices mod k), and fixes elements not in $\{\tau(a_1), \ldots, \tau(a_k)\}$. $\square$

Corollary 11.5.1 Conjugacy classes in S_n correspond to partitions

Two permutations $\sigma, \tau \in S_n$ are conjugate if and only if they have the same cycle type (i.e., the same multiset of cycle lengths). Thus conjugacy classes in S_n correspond to partitions of n.

Example 11.5.1 (Conjugacy classes in S_3 and S_4)

For $n = 3$, partitions of 3 are: $1 + 1 + 1$ (identity, 1 element), $2 + 1$ (transpositions, 3 elements), 3 (3-cycles, 2 elements). Class equation: $6 = 1 + 3 + 2$.
For $n = 4$, the 5 partitions give: identity (1), transpositions (6), double transpositions $(ij)(k\ell)$ (3), 3-cycles (8), 4-cycles (6). Class equation: $24 = 1 + 6 + 3 + 8 + 6$.

Note:-

This helps find normal subgroups: a normal subgroup $H \trianglelefteq G$ must be a union of conjugacy classes (since $gHg^{-1} = H$), must include $\{e\}$, and $|H|$ must divide $|G|$. For S_4: the only non-trivial unions of classes whose sizes sum to a divisor of 24 and that form a subgroup are $\{e, (12)(34), (13)(24), (14)(23)\} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (the Klein four-group, of order 4) and A_4 (of order 12). So the normal subgroups of S_4 are $\{e\}$, V_4, A_4, and S_4.

11.5.2 The Alternating Group and Simplicity of A_5

Recall from **Lecture 3**: the sign homomorphism $\text{sgn} : S_n \to \{\pm 1\}$ sends $\sigma \mapsto (-1)^{\#\text{inversions}}$. The alternating group is $A_n = \text{Ker}(\text{sgn})$, a normal subgroup of index 2 in S_n. A k-cycle has sign $(-1)^{k-1}$, so $\sigma \in A_n$ iff its cycle decomposition has an even number of even-length cycles.

Lemma 11.5.1 A_n is generated by 3-cycles

For $n \geqslant 3$, every element of A_n is a product of 3-cycles.

Proof: By induction. For $A_3 = \{e, (123), (132)\} \subset S_3$, the claim is clear.

Assume A_{n-1} is generated by 3-cycles. Let $\sigma \in A_n$. If $\sigma(n) = n$, then $\sigma \in A_{n-1}$ and we are done by induction. Otherwise, let $i = \sigma(n) \neq n$ and j be any element distinct from i and n. Then $\tau = (j\,i\,n) \cdot \sigma \in A_n$ and $\tau(n) = n$, so τ is a product of 3-cycles by induction. Thus $\sigma = (j\,i\,n)^{-1}\tau = (n\,i\,j) \cdot \tau$ is also a product of 3-cycles. $\square$

Proposition 11.5.2 Conjugacy class splitting in A_n

Let $C \subset S_n$ be a conjugacy class with $C \subseteq A_n$. Then either:

(a) C is a single conjugacy class in A_n, or

(b) C splits into exactly 2 conjugacy classes of equal size in A_n.

Case (a) occurs iff there exists an odd permutation τ commuting with some (hence every) $\sigma \in C$, i.e., $\mathcal{Z}(\sigma) \not\subset A_n$. Case (b) occurs iff the cycle lengths of σ are all odd and distinct.

Example 11.5.2 (Conjugacy classes of A_5)

As elements of S_5: $A_5 = \{e\} \cup \{(ij)(k\ell)\} \cup \{3\text{-cycles}\} \cup \{5\text{-cycles}\}$, with sizes 1, 15, 20, 24 in S_5.
In A_5: the double transpositions $(ij)(k\ell)$ form a single class (since $(ij) \in \mathcal{Z}((ij)(k\ell))$, and (ij) is odd). The 3-cycles form a single class (since $(45) \in \mathcal{Z}((123))$). The 5-cycles split into two classes of size 12 each (all cycle lengths are odd and distinct). The class equation of A_5 is $60 = 1 + 15 + 20 + 12 + 12$.

Theorem 11.5.1 Simplicity of A_5

The alternating group A_5 is simple: its only normal subgroups are $\{e\}$ and A_5 itself.

Proof: A normal subgroup $H \trianglelefteq A_5$ must be a union of conjugacy classes of A_5 containing $\{e\}$, with $|H|$ dividing 60. The class sizes are $1, 15, 20, 12, 12$. We need a non-trivial subset of $\{1, 15, 20, 12, 12\}$ including the 1 that sums to a divisor of 60. The divisors of 60 are $1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$. No non-trivial sub-sum $1 + (\text{subset of } \{15, 20, 12, 12\})$ equals any divisor of 60 other than 1 or 60. So $H = \{e\}$ or $H = A_5$. $\square$

Theorem 11.5.2 Simplicity of A_n for $n \geqslant 5$

For all $n \geqslant 5$, A_n is simple.

Proof: Let $H \trianglelefteq A_n$, $H \neq \{e\}$. We show H contains a 3-cycle, hence (by conjugation) all 3-cycles, hence $H = A_n$ by the lemma.

Let $\sigma \in H$, $\sigma \neq e$. Replacing σ by a power, we may assume σ has prime order p, so σ is a product of disjoint p-cycles.

Case $p \geqslant 5$: Write $\sigma = (i_1 \cdots i_p)\tau$. Let $g = (i_2\, i_3\, i_4)$ (a 3-cycle, which is in A_n). Then $g\sigma g^{-1}\sigma^{-1} \in H$ (since H is normal). Compute: $g\sigma g^{-1}$ permutes the entries of the first cycle by conjugation, giving a 5-cycle $(i_2\, i_4\, i_5 \cdots)$. The commutator $g\sigma g^{-1}\sigma^{-1}$ is a nontrivial element of H with smaller support. Iterate to reduce to a 3-cycle.

Case $p = 3$: If σ is a single 3-cycle, done. Otherwise $\sigma = (i_1\, i_2\, i_3)(i_4\, i_5\, i_6)\tau$. Let $g = (i_4\, i_3\, i_2)$. Then $g\sigma g^{-1}\sigma^{-1} = (i_1\, i_5\, i_2\, i_4\, i_3)$, a 5-cycle in H. Back to previous case.

Case $p = 2$: $\sigma = (i_1\, i_2)(i_3\, i_4)$. Since $n \geqslant 5$, pick $i_5 \notin \{i_1, i_2, i_3, i_4\}$. Let $g = (i_5\, i_3\, i_1)$. Then $g\sigma g^{-1}\sigma^{-1} = (i_1\, i_5\, i_2\, i_4\, i_3)$, a 5-cycle in H. ☐

Corollary 11.5.2 Normal subgroups of S_n

For $n \geqslant 5$, the only normal subgroups of S_n are $\{e\}$, A_n, and S_n.

Proof: Let $H \trianglelefteq S_n$. Then $H \cap A_n \trianglelefteq A_n$. Since A_n is simple, either $H \cap A_n = \{e\}$ or $H \cap A_n = A_n$. In the first case, $|H| \leqslant |S_n/A_n| = 2$, so $H = \{e\}$ or $|H| = 2$. If $|H| = 2$, then $H = \{e, \sigma\}$ with σ an odd involution. But $\tau\sigma\tau^{-1} = \sigma$ for all τ would force σ to commute with everything, and for $n \geqslant 3$ the center of S_n is trivial. In the second case, $A_n \subseteq H$, so $H = A_n$ or $H = S_n$. ☐

Exercise 11.5.1 Group actions, conjugacy, and the Sylow theorems exercises

(1) Let G be a group of order pq where $p < q$ are distinct primes and $p \nmid q - 1$. Prove that G is cyclic.

(2) Compute the class equation of S_5. Use it to determine all normal subgroups of S_5.

(3) Use Burnside's lemma to count the number of distinct necklaces made from 6 beads of 2 colors, where two necklaces are the same if one can be rotated to obtain the other.

(4) Let G be a finite group acting on a finite set S, and let $N = \#\text{orbits}$. Prove that $\sum_{s \in S} |\mathrm{Stab}(s)| = N \cdot |G|$.

(5) Prove that any group of order 12 has a normal Sylow 3-subgroup or a normal Sylow 2-subgroup.

Solution

(1) By Sylow: $s_q \mid p$ and $s_q \equiv 1 \pmod{q}$. Since $p < q$, the only possibility is $s_q = 1$. Similarly, $s_p \mid q$ and $s_p \equiv 1 \pmod{p}$, so $s_p \in \{1, q\}$. Since $p \nmid q - 1$, we cannot have $s_p = q$ (as $q \equiv 1 \pmod{p}$ would be required). So $s_p = 1$. Both Sylow subgroups are normal, and $G \cong \mathbb{Z}/p \times \mathbb{Z}/q \cong \mathbb{Z}/pq$.
(2) Partitions of 5 and their class sizes: 1^5 (1), $2 + 1^3$ (10), $2^2 + 1$ (15), $3 + 1^2$ (20), $3 + 2$ (20), $4 + 1$ (30), 5 (24). Class equation: $120 = 1 + 10 + 15 + 20 + 20 + 30 + 24$. A normal subgroup is a union of classes including $\{e\}$ with order dividing 120. The only options: 1 ($\{e\}$), $1 + 15 + 20 + 24 = 60$ (A_5), and 120 (S_5).
(3) The rotation group is $G = \mathbb{Z}/6$ acting on $S = \{\text{2-colorings of 6 positions}\}$, $|S| = 2^6 = 64$. Fixed points of rotation by k positions: a coloring is fixed iff it has period dividing $\gcd(k, 6)$. Rotation by 0 (identity): $2^6 = 64$. By 1: period 1, so monochromatic, $2^1 = 2$. By 2: period 2 (or 1), so $2^{\gcd(2,6)} = 2^2 = 4$. By 3: $2^{\gcd(3,6)} = 2^3 = 8$. By 4: same as by 2, so 4. By 5: same as by 1, so 2. Total: $\frac{1}{6}(64 + 2 + 4 + 8 + 4 + 2) = \frac{84}{6} = 14$.
(4) This is the double-counting argument from Burnside's proof. $\sum_{s \in S} |\mathrm{Stab}(s)| = |\{(g, s) : g \cdot s = s\}| = \sum_{g \in G} |S^g| = N \cdot |G|$ by Burnside's lemma.
(5) Let $|G| = 12 = 2^2 \cdot 3$. By Sylow (3): $s_3 \mid 4$ and $s_3 \equiv 1 \pmod{3}$, so $s_3 \in \{1, 4\}$. If $s_3 = 1$, the Sylow 3-subgroup is normal and we are done.
If $s_3 = 4$: G acts on the set of 4 Sylow 3-subgroups by conjugation. This gives a homomorphism $\varphi : G \to S_4$. The kernel $K = \mathrm{Ker}(\varphi)$ is a normal subgroup with $K \subseteq \bigcap$normalizers. Since the action

is non-trivial ($s_3 = 4$ and the Sylow subgroups are conjugate), $K \neq G$. So $|G/K|$ divides $|S_4| = 24$ and $|G/K| > 1$.
The 4 Sylow 3-subgroups contribute $4 \cdot 2 = 8$ elements of order 3 (they pairwise intersect trivially since each has prime order). The remaining $12 - 8 = 4$ elements must include e and 3 elements of order dividing 4. These 4 elements form the unique Sylow 2-subgroup (there is room for only one), so $s_2 = 1$ and the Sylow 2-subgroup is normal.

11.5.3 Proofs of the Sylow Theorems

The proofs use the action of G on certain subsets by left multiplication.

Lemma 11.5.2 Combinatorial lemma

Given $n = p^e m$ with $p \nmid m$, we have $p \nmid \binom{n}{p^e}$.

Proof: Write $\binom{n}{p^e} = \frac{n(n-1)\cdots(n-p^e+1)}{p^e(p^e-1)\cdots 1} = \prod_{k=0}^{p^e-1} \frac{n-k}{p^e-k}$. For each k, the highest power of p dividing $n-k = p^e m - k$ equals the highest power of p dividing k (when $0 < k < p^e$), and for $k = 0$ both numerator and denominator contribute p^e. So the powers of p in numerator and denominator cancel term by term, and $p \nmid \binom{n}{p^e}$. $\square$

Lemma 11.5.3 Stabilizer divides subset size

Let $U \subseteq G$ be any subset, and let G act on $\mathcal{P}(G) = \{\text{all subsets of } G\}$ by left multiplication: $g \cdot U = gU$. Then $|\mathrm{Stab}([U])|$ divides $|U|$.

Proof: Let $H = \mathrm{Stab}(U) = \{g \in G : gU = U\}$. Then H acts on U by left multiplication ($hU = U$ for $h \in H$, so $hu \in U$ for $u \in U$). The orbits of this action are right cosets of H: $\mathcal{O}_u = \{hu : h \in H\} = Hu$. Each orbit has size $|H|$. Since U is a disjoint union of such orbits, $|H|$ divides $|U|$. $\square$

Proof of Sylow's First Theorem (existence): Let $S = \{U \in \mathcal{P}(G) : |U| = p^e\}$, the set of all subsets of G of size p^e. Then $|S| = \binom{n}{p^e}$, and G acts on S by left multiplication $g \cdot U = gU$.

By the combinatorial lemma, $p \nmid |S|$. Since S partitions into orbits under this action, there exists an orbit $\mathcal{O}_U$ with $p \nmid |\mathcal{O}_U|$. Since p^e divides $|G| = |\mathcal{O}_U| \cdot |\mathrm{Stab}(U)|$ and $p \nmid |\mathcal{O}_U|$, we get $p^e \mid |\mathrm{Stab}(U)|$. But by the second lemma, $|\mathrm{Stab}(U)|$ divides $|U| = p^e$. So $|\mathrm{Stab}(U)| = p^e$, and $\mathrm{Stab}(U)$ is a Sylow p-subgroup. $\square$

Definition 11.5.1: Normalizer

The normalizer of a subgroup $H \leqslant G$ is $N(H) = \{g \in G : gHg^{-1} = H\}$. This is a subgroup of G, and $H \trianglelefteq K$ iff $K \subseteq N(H)$. In particular, H is normal in G iff $N(H) = G$. The number of subgroups conjugate to H is $|G/N(H)|$.

Proof of Sylow's Second Theorem (conjugacy): Let H be a Sylow p-subgroup and $K \leqslant G$ any p-subgroup. Let $C = G/H = \{gH : g \in G\}$ be the set of left cosets. Then G acts on C by left multiplication, transitively, with $p \nmid |C| = m$.

Restrict to the K-action on C. The orbits have size dividing $|K|$, hence each orbit size is a power of p. Since $|C| = m$ and $p \nmid m$, there must be an orbit of size 1, i.e., a fixed point $c_0 = [g_0 H]$. This means $K \subseteq \mathrm{Stab}(c_0) = g_0 H g_0^{-1}$, so K is contained in the conjugate Sylow subgroup $H' = g_0 H g_0^{-1}$.

In particular, if $|K| = p^e$ (i.e., K is also Sylow), then $K \subseteq H'$ and $|K| = |H'|$ forces $K = H' = g_0 H g_0^{-1}$. $\square$

Proof of Sylow's Third Theorem ($s_p \equiv 1 \bmod p$): Let H be a Sylow p-subgroup. Restrict the conjugation action of G on {Sylow p-subgroups} to the H-action. Since $hHh^{-1} = H$ for all $h \in H$, the orbit of H itself has size 1.

For any other Sylow subgroup $H' \neq H$: the orbit of H' under H-conjugation has size $|H/(H \cap N(H'))|$. If the orbit size were 1, then $H \subseteq N(H')$, making H and H' both Sylow p-subgroups of $N(H')$. By Sylow's second theorem applied within $N(H')$, they would be conjugate in $N(H')$. But H' is normal in $N(H')$ (by definition), so the only conjugate of H' in $N(H')$ is H' itself, forcing $H = H'$. Contradiction.

So every orbit other than $\{H\}$ has size divisible by p. Thus $s_p = 1 + (\text{sum of multiples of } p) \equiv 1 \pmod p$.

For $s_p \mid m$: by Sylow (2), the conjugation action is transitive, so $s_p = |G/N(H)|$. Since $H \subseteq N(H)$, we have $|H| \mid |N(H)|$, so $s_p = |G|/|N(H)|$ divides $|G|/|H| = m$. $\square$

Corollary 11.5.3 Cauchy's theorem

If p is prime and $p \mid |G|$, then G contains an element of order p.

Proof: By Sylow (1), G has a subgroup H of order $p^e \geqslant p$. Pick $g \in H$, $g \neq e$. The order of g divides $|H| = p^e$, so $\operatorname{ord}(g) = p^k$ for some $1 \leqslant k \leqslant e$. Then $g^{p^{k-1}}$ has order p. $\square$

11.6 Semidirect Products

When $N \trianglelefteq G$ and we have a complement H (a subgroup with $N \cap H = \{e\}$ and $NH = G$), the group G is determined by N, H, and the conjugation action of H on N.

Definition 11.6.1: Semidirect product

Given groups N and H and a homomorphism $\varphi : H \to \operatorname{Aut}(N)$, the semidirect product $N \rtimes_\varphi H$ is defined as:

- As a set: $N \times H$.
- Group law: $(n_1, h_1) \cdot (n_2, h_2) = (n_1\, \varphi(h_1)(n_2),\ h_1 h_2)$.

Note:-

When φ is trivial ($\varphi(h) = \mathrm{id}_N$ for all h), the semidirect product reduces to the direct product $N \times H$. The semidirect product is the direct product "twisted" by the action φ.

Proposition 11.6.1 Recognition criterion

If $N, H \leqslant G$ with N normal, $N \cap H = \{e\}$, and every element of G is uniquely $g = nh$ ($n \in N$, $h \in H$), then $G \cong N \rtimes_\varphi H$ where $\varphi(h)(n) = hnh^{-1}$.

Proof: Conjugation by h preserves N (since N is normal), defining $\varphi : H \to \operatorname{Aut}(N)$. The map $(n, h) \mapsto nh$ is a bijection by hypothesis. It is a homomorphism: $(n_1h_1)(n_2h_2) = n_1(h_1n_2h_1^{-1})h_1h_2 = (n_1\varphi(h_1)(n_2))(h_1h_2)$. $\square$

Corollary 11.6.1 Direct product criterion

If both N and H are normal in G, $N \cap H = \{e\}$, and $|G| = |N| \cdot |H|$, then $G \cong N \times H$.

Proof: Every coset of N contains a unique element of H, so every $g \in G$ is uniquely $g = nh$. Since both are normal, the commutator $nhn^{-1}h^{-1} \in N \cap H = \{e\}$ for all $n \in N$, $h \in H$ (it lies in N since N is normal, and in H since H is normal). So $nh = hn$ and the action φ is trivial. $\square$

Example 11.6.1 (Dihedral groups as semidirect products)

$S_3 \cong \mathbb{Z}/3 \rtimes \mathbb{Z}/2$ where $\mathbb{Z}/2$ acts on $\mathbb{Z}/3$ by inversion: $\varphi(1)(\sigma) = \sigma^{-1}$. More generally, $D_n \cong \mathbb{Z}/n \rtimes \mathbb{Z}/2$ with the same inversion action. The normal subgroup is the rotation subgroup, and the complement is generated by any reflection.

11.6.1 Classification of Groups of Order 12

Example 11.6.2 (The 5 groups of order 12)

Let $|G| = 12 = 2^2 \cdot 3$. Sylow gives $s_3 \in \{1, 4\}$ and $s_2 \in \{1, 3\}$. Let H denote a Sylow 2-subgroup ($|H| = 4$) and K a Sylow 3-subgroup ($|K| = 3$).
At least one of H or K is normal (Exercise 5 from before). We systematically enumerate:
Both normal ($s_3 = s_2 = 1$): $G \cong H \times K$. With $H \cong \mathbb{Z}/4$: $G \cong \mathbb{Z}/12$. With $H \cong \mathbb{Z}/2 \times \mathbb{Z}/2$: $G \cong \mathbb{Z}/2 \times \mathbb{Z}/6$.
K normal, H not ($s_3 = 1$, $s_2 = 3$): $G \cong K \rtimes H$. The action $\varphi : H \to \mathrm{Aut}(\mathbb{Z}/3) \cong \mathbb{Z}/2$ must be nontrivial.
If $H \cong \mathbb{Z}/4$: the generator x acts by inversion on K, with x^2 acting trivially. This gives a non-abelian group $\mathbb{Z}/3 \rtimes \mathbb{Z}/4$, generated by x, y with $x^4 = y^3 = e$, $xy = y^2x$.
If $H \cong \mathbb{Z}/2 \times \mathbb{Z}/2$: the kernel of φ is $\mathbb{Z}/2$, giving $G \cong D_6 \cong S_3 \times \mathbb{Z}/2$.
H normal, K not ($s_3 = 4$): Only possible with $H \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (since $\mathrm{Aut}(\mathbb{Z}/4) \cong \mathbb{Z}/2$ has no element of order 3). The action $\varphi : \mathbb{Z}/3 \to \mathrm{Aut}((\mathbb{Z}/2)^2) \cong S_3$ embeds $\mathbb{Z}/3$ as 3-cycles permuting the three nonzero elements. This gives $G \cong A_4$.
In total: $\mathbb{Z}/12$, $\mathbb{Z}/2 \times \mathbb{Z}/6$, A_4, D_6, and $\mathbb{Z}/3 \rtimes \mathbb{Z}/4$.

Example 11.6.3 (Groups of order 21)

Let $|G| = 21 = 3 \cdot 7$. Sylow: $s_7 \mid 3$ and $s_7 \equiv 1 \pmod 7$, so $s_7 = 1$. The Sylow 7-subgroup $K \cong \mathbb{Z}/7$ is normal. For the Sylow 3-subgroup: $s_3 \mid 7$ and $s_3 \equiv 1 \pmod 3$, so $s_3 \in \{1, 7\}$.
If $s_3 = 1$: $G \cong \mathbb{Z}/3 \times \mathbb{Z}/7 \cong \mathbb{Z}/21$.
If $s_3 = 7$: $G \cong \mathbb{Z}/7 \rtimes \mathbb{Z}/3$. We need a nontrivial $\varphi : \mathbb{Z}/3 \to \mathrm{Aut}(\mathbb{Z}/7) \cong \mathbb{Z}/6$. The unique subgroup of order 3 in $\mathbb{Z}/6$ corresponds to the automorphism $y \mapsto y^2$ of $\mathbb{Z}/7$. So G is generated by x, y with $x^3 = y^7 = e$ and $xyx^{-1} = y^2$. This is the unique non-abelian group of order 21.

Lecture 12.

Generators, Presentations, and Cayley Graphs

We now study how to describe groups in terms of generators and relations, and how to visualize their structure via Cayley graphs. This connects algebra to combinatorics and geometry, and leads naturally to the braid group and $\mathrm{SL}_2(\mathbb{Z})$.

12.1 Free Groups and Presentations

12.1.1 The Free Group

Recall from **Lecture 3**: the free group F_n on n generators $a_1, \ldots, a_n$ consists of all reduced words $a_{i_1}^{m_1} a_{i_2}^{m_2} \cdots a_{i_k}^{m_k}$ (where $k \geqslant 0$, $i_j \in \{1, \ldots, n\}$, $i_j \neq i_{j+1}$, and $m_j \in \mathbb{Z} \setminus \{0\}$), with multiplication by concatenation followed by reduction (canceling adjacent inverse pairs $a_i a_i^{-1}$ and combining adjacent powers of the same generator).

F_n is the "largest" group with n generators: every other group generated by n elements is a quotient of F_n. If G is generated by $g_1, \ldots, g_n \in G$, define a surjective homomorphism $\varphi : F_n \twoheadrightarrow G$ by $a_i \mapsto g_i$.

Definition 12.1.1: Finitely presented group

A group G is finitely presented if the kernel of φ is the smallest normal subgroup of F_n containing some finite set $\{r_1, \ldots, r_k\} \subset F_n$, i.e., $\mathrm{Ker}(\varphi) = \langle\!\langle r_1, \ldots, r_k \rangle\!\rangle$ (the normal closure: the subgroup generated by all conjugates $x r_i x^{-1}$). We write

$$G \cong \langle a_1, \ldots, a_n \mid r_1, \ldots, r_k \rangle$$

where the a_i are generators and the r_i are relations.

Example 12.1.1 (Presentations of familiar groups)

- $\mathbb{Z}^n \cong \langle a_1, \ldots, a_n \mid a_i a_j a_i^{-1} a_j^{-1}\ \forall i, j \rangle$ (all generators commute).
- $S_3 \cong \langle s_1, s_2 \mid s_1^2,\ s_2^2,\ (s_1 s_2)^3 \rangle$ where $s_1 = (12)$, $s_2 = (23)$.
- $S_n \cong \langle s_1, \ldots, s_{n-1} \mid s_i^2 = 1,\ s_i s_j = s_j s_i\ (|i - j| \geqslant 2),\ s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1} \rangle$ where $s_i = (i\ \ i{+}1)$.

Note:-

Given generators $g_1, \ldots, g_n \in G$, finding relations $r_1, \ldots, r_k$ such that these are a complete set (i.e., they generate the full kernel) is nontrivial. If the relations hold in G, then φ induces a surjection $\langle a_i \mid r_j \rangle \twoheadrightarrow G$. This is an isomorphism iff the relations generate the entire kernel of φ.

12.1.2 The Word Problem

Definition 12.1.2: Word problem

Given a presentation $G = \langle a_1, \ldots, a_n \mid r_1, \ldots, r_k \rangle$, the word problem asks: given a word w in the generators, does w represent the identity in G?

Note:-

For groups where we can "calculate" (e.g., matrix groups, permutation groups), the word problem is trivially solvable: just evaluate the word. But for abstractly presented groups, this can be undecidable. Novikov (1955) and Boone (1959) independently showed there exist finitely presented groups with unsolvable word problem. Two useful tools for groups where the word problem is solvable: Cayley graphs and normal forms.

12.2 Cayley Graphs

Definition 12.2.1: Cayley graph

Given a group G with generators $g_1, \ldots, g_n$, the Cayley graph of G has:

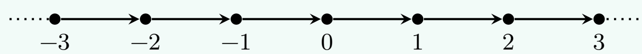
- Vertices: the elements of G.
- Edges: for each $s \in G$ and generator g_i, a directed edge from s to sg_i, labeled g_i.

Note:-

G acts on its Cayley graph by left multiplication: g sends vertex s to gs and edge $(s \to sg_i)$ to $(gs \to gsg_i)$. This action is transitive on vertices and preserves the graph structure (it is a graph automorphism). The graph is highly symmetric.

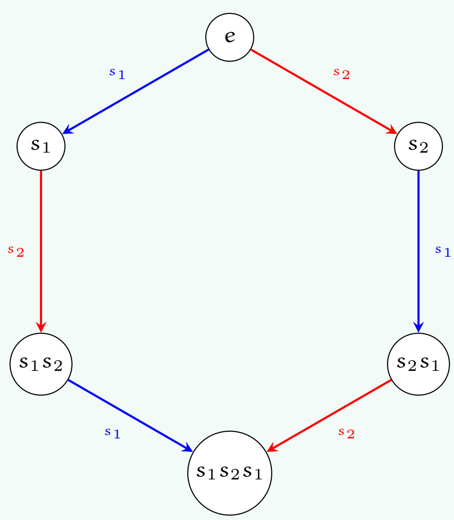
Example 12.2.1 (Cayley graph of $\mathbb{Z}$)

With generator 1, the Cayley graph is the integer number line:



Example 12.2.2 (Cayley graph of S_3)

With generators $s_1 = (12)$ and $s_2 = (23)$: the 6 vertices form a hexagonal graph. The relation $(s_1s_2)^3 = e$ corresponds to the outer hexagonal cycle closing up.



Since any word in s_1, s_2 with $s_1^2 = s_2^2 = e$ reduces to an alternating sequence $(\cdots s_1s_2s_1\cdots)$, we verify $S_3 \cong \langle s_1, s_2 \mid s_1^2, s_2^2, (s_1s_2)^3\rangle$.

Example 12.2.3 (Cayley graph of S_4)

With generators $s_i = (i \;\; i{+}1)$ for $i = 1, 2, 3$: the 24 vertices form the edge graph of a truncated octahedron (a permutohedron). The faces correspond to the relations: hexagonal faces to $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$ and square faces to $s_1 s_3 = s_3 s_1$.

Definition 12.2.2: Word length

The word length of $g \in G$ (with respect to generators $g_1, \ldots, g_n$) is the shortest distance from e to g in the Cayley graph, i.e., the minimal number of generators or their inverses needed to express g.

Note:-

For S_n with generators $s_i = (i \;\; i{+}1)$, the word length of a permutation σ equals the number of inversions $|\{(i,j) : i < j,\ \sigma(i) > \sigma(j)\}|$. The longest element is $\sigma = \left(\begin{smallmatrix} 1 & 2 & \cdots & n \\ n & n-1 & \cdots & 1 \end{smallmatrix}\right)$ with word length $\binom{n}{2}$.

12.2.1 Growth Rate

For infinite groups, the Cayley graph encodes asymptotic information about the group.

Definition 12.2.3: Growth rate

Given generators $g_1, \ldots, g_n$ of an infinite group G, the growth function counts $B(N) = |\{g \in G : \text{word length of } g \leqslant N\}|$. The group has polynomial growth if $B(N) \sim N^d$ for some d, and exponential growth if $B(N) \sim c^N$ for some $c > 1$.

Note:-

The growth type (polynomial vs exponential) is independent of the choice of generators: changing generators changes word lengths by at most a bounded factor (each new generator is a bounded-length word in the old generators and vice versa). Finitely generated abelian groups have polynomial growth. Free groups have exponential growth. Gromov's theorem (1981) characterizes polynomial growth: a finitely generated group has polynomial growth iff it is virtually nilpotent (has a nilpotent subgroup of finite index).

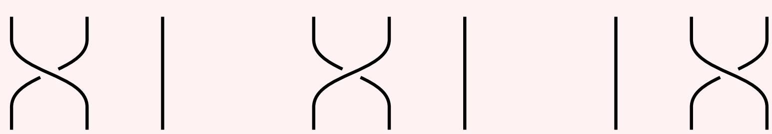
12.3 The Braid Group

Definition 12.3.1: Braid group

The braid group B_n on n strands has presentation

$$B_n = \langle s_1, \ldots, s_{n-1} \mid s_i s_j = s_j s_i \ (|i-j| \geqslant 2),\ s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1} \rangle.$$

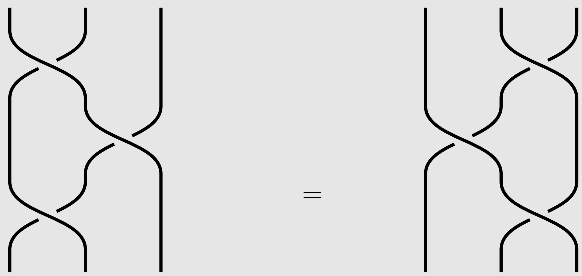
This is the same as the presentation of S_n but without the relations $s_i^2 = 1$. The generator s_i represents crossing strand i over strand $i+1$.



s_1 $\qquad$ s_1^{-1} $\qquad$ s_2

Note:-

There is a natural surjection $B_n \twoheadrightarrow S_n$ sending each generator s_i to the transposition $(i\ \ i{+}1)$. The kernel is the pure braid group P_n. Unlike S_n, the braid group is infinite: $s_i^2 \neq e$ in B_n.
Braids are best understood as diagrams: n strands running from top to bottom, where s_i crosses strand i over strand $i+1$, and s_i^{-1} crosses strand $i+1$ over strand i. Composition is vertical stacking. Two braid diagrams represent the same element iff one can be deformed into the other (isotopy). The braid relation $s_1s_2s_1 = s_2s_1s_2$ (the Yang–Baxter equation) says these two braids are isotopic:



$s_1s_2s_1$ $\qquad\qquad$ $s_2s_1s_2$

Note:-

Markov's theorem connects braids to knot theory: every knot or link in $\mathbb{R}^3$ can be represented as the closure of a braid, and two braids give isotopic links iff they are related by conjugation in B_n and stabilization $\sigma \in B_n \sim \sigma s_n^{\pm 1} \in B_{n+1}$.

Note:-

B_n is much larger than S_n, but its algorithmic aspects can be approached by the same methods, using permutation braids (braids where any two strands cross at most once, in bijection with S_n). Let Δ be the longest permutation braid. Since its shortest word can start/end with any s_i, and $s_i^{\pm 1}\Delta$ is still a permutation braid, every element of B_n can be written as $g = \Delta^k P_1 \cdots P_r$ where the P_j are permutation braids satisfying a "left greedy" condition. This is the Garside normal form (Garside 1969, Thurston–ElRifai–Morton early 1990s), which solves the word problem in B_n.

12.4 $\mathrm{SL}_2(\mathbb{Z})$ and $\mathrm{PSL}_2(\mathbb{Z})$

Definition 12.4.1: $\mathrm{SL}_2(\mathbb{Z})$

$\mathrm{SL}_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z},\ ad - bc = 1 \right\}$ is the group of 2×2 integer matrices with determinant 1. Its quotient by the center $\{\pm I\}$ is $\mathrm{PSL}_2(\mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})/\{\pm I\}$.

Proposition 12.4.1 Generators of $\mathrm{SL}_2(\mathbb{Z})$

$\mathrm{SL}_2(\mathbb{Z})$ is generated by $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Proof: Given $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$, we express M in terms of S and T by repeatedly reducing the entries.
If $c = 0$: then $ad = 1$, so $a = d = \pm 1$. Thus $M = \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} = \pm T^n$, which is T^n or S^2T^n (since $S^2 = -I$).
If $c \neq 0$: apply the Euclidean algorithm to reduce $\max(|a|, |c|)$.

- If $|a| \geqslant |c|$: write $a = nc + r$ with $|r| < |c|$. Then $T^{-n}M = \begin{pmatrix} a - nc & b - nd \\ c & d \end{pmatrix} = \begin{pmatrix} r & * \\ c & d \end{pmatrix}$, reducing $\max(|a|, |c|)$.
- If $|a| < |c|$: apply $S^{-1}M = \begin{pmatrix} c & d \\ -a & -b \end{pmatrix}$, swapping a and c (up to sign), then return to the previous case.

After finitely many steps, the product of M with some word in S and T has $c = 0$, hence is $\pm T^n$. Inverting gives M as a word in S, T. □

Note:-

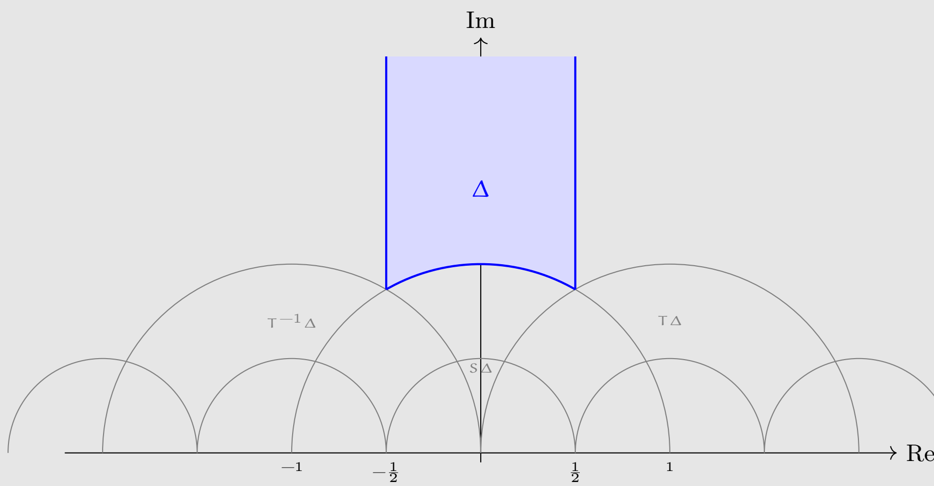
Alternative generators: S and $R = ST = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$ have finite order: $S^4 = I$ and $R^6 = I$ (so their images in $\mathrm{PSL}_2(\mathbb{Z})$ have orders 2 and 3). Also $T' = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = (TST)^{-1} = STS^{-1}$, showing T and T' are conjugate.

Theorem 12.4.1 Presentation of $\mathrm{PSL}_2(\mathbb{Z})$

$\mathrm{PSL}_2(\mathbb{Z}) \cong \langle S, R \mid S^2, R^3 \rangle \cong \mathbb{Z}/2 * \mathbb{Z}/3$, the free product of $\mathbb{Z}/2$ and $\mathbb{Z}/3$.

Note:-

This can be proved geometrically. $\mathrm{PSL}_2(\mathbb{Z})$ acts on the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$ by Möbius transformations: $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az+b}{cz+d}$. Under this action, S sends $z \mapsto -1/z$ and T sends $z \mapsto z+1$. The region $\Delta = \{z \in \mathbb{H} : |z| \geqslant 1, |\mathrm{Re}(z)| \leqslant 1/2\}$ is a fundamental domain: its translates under $\mathrm{PSL}_2(\mathbb{Z})$ tile $\mathbb{H}$ without overlap.



The regions adjacent to Δ are $T^{\pm 1}(\Delta)$ and $S(\Delta)$, and the tiling structure is the Cayley graph of $\mathrm{PSL}_2(\mathbb{Z})$ with generators S, T. That all regions can be reached from Δ in finitely many steps is equivalent to S, T generating $\mathrm{PSL}_2(\mathbb{Z})$.

Exercise 12.4.1 Generators, presentations, and Cayley graphs exercises

(1) Verify the presentation $S_3 \cong \langle s_1, s_2 \mid s_1^2, s_2^2, (s_1 s_2)^3 \rangle$ by checking that the 6 reduced words $e, s_1, s_2, s_1s_2, s_2s_1, s_1s_2s_1$ are all distinct and form a group under the given relations.

(2) Show that the braid group B_2 is isomorphic to $\mathbb{Z}$. What is B_3?

(3) Compute S^2, S^4, $(ST)^3$, and $(ST)^6$ in $\mathrm{SL}_2(\mathbb{Z})$. Use this to verify $S^4 = I$ and $R^6 = I$ where $R = ST$.

(4) Express $M = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$ as a word in S and T using the Euclidean algorithm from the proof.

Solution

(1) The relations $s_1^2 = s_2^2 = e$ and $(s_1s_2)^3 = e$ (i.e., $s_1s_2s_1 = s_2s_1s_2$) mean any word reduces to alternating sequences of bounded length. The 6 reduced words are: $e, s_1, s_2, s_1s_2, s_2s_1, s_1s_2s_1$. Compute the multiplication table using the relations: e.g., $(s_1s_2)(s_2s_1) = s_1(s_2^2)s_1 = s_1^2 = e$, so s_1s_2 and s_2s_1 are inverses. Check $s_1s_2s_1 = s_2s_1s_2$ confirms the 6 elements close under multiplication. Since $|S_3| = 6$ and the surjection $\langle s_1, s_2 \mid \cdots \rangle \twoheadrightarrow S_3$ has at most 6 elements in the domain, it is an isomorphism.
(2) $B_2 = \langle s_1 \rangle$ with no relations (since there are no pairs $|i-j| \geqslant 2$ and no braid relations). So $B_2 \cong \mathbb{Z}$. For $B_3 = \langle s_1, s_2 \mid s_1s_2s_1 = s_2s_1s_2 \rangle$: this is an infinite non-abelian group. Its center is $\mathbb{Z}$, generated by $(s_1s_2)^3 = (s_1s_2s_1)^2$, and $B_3/Z(B_3) \cong \mathrm{PSL}_2(\mathbb{Z})$.
(3) $S^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$. So $S^4 = I$.
$R = ST = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$. $R^2 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$. $R^3 = R \cdot R^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$. So $R^6 = I$.
In $\mathrm{PSL}_2(\mathbb{Z})$: $\bar{S}^2 = \bar{R}^3 = \bar{I}$, confirming orders 2 and 3.
(4) Start: $M = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$. Since $|a| = 3 > |c| = 1$, write $3 = 3 \cdot 1 + 0$. Apply $T^{-3}M = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}$.
Now $|a| = 0 < |c| = 1$, apply S^{-1}: $S^{-1}\begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = T^2$.
So $S^{-1}T^{-3}M = T^2$, giving $M = T^3ST^2$.
Check: $T^3ST^2 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$. Confirmed.

Proof of the presentation: We show $\mathrm{PSL}_2(\mathbb{Z}) \cong \langle S, R \mid S^2, R^3 \rangle$. We have $S^2 = -I$ and $R^3 = -I$ in $\mathrm{SL}_2(\mathbb{Z})$, so $\bar{S}$ and $\bar{R}$ have orders 2 and 3 in $\mathrm{PSL}_2(\mathbb{Z})$. The relations $S^2 = R^3 = e$ reduce any word in $S^{\pm 1}, R^{\pm 1}$ to the form $\cdots SR^{\pm 1}SR^{\pm 1} \cdots$ (alternating, since the word can start and end with either S or $R^{\pm 1}$).

Map $F_2 = \langle s, r \rangle \to \mathrm{PSL}_2(\mathbb{Z})$ by $r \mapsto R$, $s \mapsto S$. This induces a surjection $\langle s, r \mid s^2, r^3 \rangle \twoheadrightarrow \mathrm{PSL}_2(\mathbb{Z})$. The kernel consists of all words $\cdots sr^{\pm 1}sr^{\pm 1} \cdots$ that simplify to e in $\mathrm{PSL}_2(\mathbb{Z})$, i.e., to $\pm I$ in $\mathrm{SL}_2(\mathbb{Z})$.

Assume some word w in S and $R^{\pm 1}$ (alternating) simplifies to $\pm I$. If w starts and ends with S, conjugate by S to get a shorter word $w' = SwS$ with $w' = \pm I$ as well. If w starts with $R^{\pm 1}$, conjugate by $R^{\mp 1}$ to get another word that doesn't. Iterating, we eventually get a word $SR^{\pm 1} \cdots SR^{\pm 1} = \pm I$.

But $SR = -T = -\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $SR^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. So a product $(SR^{\varepsilon_1})(SR^{\varepsilon_2}) \cdots (SR^{\varepsilon_k})$ with each $\varepsilon_i = \pm 1$ equals $(-1)^j$ (where j counts the factors with $\varepsilon_i = +1$) times a product of the nonnegative matrices $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Any nonempty product of these nonnegative matrices has entry sum $\geqslant 3 > 2$ (one can check the sum is non-decreasing and the initial matrices already have sum 3), so it can never equal $\pm I$ (which has entry sum 2). Thus no nontrivial alternating word equals $\pm I$, and the kernel is trivial. □

Note:-

Alternative generators and presentations: using $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $T' = (TST)^{-1} = STS^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$ (which are conjugate), we get $\mathrm{SL}_2(\mathbb{Z}) \cong \langle T, T' \mid (TT')^6 = 1, TT'T = T'TT' \rangle$. The relation $TT'T = T'TT'$ is precisely the braid relation.

Note:-

This connects to our discussion of braid groups: the center of $B_3 = \langle s_1, s_2 \mid s_1 s_2 s_1 = s_2 s_1 s_2 \rangle$ is generated by $\Delta^2 = (s_1 s_2)^3$ and is isomorphic to $\mathbb{Z}$. The map $s_1 \mapsto T$, $s_2 \mapsto T'$ gives $1 \to \mathbb{Z} = \langle \Delta^2 \rangle \hookrightarrow B_3 \twoheadrightarrow PSL_2(\mathbb{Z}) \to 1$, a central extension.

Lecture 13.

Representation Theory

Representation theory studies groups via their actions on vector spaces, i.e., homomorphisms $G \to \mathrm{GL}(V)$. This converts abstract group theory into linear algebra, where we have powerful tools (eigenvalues, traces, diagonalization). We work over $k = \mathbb{C}$ throughout (or more generally, any algebraically closed field of characteristic 0).

13.1 Representations and Subrepresentations

Definition 13.1.1: Representation

A representation of a group G is a vector space V (usually finite-dimensional over $\mathbb{C}$) together with a group homomorphism $\rho : G \to \mathrm{GL}(V)$. Equivalently, a G-action on V by linear operators: $G \times V \to V$, $(g, v) \mapsto g \cdot v$, with each $g : V \to V$ linear.

Definition 13.1.2: Subrepresentation and irreducibility

- A subrepresentation is a subspace $W \subseteq V$ that is G-invariant: $gW = W$ for all $g \in G$.
- A representation V is irreducible if it has no nontrivial subrepresentations (i.e., the only G-invariant subspaces are $\{0\}$ and V itself).

Example 13.1.1 (Representations of cyclic groups)

If $G = \mathbb{Z}/n$, a representation is a vector space V with a linear operator $\varphi = \rho(1) : V \to V$ satisfying $\varphi^n = \mathrm{id}_V$. Since the minimal polynomial of φ divides $x^n - 1 = \prod_{k=0}^{n-1}(x - \lambda_k)$ where $\lambda_k = e^{2\pi i k/n}$, and this polynomial has distinct roots, φ is diagonalizable. The eigenspaces $V_{\lambda_k} = \{v : \varphi(v) = \lambda_k v\}$ are subrepresentations, and $V = \bigoplus_k V_{\lambda_k}$.
Each 1-dimensional subspace $\mathbb{C}v$ with $\varphi(v) = \lambda_k v$ is an irreducible representation, corresponding to the homomorphism $\mathbb{Z}/n \to \mathbb{C}^* = \mathrm{GL}_1(\mathbb{C})$ sending $1 \mapsto \lambda_k = e^{2\pi i k/n}$. There are exactly n such irreducible representations, and every representation of $\mathbb{Z}/n$ decomposes as a direct sum of copies of these.

Proposition 13.1.1 Representations of finite abelian groups

If $G \cong \mathbb{Z}/m_1 \times \cdots \times \mathbb{Z}/m_r$ is a finite abelian group and V is a representation of G over $\mathbb{C}$, then V splits into a direct sum of 1-dimensional (irreducible) subrepresentations.

Proof: The G-action is equivalent to commuting operators $\varphi_1, \ldots, \varphi_r : V \to V$ with $\varphi_i^{m_i} = \mathrm{id}$. Each φ_i is diagonalizable (as above). Commuting diagonalizable operators are simultaneously diagonalizable: the eigenspaces of φ_1 are invariant under all φ_j (since they commute), and the restriction of φ_j to an eigenspace of φ_1 is again of finite order, hence diagonalizable. Proceed by induction on r. The simultaneous eigenspaces are 1-dimensional subrepresentations. $\square$

Definition 13.1.3: Dual group

For a finite abelian group G, the dual group is $\hat{G} = \mathrm{Hom}(G, \mathbb{C}^*)$, the group of all 1-dimensional representations (characters in the abelian sense) under pointwise multiplication. For $G = \mathbb{Z}/n$, $\hat{G} \cong \mathbb{Z}/n$ (but not canonically). More generally, $\hat{G} \cong G$ for any finite abelian group (non-canonically). This completes the classification of complex representations of finite abelian groups.

13.1.1 Homomorphisms and Constructions

Definition 13.1.4: Homomorphism of representations

Given representations V, W of G, a homomorphism (or G-equivariant map, or intertwining operator) is a linear map $\varphi : V \to W$ satisfying $\varphi(g \cdot v) = g \cdot \varphi(v)$ for all $g \in G$, $v \in V$. Equivalently, $g \circ \varphi = \varphi \circ g$ as operators. We write $\mathrm{Hom}_G(V, W)$ for the space of such maps.

Note:-

If $\varphi \in \mathrm{Hom}_G(V, W)$, then $\mathrm{Ker}(\varphi)$ and $\mathrm{Im}(\varphi)$ are subrepresentations of V and W respectively (since $v \in \mathrm{Ker}(\varphi) \implies \varphi(gv) = g\varphi(v) = 0$, so $gv \in \mathrm{Ker}(\varphi)$).

Proposition 13.1.2 Constructions with representations

Given representations V, W of G:

(i) $V \oplus W$ is a representation: $g \cdot (v, w) = (gv, gw)$.

(ii) $V \otimes W$ is a representation: $g \cdot (v \otimes w) = gv \otimes gw$, extended by linearity.

(iii) The dual $V^* = \mathrm{Hom}(V, \mathbb{C})$ is a representation: $(g \cdot \ell)(v) = \ell(g^{-1}v)$. This is the contragredient representation.

(iv) $\mathrm{Hom}(V, W)$ is a representation: $(g \cdot \varphi)(v) = g \cdot \varphi(g^{-1}v)$, i.e., $g \cdot \varphi = g \circ \varphi \circ g^{-1}$. Note $\mathrm{Hom}(V, W)^G = \mathrm{Hom}_G(V, W)$ (the G-invariant maps are exactly the G-equivariant ones).

(v) Under the isomorphism $\mathrm{Hom}(V, W) \cong V^* \otimes W$, the G-action is $g \cdot (\ell \otimes w) = (g \cdot \ell) \otimes (g \cdot w)$, which is consistent with the above.

(vi) If $W \subseteq V$ is a subrepresentation, then V/W is a representation: $g \cdot (v + W) = gv + W$.

13.2 Maschke's Theorem and Complete Reducibility

Lemma 13.2.1 Existence of G-invariant inner product

If V is a $\mathbb{C}$-representation of a finite group G, there exists a positive-definite Hermitian inner product H on V that is G-invariant: $H(gv, gw) = H(v, w)$ for all $g \in G$, $v, w \in V$. In other words, all operators $g : V \to V$ are unitary with respect to H.

Proof: Start with any Hermitian inner product H_0 on V, and average over G:

$$H(v, w) = \frac{1}{|G|} \sum_{g \in G} H_0(gv, gw).$$

Then H is still Hermitian and positive-definite (a positive combination of positive-definite forms), and G-invariant: $H(hv, hw) = \frac{1}{|G|} \sum_g H_0(ghv, ghw) = \frac{1}{|G|} \sum_{g'} H_0(g'v, g'w) = H(v, w)$ (reindexing $g' = gh$). $\square$

Theorem 13.2.1 Maschke's Theorem

Let V be a representation of a finite group G over $\mathbb{C}$ (or any field k with $\mathrm{char}(k) \nmid |G|$), and let $W \subseteq V$ be a subrepresentation. Then there exists a complementary subrepresentation $U \subseteq V$ with $V = W \oplus U$.

Proof 1 (via invariant inner product): Equip V with a G-invariant Hermitian inner product H (by the lemma). Since each g is unitary, $g(W^\perp) = W^\perp$ (if $w \in W$ and $v \in W^\perp$, then $H(gv, w) = H(v, g^{-1}w) = 0$ since $g^{-1}w \in W$). So $U = W^\perp$ is a G-invariant complement. $\square$

Proof 2 (via averaging projection): Choose any complementary subspace U_0 with $V = W \oplus U_0$ (not necessarily G-invariant). Let $\pi_0 : V \to W$ be the projection onto W with kernel U_0 ($\pi_0|_W = \mathrm{id}$, $\pi_0|_{U_0} = 0$). Average:

$$\pi(v) = \frac{1}{|G|} \sum_{g \in G} g\, \pi_0(g^{-1}v).$$

Then $\pi : V \to W$ is G-equivariant ($g\pi g^{-1} = \pi$), and $\pi|_W = \mathrm{id}$ (since $g\pi_0(g^{-1}w) = g(g^{-1}w) = w$ for $w \in W$, as $g^{-1}w \in W$). So π is a G-equivariant projection onto W, and $U = \mathrm{Ker}(\pi)$ is a G-invariant complement. $\square$

Note:-

Maschke's theorem fails when $\mathrm{char}(k)$ divides $|G|$ (modular representation theory), and also fails for infinite groups in general: the representation of $G = \mathbb{Z}$ (or $\mathbb{R}$) on $\mathbb{C}^2$ via $t \mapsto \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ has the invariant subspace $\mathbb{C} \times \{0\}$ but no complementary invariant subspace.

Corollary 13.2.1 Complete reducibility

Any finite-dimensional representation of a finite group G (over $\mathbb{C}$) decomposes as a direct sum of irreducible representations.

13.3 Schur's Lemma

Theorem 13.3.1 Schur's Lemma

Let V, W be irreducible representations of G, and $\varphi : V \to W$ a homomorphism of representations. Then:

(i) Either $\varphi = 0$ or φ is an isomorphism.

(ii) Over $k = \mathbb{C}$: if V is irreducible and $\varphi : V \to V$ is a homomorphism of representations (i.e., $\varphi \in \mathrm{Hom}_G(V, V)$), then $\varphi = \lambda \cdot \mathrm{id}$ for some $\lambda \in \mathbb{C}$.

Proof: (i) $\mathrm{Ker}(\varphi)$ is a subrepresentation of V. Since V is irreducible, either $\mathrm{Ker}(\varphi) = V$ ($\varphi = 0$) or $\mathrm{Ker}(\varphi) = \{0\}$ (φ injective). Similarly, $\mathrm{Im}(\varphi)$ is a subrepresentation of W, so either $\mathrm{Im}(\varphi) = \{0\}$ ($\varphi = 0$) or $\mathrm{Im}(\varphi) = W$ (φ surjective).

(ii) Over $\mathbb{C}$, φ has an eigenvalue λ. Then $\varphi - \lambda \cdot \mathrm{id}$ is also a G-homomorphism $V \to V$ with nontrivial kernel. By (i), $\varphi - \lambda \cdot \mathrm{id} = 0$. $\square$

Example 13.3.1 (Center acts by scalars)

Let V be an irreducible representation of G, and $h \in Z(G)$. Since h commutes with all $g \in G$, the map $\rho(h) : V \to V$ commutes with all $\rho(g)$, i.e., $\rho(h) \in \mathrm{Hom}_G(V, V)$. By Schur, $\rho(h) = \lambda_h \cdot \mathrm{id}$ for some $\lambda_h \in \mathbb{C}$. So every central element acts as a scalar on every irreducible representation. In particular, irreducible representations of abelian groups are 1-dimensional (giving another proof of the earlier result).

13.3.1 Irreducible Representations of S_3

Example 13.3.2 (The three irreducible representations of S_3)

- The trivial representation $U = \mathbb{C}$: $\sigma \cdot z = z$ for all $\sigma \in S_3$.
- The alternating (sign) representation $U' = \mathbb{C}$: $\sigma \cdot z = \operatorname{sgn}(\sigma)z$.
- The standard representation V: start with the permutation representation $\mathbb{C}^3$ (basis e_1, e_2, e_3, with $\sigma \cdot e_i = e_{\sigma(i)}$). This has an invariant subspace $\operatorname{span}(e_1 + e_2 + e_3) \cong U$. The complement $V = \{(z_1, z_2, z_3) \in \mathbb{C}^3 : z_1 + z_2 + z_3 = 0\}$ has $\dim V = 2$ and is irreducible.

Irreducibility of V**:** Let $\tau = (123)$ and $\sigma = (12)$. Then τ acts on V with eigenvalues $\lambda = e^{2\pi i/3}$ and $\lambda^2 = e^{4\pi i/3}$ (since $\tau^3 = \mathrm{id}$ and $\tau|_V \neq \mathrm{id}$). If v is a λ-eigenvector, then $\sigma(v)$ is a λ^2-eigenvector (since $\tau(\sigma v) = \sigma(\tau^{-1}v) = \sigma(\lambda^2 v) = \lambda^2\sigma v$, using $\sigma\tau\sigma^{-1} = \tau^{-1}$). Since $\lambda \neq \lambda^2$, v and $\sigma(v)$ are linearly independent, so they span V. Any invariant subspace of V must contain an eigenvector of τ (since τ is diagonalizable on V), hence contains v or $\sigma(v)$, and then contains $\operatorname{span}(v, \sigma(v)) = V$. So V is irreducible. □

Proposition 13.3.1 These are all irreducibles of S_3

The representations U, U', V are the only irreducible representations of S_3 over $\mathbb{C}$, and any representation W of S_3 decomposes as $W \cong U^{\oplus a} \oplus U'^{\oplus b} \oplus V^{\oplus c}$ for unique $a, b, c \geqslant 0$.

Note:-

We will prove uniqueness (and a method to find a, b, c) using characters below. The key dimension check: $\dim U^2 + \dim U'^2 + \dim V^2 = 1 + 1 + 4 = 6 = |S_3|$, which we will show is a necessary condition.

13.4 Characters

Definition 13.4.1: Character

The character of a representation V is the function $\chi_V : G \to \mathbb{C}$ defined by $\chi_V(g) = \operatorname{Tr}(\rho(g))$, the trace of the linear operator $g : V \to V$.

Note:-

Since trace is conjugation-invariant ($\operatorname{Tr}(ABA^{-1}) = \operatorname{Tr}(B)$), the character only depends on the conjugacy class of g. That is, χ_V is a class function.

Definition 13.4.2: Class function

A class function is a function $f : G \to \mathbb{C}$ satisfying $f(ghg^{-1}) = f(h)$ for all $g, h \in G$.

Proposition 13.4.1 Properties of characters

Given representations V, W of G:

(i) $\chi_{V\oplus W}(g) = \chi_V(g) + \chi_W(g)$.

(ii) $\chi_{V\otimes W}(g) = \chi_V(g) \cdot \chi_W(g)$.

(iii) $\chi_{V^*}(g) = \overline{\chi_V(g)}$ (since g acts by ${}^t(g^{-1})$, and eigenvalues of g are roots of unity, so eigenvalues of g^{-1} are their conjugates).

(iv) $\chi_{\mathrm{Hom}(V,W)}(g) = \overline{\chi_V(g)} \cdot \chi_W(g)$.

(v) $\chi_V(e) = \dim V$.

(vi) For a permutation representation on a set S: $\chi_V(g) = |\{s \in S : g \cdot s = s\}| = |S^g|$.

(vii) $\chi_{\bigwedge^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 - \chi_V(g^2))$.

13.4.1 Symmetric Polynomials (Digression)

The connection between eigenvalues and characters motivates a brief digression. To store information about n complex numbers (the eigenvalues of $g : V \to V$), up to reordering, it suffices to specify the coefficients of the polynomial $\prod(x - \lambda_i)$. These coefficients are the elementary symmetric polynomials $\sigma_k(z_1, \dots, z_n) = \sum_{i_1 < \cdots < i_k} z_{i_1} \cdots z_{i_k}$.

Theorem 13.4.1 Fundamental theorem of symmetric polynomials

The subring of symmetric polynomials $\mathbb{C}[z_1, \dots, z_n]^{S_n}$ is isomorphic to the polynomial algebra $\mathbb{C}[\sigma_1, \dots, \sigma_n]$. That is, every symmetric polynomial is uniquely a polynomial expression in the elementary symmetric polynomials.

Note:-

Alternatively, one can use the power sums $\tau_k = \sum z_i^k$. Then $\mathbb{C}[z_1, \dots, z_n]^{S_n} \cong \mathbb{C}[\tau_1, \dots, \tau_n]$ as well. In particular, specifying the unordered tuple $\{z_1, \dots, z_n\}$ is equivalent to specifying $\sum z_i, \sum z_i^2, \dots, \sum z_i^n$. For representations, this means specifying all eigenvalues of g is equivalent to specifying $\mathrm{Tr}(g), \mathrm{Tr}(g^2), \dots, \mathrm{Tr}(g^n)$. But since G is a group, $\mathrm{Tr}(g^k)$ is just $\chi_V(g^k)$—which is already determined by χ_V. So the character determines the eigenvalues.

13.5 Character Orthogonality

13.5.1 The Inner Product on Class Functions

Proposition 13.5.1 Averaging projection

If V is a representation of G, the map $\varphi_V = \frac{1}{|G|}\sum_{g \in G} g : V \to V$ is a projection onto $V^G = \{v \in V : gv = v\ \forall g\}$, the space of G-invariant vectors. In particular, $\dim V^G = \mathrm{Tr}(\varphi_V) = \frac{1}{|G|}\sum_{g \in G} \chi_V(g)$.

Note:-

For representations V, W: $\mathrm{Hom}_G(V, W) = \mathrm{Hom}(V, W)^G$, so $\dim \mathrm{Hom}_G(V, W) = \frac{1}{|G|}\sum_{g \in G} \chi_{\mathrm{Hom}(V,W)}(g) = \frac{1}{|G|}\sum_{g \in G} \overline{\chi_V(g)}\chi_W(g)$. If V and W are irreducible, Schur's lemma gives $\dim \mathrm{Hom}_G(V, W) = \begin{cases} 1 & V \cong W \\ 0 & V \not\cong W \end{cases}$.

Definition 13.5.1: Hermitian inner product on class functions

Define $H : \{\text{class functions}\} \times \{\text{class functions}\} \to \mathbb{C}$ by

$$H(\alpha, \beta) = \frac{1}{|G|}\sum_{g \in G} \overline{\alpha(g)}\beta(g).$$

Theorem 13.5.1 Character orthogonality

The characters of the irreducible representations of G are orthonormal with respect to H. That is, if $V_1, \dots, V_k$ are the distinct irreducible representations of G, then

$$H(\chi_{V_i}, \chi_{V_j}) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_{V_i}(g)} \chi_{V_j}(g) = \delta_{ij}.$$

In particular, the characters of distinct irreducible representations are linearly independent as class functions.

Corollary 13.5.1 Number of irreducibles

The number of irreducible representations of G is at most the number of conjugacy classes of G (since the irreducible characters are linearly independent class functions, and the space of class functions has dimension = number of conjugacy classes). In fact, equality holds: the number of irreducible representations equals the number of conjugacy classes.

Corollary 13.5.2 Character determines representation

Every representation of G is completely determined (up to isomorphism) by its character. If $W \cong \bigoplus V_i^{\oplus a_i}$, then $\chi_W = \sum a_i \chi_{V_i}$, and $a_i = H(\chi_{V_i}, \chi_W)$.

Corollary 13.5.3 Irreducibility criterion

A representation W is irreducible if and only if $H(\chi_W, \chi_W) = 1$. More generally, $H(\chi_W, \chi_W) = \sum a_i^2$ where $W \cong \bigoplus V_i^{\oplus a_i}$, so the inner product counts the "number of irreducible summands" (with multiplicity squared).

13.5.2 The Regular Representation

Definition 13.5.2: Regular representation

The regular representation R of G is the vector space $\mathbb{C}[G]$ with basis $\{e_g\}_{g \in G}$, on which G acts by left multiplication: $g \cdot e_h = e_{gh}$.

Proposition 13.5.2 Decomposition of the regular representation

In the regular representation, each irreducible V_i appears with multiplicity $\dim V_i$: $R \cong \bigoplus_i V_i^{\oplus \dim V_i}$.

Proof: $\chi_R(g) = |\{h \in G : gh = h\}| = \begin{cases} |G| & g = e \\ 0 & g \neq e \end{cases}$ (permutation representation where only e has fixed points). Then $a_i = H(\chi_{V_i}, \chi_R) = \frac{1}{|G|} \sum_g \overline{\chi_{V_i}(g)} \chi_R(g) = \frac{1}{|G|} \overline{\chi_{V_i}(e)} \cdot |G| = \chi_{V_i}(e) = \dim V_i$. □

Corollary 13.5.4 Dimension formula

$\sum_i (\dim V_i)^2 = |G|$, where the sum is over all irreducible representations.

Proof: $\dim R = |G| = \sum_i a_i \dim V_i = \sum_i (\dim V_i)^2$. □

13.5.3 Column Orthogonality

The row orthogonality relations (character orthonormality) have a dual version: orthogonality of the columns of the character table.

Proposition 13.5.3 Column orthogonality relations

Let $C_1, \dots, C_k$ be the conjugacy classes of G, with representatives $g_1, \dots, g_k$ and sizes $|C_1|, \dots, |C_k|$. Then

$$\sum_{i=1}^{k} \chi_{V_i}(g_s)\,\overline{\chi_{V_i}(g_t)} = \frac{|G|}{|C_s|}\delta_{st}.$$

Proof: Consider the $k \times k$ character table matrix X with entries $X_{ij} = \chi_{V_i}(g_j)$. Row orthonormality states $\frac{1}{|G|}\sum_j |C_j| X_{ij}\overline{X_{\ell j}} = \delta_{i\ell}$. In matrix form, $XD\overline{X}^T = I$ where $D = \mathrm{diag}(|C_1|/|G|, \dots, |C_k|/|G|)$. Since X is square and invertible, $X^{-1} = D\overline{X}^T$. Taking the transpose: $(X^T)^{-1} = \overline{X}D$, so $X^T\overline{X}D = I$, giving $X^T\overline{X} = D^{-1}$. Entry-by-entry, this is the column orthogonality relation. □

Note:-

Column orthogonality provides an independent check on computed character tables. It also has a group-theoretic interpretation: two elements $g, h \in G$ are conjugate if and only if $\chi_V(g) = \chi_V(h)$ for every representation V. Characters separate conjugacy classes.

13.6 Character Tables

13.6.1 Character Table of S_3

S_3 has 3 conjugacy classes: $\{e\}$, $\{(12), (13), (23)\}$, $\{(123), (132)\}$. The 3 irreducible representations have dimensions satisfying $d_1^2 + d_2^2 + d_3^2 = 6$, so $d_i \in \{1, 1, 2\}$.

	e	(12)	(123)
	1	3	2
U (trivial)	1	1	1
U' (sign)	1	-1	1
V (standard)	2	0	-1

The character of V is obtained from the permutation representation: $\chi_{U\oplus V}(g) = |S^g|$, giving $\chi_{U\oplus V} = (3, 1, 0)$, so $\chi_V = (2, 0, -1)$. Check: $H(\chi_V, \chi_V) = \frac{1}{6}(4 + 0 + 2) = 1$, confirming irreducibility.

13.6.2 Character Table of S_4

S_4 has 5 conjugacy classes (partitions of 4): $\{e\}$ (1), transpositions (6), double transpositions (3), 3-cycles (8), 4-cycles (6). The dimension formula $\sum d_i^2 = 24$ with 5 irreducibles gives $d_i \in \{1, 1, 2, 3, 3\}$.

Three known irreducibles: U (trivial, dim 1), U' (sign, dim 1), V (standard, dim 3: permutation rep on $\mathbb{C}^4$ minus trivial summand). The tensor product $V' = V \otimes U'$ (twist standard by sign) has $\chi_{V'}(g) = \chi_V(g) \cdot \mathrm{sgn}(g)$ and is irreducible ($H(\chi_{V'}, \chi_{V'}) = 1$) and distinct from V. This gives 4 of the 5; we need one more of dimension 2 (since $1 + 1 + 9 + 9 = 20$, and $24 - 20 = 4 = 2^2$).

For the missing W (dim 2): use $V \otimes V$. We have $\chi_{V\otimes V} = \chi_V^2 = (9, 1, 1, 0, 1)$. Compute inner products with known characters to find $V \otimes V \cong U \oplus V \oplus V' \oplus W$. Then $\chi_W = \chi_{V\otimes V} - \chi_U - \chi_V - \chi_{V'} = (2, 0, 2, -1, 0)$. Check: $H(\chi_W, \chi_W) = \frac{1}{24}(4 + 0 + 12 + 8 + 0) = 1$.

The complete character table:

	e	(12)	(12)(34)	(123)	(1234)
	1	6	3	8	6
U	1	1	1	1	1
U'	1	-1	1	1	-1
V	3	1	-1	0	-1
V'	3	-1	-1	0	1
W	2	0	2	-1	0

Note:-

The representation W has $\chi_W((12)(34)) = 2 = \dim W$, so double transpositions act trivially. The normal subgroup $H = \{e, (12)(34), (13)(24), (14)(23)\} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ acts trivially on W, so W factors through $S_4/H \cong S_3$. In fact, W is the standard representation of S_3 "pulled back" to S_4 via the quotient map.

13.6.3 Character Table of A_4

A_4 has 4 conjugacy classes: $\{e\}$ (1), $\{(12)(34), (13)(24), (14)(23)\}$ (3), $\{(123), (134), (243), (142)\}$ (4), $\{(132), (143), (234), (124)\}$ (4). Note the 3-cycles split into two classes (since A_4 has no odd permutations to conjugate them). The dimension formula $\sum d_i^2 = 12$ gives $d_i \in \{1, 1, 1, 3\}$.

The three 1-dimensional representations correspond to $\mathrm{Hom}(A_4, \mathbb{C}^*) \cong \mathrm{Hom}(A_4/H, \mathbb{C}^*) \cong \widehat{\mathbb{Z}/3} = \{1, \omega, \omega^2\}$ where $\omega = e^{2\pi i/3}$, and the 3-dimensional irreducible is the restriction of the standard representation of S_4.

	e	$(12)(34)$	(123)	(132)
	1	3	4	4
U	1	1	1	1
U'	1	1	ω	ω^2
U''	1	1	ω^2	ω
V	3	-1	0	0

Exercise 13.6.1 Representation theory exercises

(1) Decompose the permutation representation $\mathbb{C}^3$ of S_3 into irreducibles and verify using the character inner product.

(2) Let V be the standard representation of S_3. Decompose $V \otimes V$, $\mathrm{Sym}^2(V)$, and $\bigwedge^2 V$ into irreducibles.

(3) Compute the character table of $\mathbb{Z}/2 \times \mathbb{Z}/2$ (the Klein four-group). How many irreducible representations are there, and what are their dimensions?

(4) Using the character table of S_4, verify that $V \otimes V' \cong U' \oplus W \oplus V \oplus V'$ (where V, V', W are as in the table).

(5) Prove that for any representation V of a finite group G, $\frac{1}{|G|} \sum_{g \in G} \chi_V(g) = \dim V^G$.

Solution

(1) The permutation representation has character $\chi(g) = |S^g|$: $\chi(e) = 3$, $\chi((12)) = 1$, $\chi((123)) = 0$. Compute: $H(\chi_U, \chi) = \frac{1}{6}(3 + 3 + 0) = 1$, $H(\chi_{U'}, \chi) = \frac{1}{6}(3 - 3 + 0) = 0$, $H(\chi_V, \chi) = \frac{1}{6}(6 + 0 + 0) = 1$. So $\mathbb{C}^3 \cong U \oplus V$.
(2) $\chi_{V \otimes V}(g) = \chi_V(g)^2$: values $(4, 0, 1)$. Inner products: $H(\chi_U, \chi_{V \otimes V}) = \frac{1}{6}(4 + 0 + 2) = 1$, $H(\chi_{U'}, \chi_{V \otimes V}) = \frac{1}{6}(4 + 0 + 2) = 1$, $H(\chi_V, \chi_{V \otimes V}) = \frac{1}{6}(8 + 0 - 2) = 1$. So $V \otimes V \cong U \oplus U' \oplus V$.
For $\mathrm{Sym}^2(V)$: $\chi_{\mathrm{Sym}^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 + \chi_V(g^2))$. Values: $\frac{1}{2}(4 + 2, 0 + 2, 1 + (-1)) = (3, 1, 0)$. So $\chi_{\mathrm{Sym}^2 V} = \chi_U + \chi_V$, giving $\mathrm{Sym}^2 V \cong U \oplus V$.
For $\bigwedge^2 V$: $\chi_{\bigwedge^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 - \chi_V(g^2))$. Values: $\frac{1}{2}(4 - 2, 0 - 2, 1 + 1) = (1, -1, 1) = \chi_{U'}$. So $\bigwedge^2 V \cong U'$. (This makes sense: the determinant map $\bigwedge^2 V \to \mathbb{C}$ of a 2-dimensional rep is the sign character.)
(3) $\mathbb{Z}/2 \times \mathbb{Z}/2$ is abelian with 4 elements, so it has 4 conjugacy classes and 4 one-dimensional irreducibles. The dual group is $\widehat{\mathbb{Z}/2 \times \mathbb{Z}/2} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$. Writing elements as $\{(0, 0), (1, 0), (0, 1), (1, 1)\}$, the 4 characters are $\chi_{ab}(x, y) = (-1)^{ax+by}$ for $(a, b) \in \{0, 1\}^2$.
(4) $\chi_{V \otimes V'} = \chi_V \cdot \chi_{V'}$: values $(9, -1, 1, 0, -1)$. Compute H with each irreducible to find multiplicities. This yields $V \otimes V' \cong U' \oplus W \oplus V \oplus V'$ (verify the character sum matches).
(5) The averaging operator $\varphi = \frac{1}{|G|} \sum_g g$ projects onto V^G. Taking traces: $\mathrm{Tr}(\varphi) = \frac{1}{|G|} \sum_g \mathrm{Tr}(g) =$

$\frac{1}{|G|}\sum_g \chi_V(g)$. Since φ is a projection ($\varphi^2 = \varphi$), its trace equals its rank: $\mathrm{Tr}(\varphi) = \dim \mathrm{Im}(\varphi) = \dim V^G$.

13.7 Characters Form an Orthonormal Basis

We proved that the irreducible characters $\chi_{V_1}, \dots, \chi_{V_k}$ are orthonormal with respect to H. This shows they are linearly independent, giving $k \leqslant$ (number of conjugacy classes). We now prove they also *span* the space of class functions, completing the proof that k equals the number of conjugacy classes.

13.7.1 The Generalized Projection Operator

The averaging projection $\varphi_V = \frac{1}{|G|}\sum_{g\in G} g$ onto V^G can be generalized. Given any function $\alpha : G \to \mathbb{C}$ and any representation V of G, define

$$\varphi_{\alpha,V} = \frac{1}{|G|}\sum_{g\in G} \alpha(g)\cdot g : V \to V.$$

This is a "weighted average" of the group elements acting on V.

Proposition 13.7.1 Equivariance of φ_α

If $\alpha : G \to \mathbb{C}$ is a class function, then $\varphi_{\alpha,V}$ is G-equivariant, i.e., $\varphi_{\alpha,V} \in \mathrm{Hom}_G(V, V)$.

Proof: We must show $h \cdot \varphi_\alpha(v) = \varphi_\alpha(h \cdot v)$ for all $h \in G$, $v \in V$. For the left side:

$$h \cdot \varphi_\alpha(v) = \frac{1}{|G|}\sum_{g\in G} \alpha(g)\cdot hg(v).$$

Relabeling $k = hg$ (so $g = h^{-1}k$):

$$h \cdot \varphi_\alpha(v) = \frac{1}{|G|}\sum_{k\in G} \alpha(h^{-1}k)\cdot k(v).$$

For the right side:

$$\varphi_\alpha(hv) = \frac{1}{|G|}\sum_{g\in G} \alpha(g)\cdot g(hv) = \frac{1}{|G|}\sum_{g\in G} \alpha(g)\cdot gh(v).$$

Relabeling $k = gh$ (so $g = kh^{-1}$):

$$\varphi_\alpha(hv) = \frac{1}{|G|}\sum_{k\in G} \alpha(kh^{-1})\cdot k(v).$$

These are equal if and only if $\alpha(h^{-1}k) = \alpha(kh^{-1})$ for all $h, k \in G$, which holds precisely when α is a class function (since $kh^{-1} = k(h^{-1}k)k^{-1}$, so $h^{-1}k$ and kh^{-1} are conjugate). $\square$

Lecture 14.

The Representation Ring; Characters of S_5 and A_5

We proved that the irreducible characters $\chi_{V_1}, \ldots, \chi_{V_k}$ are orthonormal with respect to the Hermitian inner product $H(\alpha, \beta) = \frac{1}{|G|} \sum_{g \in G} \overline{\alpha(g)} \beta(g)$ on the space of class functions $G \to \mathbb{C}$. We also showed that the number of irreducible representations is at most the number of conjugacy classes. We now prove that these characters in fact form a basis for the space of class functions, completing the proof that the number of irreducible representations equals the number of conjugacy classes.

14.1 Characters as an Orthonormal Basis

The key tool is a generalization of the averaging projection.

Proposition 14.1.1 Generalized averaging operator

Given any class function $\alpha : G \to \mathbb{C}$ and any representation V of G, define

$$\varphi_{\alpha,V} = \frac{1}{|G|} \sum_{g \in G} \alpha(g)\, g : V \to V.$$

Then $\varphi_{\alpha,V}$ is G-equivariant (i.e. $\varphi_{\alpha,V} \in \mathrm{End}_G(V)$).

Proof: For any $h \in G$,

$$\varphi_{\alpha,V}(hv) = \frac{1}{|G|} \sum_{g \in G} \alpha(g)\, g\, h\, v = \frac{1}{|G|} \sum_{g \in G} \alpha(hg'h^{-1})(hg'h^{-1})h\, v = \frac{1}{|G|} \sum_{g' \in G} \alpha(g')\, h\, g' v = h \cdot \varphi_{\alpha,V}(v),$$

where we substituted $g = hg'h^{-1}$ (so $g' = h^{-1}gh$) and used that α is a class function. □

Note:-

When $\alpha = 1$ (the constant function), $\varphi_{1,V} = \frac{1}{|G|} \sum_g g$ is the projection onto V^G that we used previously.

Theorem 14.1.1 Characters form an orthonormal basis

The characters $\chi_{V_1}, \ldots, \chi_{V_k}$ of the irreducible representations of G form an orthonormal basis for the space of class functions $G \to \mathbb{C}$ with respect to H. In particular, the number of irreducible representations equals the number of conjugacy classes.

Proof: We already proved orthonormality. It remains to show spanning: if $H(\alpha, \chi_{V_i}) = 0$ for all i, then $\alpha = 0$.

Given any class function α and any irreducible representation V_i of dimension n_i, the operator $\varphi_{\overline{\alpha},V_i} : V_i \to V_i$ is G-equivariant by the proposition above (since $\overline{\alpha}$ is also a class function). By Schur's lemma, $\varphi_{\overline{\alpha},V_i} = \lambda \cdot \mathrm{id}_{V_i}$ for some $\lambda \in \mathbb{C}$. Taking traces:

$$\lambda = \frac{1}{n_i} \mathrm{Tr}(\varphi_{\overline{\alpha},V_i}) = \frac{1}{n_i} \cdot \frac{1}{|G|} \sum_{g \in G} \overline{\alpha(g)} \chi_{V_i}(g) = \frac{1}{n_i} H(\alpha, \chi_{V_i}).$$

So if $H(\alpha, \chi_{V_i}) = 0$ for all irreducible V_i, then $\varphi_{\overline{\alpha},V_i} = 0$ for all irreducible V_i. By complete reducibility, $\varphi_{\overline{\alpha},V} = 0$ for every representation V of G (since $\varphi_{\overline{\alpha},\cdot}$ respects direct sums). In particular, taking $V = R$ the regular representation with basis $\{e_g\}_{g\in G}$:

$$\varphi_{\overline{\alpha},R}(e_e) = \frac{1}{|G|}\sum_{g\in G}\overline{\alpha(g)}\,e_g = 0.$$

Since the e_g are linearly independent, $\overline{\alpha(g)} = 0$ for all $g \in G$, hence $\alpha = 0$. $\square$

14.1.1 Projection onto Isotypic Components

The proof above also yields an explicit projection formula. For V_i, V_j irreducible, consider the class function $\alpha = \overline{\chi_{V_i}}$. Then

$$\varphi_{\overline{\chi_{V_i}},V_j} = \frac{1}{n_j}H(\chi_{V_i},\chi_{V_j})\cdot \mathrm{id}_{V_j} = \begin{cases}\frac{1}{n_i}\cdot \mathrm{id}_{V_i} & \text{if } i=j,\\ 0 & \text{if } i\neq j.\end{cases}$$

Proposition 14.1.2 Isotypic projection formula

If V is any representation of G with decomposition $V = \bigoplus_i V_i^{\oplus a_i}$, then the map

$$\varphi_{V_i} = \frac{n_i}{|G|}\sum_{g\in G}\overline{\chi_{V_i}(g)}\,g : V \to V$$

where $n_i = \dim V_i$, is the projection onto the isotypic component $V_i^{\oplus a_i}$ (i.e. the identity on $V_i^{\oplus a_i}$ and zero on all other summands).

Note:-

The case $V_i = U$ (trivial representation, $n_i = 1$, $\chi_U = 1$) recovers the averaging projection $\varphi_U = \frac{1}{|G|}\sum_g g$ onto V^G.

14.2 The Representation Ring

What algebraic structure do the representations of a fixed finite group G carry? The set of isomorphism classes of finite-dimensional complex representations of G has two natural operations: direct sum $\oplus$ and tensor product $\otimes$. These are commutative, associative, and tensor distributes over direct sum: $(U \oplus V)\otimes W \cong (U\otimes W)\oplus(V\otimes W)$. This is almost a ring, except that we lack additive inverses.

Definition 14.2.1: Representation ring

Let $\hat{R}$ be the free abelian group on isomorphism classes $[V]$ of finite-dimensional complex representations of G, i.e. formal $\mathbb{Z}$-linear combinations $\sum a_i[V_i]$. Define $R(G)$ to be the quotient of $\hat{R}$ by the subgroup generated by all elements $[V]+[W]-[V\oplus W]$. Then $R(G)$ is a commutative ring with:

- Addition: $[V]+[W] = [V\oplus W]$ (and now we can subtract: $[V]-[W]$ makes sense).
- Multiplication: $[V]\cdot[W] = [V\otimes W]$.
- Identity: $[U]$, the class of the trivial representation.

This is the representation ring (or Green ring) of G.

By complete reducibility and uniqueness of decomposition into irreducibles, every element of $R(G)$ can be written uniquely as $\sum_{i=1}^k a_i[V_i]$ where $V_1,\dots,V_k$ are the irreducible representations and $a_i \in \mathbb{Z}$. So $(R(G),+)\cong \mathbb{Z}^k$ as an abelian group.

General elements of $R(G)$ with some $a_i < 0$ are called virtual representations. The actual representations (those with all $a_i \geqslant 0$) form a cone inside $R(G)$.

Proposition 14.2.1 The character map is a ring homomorphism

The character map $V \mapsto \chi_V$ extends by linearity to a ring homomorphism

$$\Psi : R(G) \to \mathbb{C}_{\text{class}}(G), \qquad \sum a_i[V_i] \mapsto \sum a_i \chi_{V_i},$$

where $\mathbb{C}_{\text{class}}(G)$ denotes the ring of class functions $G \to \mathbb{C}$ under pointwise addition and multiplication. This is a ring homomorphism because $\chi_{V \oplus W} = \chi_V + \chi_W$ and $\chi_{V \otimes W} = \chi_V \cdot \chi_W$.

The image of Ψ is the lattice of virtual characters $\Lambda = \{\sum a_i \chi_{V_i} : a_i \in \mathbb{Z}\} \subset \mathbb{C}_{\text{class}}(G)$. After extending scalars, our basis theorem gives:

Corollary 14.2.1

The induced map $\Psi_{\mathbb{C}} : R(G) \otimes_{\mathbb{Z}} \mathbb{C} \xrightarrow{\sim} \mathbb{C}_{\text{class}}(G)$ is an isomorphism of $\mathbb{C}$-algebras.

Note:-

There are theorems of Artin and Brauer that describe the lattice Λ of virtual characters inside $\mathbb{C}_{\text{class}}(G)$:

- (Artin) Every character of a representation of G is a $\mathbb{Q}$-linear combination of characters induced from cyclic subgroups of G.
- (Brauer) Every character of a representation of G is a $\mathbb{Z}$-linear combination of characters of 1-dimensional representations induced from "elementary" subgroups of G, where an elementary subgroup is a product $C \times H$ with C cyclic of order coprime to p and H a p-group, for some prime p.

See Serre, Linear Representations of Finite Groups, for proofs.

14.3 Character Table of S_5

S_5 has 7 conjugacy classes, determined by cycle type:

cycle type	1^5	$2 \cdot 1^3$	$2^2 \cdot 1$	$3 \cdot 1^2$	$3 \cdot 2$	$4 \cdot 1$	5
representative	e	(12)	(12)(34)	(123)	(123)(45)	(1234)	(12345)
class size	1	10	15	20	20	30	24

So there are 7 irreducible representations. Since $|S_5| = 120 = \sum d_i^2$, we need to find dimensions summing in squares to 120.

We start from known representations:

- U: trivial, $\dim = 1$.
- U': sign (alternating), $\dim = 1$.
- V: standard representation. The permutation representation $\mathbb{C}^5$ decomposes as $U \oplus V$ where $V = \{(z_1, \ldots, z_5) : \sum z_i = 0\}$, so $\dim V = 4$. Its character is $\chi_V(\sigma) = |\text{Fix}(\sigma)| - 1$.
- $V' = V \otimes U'$: another 4-dimensional irreducible with $\chi_{V'} = \chi_V \cdot \text{sgn}$.

These account for $1 + 1 + 16 + 16 = 34$, leaving $120 - 34 = 86 = d_5^2 + d_6^2 + d_7^2$. Since we need three more squares summing to 86, the only possibility is $5^2 + 5^2 + 6^2 = 86$. So the remaining dimensions are 5, 5, 6.

14.3.1 Finding the Remaining Irreducibles via Tensor Products

The most effective method is to analyze $V \otimes V$ ($\dim = 16$), or rather its two pieces: $\text{Sym}^2 V$ ($\dim = 10$) and $\bigwedge^2 V$ ($\dim = 6$).

If $g : V \to V$ has eigenvalues $\lambda_1, \ldots, \lambda_r$, then $\mathrm{Sym}^2 V$ has eigenvalues $\lambda_i\lambda_j$ for $1 \leqslant i \leqslant j \leqslant r$, and $\bigwedge^2 V$ has eigenvalues $\lambda_i\lambda_j$ for $1 \leqslant i < j \leqslant r$. The resulting character formulas are:

$$\chi_{\bigwedge^2 V}(g) = \frac{1}{2}\big(\chi_V(g)^2 - \chi_V(g^2)\big), \qquad \chi_{\mathrm{Sym}^2 V}(g) = \frac{1}{2}\big(\chi_V(g)^2 + \chi_V(g^2)\big).$$

We compute these for the standard representation V of S_5, whose character is $\chi_V(\sigma) = |\mathrm{Fix}(\sigma)| - 1$. To apply the formulas, we need $\chi_V(g^2)$. Squaring cycle types: $(12)^2 = e$, $((12)(34))^2 = e$, $(123)^2 = (132)$ (a 3-cycle), $((123)(45))^2 = (132)$, $(1234)^2 = (13)(24)$ (double transposition), $(12345)^2 = (13524)$ (a 5-cycle).

	e	(12)	(123)	(1234)	(12345)	(12)(34)	(123)(45)
class size	1	10	20	30	24	15	20
χ_V	4	2	1	0	−1	0	−1
$\chi_V(g^2)$	4	4	1	0	−1	4	1
$\bigwedge^2 V$	6	0	0	0	1	−2	0
$\mathrm{Sym}^2 V$	10	4	1	0	0	2	1

We check irreducibility of $\bigwedge^2 V$:

$$H(\chi_{\bigwedge^2 V}, \chi_{\bigwedge^2 V}) = \frac{1}{120}\big(1 \cdot 36 + 10 \cdot 0 + 20 \cdot 0 + 30 \cdot 0 + 24 \cdot 1 + 15 \cdot 4 + 20 \cdot 0\big) = \frac{120}{120} = 1.$$

So $\bigwedge^2 V$ is irreducible.

For $\mathrm{Sym}^2 V$:

$$H(\chi_{\mathrm{Sym}^2 V}, \chi_{\mathrm{Sym}^2 V}) = \frac{1}{120}\big(100 + 160 + 20 + 0 + 0 + 60 + 20\big) = \frac{360}{120} = 3.$$

So $\mathrm{Sym}^2 V$ splits into 3 irreducible summands. Computing inner products with known characters:

$$H(\chi_U, \chi_{\mathrm{Sym}^2 V}) = \frac{1}{120}(10+40+20+0+0+30+20) = 1, \quad H(\chi_V, \chi_{\mathrm{Sym}^2 V}) = \frac{1}{120}(40+80+20+0+0+0-20) = 1,$$

while $H(\chi_{U'}, \chi_{\mathrm{Sym}^2 V}) = 0$ and $H(\chi_{V'}, \chi_{\mathrm{Sym}^2 V}) = 0$. So $\mathrm{Sym}^2 V = U \oplus V \oplus W$ where W is a new irreducible of dimension $10 - 1 - 4 = 5$. Its character is

$$\chi_W = \chi_{\mathrm{Sym}^2 V} - \chi_U - \chi_V = (5, 1, -1, -1, 0, 1, 1).$$

Finally, $W' = W \otimes U'$ gives another 5-dimensional irreducible with $\chi_{W'} = \chi_W \cdot \mathrm{sgn} = (5, -1, -1, 1, 0, 1, -1)$.

14.3.2 The Complete Character Table

	e	(12)	(123)	(1234)	(12345)	(12)(34)	(123)(45)
	1	10	20	30	24	15	20
U	1	1	1	1	1	1	1
U'	1	−1	1	−1	1	1	−1
V	4	2	1	0	−1	0	−1
$V' = V \otimes U'$	4	−2	1	0	−1	0	1
$\bigwedge^2 V$	6	0	0	0	1	−2	0
W	5	1	−1	−1	0	1	1
$W' = W \otimes U'$	5	−1	−1	1	0	1	−1

Check: $\sum d_i^2 = 1 + 1 + 16 + 16 + 36 + 25 + 25 = 120 = |S_5|$. The column sums of $|\chi|^2$ weighted by class size also verify the orthogonality relations.

Note:-

The standard representation V of S_n and its exterior powers $\bigwedge^k V$, for $0 \leqslant k \leqslant n-1$, are all irreducible. This is a general fact; see Fulton–Harris, §3.2.

14.4 Character Table of A_5

A_5 has 5 conjugacy classes. Since A_5 contains no odd permutations, conjugacy classes in S_5 can split when restricted to A_5: a class of cycle type λ in S_5 remains a single class in A_5 if some odd permutation commutes with a representative, and splits into two equal-sized classes otherwise. For S_5, the 5-cycles split (the centralizer of a 5-cycle in S_5 has order 5 and consists entirely of even permutations), while the other even conjugacy classes do not split. So the conjugacy classes of A_5 are:

representative	e	(123)	(12345)	(12354)	(12)(34)
class size	1	20	12	12	15

The approach is to restrict irreducible representations of S_5 to A_5 and check which remain irreducible. Since every element of A_5 is even, $U'|_{A_5}$ is trivial. So for any S_5-representation X, the restrictions of X and $X \otimes U'$ to A_5 are isomorphic. In particular, $V|_{A_5} \cong V'|_{A_5}$ and $W|_{A_5} \cong W'|_{A_5}$.

The restricted characters, using the column ordering $(e, (123), (12345), (12354), (12)(34))$ with class sizes $(1, 20, 12, 12, 15)$, are:

	e	(123)	(12345)	(12354)	(12)(34)
	1	20	12	12	15
U	1	1	1	1	1
V	4	1	-1	-1	0
W	5	-1	0	0	1
$\bigwedge^2 V$	6	0	1	1	-2

Computing inner products over A_5: $H(\chi_V, \chi_V) = \frac{1}{60}(16{+}20{+}12{+}12) = 1$ and $H(\chi_W, \chi_W) = \frac{1}{60}(25{+}20{+}15) = 1$, so V and W remain irreducible. However,

$$H(\chi_{\bigwedge^2 V}, \chi_{\bigwedge^2 V}) = \frac{1}{60}(36 + 0 + 12 + 12 + 60) = 2,$$

so $\bigwedge^2 V$ splits as a direct sum of two distinct irreducibles when restricted to A_5. Moreover, $\bigwedge^2 V$ contains no copies of U, V, or W (all inner products vanish), so $\bigwedge^2 V|_{A_5} = Y \oplus Z$ for two new irreducible representations of A_5.

From the dimension formula $\sum d_i^2 = 60$ and $1 + 16 + 25 + d_4^2 + d_5^2 = 60$, we get $d_4^2 + d_5^2 = 18$ with $d_4 + d_5 = 6$, giving $\dim Y = \dim Z = 3$.

14.4.1 Determining Y and Z

We know $\chi_Y + \chi_Z = (6,\ 0,\ 1,\ 1,\ -2)$. Write $\chi_Y - \chi_Z = (0,\ b,\ c,\ d,\ f)$ (since both have dimension 3, the value at e cancels). Orthogonality of $\chi_Y - \chi_Z$ against χ_U, χ_V, and χ_W gives a linear system whose solution is $b = 0$, $f = 0$, and $d = -c$. So

$$\chi_Y - \chi_Z = (0,\ 0,\ a,\ -a,\ 0)$$

for some a. Since $H(\chi_Y - \chi_Z, \chi_Y - \chi_Z) = H(\chi_Y, \chi_Y) + H(\chi_Z, \chi_Z) = 2$, we get

$$\frac{1}{60}(12a^2 + 12a^2) = 2 \implies a^2 = 5 \implies a = \pm\sqrt{5}.$$

Taking $a = \sqrt{5}$ (the choice of sign corresponds to labeling Y vs. Z):

	e	(123)	(12345)	(12354)	(12)(34)
	1	20	12	12	15
U	1	1	1	1	1
V	4	1	-1	-1	0
W	5	-1	0	0	1
Y	3	0	$\frac{1+\sqrt{5}}{2}$	$\frac{1-\sqrt{5}}{2}$	-1
Z	3	0	$\frac{1-\sqrt{5}}{2}$	$\frac{1+\sqrt{5}}{2}$	-1

The golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ appears naturally. The two representations Y and Z are exchanged by the outer automorphism of A_5 given by conjugation by any transposition in S_5.

Note:-

The group A_5 is the group of rotational symmetries of the regular icosahedron in $\mathbb{R}^3$. The inclusion $A_5 \hookrightarrow SO(3) \subset GL_3(\mathbb{R}) \subset GL_3(\mathbb{C})$ gives a 3-dimensional real representation, which must be Y or Z. The irrational character values have a geometric consequence: there is no regular icosahedron (or dodecahedron) in $\mathbb{R}^3$ whose vertices all have rational coordinates. If such a polyhedron existed, the representation $A_5 \hookrightarrow GL_3(\mathbb{R})$ would factor through $GL_3(\mathbb{Q})$, forcing all character values to be rational—contradicting $\chi_Y((12345)) = \frac{1+\sqrt{5}}{2} \notin \mathbb{Q}$. This holds in any coordinate system, not just an orthonormal one.

Exercise 14.4.1 Exercises

(1) Let V be the standard representation of S_5. Verify directly that $\bigwedge^3 V \cong V'$ (the 4-dimensional irreducible with character $\chi_V \cdot \mathrm{sgn}$) by computing $\chi_{\bigwedge^3 V}$ using the formula $\chi_{\bigwedge^3 V}(g) = \frac{1}{6}\big(\chi_V(g)^3 - 3\chi_V(g)\chi_V(g^2) + 2\chi_V(g^3)\big)$.

(2) Verify column orthogonality for the character table of A_5: for conjugacy classes $C_i \neq C_j$, $\sum_\rho \chi_\rho(C_i)\overline{\chi_\rho(C_j)} = 0$.

(3) Using the character table of S_5, decompose the representation $\bigwedge^2 V \otimes U'$ and $V \otimes W$ into irreducibles.

(4) Prove that $\dim R(G) \otimes_{\mathbb{Z}} \mathbb{Q} = k$ where k is the number of conjugacy classes, and that the multiplication table for the basis $[V_1], \ldots, [V_k]$ of $R(G)$ has nonneg integer structure constants.

(5) Show that the determinant map $\det : S_n \to \{+1, -1\}$ gives $\bigwedge^{n-1} V \cong U'$ for the standard representation V of S_n (for any n).

Solution

(1) We need $\chi_V(g^3)$ for each conjugacy class. Cubing: $(12)^3 = (12)$, $(123)^3 = e$, $(1234)^3 = (1432)$ (a 4-cycle), $(12345)^3 = (14253)$ (a 5-cycle), $((12)(34))^3 = (12)(34)$, $((123)(45))^3 = (45)$ (a transposition). So $\chi_V(g^3) = (4, 2, 4, 0, -1, 0, 2)$.
Applying the formula with $\chi_V = (4, 2, 1, 0, -1, 0, -1)$, $\chi_V(g^2) = (4, 4, 1, 0, -1, 4, 1)$, $\chi_V(g^3) = (4, 2, 4, 0, -1, 0, 2)$:

$$\chi_{\bigwedge^3 V}(g) = \frac{1}{6}\big(\chi^3 - 3\chi \cdot \chi(g^2) + 2\chi(g^3)\big).$$

Computing entry by entry (in the column order $e, (12), (123), (1234), (12345), (12)(34), (123)(45)$):

$$\frac{1}{6}(64 - 48 + 8,\ 8 - 24 + 4,\ 1 - 3 + 8,\ 0 - 0 + 0,\ -1 - 3 - 2,\ 0 - 0 + 0,\ -1 + 3 + 4)$$

$$= \frac{1}{6}(24, -12, 6, 0, -6, 0, 6)$$
$$= (4, -2, 1, 0, -1, 0, 1) = \chi_{V'}.$$

So $\bigwedge^3 V \cong V'$.
(2) Taking columns $C_1 = \{e\}$ and $C_2 = \{(123)\}$: $\sum_\rho \chi_\rho(e)\overline{\chi_\rho((123))} = 1 \cdot 1 + 4 \cdot 1 + 5 \cdot (-1) + 3 \cdot 0 + 3 \cdot 0 = 0$. Similarly for other pairs.
(3) $\chi_{\bigwedge^2 V \otimes U'} = \chi_{\bigwedge^2 V} \cdot \mathrm{sgn} = (6, 0, 0, 0, 1, -2, 0)$. This equals $\chi_{\bigwedge^2 V}$ (since sgn is $+1$ on even permutations and -1 on odd, and the only nonzero odd-permutation values are at (1234) which is already 0). So $\bigwedge^2 V \otimes U' \cong \bigwedge^2 V$.
For $V \otimes W$: $\chi_{V \otimes W} = \chi_V \cdot \chi_W = (20, 2, -1, 0, 0, 0, -1)$. Computing inner products:

$$H(\chi_U, \chi_{V \otimes W}) = \frac{1}{120}(20 + 20 - 20 + 0 + 0 + 0 - 20) = 0,$$

$$H(\chi_{U'}, \chi_{V\otimes W}) = \frac{1}{120}(20 - 20 - 20 + 0 + 0 + 0 + 20) = 0,$$

$$H(\chi_{V}, \chi_{V\otimes W}) = \frac{1}{120}(80 + 40 - 20 + 0 + 0 + 0 + 20) = 1,$$

$$H(\chi_{V'}, \chi_{V\otimes W}) = \frac{1}{120}(80 - 40 - 20 + 0 + 0 + 0 - 20) = 0,$$

$$H(\chi_{\bigwedge^2 V}, \chi_{V\otimes W}) = \frac{1}{120}(120 + 0 + 0 + 0 + 0 + 0 + 0) = 1,$$

$$H(\chi_{W}, \chi_{V\otimes W}) = \frac{1}{120}(100 + 20 + 20 + 0 + 0 + 0 - 20) = 1,$$

$$H(\chi_{W'}, \chi_{V\otimes W}) = \frac{1}{120}(100 - 20 + 20 + 0 + 0 + 0 + 20) = 1.$$

So $V \otimes W \cong V \oplus \bigwedge^2 V \oplus W \oplus W'$ (dimensions $4 + 6 + 5 + 5 = 20$).

(4) Since the irreducible characters $\chi_{V_1}, \dots, \chi_{V_k}$ form a $\mathbb{Z}$-basis for the lattice $\Lambda \subset \mathbb{C}_{\text{class}}(G)$, $\dim_{\mathbb{Q}}(R(G) \otimes \mathbb{Q}) = k$. For structure constants: $[V_i] \cdot [V_j] = [V_i \otimes V_j] = \sum_\ell n_{ij}^\ell [V_\ell]$ where $n_{ij}^\ell = H(\chi_{V_\ell}, \chi_{V_i}\chi_{V_j})$. Since $\chi_{V_i}\chi_{V_j} = \chi_{V_i \otimes V_j}$ is the character of an actual representation, each $n_{ij}^\ell \geqslant 0$.

(5) The character of $\bigwedge^{n-1} V$ on $\sigma \in S_n$ equals det of the matrix of σ on V restricted to the $(n-1)$-fold exterior power, which is just $\det(\sigma|_V) = \operatorname{sgn}(\sigma)$. More precisely, $\bigwedge^{n-1} V$ is 1-dimensional (since $\dim V = n-1$), and σ acts by $\det(\sigma|_V) = \operatorname{sgn}(\sigma)$ (the determinant of a permutation matrix restricted to the hyperplane $\sum x_i = 0$ equals the sign).

Lecture 15.

Restriction and Induction of Representations

Throughout, G is a finite group and $H \leqslant G$ a subgroup.

15.1 Restriction

Definition 15.1.1: Restriction

Given a representation V of G, the restriction of V to H, denoted $\mathrm{Res}_H^G V$, is V viewed as a representation of H (forget the action of elements outside H).

This defines a functor $\mathrm{Res}_H^G : \mathrm{Rep}(G) \to \mathrm{Rep}(H)$, where the categories have objects = finite-dimensional complex representations and morphisms = G-equivariant (resp. H-equivariant) linear maps.

An irreducible representation of G need not remain irreducible upon restriction to H. The problem we address next is: can we go in the opposite direction?

15.2 Induced Representations

Suppose V is a representation of G and $W \subset V$ is a subspace that is invariant under H (so W is a subrepresentation of $\mathrm{Res}_H^G V$), but not necessarily under all of G. For $g \in G$, the subspace $gW \subset V$ depends only on the coset gH, and gW is a representation of the conjugate subgroup gHg^{-1} via $H \xrightarrow{\rho} \mathrm{GL}(W)$, $gHg^{-1} \xrightarrow{\sim} \mathrm{GL}(gW)$ (conjugation by g).

The simplest scenario is when

$$V = \bigoplus_{\sigma \in G/H} \sigma W$$

as a direct sum of vector spaces, where σ ranges over a set of coset representatives. In this case, $\dim V = [G : H] \cdot \dim W$, and the entire representation V is completely determined by the H-representation W.

Definition 15.2.1: Induced representation

A representation V of G, together with a subspace $W \subset V$ invariant under H (i.e. W is a subrepresentation of $\mathrm{Res}_H^G V$), is said to be induced by $W \in \mathrm{Rep}(H)$ if

$$V = \bigoplus_{\sigma \in G/H} \sigma W.$$

We write $V = \mathrm{Ind}_H^G W$. Concretely: fix coset representatives $\sigma_1, \dots, \sigma_k$ of G/H (where $k = [G : H]$). Then every $v \in V$ can be written uniquely as $v = \sigma_1 w_1 + \cdots + \sigma_k w_k$ with $w_i \in W$, and G acts by permuting the summands: if $g\sigma_i = \sigma_j h$ for some $h \in H$, then $g(\sigma_i w) = \sigma_j(hw)$.

Theorem 15.2.1 Existence and uniqueness of induced representations

Given a representation W of H, the induced representation $V = \mathrm{Ind}_H^G W$ exists and is unique up to isomorphism of G-representations.

Proof: Uniqueness. Suppose V is a representation of G and $W \subset V$ is H-invariant with $V = \bigoplus_{i=1}^{k} \sigma_i W$. For any $g \in G$, there is a unique j such that $g\sigma_i \in \sigma_j H$, i.e. $h = \sigma_j^{-1} g \sigma_i \in H$, and necessarily $g(\sigma_i w) = \sigma_j(hw)$. This determines the G-action on V uniquely from the H-action on W.

Existence. Build $V = \bigoplus_{i=1}^{k} \sigma_i W$ where the σ_i are formal symbols (i.e. V consists of k copies of W as a vector space). Define $g \in G$ to act by: given $g\sigma_i = \sigma_j h$, set $g(\sigma_i w) = \sigma_j(hw)$. This is well-defined (independent of the choice of coset representatives) and defines a group homomorphism $G \to \mathrm{GL}(V)$. □

15.2.1 Examples

Example 15.2.1 (Permutation representation from induction)

The permutation representation of G on the coset space G/H equals $\mathrm{Ind}_H^G U$, where U is the trivial representation of H. Indeed, the permutation representation has basis $\{e_{\sigma H}\}_{\sigma \in G/H}$, the element e_H spans a 1-dimensional subspace fixed by H, and $gW = \mathrm{span}(e_{gH})$, so $V = \bigoplus_{gH \in G/H} gW$.

Example 15.2.2 (Regular representation from induction)

The regular representation of G is $\mathrm{Ind}_H^G R_H$, where R_H is the regular representation of H. Here $W = \mathrm{span}\{e_h : h \in H\} \subset V = \mathrm{span}\{e_g : g \in G\}$.

15.2.2 Properties of Induction

Proposition 15.2.1 Basic properties

Let $H \leqslant G$, and let W, W' be representations of H and U a representation of G.

(i) $\mathrm{Ind}_H^G(W \oplus W') \cong \mathrm{Ind}_H^G(W) \oplus \mathrm{Ind}_H^G(W')$.

(ii) $\mathrm{Ind}_H^G(\mathrm{Res}_H^G(U) \otimes W) \cong U \otimes \mathrm{Ind}_H^G(W)$.

(iii) In particular, $\mathrm{Ind}_H^G(\mathrm{Res}_H^G(U)) \cong U \otimes \mathrm{Ind}_H^G(U_H)$, where U_H is the trivial representation of H. The right side is U tensored with the permutation representation on G/H.

Proof of (ii): Write $\mathrm{Ind}_H^G(W) = \bigoplus_{\sigma \in G/H} \sigma W$. Then

$$U \otimes \mathrm{Ind}_H^G(W) = \bigoplus_{\sigma \in G/H} (U \otimes \sigma W) = \bigoplus_{\sigma \in G/H} \sigma(U \otimes W),$$

where $U \otimes W \subset U \otimes \mathrm{Ind}_H^G(W)$ is invariant under H, and as an H-representation it is $\mathrm{Res}_H^G(U) \otimes W$. □

Note:-

In general, $\mathrm{Ind}(W \otimes W') \not\cong \mathrm{Ind}(W) \otimes \mathrm{Ind}(W')$.

15.2.3 Character of an Induced Representation

Proposition 15.2.2 Character formula for induction

Let W be a representation of $H \leqslant G$. Then

$$\chi_{\mathrm{Ind}_H^G W}(g) = \frac{1}{|H|} \sum_{\substack{s \in G \\ s^{-1} g s \in H}} \chi_W(s^{-1} g s).$$

Proof: Choose coset representatives $\sigma_1, \dots, \sigma_k$ of G/H. The element $g \in G$ maps $\sigma_i W$ to $\sigma_j W$ where j is determined by $g\sigma_i \in \sigma_j H$. If $i \neq j$, this contributes nothing to the trace. If $i = j$, then $h = \sigma_i^{-1} g \sigma_i \in H$ and g acts on $\sigma_i W$ as $\sigma_i h \sigma_i^{-1}$ composed with the identification $\sigma_i W \cong W$, so the trace on $\sigma_i W$ is $\mathrm{Tr}(h|_W) = \chi_W(\sigma_i^{-1} g \sigma_i)$. Summing over all σ_i:

$$\chi_{\mathrm{Ind}\, W}(g) = \sum_{\substack{i \\ \sigma_i^{-1} g \sigma_i \in H}} \chi_W(\sigma_i^{-1} g \sigma_i).$$

Since each coset $\sigma_i H$ has $|H|$ elements, and the condition $\sigma_i^{-1} g \sigma_i \in H$ is the same for any representative of the coset, we can rewrite this as a sum over all $s \in G$ with the normalization factor $1/|H|$. □

15.3 Frobenius Reciprocity

Theorem 15.3.1 Frobenius reciprocity

Let U be a representation of G and W a representation of $H \leqslant G$. Then every H-equivariant map $\varphi : W \to \mathrm{Res}_H^G(U)$ extends uniquely to a G-equivariant map $\tilde{\varphi} : \mathrm{Ind}_H^G(W) \to U$. That is,

$$\mathrm{Hom}_H(W, \mathrm{Res}_H^G U) \cong \mathrm{Hom}_G(\mathrm{Ind}_H^G W, U).$$

Proof: Choose coset representatives $\sigma_1, \dots, \sigma_k \in G$ and write $V = \mathrm{Ind}_H^G W = \bigoplus_{i=1}^k \sigma_i W$.

Uniqueness. Given $\varphi : W \to \mathrm{Res}(U)$, if $\tilde{\varphi} : V \to U$ is G-equivariant with $\tilde{\varphi}|_W = \varphi$, then for any $\sigma_i w \in \sigma_i W$:

$$\tilde{\varphi}(\sigma_i w) = \sigma_i \cdot \tilde{\varphi}(w) = \sigma_i \cdot \varphi(w),$$

where the last σ_i acts via the G-action on U. This determines $\tilde{\varphi}$ on each summand, hence on all of V.

Existence and G-equivariance. Define $\tilde{\varphi}(\sigma_i w) = \sigma_i \cdot \varphi(w)$. To verify G-equivariance: let $g \in G$ act on V by sending $\sigma_i W$ to $\sigma_j W$ where $g\sigma_i = \sigma_j h$ for $h \in H$. For $\sigma_i w \in \sigma_i W$:

$$\tilde{\varphi}(g(\sigma_i w)) = \tilde{\varphi}(\sigma_j(hw)) = \sigma_j \cdot \varphi(hw) = \sigma_j \cdot h \cdot \varphi(w) = g \cdot \sigma_i \cdot \varphi(w) = g \cdot \tilde{\varphi}(\sigma_i w),$$

using H-equivariance of φ in the third equality and $\sigma_j h = g\sigma_i$ in the fourth. So $\tilde{\varphi}$ is G-equivariant.

Conversely, given $\tilde{\varphi} \in \mathrm{Hom}_G(V, U)$, its restriction to $W \subset V$ is H-equivariant. These two maps are inverse to each other. □

Corollary 15.3.1 Frobenius reciprocity (character form)

For any representation U of G and W of H,

$$\langle \chi_{\mathrm{Ind}_H^G W}, \chi_U \rangle_G = \langle \chi_W, \chi_{\mathrm{Res}_H^G U} \rangle_H.$$

Note:-

Thus, if U is an irreducible representation of G and W an irreducible representation of H, the number of times W appears in $\mathrm{Res}_H^G(U)$ equals the number of times U appears in $\mathrm{Ind}_H^G(W)$.

15.4 Dimension Formulas and Irreducibility

Proposition 15.4.1 Dimension of induced representations

$\dim \mathrm{Ind}_H^G W = [G : H] \cdot \dim W$. This is immediate from the construction $V = \bigoplus_{\sigma \in G/H} \sigma W$.

Note:-

An induced representation is almost never irreducible. For instance, $\mathrm{Ind}_{\{e\}}^{G}(\mathbb{C}) = \mathbb{C}[G]$ (the regular representation), which decomposes into all irreducibles of G. Frobenius reciprocity gives a criterion: $\mathrm{Ind}_H^G W$ is irreducible iff $H(\chi_{\mathrm{Ind}\,W}, \chi_{\mathrm{Ind}\,W}) = 1$, which by Frobenius equals $\langle \chi_{\mathrm{Ind}\,W}, \chi_{\mathrm{Ind}\,W}\rangle_G = \langle \chi_W, \chi_{\mathrm{Res}\,\mathrm{Ind}\,W}\rangle_H$. Analyzing $\mathrm{Res}_H^G \mathrm{Ind}_H^G W$ leads to Mackey's formula, which we state without proof.

Proposition 15.4.2 Mackey's irreducibility criterion

Let W be an irreducible representation of $H \leqslant G$. Then $\mathrm{Ind}_H^G W$ is irreducible if and only if for every $g \in G \setminus H$, the representations W and W^g of $H \cap gHg^{-1}$ have no common irreducible constituent, where W^g denotes the conjugate representation $W^g(h) = W(g^{-1}hg)$ restricted to $H \cap gHg^{-1}$.

Example 15.4.1 (Induction from normal subgroups)

If $H \trianglelefteq G$ and W is an irreducible representation of H, then $\mathrm{Ind}_H^G W$ is irreducible iff for all $g \in G \setminus H$, the conjugate W^g is not isomorphic to W as an H-representation. In other words, W is not fixed by any nontrivial element of G/H acting on the set of irreducible representations of H by conjugation.

Example 15.4.2 (Inducing from index-2 subgroups)

Let $[G : H] = 2$ and W an irreducible representation of H. Since H is normal of index 2, there is a single nontrivial coset representative g. Then $\mathrm{Ind}_H^G W$ is irreducible iff $W \not\cong W^g$, and splits as a direct sum of two irreducibles (each of dimension $\dim W$) when $W \cong W^g$.
For example, let $G = S_3$ and $H = A_3 \cong \mathbb{Z}/3$. The nontrivial irreducibles of A_3 are the characters $\chi_\omega : (123) \mapsto \omega$ and $\chi_{\omega^2} : (123) \mapsto \omega^2$ (where $\omega = e^{2\pi i/3}$). Conjugation by (12) swaps $\chi_\omega \leftrightarrow \chi_{\omega^2}$, so $\chi_\omega \not\cong \chi_\omega^g$ and $\mathrm{Ind}_{A_3}^{S_3} \chi_\omega$ is irreducible. Since $\dim = [S_3 : A_3] \cdot 1 = 2$, this is the standard representation V of S_3.

15.5 Example: $S_4 \supset S_3$

Example 15.5.1 (Restriction and induction between S_4 and S_3)

Let $G = S_4$ with irreducibles U_4, U_4', V_4, V_4', W (dimensions 1, 1, 3, 3, 2), and $H = S_3$ with irreducibles U_3, U_3', V_3 (dimensions 1, 1, 2). We compute restrictions using character tables (restricting the character of a representation of S_4 to elements of $S_3 \hookrightarrow S_4$):

- $\mathrm{Res}(U_4) = U_3$ and $\mathrm{Res}(U_4') = U_3'$ (trivial and sign restrict to trivial and sign).
- $\mathrm{Res}(V_4) = V_3 \oplus U_3$. Indeed, the permutation representation $\mathbb{C}^4$ restricts as $\mathbb{C}^3 \oplus \mathbb{C}$ (the 4th coordinate is fixed by S_3), so $\mathrm{Res}(V_4 \oplus U_4) = V_3 \oplus U_3 \oplus U_3$.
- $\mathrm{Res}(V_4') = V_3 \oplus U_3'$ (since $V_4' = V_4 \otimes U_4'$ and $V_3 \otimes U_3' \cong V_3$).
- $\mathrm{Res}(W) = V_3$. The representation W factors through $S_4/\{e, (12)(34), (13)(24), (14)(23)\} \cong S_3$, and W as a representation of this quotient is the standard representation V_3.

By Frobenius reciprocity, we can read off the induced representations:

- $\mathrm{Ind}_{S_3}^{S_4}(V_3)$ contains V_4, V_4', W each with multiplicity 1, giving $\mathrm{Ind}(V_3) = V_4 \oplus V_4' \oplus W$ (dim $= 3 + 3 + 2 = 8 = [S_4 : S_3] \cdot 2$).
- $\mathrm{Ind}_{S_3}^{S_4}(U_3) = U_4 \oplus V_4$ (dim $= 1 + 3 = 4 = [S_4 : S_3] \cdot 1$).
- $\mathrm{Ind}_{S_3}^{S_4}(U_3') = U_4' \oplus V_4'$ (dim $= 1 + 3 = 4$).

Exercise 15.5.1 Exercises

(1) Let $H = \{e\} \leqslant G$. Show that $\operatorname{Ind}_{\{e\}}^G W \cong W \otimes R$ where R is the regular representation of G and W is viewed as a vector space with trivial G-action on the second factor.

(2) Let $H = A_n \leqslant S_n$ for $n \geqslant 2$. Show that $\operatorname{Ind}_{A_n}^{S_n}(U)$, where U is the trivial representation of A_n, is isomorphic to $U_{S_n} \oplus U'_{S_n}$ (the trivial plus the sign representation of S_n).

(3) Using Frobenius reciprocity and the character table of S_5, compute $\operatorname{Ind}_{S_4}^{S_5}(V_4)$ (where V_4 is the standard representation of S_4, embedded as the subgroup fixing 5).

(4) Prove that the character formula for induction can alternatively be written as $\chi_{\operatorname{Ind} W}(g) = \sum_{i:\sigma_i^{-1} g\sigma_i \in H} \chi_W(\sigma_i^{-1} g\sigma_i)$ where $\sigma_1, \ldots, \sigma_k$ are coset representatives.

(5) (Transitivity of induction) If $K \leqslant H \leqslant G$, show $\operatorname{Ind}_H^G \operatorname{Ind}_K^H W \cong \operatorname{Ind}_K^G W$.

Solution

(1) $\operatorname{Ind}_{\{e\}}^G W = \bigoplus_{g\in G} gW$, a direct sum of $|G|$ copies of W, with G permuting the summands. This is exactly $W \otimes R$ where the G-action is on the R factor.
(2) $\operatorname{Ind}_{A_n}^{S_n}(U) = \bigoplus_{\sigma \in S_n/A_n} \sigma U$. Since $[S_n : A_n] = 2$, this is 2-dimensional with basis e_1, e_2 indexed by the two cosets. Even permutations fix both; odd permutations swap them. The invariant subspace $\operatorname{span}(e_1 + e_2) \cong U_{S_n}$, and the anti-invariant $\operatorname{span}(e_1 - e_2) \cong U'_{S_n}$.
(3) By Frobenius reciprocity, the multiplicity of an S_5-irreducible X in $\operatorname{Ind}_{S_4}^{S_5}(V_4)$ equals $\langle \chi_{V_4}, \chi_{\operatorname{Res}(X)}\rangle_{S_4}$. We restrict each S_5-irreducible to S_4 (the subgroup fixing 5) by evaluating its character on elements of cycle types $1^5, 2{\cdot}1^3, 2^2{\cdot}1, 3{\cdot}1^2, 4{\cdot}1$. The S_4 inner product with $\chi_{V_4} = (3, 1, -1, 0, -1)$ (class sizes 1, 6, 3, 8, 6) gives:

$$\begin{aligned}
\langle \operatorname{Res}(U), \chi_{V_4}\rangle &= \tfrac{1}{24}(3 + 6 - 3 + 0 - 6) = 0,\\
\langle \operatorname{Res}(U'), \chi_{V_4}\rangle &= \tfrac{1}{24}(3 - 6 - 3 + 0 + 6) = 0,\\
\langle \operatorname{Res}(V), \chi_{V_4}\rangle &= \tfrac{1}{24}(12 + 12 + 0 + 0 + 0) = 1,\\
\langle \operatorname{Res}(V'), \chi_{V_4}\rangle &= \tfrac{1}{24}(12 - 12 + 0 + 0 + 0) = 0,\\
\langle \operatorname{Res}(\textstyle\bigwedge^2 V), \chi_{V_4}\rangle &= \tfrac{1}{24}(18 + 0 + 6 + 0 + 0) = 1,\\
\langle \operatorname{Res}(W), \chi_{V_4}\rangle &= \tfrac{1}{24}(15 + 6 - 3 + 0 + 6) = 1,\\
\langle \operatorname{Res}(W'), \chi_{V_4}\rangle &= \tfrac{1}{24}(15 - 6 - 3 + 0 - 6) = 0.
\end{aligned}$$

So $\operatorname{Ind}_{S_4}^{S_5}(V_4) = V \oplus \bigwedge^2 V \oplus W$ (dimensions $4 + 6 + 5 = 15 = [S_5 : S_4] \cdot 3$).
(4) The sum $\sum_{i:\sigma_i^{-1} g\sigma_i \in H} \chi_W(\sigma_i^{-1} g\sigma_i)$ counts each valid conjugate exactly once per coset. In the formula $\frac{1}{|H|}\sum_{s\in G, s^{-1}gs\in H}$, replacing s by $\sigma_i h$ for $h \in H$: the condition $s^{-1}gs \in H$ becomes $h^{-1}(\sigma_i^{-1} g\sigma_i)h \in H$, which holds iff $\sigma_i^{-1} g\sigma_i \in H$. The $|H|$ values of h give $\chi_W(h^{-1}(\sigma_i^{-1} g\sigma_i)h) = \chi_W(\sigma_i^{-1} g\sigma_i)$ (class function), contributing $|H|$ times the same value. Dividing by $|H|$ recovers the coset-representative sum.
(5) Write coset representatives for G/K by combining those for G/H and H/K: if $\{\sigma_i\}$ represents G/H and $\{\tau_j\}$ represents H/K, then $\{\sigma_i\tau_j\}$ represents G/K. The direct sum decompositions are compatible: $\operatorname{Ind}_K^G W = \bigoplus_{i,j} \sigma_i\tau_j W = \bigoplus_i \sigma_i(\bigoplus_j \tau_j W) = \bigoplus_i \sigma_i(\operatorname{Ind}_K^H W) = \operatorname{Ind}_H^G(\operatorname{Ind}_K^H W)$.

Lecture 16.

Real and Quaternionic Representations

We have studied actions of finite groups on complex vector spaces. Now we consider the real case and the relationship between real and complex representations.

16.1 Real Representations

Let G be a finite group. A real representation of G is a real vector space V_0 together with a group homomorphism $G \to \mathrm{GL}(V_0)$ (automorphisms as an $\mathbb{R}$-vector space).

The averaging trick works over $\mathbb{R}$: starting from any inner product on V_0 and averaging over G, we obtain a G-invariant (positive-definite, symmetric) inner product. So if $W_0 \subset V_0$ is a G-invariant subspace, then $W_0^\perp$ is also G-invariant, and every real representation of a finite group is completely reducible.

However, Schur's lemma in its strong form fails over $\mathbb{R}$. For instance, $\mathbb{Z}/n$ acts on $\mathbb{R}^2$ by rotation by $2\pi/n$, which is irreducible (for $n \geqslant 3$) but has nontrivial endomorphisms: any rotation commuting with rotation by $2\pi/n$ works, so $\mathrm{End}_G(\mathbb{R}^2) \cong \mathbb{C}$ (not just $\mathbb{R}$).

16.2 Complexification

The main tool for relating real and complex representations is complexification.

Definition 16.2.1: Complexification

Given a real representation V_0 of G, its complexification is the complex vector space $V = V_0 \otimes_{\mathbb{R}} \mathbb{C} = V_0 \oplus iV_0$, with G acting by $g(v + iw) = gv + i\,gw$ for $v, w \in V_0$. This defines a functor $\{\text{real reps of } G\} \to \{\text{complex reps of } G\}$.

Definition 16.2.2: Real complex representation

A complex representation V of G is called real if there exists a real representation V_0 such that $V \cong V_0 \otimes_{\mathbb{R}} \mathbb{C}$.

Proposition 16.2.1 Necessary condition

If $V = V_0 \otimes_{\mathbb{R}} \mathbb{C}$ is real, then χ_V takes values in $\mathbb{R}$. (The trace of a matrix with real entries is real.)

The converse is also true for irreducible representations, as we will show.

16.3 Characterizing Real Representations

Proposition 16.3.1 Antilinear involution criterion

A complex representation V of G is real if and only if there exists a G-equivariant, complex antilinear map $\tau : V \to V$ (i.e. $\tau(\lambda v) = \bar{\lambda}\tau(v)$ for all $\lambda \in \mathbb{C}$) with $\tau^2 = \mathrm{id}$.

Proof: ($\Rightarrow$) If $V = V_0 \otimes_{\mathbb{R}} \mathbb{C}$, define $\tau(v + iw) = v - iw$ for $v, w \in V_0$ (complex conjugation with respect to the real form V_0). Then τ is antilinear, $\tau^2 = \mathrm{id}$, and G-equivariant since g preserves V_0.

($\Leftarrow$) Given such a τ, every $v \in V$ decomposes as

$$v = \underbrace{\frac{v + \tau(v)}{2}}_{\in V^\tau} + \underbrace{\frac{v - \tau(v)}{2}}_{\in V^{-\tau}}$$

where $V^\tau = \ker(\tau - \mathrm{id})$ is the $+1$-eigenspace and $V^{-\tau} = \ker(\tau + \mathrm{id})$ is the -1-eigenspace. Since τ is antilinear, $\tau(iv) = -i\tau(v)$, so $V^{-\tau} = iV^\tau$. Thus V^τ is an $\mathbb{R}$-subspace (not a $\mathbb{C}$-subspace!) with $V = V^\tau \oplus iV^\tau = V^\tau \otimes_{\mathbb{R}} \mathbb{C}$. Since τ is G-equivariant, $V_0 := V^\tau$ is preserved by G, giving a real representation with $V = V_0 \otimes_{\mathbb{R}} \mathbb{C}$. $\square$

16.4 The Frobenius–Schur Indicator

For irreducible representations, we can determine whether a representation is real, quaternionic, or neither using the bilinear form it carries.

> **Proposition 16.4.1** Invariant bilinear form from self-duality
>
> Let V be an irreducible complex representation of G with χ_V real-valued. Then $\chi_V = \overline{\chi_V} = \chi_{V^*}$, so $V \cong V^*$. By Schur's lemma, $\dim \mathrm{Hom}_G(V, V^*) = 1$, so there is a G-invariant bilinear form $B : V \times V \to \mathbb{C}$, unique up to scaling, and nondegenerate.

Since $B \in (V \otimes V)^* \cong \mathrm{Sym}^2 V^* \oplus \bigwedge^2 V^*$, and both the symmetric and skew-symmetric parts are also G-invariant, uniqueness forces exactly one of them to be zero. So B is either symmetric or skew-symmetric. This dichotomy distinguishes real from quaternionic representations.

> **Proposition 16.4.2** Real iff symmetric form
>
> An irreducible complex representation V of a finite group G is real if and only if V carries a G-invariant nondegenerate symmetric bilinear form $B : V \times V \to \mathbb{C}$.

Proof: ($\Rightarrow$) If $V = V_0 \otimes_{\mathbb{R}} \mathbb{C}$, then V_0 carries a G-invariant real inner product B_0 (by the averaging argument). Extend $\mathbb{C}$-bilinearly:

$$B(v_1 + iw_1,\, v_2 + iw_2) = B_0(v_1, v_2) + iB_0(w_1, v_2) + iB_0(v_1, w_2) - B_0(w_1, w_2).$$

This is a nondegenerate, symmetric, G-invariant $\mathbb{C}$-bilinear form on V.

($\Leftarrow$) Given B, the form determines a G-equivariant isomorphism $\varphi : V \xrightarrow{\sim} V^*$. Choose a G-invariant Hermitian inner product H on V (by averaging), which gives a G-equivariant, antilinear isomorphism $V \xrightarrow{\sim} V^*$ (via $v \mapsto H(-, v)$). Composing one with the inverse of the other yields a G-equivariant, antilinear map $\tau : V \to V$, characterized by $H(\tau(v), w) = B(v, w)$.

Now $\tau^2 : V \to V$ is $\mathbb{C}$-linear and G-equivariant, so by Schur's lemma, $\tau^2 = \lambda \cdot \mathrm{id}$ for some $\lambda \in \mathbb{C}$. We show $\lambda \in \mathbb{R}_{>0}$:

$$H(\tau^2(v), v) = B(\tau(v), v) = B(v, \tau(v)) = H(\tau(v), \tau(v)) \geqslant 0$$

where the second equality uses symmetry of B. So $\lambda \geqslant 0$, and $\lambda > 0$ since $\tau \neq 0$. Replacing τ by $\lambda^{-1/2}\tau$ (which is still antilinear and G-equivariant), we arrange $\tau^2 = \mathrm{id}$. By the antilinear involution criterion, V is real. $\square$

16.5 Quaternionic Representations

In the other case, the invariant bilinear form B is skew-symmetric. The same argument produces a G-equivariant antilinear map $J : V \to V$ with $J^2 = -\mathrm{id}$.

Definition 16.5.1: Quaternionic structure

A quaternionic structure on a complex vector space V is a $\mathbb{C}$-antilinear map $J : V \to V$ with $J^2 = -\mathrm{id}$. A representation with such a G-equivariant J is called a quaternionic representation.

Such a J makes V into a module over the quaternions $\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}$, the division algebra (noncommutative ring where every nonzero element has a multiplicative inverse) with $i^2 = j^2 = k^2 = ijk = -1$. Concretely, $\mathbb{H} = \mathbb{C} \oplus \mathbb{C}j$ with $ji = -ij$, $j^2 = -1$, so an $\mathbb{H}$-module is a $\mathbb{C}$-vector space with an antilinear map J ("right multiplication by j") satisfying $J^2 = -\mathrm{id}$.

Note:-

A quaternionic representation has χ_V real-valued (since $V \cong V^*$) but does not come from a real representation. The three cases for an irreducible V with real character are:

- V admits a G-invariant nondegenerate symmetric bilinear form $\iff$ V is real $\iff$ $V = V_0 \otimes_{\mathbb{R}} \mathbb{C}$.
- V admits a G-invariant nondegenerate skew-symmetric bilinear form $\iff$ V is quaternionic $\iff$ V carries a G-equivariant antilinear J with $J^2 = -\mathrm{id}$.

And if χ_V is not real-valued, then $V \not\cong V^*$ and V is neither real nor quaternionic.

16.6 The Frobenius–Schur Indicator

The symmetric/skew dichotomy can be detected numerically.

Theorem 16.6.1 Frobenius–Schur indicator

For an irreducible representation V of a finite group G,

$$\nu(V) := \frac{1}{|G|}\sum_{g \in G} \chi_V(g^2) = \begin{cases} 1 & \text{if } V \text{ is real,} \\ 0 & \text{if } \chi_V \notin \mathbb{R}, \\ -1 & \text{if } V \text{ is quaternionic.} \end{cases}$$

Proof: Using the character formulas $\chi_{\mathrm{Sym}^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 + \chi_V(g^2))$ and $\chi_{\bigwedge^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 - \chi_V(g^2))$, we get

$$\nu(V) = \frac{1}{|G|}\sum_{g} \chi_V(g^2) = \dim(\mathrm{Sym}^2 V)^G - \dim(\bigwedge^2 V)^G.$$

Now $(\mathrm{Sym}^2 V)^G$ counts the multiplicity of the trivial representation in $\mathrm{Sym}^2 V$, and similarly for $\bigwedge^2 V$. A G-invariant bilinear form on V (a G-fixed element of $(V \otimes V)^*$) is either symmetric or skew-symmetric.

If $V \not\cong V^*$, then $\mathrm{Hom}_G(V, V^*) = 0$, so no invariant bilinear form exists. Both dimensions are 0, giving $\nu(V) = 0$.

If $V \cong V^*$ (i.e., χ_V real-valued), then $\dim \mathrm{Hom}_G(V, V^*) = 1$ by Schur. The unique form is either symmetric or skew, so exactly one of the two dimensions equals 1. The form is symmetric iff V is real ($\nu = 1 - 0 = 1$), and skew iff V is quaternionic ($\nu = 0 - 1 = -1$). □

Proposition 16.6.1 Quaternionic representations have even dimension

If V is quaternionic, then $\dim_{\mathbb{C}} V$ is even, since J with $J^2 = -\mathrm{id}$ makes V into an $\mathbb{H}$-module and $\dim_{\mathbb{C}} V = 2\dim_{\mathbb{H}} V$. In particular, an irreducible representation with real character and odd dimension is automatically real.

16.7 Examples

Example 16.7.1 (Standard representation of S_3)

The 2-dimensional irreducible V of $S_3 \cong D_3$ is real: S_3 acts on $V_0 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : \sum x_i = 0\} \cong \mathbb{R}^2$ by rotations and reflections, and $V = V_0 \otimes_{\mathbb{R}} \mathbb{C}$.
Alternatively: $V^* \cong V$ (since χ_V is real-valued), and $\bigwedge^2 V^* \cong U'$ contains no trivial summand, so the G-invariant bilinear form must be in $\mathrm{Sym}^2 V^*$. Since $\mathrm{Sym}^2 V^* \cong U \oplus V$ contains a trivial summand, an invariant symmetric form exists, confirming V is real.

Example 16.7.2 (The quaternion group Q_8)

The quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$ acts faithfully on $\mathbb{C}^2$ by

$$\pm 1 \mapsto \pm I, \quad \pm i \mapsto \pm \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \pm j \mapsto \pm \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \pm k \mapsto \pm \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

The character takes values $\chi(\pm 1) = \pm 2$ and $\chi = 0$ on all other elements—all real. However, this representation is <u>not</u> real: it is quaternionic.
To see this cannot come from a real representation: if Q_8 acted faithfully on $\mathbb{R}^2$, the invariant inner product would give $Q_8 \hookrightarrow O(2)$. But $O(2)$ has only 2 elements squaring to $-I$ (rotations by $\pm 90°$), while Q_8 needs 6 elements ($\pm i, \pm j, \pm k$) with square -1. Contradiction.
The quaternionic structure is: view $\mathbb{C}^2 = \mathbb{H}$ via $(z_1, z_2) \leftrightarrow z_1 + jz_2$, and the above matrices correspond to left multiplication by $\pm 1, \pm i, \pm j, \pm k$. The antilinear map $J : V \to V$ with $J^2 = -\mathrm{id}$ is right multiplication by j, which commutes with left multiplication.

Exercise 16.7.1 Exercises

(1) Classify each irreducible representation of S_4 as real, quaternionic, or complex-type. (Hint: all characters of S_4 are real-valued, so each is either real or quaternionic. Use the dimension criterion: a quaternionic representation must have even dimension.)

(2) Show that if V is a quaternionic representation of G (carrying an antilinear J with $J^2 = -\mathrm{id}$), then $\dim_{\mathbb{C}} V$ is even.

(3) Prove the Frobenius–Schur indicator formula: for an irreducible representation V,

$$\frac{1}{|G|} \sum_{g \in G} \chi_V(g^2) = \begin{cases} 1 & \text{if } V \text{ is real,} \\ 0 & \text{if } \chi_V \notin \mathbb{R}, \\ -1 & \text{if } V \text{ is quaternionic.} \end{cases}$$

(Hint: $\frac{1}{|G|} \sum_g \chi_V(g^2) = \dim(\mathrm{Sym}^2 V)^G - \dim(\bigwedge^2 V)^G$, and exactly one of these dimensions is 1 when $V \cong V^*$.)

(4) Using the Frobenius–Schur indicator, verify that the 2-dimensional irreducible of Q_8 is quaternionic.

(5) Show that the representations Y and Z of A_5 (the 3-dimensional irreducibles with golden-ratio character values) are real representations. (Their characters are real, and they have odd dimension.)

Solution

(1) All characters of S_4 are real-valued (check the character table). A quaternionic representation has $J : V \to V$ with $J^2 = -\mathrm{id}$, and J pairs basis vectors into orbits of size 2, so $\dim V$ must be even.

The irreducibles of S_4 have dimensions 1, 1, 2, 3, 3. The odd-dimensional ones (U, U', V_4, V_4') are automatically real. For W (dim = 2): compute the Frobenius–Schur indicator $\frac{1}{24}\sum \chi_W(g^2)$. We have $\chi_W = (2, 0, 2, -1, 0)$ with class sizes $(1, 6, 3, 8, 6)$. Squaring: $e \to e, (12) \to e, (12)(34) \to e, (123) \to (132), (1234) \to (13)(24)$. So $\chi_W(g^2) = (2, 2, 2, -1, 2)$. The indicator is $\frac{1}{24}(2 + 12 + 6 - 8 + 12) = 1$. So W is also real.

(2) The antilinear map J gives an $\mathbb{R}$-linear map $V \to V$ whose square is $-\mathrm{id}$. Viewing V as $\mathbb{R}^{2n}$ (where $\dim_{\mathbb{C}} V = n$), this gives a real matrix M with $M^2 = -I$. The characteristic polynomial $\det(M - tI)$ has degree $2n$ and all roots are $\pm i$, which come in conjugate pairs. More directly: J makes V into an $\mathbb{H}$-module, and $\dim_{\mathbb{H}} V = \frac{1}{2}\dim_{\mathbb{C}} V$ must be a positive integer.

(3) We have $\frac{1}{|G|}\sum_g \chi_V(g^2) = \frac{1}{|G|}\sum_g \mathrm{Tr}(\rho(g^2))$. Using the decomposition $V \otimes V = \mathrm{Sym}^2 V \oplus \bigwedge^2 V$ and the identities $\chi_{\mathrm{Sym}^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 + \chi_V(g^2))$, $\chi_{\bigwedge^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 - \chi_V(g^2))$, we get the indicator $= \dim(\mathrm{Sym}^2 V)^G - \dim(\bigwedge^2 V)^G$. When $V \cong V^*$, the unique invariant bilinear form is either symmetric (contributing 1 to Sym^2) or skew (contributing 1 to $\bigwedge^2$). When $V \not\cong V^*$, neither $\mathrm{Sym}^2 V$ nor $\bigwedge^2 V$ contains a trivial summand, giving 0.

(4) Q_8 has 5 conjugacy classes: $\{1\}, \{-1\}, \{\pm i\}, \{\pm j\}, \{\pm k\}$ of sizes 1, 1, 2, 2, 2. For the 2-dimensional rep: $\chi = (2, -2, 0, 0, 0)$. Squaring: $1^2 = 1$, $(-1)^2 = 1$, $(\pm i)^2 = -1$, $(\pm j)^2 = -1$, $(\pm k)^2 = -1$. So $\chi(g^2) = (2, 2, -2, -2, -2)$. Indicator: $\frac{1}{8}(2 + 2 - 4 - 4 - 4) = -1$. Quaternionic.

(5) Both Y and Z have real characters and odd dimension 3. Since a quaternionic representation must have even complex dimension (exercise 2), odd-dimensional representations with real characters are automatically real.

Lecture 17.

Further Topics in Algebra

This chapter collects three subjects that extend the algebra developed in earlier lectures: a deeper study of commutative rings (building toward unique factorization), Galois theory (connecting field extensions to group theory), and an introduction to Lie groups and Lie algebras (the continuous analogues of finite groups and their representations). These topics go beyond the Math 55a syllabus but follow naturally from the material covered so far.

I. Commutative Ring Theory

17.1 Integral Domains and Divisibility

We work with commutative rings with identity throughout.

Definition 17.1.1: Integral domain

A commutative ring R (with $1 \neq 0$) is an integral domain if it has no zero divisors: $ab = 0 \implies a = 0$ or $b = 0$. Equivalently, the cancellation law holds: $ab = ac$ and $a \neq 0$ implies $b = c$.

Example 17.1.1

The rings $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}[i]$, and $k[x]$ (for k a field) are integral domains. The ring $\mathbb{Z}/6$ is not ($2 \cdot 3 = 0$). A field is an integral domain; $\mathbb{Z}/p$ is a field (hence integral domain) for p prime.

Definition 17.1.2: Units, irreducibles, primes

Let R be an integral domain.

- A unit is an element $u \in R$ with a multiplicative inverse: $uv = 1$ for some $v \in R$. The units form a group $R^\times$.
- Two elements a, b are associates if $a = ub$ for some unit u.
- A nonzero non-unit $p \in R$ is irreducible if $p = ab$ implies a or b is a unit.
- A nonzero non-unit $p \in R$ is prime if $p \mid ab$ implies $p \mid a$ or $p \mid b$.

Proposition 17.1.1 Prime implies irreducible

In any integral domain, every prime element is irreducible.

Proof: Suppose p is prime and $p = ab$. Then $p \mid ab$, so $p \mid a$ or $p \mid b$. Say $p \mid a$, so $a = pc$. Then $p = pcb$, and canceling p gives $1 = cb$, so b is a unit. $\square$

Note:-

The converse (irreducible $\implies$ prime) holds in PIDs and UFDs but fails in general. In $\mathbb{Z}[\sqrt{-5}]$, the element 2 is irreducible but not prime: $2 \mid (1+\sqrt{-5})(1-\sqrt{-5}) = 6$ but $2 \nmid (1+\sqrt{-5})$ and $2 \nmid (1-\sqrt{-5})$.

17.2 Euclidean Domains

Definition 17.2.1: Euclidean domain

An integral domain R is a Euclidean domain (ED) if there exists a function $N : R \setminus \{0\} \to \mathbb{Z}_{\geqslant 0}$ (the Euclidean norm) such that for any $a, b \in R$ with $b \neq 0$, there exist $q, r \in R$ with

$$a = bq + r, \qquad \text{where } r = 0 \text{ or } N(r) < N(b).$$

Example 17.2.1 (Euclidean domains)

- $\mathbb{Z}$ with $N(a) = |a|$.
- $k[x]$ (for k a field) with $N(f) = \deg(f)$.
- The Gaussian integers $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ with $N(a + bi) = a^2 + b^2$.

- The Eisenstein integers $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$ where $\omega = e^{2\pi i/3}$, with $N(a + b\omega) = a^2 - ab + b^2$.

17.3 Principal Ideal Domains

Definition 17.3.1: Principal ideal domain

An integral domain R is a principal ideal domain (PID) if every ideal $I \subseteq R$ is principal, i.e., $I = (a) = aR$ for some $a \in R$.

Theorem 17.3.1 Every Euclidean domain is a PID

If R is a Euclidean domain with norm N, then R is a PID.

Proof: Let $I \subseteq R$ be a nonzero ideal. Among all nonzero elements of I, choose b with $N(b)$ minimal. We claim $I = (b)$. Clearly $(b) \subseteq I$. For any $a \in I$, write $a = bq + r$ with $r = 0$ or $N(r) < N(b)$. Since $r = a - bq \in I$ and $N(b)$ was minimal, we must have $r = 0$, so $a = bq \in (b)$. □

Note:-

The converse fails: there exist PIDs that are not Euclidean domains. The ring $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$ is a PID but not an ED (a result of Motzkin, 1949).

Theorem 17.3.2 In a PID, irreducible $\iff$ prime

Let R be a PID and $p \in R$ a nonzero non-unit. Then p is irreducible if and only if p is prime. Equivalently, the ideal (p) is prime if and only if it is maximal among proper principal ideals.

Proof: ($\Leftarrow$) Already proved: prime $\implies$ irreducible in any integral domain.

($\Rightarrow$) Suppose p is irreducible and $p \mid ab$. Consider the ideal (p, a). Since R is a PID, $(p, a) = (d)$ for some d. Then $d \mid p$, so either d is a unit (hence $(p, a) = R$, so $1 = rp + sa$ for some r, s, and $b = rpb + sab$; since $p \mid ab$, we get $p \mid b$) or d is an associate of p (hence $p \mid a$). □

17.4 Unique Factorization Domains

Definition 17.4.1: Unique factorization domain

An integral domain R is a unique factorization domain (UFD) if:

(i) Every nonzero non-unit $a \in R$ can be written as a product of irreducibles: $a = p_1 p_2 \cdots p_n$.

(ii) This factorization is unique up to order and associates: if $a = p_1 \cdots p_n = q_1 \cdots q_m$, then $n = m$ and (after reordering) each p_i is an associate of q_i.

Theorem 17.4.1 Every PID is a UFD

Proof (sketch): Existence of factorization follows from the ascending chain condition on principal ideals in a PID: if a is not irreducible, write $a = bc$ with neither b nor c a unit. Then $(a) \subsetneq (b)$ and $(a) \subsetneq (c)$. Repeat for b and c. This process terminates because an infinite strictly ascending chain $(a_1) \subsetneq (a_2) \subsetneq \cdots$ would generate an ideal $\bigcup (a_i)$ that is not principal, contradicting the PID hypothesis.

Uniqueness follows from the fact that irreducibles are prime in a PID: if $p_1 \cdots p_n = q_1 \cdots q_m$, then $p_1 \mid q_1 \cdots q_m$, so $p_1 \mid q_j$ for some j; since q_j is irreducible, p_1 and q_j are associates. Cancel and proceed by induction. □

The chain of implications is:

$$\text{Euclidean domain} \implies \text{PID} \implies \text{UFD} \implies \text{integral domain.}$$

None of these implications reverse.

Example 17.4.1 (Failure of unique factorization)
In $\mathbb{Z}[\sqrt{-5}] = \{a+b\sqrt{-5} : a, b \in \mathbb{Z}\}$, we have $6 = 2\cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$. The norm $N(a+b\sqrt{-5}) = a^2 + 5b^2$ is multiplicative: $N(\alpha\beta) = N(\alpha)N(\beta)$. Since $N(2) = 4$, $N(3) = 9$, $N(1 \pm \sqrt{-5}) = 6$, and no element has norm 2 or 3, each of these four elements is irreducible. But 2 and $1 + \sqrt{-5}$ are not associates (they have different norms). So $\mathbb{Z}[\sqrt{-5}]$ is not a UFD.

17.5 Noetherian Rings

Definition 17.5.1: Noetherian ring

A commutative ring R is Noetherian if it satisfies the ascending chain condition (ACC) on ideals: every ascending chain $I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$ of ideals eventually stabilizes ($I_n = I_{n+1} = \cdots$ for some n). Equivalently, every ideal of R is finitely generated.

Note:-
Every PID is Noetherian (every ideal is generated by one element). Every field is Noetherian. The ring of integers $\mathbb{Z}$ is Noetherian. However, the polynomial ring $k[x_1, x_2, \dots]$ in infinitely many variables is not Noetherian: $(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \cdots$.

Theorem 17.5.1 Hilbert Basis Theorem
If R is a Noetherian ring, then so is the polynomial ring $R[x]$.

Proof: Let $I \subseteq R[x]$ be an ideal. For each $n \geqslant 0$, let $L_n \subseteq R$ be the set of leading coefficients of polynomials in I of degree $\leqslant n$, together with 0. Each L_n is an ideal of R, and $L_0 \subseteq L_1 \subseteq L_2 \subseteq \cdots$. Since R is Noetherian, this chain stabilizes at some L_N.

For each $n \leqslant N$, the ideal L_n is finitely generated (since R is Noetherian), say by leading coefficients of polynomials $f_{n,1}, \dots, f_{n,k_n} \in I$ of degree $\leqslant n$. We claim these finitely many polynomials generate I.

Given $g \in I$ of degree d, if $d > N$, the leading coefficient of g lies in $L_N = L_d$, so we can subtract an $R[x]$-linear combination of the $f_{N,j}$ (multiplied by appropriate powers of x) to reduce the degree. If $d \leqslant N$, the leading coefficient lies in L_d, and we subtract a combination of the $f_{d,j}$ to reduce the degree further. This process terminates, producing g as a combination of the chosen generators. □

Corollary 17.5.1
For any Noetherian ring R and any $n \geqslant 1$, the polynomial ring $R[x_1, \dots, x_n]$ is Noetherian. In particular, $k[x_1, \dots, x_n]$ is Noetherian for any field k.

Theorem 17.5.2 Gauss's Lemma and polynomial UFDs
If R is a UFD, then so is $R[x]$. In particular, $\mathbb{Z}[x]$ and $k[x_1, \dots, x_n]$ are UFDs.

Proof (idea): A polynomial $\mathsf{f} \in \mathsf{R}[\mathsf{x}]$ is primitive if the gcd of its coefficients is 1. Gauss's lemma states that the product of primitive polynomials is primitive. The key step: an irreducible element of $\mathsf{R}[\mathsf{x}]$ is either an irreducible of R (viewed as a constant polynomial) or a primitive polynomial that is irreducible over the fraction field of R. Factoring out content and using unique factorization in R and in $\mathrm{Frac}(\mathsf{R})[\mathsf{x}]$ (a PID, hence UFD) gives the result. $\square$

II. Galois Theory

17.6 Field Extensions

Definition 17.6.1: Field extension

A field extension L/K is a field L containing K as a subfield. The degree $[L : K] = \dim_K L$ is the dimension of L as a K-vector space. An extension is finite if $[L : K] < \infty$.

Proposition 17.6.1 Tower law

If $K \subseteq L \subseteq M$ are fields, then $[M : K] = [M : L] \cdot [L : K]$.

Proof: If $\{e_i\}$ is an L-basis for M and $\{f_j\}$ is a K-basis for L, then $\{e_i f_j\}$ is a K-basis for M. $\square$

Definition 17.6.2: Algebraic and transcendental elements

Let L/K be a field extension and $\alpha \in L$.

- α is algebraic over K if there exists a nonzero polynomial $f \in K[x]$ with $f(\alpha) = 0$.
- α is transcendental over K if no such polynomial exists.
- The minimal polynomial of an algebraic element α is the unique monic polynomial $m_\alpha \in K[x]$ of least degree with $m_\alpha(\alpha) = 0$. It is irreducible and generates $\ker(\mathrm{ev}_\alpha : K[x] \to L)$.

Proposition 17.6.2 Simple algebraic extensions

If α is algebraic over K with minimal polynomial m_α of degree n, then $K(\alpha) \cong K[x]/(m_\alpha)$ is a field extension of degree n, with basis $\{1, \alpha, \alpha^2, \dots, \alpha^{n-1}\}$ over K.

Example 17.6.1

The minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$ is $x^2 - 2$. So $\mathbb{Q}(\sqrt{2}) \cong \mathbb{Q}[x]/(x^2 - 2)$ has degree 2 over $\mathbb{Q}$, with basis $\{1, \sqrt{2}\}$. The minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is $x^3 - 2$ (irreducible by Eisenstein at $p = 2$), giving $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$.

17.7 Splitting Fields and Normal Extensions

Definition 17.7.1: Splitting field

A splitting field of a polynomial $f \in K[x]$ is an extension L/K such that f factors completely into linear factors in $L[x]$ and L is generated over K by the roots of f. A splitting field exists for any f and is unique up to (non-unique) isomorphism.

Definition 17.7.2: Normal extension

A finite extension L/K is normal if every irreducible polynomial in $K[x]$ that has one root in L splits completely in $L[x]$. Equivalently, L is a splitting field of some polynomial over K.

Definition 17.7.3: Separable extension

A polynomial $f \in K[x]$ is separable if it has no repeated roots (in its splitting field). An algebraic element α is separable over K if its minimal polynomial is separable. An algebraic extension L/K is separable if every element of L is separable over K. In characteristic 0, every algebraic extension is separable.

17.8 The Galois Group

Definition 17.8.1: Galois extension and Galois group

A finite extension L/K is Galois if it is both normal and separable. The Galois group is

$$\mathrm{Gal}(L/K) = \mathrm{Aut}_K(L) = \{\sigma : L \xrightarrow{\sim} L \mid \sigma|_K = \mathrm{id}_K\},$$

the group of field automorphisms of L fixing K pointwise. For a Galois extension, $|\mathrm{Gal}(L/K)| = [L : K]$.

Example 17.8.1 (Galois groups of quadratic extensions)
Let $d \in \mathbb{Q}$ be a non-square. The extension $\mathbb{Q}(\sqrt{d})/\mathbb{Q}$ is Galois of degree 2. The Galois group is $\mathrm{Gal}(\mathbb{Q}(\sqrt{d})/\mathbb{Q}) = \{\mathrm{id}, \sigma\}$ where $\sigma(\sqrt{d}) = -\sqrt{d}$. So $\mathrm{Gal}(\mathbb{Q}(\sqrt{d})/\mathbb{Q}) \cong \mathbb{Z}/2$.

Example 17.8.2 (Splitting field of $x^3 - 2$)
The splitting field of x^3-2 over $\mathbb{Q}$ is $L = \mathbb{Q}(\sqrt[3]{2}, \omega)$ where $\omega = e^{2\pi i/3}$. The roots are $\sqrt[3]{2}, \omega\sqrt[3]{2}, \omega^2\sqrt[3]{2}$. We have $[L : \mathbb{Q}] = 6$ (since $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$ and $[\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}(\sqrt[3]{2})] = 2$). The Galois group is $\mathrm{Gal}(L/\mathbb{Q}) \cong S_3$: it acts faithfully on the three roots, and $|S_3| = 6 = [L : \mathbb{Q}]$.

17.9 The Fundamental Theorem of Galois Theory

Theorem 17.9.1 Fundamental theorem of Galois theory
Let L/K be a finite Galois extension with Galois group $G = \mathrm{Gal}(L/K)$. There is an inclusion-reversing bijection

$$\{\text{intermediate fields } K \subseteq E \subseteq L\} \longleftrightarrow \{\text{subgroups } H \leqslant G\}$$

given by $E \mapsto \mathrm{Gal}(L/E)$ and $H \mapsto L^H = \{x \in L : \sigma(x) = x \ \forall \sigma \in H\}$. Under this correspondence:

(i) $[L : E] = |H|$ and $[E : K] = [G : H]$.

(ii) E/K is normal (Galois) if and only if H is a normal subgroup of G, in which case $\mathrm{Gal}(E/K) \cong G/H$.

(iii) Intersections of fields correspond to subgroups generated by unions, and composites of fields correspond to intersections of subgroups.

Note:-
The proof uses the primitive element theorem (every finite separable extension is simple), Artin's lemma ($[L : L^H] = |H|$ for any finite group H of automorphisms of L), and careful bookkeeping.

Example 17.9.1 (Galois correspondence for $x^3 - 2$)
For $L = \mathbb{Q}(\sqrt[3]{2}, \omega)$ over $\mathbb{Q}$ with $G = S_3$:

- The subgroup $\langle(123)\rangle \cong \mathbb{Z}/3$ (index 2, normal) corresponds to $\mathbb{Q}(\omega)$. The quotient $S_3/\mathbb{Z}/3 \cong \mathbb{Z}/2$ gives $\mathrm{Gal}(\mathbb{Q}(\omega)/\mathbb{Q}) \cong \mathbb{Z}/2$.
- The three subgroups $\langle(12)\rangle, \langle(13)\rangle, \langle(23)\rangle \cong \mathbb{Z}/2$ (index 3, not normal) correspond to the three cubic subfields $\mathbb{Q}(\sqrt[3]{2})$, $\mathbb{Q}(\omega\sqrt[3]{2})$, $\mathbb{Q}(\omega^2\sqrt[3]{2})$. These are not Galois over $\mathbb{Q}$ (the subgroups are not normal).

17.10 Cyclotomic Extensions

Definition 17.10.1: Cyclotomic extension

The nth cyclotomic extension of $\mathbb{Q}$ is $\mathbb{Q}(\zeta_n)$ where $\zeta_n = e^{2\pi i/n}$ is a primitive nth root of unity. The minimal polynomial of ζ_n over $\mathbb{Q}$ is the nth cyclotomic polynomial $\Phi_n(x)$, which has degree $\varphi(n)$ (Euler's totient).

Proposition 17.10.1 Galois group of cyclotomic extensions

The extension $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is Galois, and

$$\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n)^\times.$$

The isomorphism sends σ_a (defined by $\sigma_a(\zeta_n) = \zeta_n^a$) to $a \bmod n$.

Note:-

This is abelian, so every subextension of $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is Galois over $\mathbb{Q}$. The Kronecker–Weber theorem states that <u>every</u> abelian extension of $\mathbb{Q}$ is contained in some cyclotomic extension.

17.11 Solvability by Radicals

Definition 17.11.1: Radical extension

A field extension L/K is a radical extension if there is a tower $K = K_0 \subset K_1 \subset \cdots \subset K_m = L$ where each $K_{i+1} = K_i(\alpha_i)$ with $\alpha_i^{n_i} \in K_i$ for some $n_i \geqslant 1$. A polynomial $f \in K[x]$ is solvable by radicals if its splitting field is contained in some radical extension of K.

Theorem 17.11.1 Galois's criterion

A polynomial $f \in \mathbb{Q}[x]$ is solvable by radicals if and only if its Galois group $\mathrm{Gal}(L/\mathbb{Q})$ (where L is the splitting field of f) is a solvable group.

Note:-

Recall that a group G is solvable if it has a subnormal series $G = G_0 \supset G_1 \supset \cdots \supset G_n = \{e\}$ with each G_i/G_{i+1} abelian. For $n \leqslant 4$, the group S_n is solvable (e.g., $S_4 \supset A_4 \supset V_4 \supset \{e\}$ where $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$). The key fact: A_5 is simple (proved in Lecture 11), so S_5 is not solvable.

Corollary 17.11.1 Insolvability of the general quintic

There is no formula involving only the four arithmetic operations and radicals that expresses the roots of a general degree-5 polynomial in terms of its coefficients. More precisely: there exist quintic polynomials $f \in \mathbb{Q}[x]$ whose Galois group is S_5, which is not solvable.

Example 17.11.1

The polynomial $f(x) = x^5 - 4x + 2 \in \mathbb{Q}[x]$ is irreducible (Eisenstein at $p = 2$) and has Galois group S_5. It has exactly three real roots and one pair of complex conjugate roots: the complex conjugation gives a transposition in $\mathrm{Gal}(L/\mathbb{Q})$, and irreducibility gives a 5-cycle. A transposition and a 5-cycle generate S_5.

Note:-

This is the Abel–Ruffini theorem (1824), with the group-theoretic formulation due to Galois (1832). For degree $\leqslant 4$, the classical formulas (quadratic, Cardano, Ferrari) exist precisely because S_2, S_3, S_4 are solvable. Galois theory unifies and explains all of this.

III. Lie Groups and Lie Algebras

17.12 Matrix Lie Groups

The finite groups studied so far (symmetric groups, dihedral groups, etc.) are discrete. Many important groups in mathematics and physics are continuous: the group of rotations $SO(3)$, the group of invertible matrices $GL_n(\mathbb{R})$, and so on. Lie theory provides the algebraic tools to study these.

Definition 17.12.1: Matrix Lie group

A matrix Lie group is a closed subgroup $G \subseteq GL_n(\mathbb{C})$ (closed in the topology inherited from $\mathbb{C}^{n\times n} \cong \mathbb{C}^{n^2}$). The "Lie" condition means G is simultaneously a group and a smooth manifold, with the group operations (multiplication and inversion) being smooth maps.

Example 17.12.1 (The classical matrix Lie groups)

- $GL_n(\mathbb{R})$, $GL_n(\mathbb{C})$: the general linear groups (invertible $n \times n$ matrices).
- $SL_n(\mathbb{R}) = \{A \in GL_n(\mathbb{R}) : \det(A) = 1\}$, $SL_n(\mathbb{C})$: the special linear groups.
- $O(n) = \{A \in GL_n(\mathbb{R}) : A^T A = I\}$: the orthogonal group. $SO(n) = O(n) \cap SL_n(\mathbb{R})$: the special orthogonal group (rotations).
- $U(n) = \{A \in GL_n(\mathbb{C}) : A^* A = I\}$: the unitary group. $SU(n) = U(n) \cap SL_n(\mathbb{C})$.
- $Sp(2n, \mathbb{R}) = \{A \in GL_{2n}(\mathbb{R}) : A^T J A = J\}$ where $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right)$: the symplectic group.

Note:-

The dimension of a Lie group (as a manifold) can be computed from its defining equations: $GL_n(\mathbb{R})$ has dimension n^2; the constraint $\det = 1$ removes 1 dimension, giving $\dim SL_n = n^2 - 1$; the constraint $A^T A = I$ gives $\frac{n(n+1)}{2}$ equations (since $A^T A$ is symmetric), so $\dim O(n) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$.

17.13 Lie Algebras

The idea: a Lie group is a curved object (a manifold), but its tangent space at the identity is a flat object (a vector space) that retains much of the group's structure. This tangent space is the Lie algebra.

Definition 17.13.1: Lie algebra (abstract)

A Lie algebra over a field k is a k-vector space $\mathfrak{g}$ equipped with a bilinear operation $[\cdot,\cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ (the Lie bracket) satisfying:

(i) Antisymmetry: $[X, X] = 0$ for all $X \in \mathfrak{g}$ (equivalently, $[X, Y] = -[Y, X]$).

(ii) The Jacobi identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$ for all $X, Y, Z \in \mathfrak{g}$.

Definition 17.13.2: Lie algebra of a matrix Lie group

For a matrix Lie group $G \subseteq GL_n(\mathbb{C})$, the Lie algebra $\mathfrak{g}$ is

$$\mathfrak{g} = \mathrm{Lie}(G) = \{X \in \mathbb{C}^{n\times n} : e^{tX} \in G \text{ for all } t \in \mathbb{R}\} = T_I G,$$

the tangent space to G at the identity. This is a real vector space with the commutator bracket $[X, Y] = XY - YX$.

Note:-

The commutator bracket satisfies the Lie algebra axioms: antisymmetry is clear ($[X, X] = X^2 - X^2 = 0$), and the Jacobi identity follows by expanding the triple brackets.

Example 17.13.1 (Classical Lie algebras)

- $\mathfrak{gl}_n(\mathbb{R}) = \mathbb{R}^{n\times n}$ (all $n \times n$ real matrices), dimension n^2.
- $\mathfrak{sl}_n(\mathbb{R}) = \{X \in \mathbb{R}^{n\times n} : \mathrm{Tr}(X) = 0\}$, dimension $n^2 - 1$. (Differentiating $\det(e^{tX}) = e^{t\,\mathrm{Tr}(X)} = 1$ gives $\mathrm{Tr}(X) = 0$.)
- $\mathfrak{so}(n) = \{X \in \mathbb{R}^{n\times n} : X^T = -X\}$ (skew-symmetric matrices), dimension $\frac{n(n-1)}{2}$. (Differentiating $(e^{tX})^T e^{tX} = I$ gives $X^T + X = 0$.)
- $\mathfrak{su}(n) = \{X \in \mathbb{C}^{n\times n} : X^* = -X,\ \mathrm{Tr}(X) = 0\}$ (skew-Hermitian traceless), dimension $n^2 - 1$.
- $\mathfrak{sp}(2n) = \{X \in \mathbb{R}^{2n\times 2n} : X^T J + JX = 0\}$, dimension $n(2n + 1)$.

17.14 The Exponential Map

Definition 17.14.1: Matrix exponential

For $X \in \mathbb{C}^{n\times n}$, the matrix exponential is

$$e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!} = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots .$$

This converges absolutely for any X. The exponential map $\exp : \mathfrak{g} \to G$ sends $X \mapsto e^X$.

Proposition 17.14.1 Properties of the matrix exponential

(i) $e^0 = I$.

(ii) $\det(e^X) = e^{\mathrm{Tr}(X)}$.

(iii) If $XY = YX$, then $e^{X+Y} = e^X e^Y$. In general, $e^{X+Y} \neq e^X e^Y$.

(iv) $\frac{d}{dt}\big|_{t=0} e^{tX} = X$, confirming that $\mathfrak{g}$ is the tangent space at the identity.

(v) The Baker–Campbell–Hausdorff formula: $e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]+\cdots}$, where the higher-order terms involve iterated brackets. This shows the group multiplication near the identity is determined by the Lie bracket.

Note:-

The exponential map is a local diffeomorphism near $0 \in \mathfrak{g}$ (by the inverse function theorem, since $d\exp_0 = \mathrm{id}$). It need not be surjective globally: for $SL_2(\mathbb{R})$, the matrix $\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$ is not in the image of exp. However, for compact connected Lie groups (e.g., $SO(n)$, $SU(n)$), exp is surjective.

17.15 Homomorphisms and Representations

Definition 17.15.1: Lie algebra homomorphism

A linear map $\varphi : \mathfrak{g} \to \mathfrak{h}$ between Lie algebras is a homomorphism if $\varphi([X, Y]) = [\varphi(X), \varphi(Y)]$ for all $X, Y \in \mathfrak{g}$.

Proposition 17.15.1 Functoriality

Every Lie group homomorphism $\Phi : G \to H$ induces a Lie algebra homomorphism $d\Phi_e : \mathfrak{g} \to \mathfrak{h}$ (the differential at the identity). This defines a functor from (connected, simply-connected) Lie groups to Lie algebras, and this functor is an equivalence of categories: a Lie algebra homomorphism $\mathfrak{g} \to \mathfrak{h}$ integrates to a unique Lie group homomorphism $\tilde{G} \to H$ (where $\tilde{G}$ is the simply-connected cover of G).

Definition 17.15.2: Representation of a Lie algebra

A representation of a Lie algebra $\mathfrak{g}$ on a vector space V is a Lie algebra homomorphism $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$, i.e., a linear map satisfying $\rho([X, Y]) = \rho(X)\rho(Y) - \rho(Y)\rho(X)$. A representation of a Lie group G on V (a smooth homomorphism $G \to GL(V)$) induces a representation of $\mathfrak{g}$ on V by differentiation.

Example 17.15.1 (The adjoint representation)

Every Lie algebra $\mathfrak{g}$ has a natural representation on itself: the adjoint representation $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ defined by $\mathrm{ad}(X)(Y) = [X, Y]$. The Jacobi identity is precisely the statement that ad is a Lie algebra homomorphism.

17.16 Structure and Classification

The structure theory of Lie algebras parallels finite group theory:

Definition 17.16.1: Solvable and semisimple Lie algebras

The derived series of a Lie algebra $\mathfrak{g}$ is $\mathfrak{g}^{(0)} = \mathfrak{g}$, $\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$. The algebra $\mathfrak{g}$ is solvable if $\mathfrak{g}^{(k)} = 0$ for some k, and semisimple if it has no nonzero solvable ideals. Every Lie algebra decomposes as $\mathfrak{g} = \mathfrak{r} \rtimes \mathfrak{s}$ where $\mathfrak{r}$ is the radical (maximal solvable ideal) and $\mathfrak{s}$ is semisimple (Levi decomposition).

Note:-

Every semisimple Lie algebra decomposes (uniquely) as a direct sum of simple Lie algebras (those with no nontrivial ideals). The simple Lie algebras over $\mathbb{C}$ are completely classified by Dynkin diagrams:

- Four infinite families: $A_n = \mathfrak{sl}_{n+1}$ $(n \geqslant 1)$, $B_n = \mathfrak{so}_{2n+1}$ $(n \geqslant 2)$, $C_n = \mathfrak{sp}_{2n}$ $(n \geqslant 3)$, $D_n = \mathfrak{so}_{2n}$ $(n \geqslant 4)$.
- Five exceptional algebras: G_2 (dim 14), F_4 (dim 52), E_6 (dim 78), E_7 (dim 133), E_8 (dim 248).

The classification proceeds via root systems: given a semisimple $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h}$ (maximal abelian subalgebra of semisimple elements), the adjoint action of $\mathfrak{h}$ on $\mathfrak{g}$ decomposes $\mathfrak{g}$ into root spaces. The resulting set of roots $\Phi \subset \mathfrak{h}^*$ is a root system, which is encoded by its Dynkin diagram. This is one of the great classification theorems in mathematics (Killing 1888, Cartan 1894).

Example 17.16.1 ($\mathfrak{sl}_2(\mathbb{C})$ — the simplest semisimple Lie algebra)

The Lie algebra $\mathfrak{sl}_2(\mathbb{C})$ has dimension 3 with standard basis

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

satisfying $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. This is the type A_1 Lie algebra. Its finite-dimensional representations are completely classified: for each $n \geqslant 0$ there is a unique irreducible representation of dimension $n+1$, realized on $\mathrm{Sym}^n(\mathbb{C}^2)$. The representation theory of $\mathfrak{sl}_2$ is the foundation for the general theory.

Exercise 17.16.1 Exercises

(1) Show that $\mathbb{Z}[\sqrt{-5}]$ is not a PID by finding an ideal that is not principal. (Hint: consider $(2, 1+\sqrt{-5})$.)

(2) Prove that $\mathbb{Z}[i]$ is a Euclidean domain with norm $N(a+bi) = a^2 + b^2$. (The key step: for any $\alpha, \beta \in \mathbb{Z}[i]$ with $\beta \neq 0$, the quotient $\alpha/\beta \in \mathbb{Q}(i)$ lies within distance $\frac{\sqrt{2}}{2} < 1$ of some Gaussian integer.)

(3) Compute $\mathrm{Gal}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})$ and draw its lattice of subgroups alongside the lattice of intermediate fields.

(4) Show that the Galois group of $x^4 - 2$ over $\mathbb{Q}$ is the dihedral group D_4 of order 8 (not S_4).

(5) Verify the Lie bracket relations $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$ for the standard basis of $\mathfrak{sl}_2(\mathbb{C})$.

(6) Show that $\mathfrak{so}(3)$ (real 3×3 skew-symmetric matrices) is isomorphic as a Lie algebra to $(\mathbb{R}^3, \times)$ where $\times$ is the cross product.

(7) Let $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{C})$. Show that the Killing form $B(X, Y) = \mathrm{Tr}(\mathrm{ad}(X)\,\mathrm{ad}(Y))$ equals $2n\,\mathrm{Tr}(XY) - 2\,\mathrm{Tr}(X)\,\mathrm{Tr}(Y)$.

Solution

(1) The ideal $I = (2, 1+\sqrt{-5})$ is not principal. If $I = (\alpha)$, then $\alpha \mid 2$ and $\alpha \mid (1+\sqrt{-5})$, so $N(\alpha) \mid N(2) = 4$ and $N(\alpha) \mid N(1+\sqrt{-5}) = 6$. Thus $N(\alpha) \mid \gcd(4, 6) = 2$. But no element of $\mathbb{Z}[\sqrt{-5}]$ has norm 2 (since $a^2 + 5b^2 = 2$ has no integer solutions). So $N(\alpha) = 1$, meaning α is a unit and $I = \mathbb{Z}[\sqrt{-5}]$. But $3 \notin I$ (checking: $3 = 2a + (1+\sqrt{-5})b$ for $a, b \in \mathbb{Z}[\sqrt{-5}]$ leads to a contradiction modulo 2), so I is proper and not principal.
(2) Given $\alpha, \beta \in \mathbb{Z}[i]$ with $\beta \neq 0$, write $\alpha/\beta = p + qi$ with $p, q \in \mathbb{Q}$. Choose integers m, n closest to p, q, so $|p - m| \leqslant 1/2$ and $|q - n| \leqslant 1/2$. Set $\gamma = m + ni$ and $r = \alpha - \beta\gamma$. Then $N(r) = N(\beta) \cdot N(\alpha/\beta - \gamma) = N(\beta)((p-m)^2 + (q-n)^2) \leqslant N(\beta)(1/4 + 1/4) = N(\beta)/2 < N(\beta)$.
(3) The extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ has degree 4 (since $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt{2})$). The splitting field of $(x^2-2)(x^2-3)$ is normal and separable, hence Galois. The Galois group has order 4 and is generated by $\sigma : \sqrt{2} \mapsto -\sqrt{2}, \sqrt{3} \mapsto \sqrt{3}$ and $\tau : \sqrt{2} \mapsto \sqrt{2}, \sqrt{3} \mapsto -\sqrt{3}$. Since $\sigma^2 = \tau^2 = \mathrm{id}$ and $\sigma\tau = \tau\sigma$, we get $\mathrm{Gal} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$. The three subgroups of order 2 correspond to the three intermediate fields $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{6})$.
(4) The splitting field of $x^4 - 2$ over $\mathbb{Q}$ is $L = \mathbb{Q}(\sqrt[4]{2}, i)$. We have $[L : \mathbb{Q}] = 8$ ($[\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}] = 4$ and $[L : \mathbb{Q}(\sqrt[4]{2})] = 2$). The Galois group has order 8 and is determined by the actions on $\sqrt[4]{2}$ and i. Let $\sigma(\sqrt[4]{2}) = i\sqrt[4]{2}$, $\sigma(i) = i$ (a rotation of the four roots) and $\tau(\sqrt[4]{2}) = \sqrt[4]{2}$, $\tau(i) = -i$ (complex conjugation). Then $\sigma^4 = \mathrm{id}$, $\tau^2 = \mathrm{id}$, and $\tau\sigma\tau = \sigma^{-1}$, giving $\mathrm{Gal}(L/\mathbb{Q}) \cong D_4$. This is a proper subgroup of S_4 (order 8 vs 24).
(5) Direct computation: $[h, e] = he - eh = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) - \left(\begin{smallmatrix} 0 & -1 \\ 0 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right) = 2e$. Similarly $[h, f] = hf - fh = \left(\begin{smallmatrix} 0 & 0 \\ -1 & 0 \end{smallmatrix}\right) - \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right) = -2f$. And $[e, f] = ef - fe = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right) - \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right) = h$.

(6) A basis for $\mathfrak{so}(3)$ is $E_1 = \begin{pmatrix} 0&0&0\\ 0&0&-1\\ 0&1&0 \end{pmatrix}$, $E_2 = \begin{pmatrix} 0&0&1\\ 0&0&0\\ -1&0&0 \end{pmatrix}$, $E_3 = \begin{pmatrix} 0&-1&0\\ 1&0&0\\ 0&0&0 \end{pmatrix}$. Computing: $[E_1, E_2] = E_3$, $[E_2, E_3] = E_1$, $[E_3, E_1] = E_2$. Under the identification $E_i \leftrightarrow e_i$ (standard basis of $\mathbb{R}^3$), these bracket relations are exactly $e_1 \times e_2 = e_3$, etc. So $\varphi(E_i) = e_i$ is a Lie algebra isomorphism $(\mathfrak{so}(3), [\cdot,\cdot]) \cong (\mathbb{R}^3, \times)$.
(7) For $\mathfrak{g} = \mathfrak{gl}_n$, $\operatorname{ad}(X)(Z) = XZ - ZX$. So $\operatorname{ad}(X)\operatorname{ad}(Y)(Z) = X(YZ - ZY) - (YZ - ZY)X = XYZ - XZY - YZX + ZYX$. Taking the trace over Z (summing over the basis E_{ij} with $Z = E_{ij}$) requires careful index manipulation. The result $B(X, Y) = 2n\,\mathrm{Tr}(XY) - 2\,\mathrm{Tr}(X)\,\mathrm{Tr}(Y)$ follows from the identities $\mathrm{Tr}(E_{ij} \cdot M) = M_{ji}$ and the combinatorial sums. In particular, B restricted to $\mathfrak{sl}_n$ (where $\mathrm{Tr} = 0$) is just $2n\,\mathrm{Tr}(XY)$, which is nondegenerate—confirming $\mathfrak{sl}_n$ is semisimple (Cartan's criterion).